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Computer Graphics and Animation

Course Title: Computer Graphics and Animation (3 Cr.)

Course Code: CACS305

Year/Semester: 111/V

Class Load: 6 Hrs. 1 Week (Theory: 3Hrs. Tutorial: 1 Hrs. Practical: 2 Hrs.)

Course Description

This course is designed to extend students' knowledge and practice in Graphics hardware, software, and applications. It also provides the knowledge of data structures for graphics, graphics languages, and models for 2D and 3D objects, clipping, hidden surface elimination, depth buffer, raster graphics, shading, and rendering.

Course objectives

Upon completion of this course, students should be able to 1. Explain basic principle of computer graphics. 2. Develop 2D and 3D computer graphics applications. 3. Specify lighting and object's materials in computer graphics programming.

Course Contents

Unit 1: Introduction

[6]

- 1.1 Advantage of Computer Graphics and Areas of Applications
- 1.2 Hardware and Software for Computer Graphics. (Hard Copy, Display Technologies),
- 1.3 Random Scan Display System, Video Controller, Random Scan Display Processor
- 1.4 Raster Graphics
- 1.5 Scan Conversion Algorithms (Line, Circle, Ellipse)
- 1.6 Area Filling (Rectangle, Ellipse), Clipping (Lines, Circle, Ellipse), Clipping Polygons

Unit 2: Two dimensional and three dimensional transformations

[7]

- 2.1 2-Dimensional transformation
- 2.2 2-D Translation, Rotation, Scaling,
- 2.3 Homogeneous Coordinates, Reflection, Shear transform
- 2.4 3-dimensional transformation,
- 2.5 3-D Translation, Rotation Scaling, Reflection, Shear.

Unit 3: Clipping

[7]

- 3.1 Window to view port transformation
- 3.2 Clipping, line clipping,
- 3.3 Cohen -Sutherland line clipping
- 3.4 Polygon clipping
- 3.5 Sutherland and Gary Hodgman polygon clipping algorithm

Unit 4: Visible Surface Determination and computer graphics algorithm

[15]

- 4.1 Image space and object space techniques
- 4.2 Hidden Surface removal-Depth comparison
- 4.3 Z-Buffer Algorithm
- 4.4 Back-Face Removal
- 4.5 The Painter's Algorithm
- 4.6 Scan-Line Algorithm
- 4.7 Light and Color and different color models (RGB, CMY, YIQ)

Unit 5: Animation and virtual reality

[10]

- 5.1 Basic Principles of Animation and Types of Animation
- 5.2 Introduction to the flash interface
- 5.3 Setting stage dimensions, working with panels, panel layouts
- 5.4 Layers & Views
- 5.5 Shaping Objects- Overview of shapes, Drawing & Modifying Shapes
- 5.6 Bitmap Images & Sounds
- 5.7 Animation -Principles, Frame by frame animation, tweening, masks
- 5.8 Introduction to virtual reality

Laboratory Work

Laboratory work should be done covering all the topics listed above and a small project work should be carried out using the concept learnt in this course using Open GL.

Reference Books:

1. Foley, J. D., A. V. Dam, S. K. Feiner, J. F. Hughes, Computer Graphics Principle and Practices, Addison Wesley Longman, Singapore Pvt. Ltd.,
2. Heam Donald, M. P. Baker, Computer Graphics, 2E, Prentice Hall of India Private Limited, New Delhi
3. Robert R & Snow D Flash CS4 Professional Bible, Wiley Publishing

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

MODEL QUESTIONS-ANSWERS

Bachelor in Computer Applications
 Course Title: Computer Graphics Animation
 Code No: CACS305
 Semester: V

Full Marks: 60
 Pass Marks: 24
 Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.

Group 'A'

10 × 1 = 10

Attempt all the questions.

1. Circle (O) the correct answer in the following questions

- i. GUI stands for.....
 - a) Graphical user Interface
 - b) Graph Use Interface
 - c) Graphical Universal Interface
 - d) None of these
- ii. DDA stands for.....
 - a) Direct differential analyzer
 - b) Data differential analyzer
 - c) Direct difference analyzer
 - d) Digital differential analyzer
- iii. _____ animation is used to animate things that are smaller than life size.
 - a) Immersive
 - b) Claymotion
 - c) Stop motion
 - d) Augmented
- iv. The rotation axis that is perpendicular to the xy plane and passes through the pivot point is known as.....
 - a) Rotation
 - b) Translation
 - c) Scaling
 - d) Shearing
- v. The basic geometric transformations are.....
 - a) Translation
 - b) Rotation
 - c) Scaling
 - d) All of the mentioned
- vi. The process of removal of hidden surfaces is termed as.....
 - a) Clipping
 - b) copying
 - c) culling
 - d) shorting
- vii. What is the primary use of clipping in computer graphics?
 - a) Adding graphics
 - b) removing objects and lines
 - c) Zooming
 - d) copying
- viii. Z-buffer algorithm is used for.....
 - a) Frame buffer removal
 - b) Hidden line removal
 - c) Rendering
 - d) Animation.
- ix. Which of the following color models are defined with three primary colors?
 - a) RGB and HSV color models
 - b) CMY and HSV color models
 - c) HSV and HLS color models
 - d) RGB and CMY color models
- x. Sutherland Hodgeman algorithm is used for.....
 - a) Line clipping
 - b) Point clipping
 - c) Polygon clipping
 - d) Hybrid clipping

Answer-Key

i. (a)	ii. (d)	iii. (c)	iv. (a)	v. (d)	vi. (a)	vii. (b)	viii. (b)	ix. (d)	x. (c)
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Group B

Attempt any six questions.

1. Explain ambient light, diffuse reflection and specular reflection with examples.

Ans: Ambient light

Ambient light means the light that is already present in a scene, before any additional lighting is added. It usually refers to natural light, either outdoors or coming through windows etc. It can also mean artificial lights such as normal room lights. An ambient light source represents a fixed-intensity and fixed-color light source that affects all objects in the scene equally.

Surface that is not exposed directly to a light source still will be visible if nearby objects are illuminated. This light is called ambient light. This is a simple way to model the combination of light reflections from various surfaces to produce a uniform illumination called the ambient light, or background light. Ambient light has no spatial or directional characteristics. The amount of ambient light incident on each object is a constant for all surfaces and over all directions.

If a surface is exposed only to ambient light, then the intensity of the diffuse reflection at any point on the surface is;

$$I = k_a I_a$$

Where, I_a is the intensity of the ambient light, and k_a is the ambient reflection coefficient.

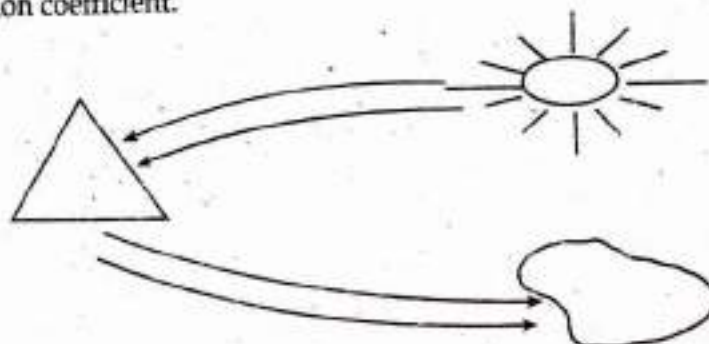


Fig: Object illuminated due to ambient light

Ambient light does not come from a specific light source and has no direction. It represents the light that is more or less everywhere in the scene, originating from multiple reflections of light at various surfaces. In a room with a lamp on a table, it will not be completely dark under the table although the lamp cannot shed its light directly under the table. The light is reflected by the surface of the table, the walls, the ceiling and the floor. Of course, the light under the table will have a lower intensity, but it will still be there with approximately the same intensity everywhere, not coming from a specific direction.

2. Diffuse reflection

Objects illuminated by ambient light are uniformly illuminated across their surfaces then the illumination is called diffuse illumination. It is the background light reflected from walls, floor and ceilings. We assume that the reflection is constant over each surface of the object and are independent of the viewing direction. This type of reflection on dull surfaces is called diffuse reflection.

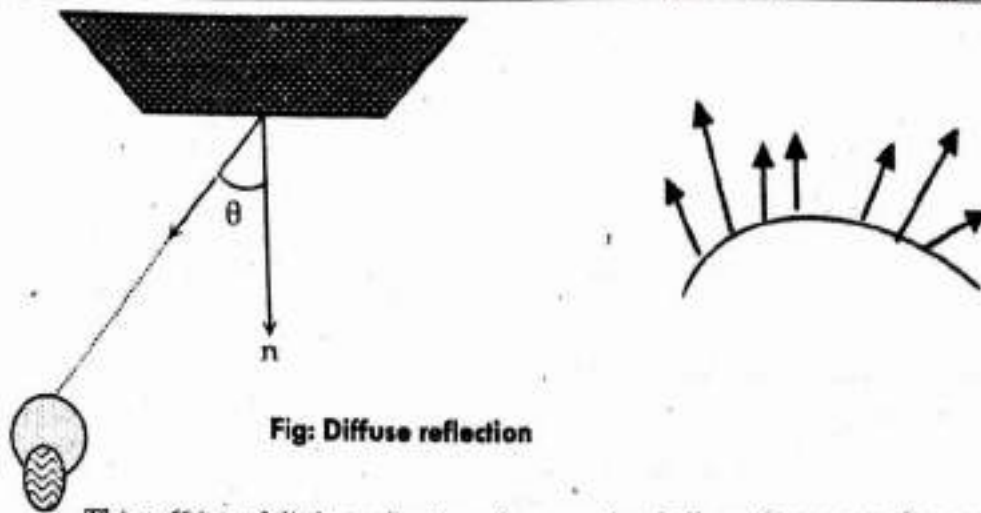


Fig: Diffuse reflection

This effect of light reflection for purely dull surfaces can be computed according to Lambert's cosine law by the illumination equation:

$$I = I_L \cdot K_d \cdot \cos\theta \dots \dots \dots (1)$$

Where, I_L is the intensity of the light hitting the surface, $0 \leq k_d \leq 1$ is the reflection coefficient of the surface or the material, and θ is the angle between the normal vector n to the surface in the considered point and the vector I pointing in the direction where the light comes from. The illumination equation (1) for diffuse reflection is only valid for angles θ between 0° and 90° . Otherwise, the light ray hits the surface from the backside so that no reflection occurs. In the case of directional light coming from a light source in infinite distance, the variable I_L in (1) has the same value everywhere. In the case of a point light source, I_L is the intensity of the light source multiplied by the attenuation factor, which depends on the distance of the point on the surface to the light source.

3. Specular reflection

Specular light is the white highlight reflection seen on smooth, shiny objects. Specular light is dependent on the direction of the light, the surface normal and the viewer location. When we look at an illuminated shiny surface, such as polished metal, a person's forehead, we see a highlight or bright spot, at certain viewing direction. Such phenomenon is called specular reflection. This phenomenon occurs as a result of total internal reflection of the incident light in a concentrated region around the specular reflection angle. Due to this phenomenon, the surface appears to be not in its original color but of white color.

Diffuse reflection on dull surfaces reflects the light into all directions. **Specular reflection** occurs on shiny surfaces. Such shiny surfaces reflect at least a portion of the light in a similar way as a mirror. In contrast to diffuse reflection, ideal specular reflection takes place only in one direction. The difference between diffuse and specular reflection is illustrated in fig below.

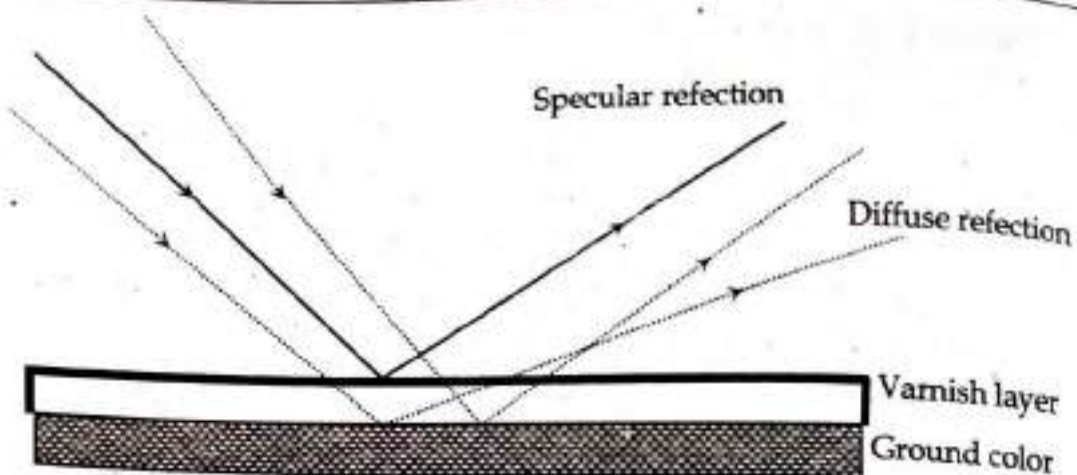


Fig: Difference between diffuse and specular reflections

Shiny surfaces very often have a very thin transparent layer, for instance varnish. When light hits the surface, part of the light penetrates the varnish layer and is reflected on the dull surface of the object. This part of the light is subject to diffuse reflection and the color of the reflected light depends strongly on the ground color of the dull surface. Another part of the light is directly reflected on the transparent layer by specular reflection. Therefore, specular reflection does usually not change the color of the light. The position of the viewer is of no importance for calculating effects coming from diffuse reflection. Whether or where the viewer can see specular reflection depends on his position.

The Phong specular reflection model described by the relation,

$$I = I_L \cdot w(\theta) \cdot \cos(\alpha)^n \dots \dots \dots (1)$$

Where, I_L is the intensity of the light. The value $0 \leq w(\theta) \leq 1$ is the fraction of the light which is directly reflected at the shiny surface. n is the specular reflection exponent of the surface. For a perfect mirror, $n = \infty$ would hold. A smaller n leads to a less focused specular reflection.

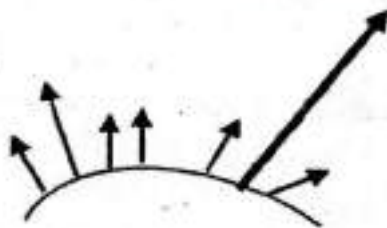


Fig: Specular reflection

2. How do a video controller and a frame buffer jointly collaborate to produce graphical display on the screen, in case of a Raster Display?

Ans: In a raster scan system, the electron beam is swept across the screen, one row at a time from top to bottom. As the electron beam moves across each row, the beam intensity is turned on and off to create a pattern of illuminated spots. Picture definition is stored in memory area called the **Refresh Buffer** or **Frame Buffer**. This memory area holds the set of intensity values for all the screen points. Stored intensity values are then retrieved from the refresh buffer and "painted" on the screen one row (scan line) at a time as shown in the following illustration. Each screen point is referred to as a **pixel (picture element)** or **pel**. At the end of each scan line, the electron beam returns to the left side of the screen to begin

displaying the next scan line. In comparisons of random scan display method it is more accurate method.

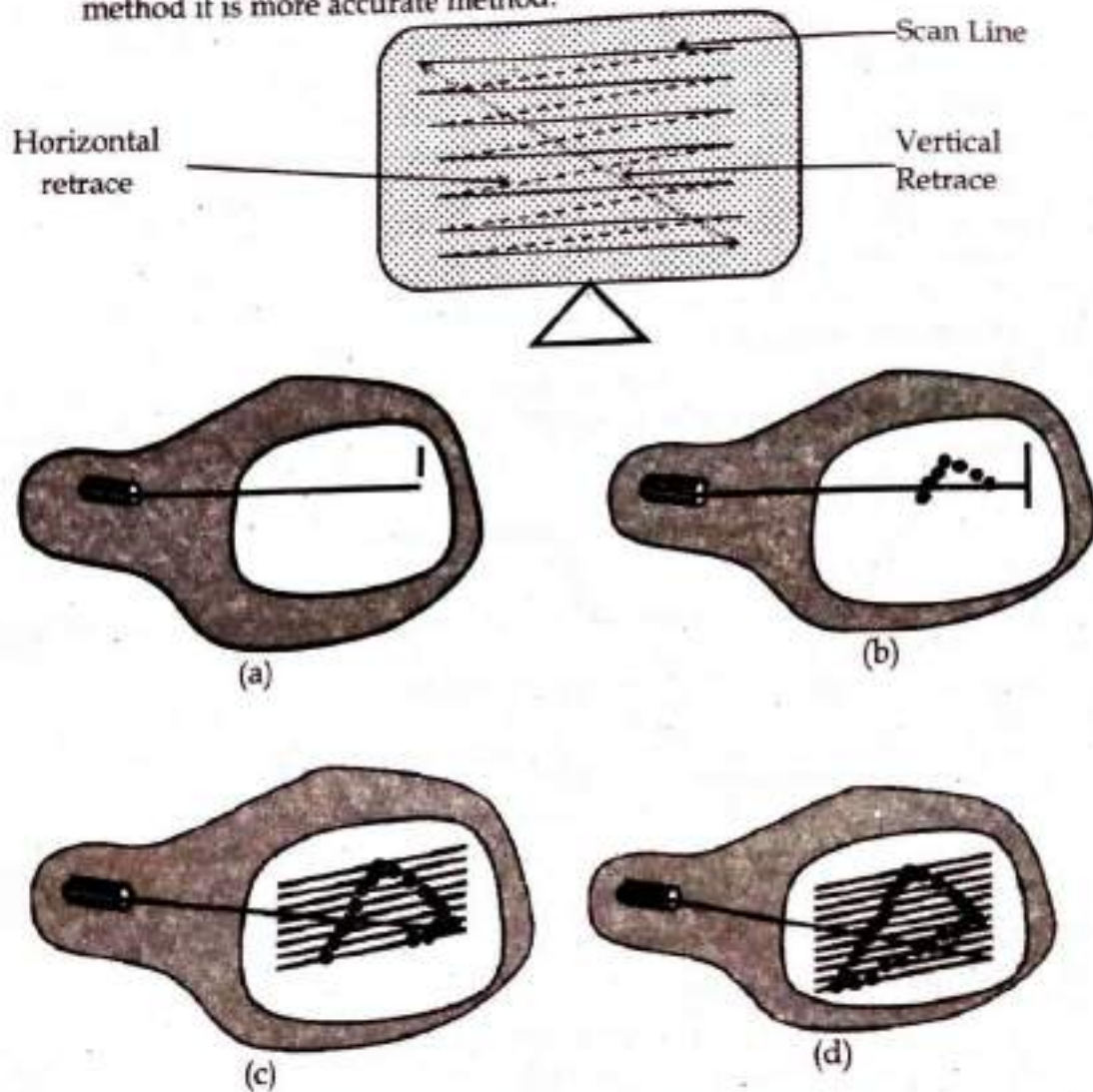


Fig: A raster-scan system displays an object as a set of points across each screen scan line

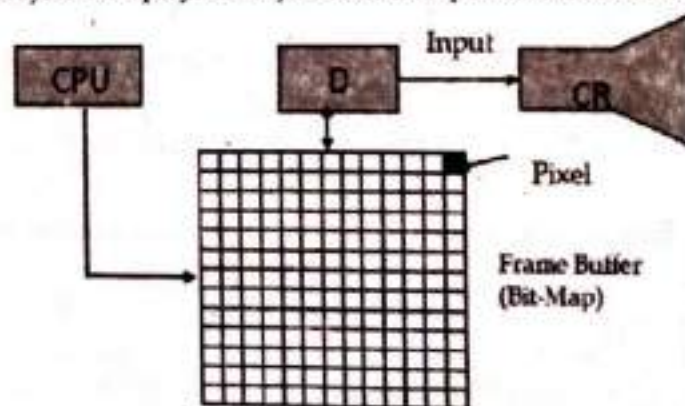


Fig: Raster Scan display system

The stored intensity value is retrieved from frame buffer and painted on the scan line at a time. The stored intensity value is retrieved from frame buffer and painted on the screen. Refreshing on Raster-Scan display is carried out at the rate of 60 or higher frames per second. 60 frames per second is also termed as 60 cycle per second usually used unit Hertz (HZ). Most of display devices are based on this technology. For example, CRT, color CRT, LCD, and LEDetc.

Architecture of Raster-Scan System

The raster graphics systems typically consist of several processing units. CPU is the main processing unit of computer systems. Besides CPU, graphics system consists of a special buffer processor called video controller or display controller. It is used to control the operation of the display device. In addition, to the video controller, raster scan systems can have other processors as co-processors which are called graphic controller or display processors. A fixed area of system memory is reserved for the frame buffer. The video controller has the direct access to the frame buffer for refreshing the screen. The video controller cycles through the frame buffer, one scan line at a time, typically at 60 times per second or higher. The contents of frame buffer are used to control the CRT beam's intensity or color. The organization of simple raster system is shown in the figure below.

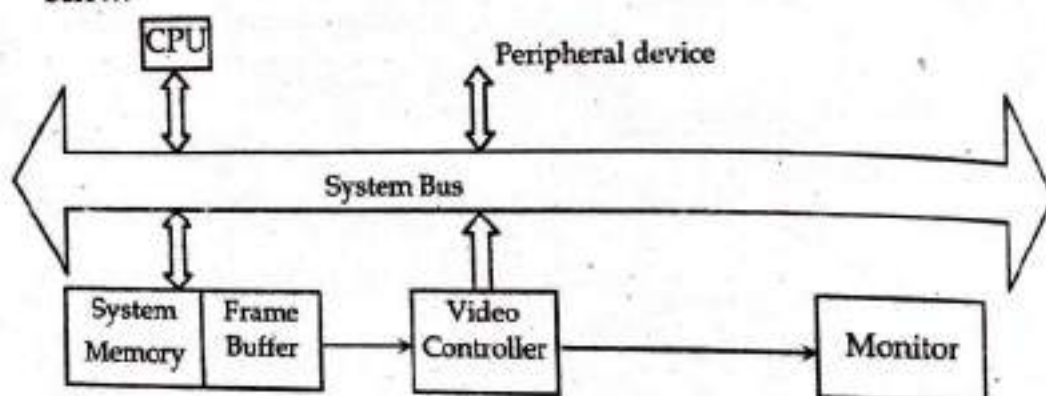


Fig: Architecture of simple Raster-scan system.

Advantages of Raster scale system

- Show Realistic pictures
- Million Unique hues can be produced
- Shadow scenes are conceivable

Disadvantages of Raster scale system

- Low Resolution
- Electron beam coordinated to whole screen not exclusively to those parts of the screen where picture is to be drawn so tedious when the drawn picture estimate is especially not as much as the whole screen.
- Expensive

3. Trace the Bresenham's Line drawing algorithm for the end points (1, 1) and (8, 5).

Solution:

$$x_1=1$$

$$y_1=1$$

$$x_2=8$$

$$y_2=5$$

$$dx = x_2 - x_1 = 8 - 1 = 7$$

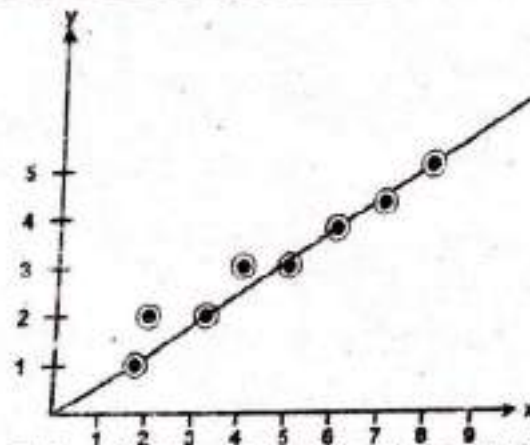
$$dy = y_2 - y_1 = 5 - 1 = 4$$

$$I_1 = 2 * \Delta y = 2 * 4 = 8$$

$$I_2 = 2 * (\Delta y - \Delta x) = 2 * (4 - 7) = -6$$

$$d = I_1 - \Delta x = 8 - 7 = 1$$

X	y	$d=d+l_1$ or l_2
1	1	$d+l_2=1+(-6)=-5$
2	2	$d+l_1=-5+8=3$
3	2	$d+l_2=3+(-6)=-3$
4	3	$d+l_1=-3+8=5$
5	3	$d+l_2=5+(-6)=-1$
6	4	$d+l_1=-1+8=7$
7	4	$d+l_2=7+(-6)=1$
8	5	



4. Derive the relation for three-dimensional translation and rotation.

Ans: Translation

Translation is used to move a point, or a set of points, linearly in space. Since now we are talking about 3D, therefore each point has 3 coordinates i.e. x , y and z . Similarly, the translation distances can also be specified in any of the 3 dimensions. These Translation Distances are given by t_x , t_y and t_z .

For any point $P(x, y, z)$ after translation we have

$P'(x', y', z')$ where

$$x' = x + t_x,$$

$$y' = y + t_y,$$

$$z' = z + t_z$$

And (t_x, t_y, t_z) is Translation vector.

Now this can be expressed as a single matrix equation.

$$P' = P + T$$

Where,

$$P = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, P' = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}, \text{ and } T = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Rotation

Rotation is the process of moving a point in space in a non-linear manner. More particularly, it involves moving the point from one position on a sphere whose center is at the origin to another position on the sphere. 3D rotation is not same as 2D rotation. In 3D rotation, we have to specify the

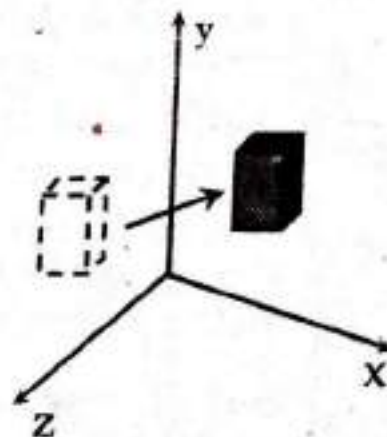


Fig: 3D Translation

angle of rotation along with the axis of rotation. We can perform 3D coordinate axes rotation as

- Z-axis rotation (Roll)
- Y-axis rotation (Yaw)
- X-axis rotation (Pitch)

Z-axis rotation (Roll)

If you look closely, you should note that when we rotate around the Z axis, the Z element of the point does not change. In fact, we can just ignore the Z - we already know what it will be after the rotation. If we ignore the Z element, then we have the same case as if we were rotating the two-dimensional point $\langle x, y \rangle$ through the angle θ . Z-axis rotation is same as the origin about the 2D for which we have the derived matrices already.

$$\begin{aligned}x' &= x \cos \theta - y \sin \theta \\y' &= x \sin \theta + y \cos \theta \\z' &= z\end{aligned}$$

Homogeneous representation is:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\text{i.e. } P' = R_z(\theta).P$$

Y-axis rotation (Yaw)

The equations for Y-axis rotation

$$\begin{aligned}x' &= x \cos \theta + z \sin \theta \\y' &= y \\z' &= z \cos \theta - x \sin \theta\end{aligned}$$

Homogeneous representation is:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 1 & 0 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\text{i.e. } P' = R_y(\theta).P$$

X-axis rotation (Pitch)

The equations for X-axis rotation

$$\begin{aligned}x' &= x \\y' &= y \cos \theta - z \sin \theta \\z' &= y \sin \theta + z \cos \theta\end{aligned}$$

Homogeneous representation is:

$$\begin{bmatrix} x' \\ y' \\ z' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

$$\text{i.e. } P' = R_x(\theta).P$$

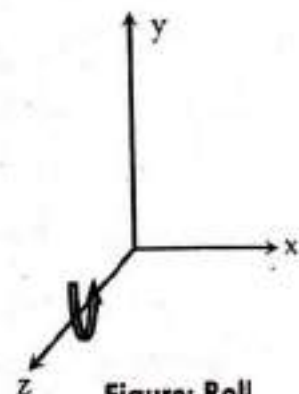


Figure: Roll

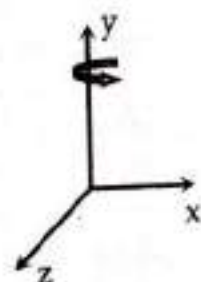


Figure: Yaw

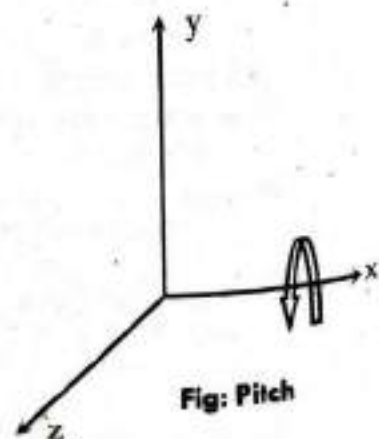


Fig: Pitch

5. Find the transformation of triangle A(1, 0), B(0, 1) and C(1,1) by rotating 90° about the origin and then translating one unit in x and y direction.

Solution:

$$\text{Object} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Composite Transformation matrix = $T(1, 1) R(90^\circ)$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 90^\circ & -\sin 90^\circ & 0 \\ \sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

We have,

Image = Transformation Matrix $\times$ Object

$$\text{Image} = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

The coordinates of the image are A'(1, 2), B'(0, 1) and C'(0, 2).

6. Calculate the points to draw a circle having radius 10 and center at (0, 0)

Solution:

We have $r=10$

Centre $(h, k) = (0, 0)$

Set $x=0, y=r=10$

Now the first pixel to be drawn = (0, 10)

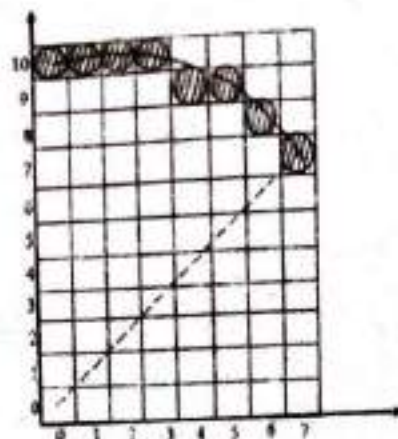
Initial decision parameter $(p) = 1 - r = 1 - 10 = -9$

Now points on the octant are given by,

K	P_k	(x_{k-1}, y_{k-1})	$2x_{k-1}$	$2y_{k-1}$
1	-9	(1, 10)	2	20
2	-6	(2, 10)	4	20
3	-1	(3, 10)	6	20
4	6	(4, 9)	8	18
5	3	(5, 9)	10	18
6	8	(6, 8)	12	16
7	5	(7, 7)	14	14

Now using 8 point symmetry we get all the points on the circle as below:

(x, y)	(0, 10)	(1, 10)	(2, 10)	(3, 10)	(4, 9)	(5, 9)	(6, 8)	(7, 7)
(-x, y)	(0, 10)	(-1, 10)	(-2, 10)	(-3, 10)	(-4, 9)	(-5, 9)	(-6, 8)	(-7, 7)
(x, -y)	(0, -10)	(1, -10)	(2, -10)	(3, -10)	(4, -9)	(5, -9)	(6, -8)	(7, -7)
(-x, -y)	(0, -10)	(-1, -10)	(-2, -10)	(-3, -10)	(-4, -9)	(-5, -9)	(-6, -8)	(-7, -7)
(y, x)	(10, 0)	(10, 1)	(10, 2)	(10, 3)	(9, 4)	(9, 5)	(8, 6)	(7, 7)
(-y, x)	(10, 0)	(-10, 1)	(-10, 2)	(-10, 3)	(-9, 4)	(-9, 5)	(-8, 6)	(-7, 7)
(y, -x)	(10, 0)	(10, -1)	(10, -2)	(10, -3)	(9, -4)	(9, -5)	(8, -6)	(7, -7)
(-y, -x)	(10, 0)	(-10, -1)	(-10, -2)	(-10, -3)	(-9, -4)	(-9, -5)	(-8, -6)	(-7, -7)



7. Differentiate between Random and Raster scan display systems

Ans: Comparison between Raster and Random (Vector) scan display systems

Raster Scan System		Random Scan System	
Resolution	It has poor or less resolution because picture definition is stored as an intensity value.	It has High Resolution because it stores picture definition as a set of line commands.	
Electron-Beam	Electron beam is directed from top to bottom and one row at a time on screen, but electron beam is directed to whole screen.	Electron beam is directed to only that part of screen where picture is required to be drawn, one line at a time so also called vector display.	
Cost	It is less expensive than Random Scan System.	It is Costlier than Raster Scan System.	
Refresh Rate	The refresh rate is independent of picture complexity. Refresh rate is 60 to 80 frames per second.	Refresh Rate depends on the number of lines to be displayed i.e. 30 to 60 times per second.	
Picture Definition	It Stores picture definition in Refresh Buffer also called Frame Buffer.	It Stores picture definition as a set of line commands called Refresh Display File.	
Line Drawing	Zig-Zag line is produced because plotted values are discrete.	Smooth line is produced because directly the line path is followed by electron beam.	
Realism in display	It contains shadow, advance shading and hidden surface technique so gives the realistic display of scenes.	It does not contain shadow and hidden surface technique so it cannot give realistic display of scenes.	
Image Drawing	It uses Pixels along scan lines for drawing an image.	It is designed for line drawing applications and uses various mathematical functions to draw.	

Group C

Attempts any two questions

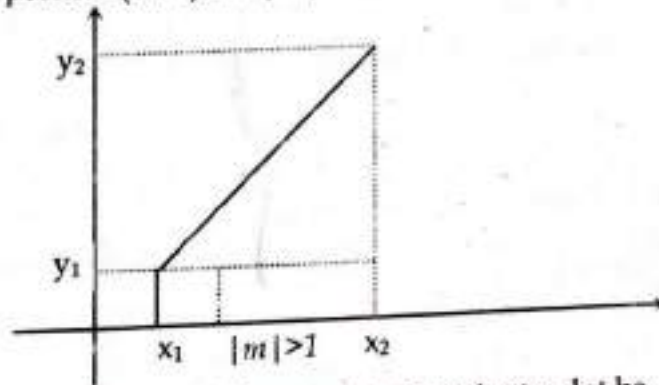
8. Explain the working details of DDA algorithm? Explain. Digitize a line with end points A(6, 12) and B(10, 5) using Bresenham's line drawing algorithm.

Ans: The digital differential analyzer (DDA) is a scan conversion line drawing algorithm based on calculating either Δx or Δy from the equation,

$$m = \frac{\Delta y}{\Delta x}$$

We sample the line at unit intervals in one co-ordinate and determine the corresponding integer values nearest to the line path for the other co-ordinates. Here following cases occur:

Case 1: If m is positive ($m > 0$) and $|m| \leq 1$,



Let (x_k, y_k) be the plotted pixel then next pixels going to plot be,

$$x_{k+1} = x_k + \Delta x$$

$$y_{k+1} = y_k + \Delta y$$

Where, Δx and Δy are small incremental distance.

We have $m = \frac{\Delta y}{\Delta x}$

$$\Delta y = m \Delta x \dots \dots \dots (1)$$

Also here $|m| \leq 1$, sampling be done along x-axis with $\Delta x = 1$.

Then,

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k + m$$

Here k takes value from starting point and increase by 1 until final end point. m can be any real value between 0 and 1.

Case 2: If m is positive ($m > 0$) and $|m| > 1$,

Let (x_k, y_k) be the plotted pixel then next pixels going to plot be,

$$x_{k+1} = x_k + \Delta x$$

$$y_{k+1} = y_k + \Delta y$$

Where Δx and Δy are small incremental distance

We have $m = \frac{\Delta y}{\Delta x}$

$$\Delta y = m \Delta x \dots \dots \dots (1)$$

Also here $|m| > 1$, sampling be done along y-axis with $\Delta y = 1$.

Then, $\Delta x = \frac{1}{m}$

$$x_{k+1} = x_k + \frac{1}{m}$$

$$y_{k+1} = y_k + 1$$

The above equations are under the assumption that the lines are processed from left to right. I.e. left end point is starting.

If the processing is from right to left

Case 3: if $|m| \leq 1$

Let (x_k, y_k) be the plotted pixel then next pixels going to plot be,

$$x_{k+1} = x_k + \Delta x$$

$$y_{k+1} = y_k + \Delta y$$

Where Δx and Δy are small incremental distance

We have $m = \frac{\Delta y}{\Delta x}$

$$\Delta y = m \Delta x \dots \dots \dots (1)$$

Also here $|m| \leq 1$, sampling be done along x-axis with $\Delta x = 1$.

Then,

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k + m$$

Case 4: If $|m| > 1$

Let (x_k, y_k) be the plotted pixel then next pixels going to plot be,

$$x_{k+1} = x_k + \Delta x$$

$$y_{k+1} = y_k + \Delta y$$

Where Δx and Δy are small incremental distance

We have $m = \frac{\Delta y}{\Delta x}$

$$\Delta y = m \Delta x \dots \dots \dots (1)$$

Also here $|m| > 1$, sampling be done along y-axis with $\Delta y = 1$.

Then,

$$x_{k+1} = x_k + \frac{1}{m}$$

$$y_{k+1} = y_k + 1$$

Therefore, in general,

$$\therefore y_{k+1} = y_k \pm m \quad \text{For } |m| < 1 \text{ and}$$

$$x_{k+1} = x_k \pm \frac{1}{m} \quad \text{For } |m| > 1$$

Digitize a line with end points (6, 12) and (10, 5) using BSA

Solution:

$$\text{Here, } \Delta y = |5 - 12| = 7$$

$$\Delta x = |10 - 6| = 4$$

Since, $|m| = \frac{\Delta y}{\Delta x} \geq 1$, so we sample at y direction i.e., at each step we simply

increment y-coordinate by 1 and find appropriate x-coordinate.

First pixel to plot is (6, 12), which is the starting endpoint.

Now, initial decision parameter, $p_0 = 2\Delta x - \Delta y = 2 \times 4 - 7 = 1$

We know that, If $p_k < 0$, then we need to set $p_{k+1} = p_k + 2\Delta x$ and plot pixel

$(x_k, y_k - 1)$

And if $p_k \geq 0$, then we need to set $p_{k+1} = p_k + 2\Delta x - 2\Delta y$ then plot pixel

$(x_k + 1, y_k - 1)$

Here, $p_0 = 1$, we have i.e., $p_k \geq 0$

$$\text{So, } p_1 = p_0 + 2\Delta x - 2\Delta y = 1 + 2 \times 4 - 2 \times 7 = -5$$

$$p_2 = p_1 + 2\Delta x = -5 + 2 \times 4 = 3$$

$$p_3 = p_2 + 2\Delta x - 2\Delta y = 3 + 2 \times 4 - 2 \times 7 = -3$$

$$p_4 = p_3 + 2\Delta x = -3 + 2 \times 4 = 5$$

$$p_5 = p_4 + 2 \Delta x - 2 \Delta y = 5 + 2 \cdot 4 - 2 \cdot 7 = -1$$

$$p_6 = p_5 + 2 \Delta x = -1 + 2 \cdot 4 = 7$$

$$p_7 = p_6 + 2 \Delta x - 2 \Delta y = 7 + 2 \cdot 4 - 2 \cdot 7 = 1$$

$$p_8 = p_7 + 2 \Delta x - 2 \Delta y = 1 + 2 \cdot 4 - 2 \cdot 7 = -5$$

$$p_9 = p_8 + 2 \Delta x = -5 + 2 \cdot 4 = 3$$

Successive pixel positions and decision parameter calculation is shown in the table below.

k	x	y	p_k	(x_{k+1}, y_{k+1})
0	6	12	1 (≥ 0)	(7, 11)
1	7	11	-5 (< 0)	(7, 10)
2	7	10	3	(8, 9)
3	8	9	-3	(8, 8)
4	8	8	5	(9, 7)
5	9	7	-1	(9, 6)
6	9	6	7	(10, 5)

9. Define window, viewport and viewing transformation. Let ABCD be the rectangular window with A(20, 20), B(90, 20), C(90, 70), D(20, 70). Find the region codes for end points and use Cohen Sutherland algorithm to clip the lines P(10, 20), Q(80, 90).

Ans: **Window**

A world-coordinate area selected for display is called a window. That is, window is the section of the 2D scene that is selected for viewing. The window defines what is to be viewed.

Viewport

An area on a display device to which a window is mapped is called a viewport. The viewport indicates where on an output device selected part will be displayed. Figure given below illustrates the mapping of a 2D picture that falls within a rectangular window onto a designated rectangular viewpoint.

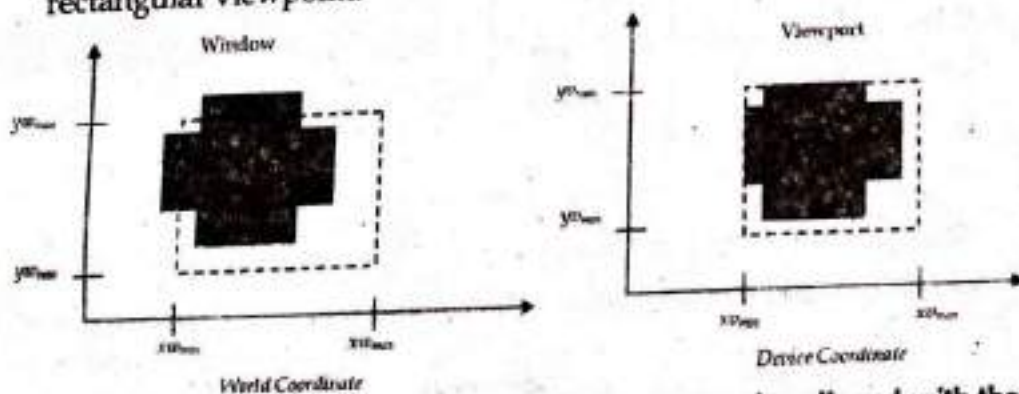


Figure: World and Viewport Coordinates, specified as rectangles aligned with the co-ordinate axes.

Window and viewport are often rectangular in standard positions, because it simplifies the transformation process and clipping process. Other shapes such as polygons, circles take longer time to process.

Viewing transformation

Viewing Transformation is the mapping of coordinates of points and lines that form the picture into appropriate coordinates on the display device. The window-to-viewport transformation can be explained in three steps as below:

1. Translate object along with window such that the lower left corner of the window is at the origin. That is, apply $T(-x_{Wmin}, -y_{Wmin})$.

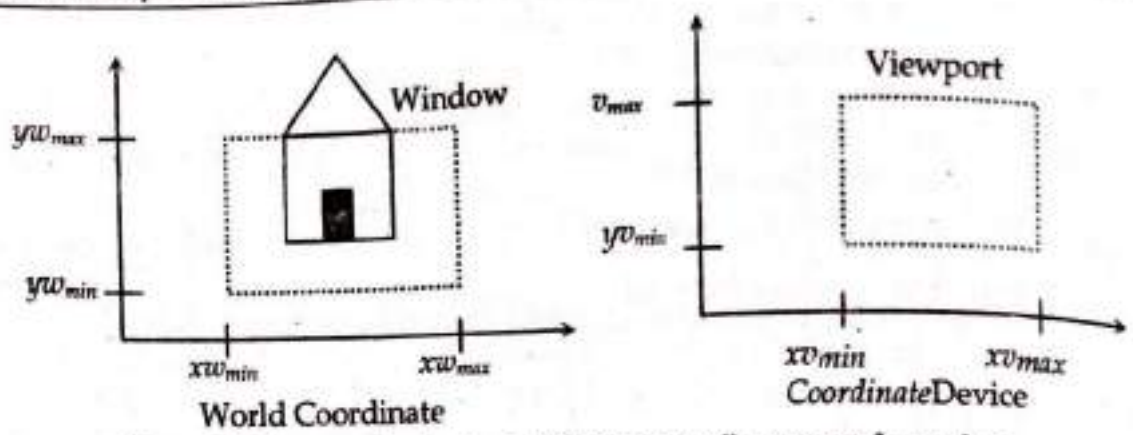


Figure: Window port and viewport before coordinate transformation

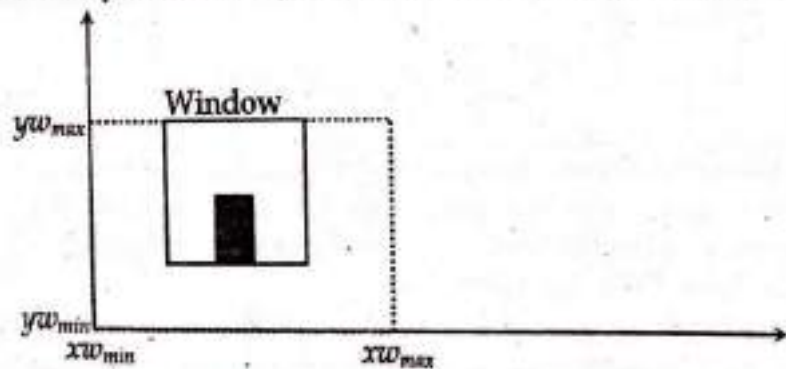
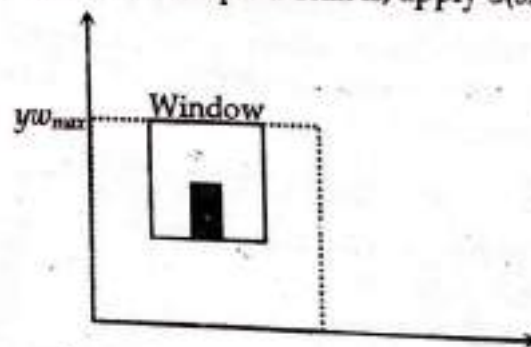


Figure: Translation of window to origin

2. Scale the object and the window such that window has the same dimension as that of a viewport. Simply we can say that we are converting the object into image and window in viewport. That is, apply $S(sx, sy)$.



3. Finally, apply another translation to move the viewport to its original position on the screen. That is, apply $T(xv_{min}, yv_{min})$.

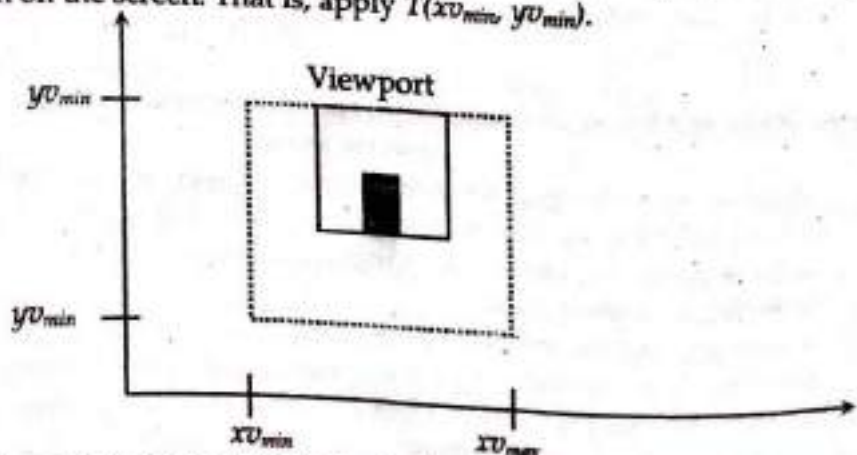


Figure: Translating the scaled window to the original position of viewport.

Therefore, net transformation (Composite Transformation),

$$T_{WV} = T(xv_{min}, yv_{min}). S(S_x, S_y). T(-xw_{min}, -yw_{min}) \dots \dots \dots (i)$$

Here, S_x and S_y are scaling factors, where;

$$S_x = \frac{xv_{max} - xv_{min}}{xw_{max} - xw_{min}} \quad S_y = \frac{yv_{max} - yv_{min}}{yw_{max} - yw_{min}}$$

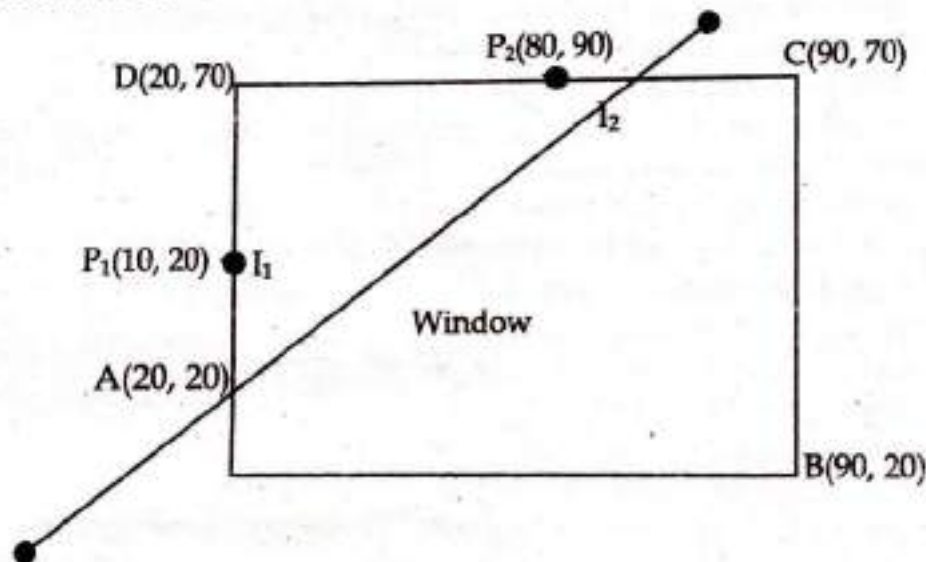
Now, writing in matrix form;

$$T_{WV} = \begin{bmatrix} 1 & 0 & xv_{min} \\ 0 & 1 & yv_{min} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -yw_{min} \\ 0 & 1 & -yw_{min} \\ 0 & 0 & 1 \end{bmatrix}$$

This is required window-to-viewpoint transformation. Let $P(x, y)$ be the world coordinate point that is mapped onto the viewport point $P'(u, v)$, then we must have:

$$P' = TWV.P$$

Numerical Part



Since the point P_1 lies left outside the window hence its outcode is 0001

Similarly the outcode of $P_2 = 1000$

The outcode for I_1 is 0001

And outcode for I_2 is 1000

The Logical AND for $P_1P_2 = 0001 \text{ AND } 1000 = 0000$.

Which is zero hence line is partially visible.

Since left edge of window is having x coordinate as 20. The intersection point's x-coordinate will also be 20 only. Now we have to find y-coordinate of intersection point. For this we will use slope intercept form of line.

i.e. let's find slope of line P_1P_2 .

$$\text{Hence, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{90 - 20}{80 - 10} = \frac{70}{70} = 1$$

So, slope of line P_1P_2 is 1

We know that the slope between two endpoints of a line is same as that of slope between one endpoint and any other point which is lying on the line.

Hence slope between points P_1 and intersection point will be,

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Where, x_1 and y_1 are related with P_1 , x_2 is known i.e. 20, y_2 we have to find.

$$\text{Hence, } m(x_2 - x_1) = y_2 - y_1$$

$$y_2 = m(x_2 - x_1) + y_1$$

$$y_2 = 0.8(20 - 10) + 30$$

$$= 0.8 * 10 + 30 = 8 + 30 = 38$$

$$y_2 = 38$$

Therefore the intersection point is (20, 38). Let's call this point as I_1 . Since I_1 point is exactly at the boundary. Its region code will be (0, 0, 0, 0). Now we have to discard the line segment $P_1 I_1$ since P_1 point is outside window. Similarly we have to find another intersection point I_2 with respect to top boundary. Now I_2 points region.

Code will be 0000. So we will discard $I_2 P_2$ since P_2 point is above the window.

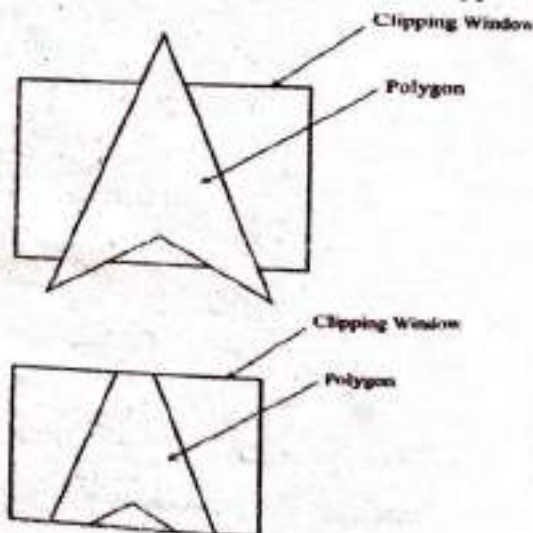
Now our line segment is $I_1 I_2$. Both I_1 and I_2 's region codes are 0000. So the line $I_1 I_2$ is completely visible and we have to display $I_1 I_2$ line.

10. **How can polygons be clipped? Why is Phong shading also called Normal Vector Interpolation scheme? Explain.**

Ans: **Polygon Clipping**

A set of connected lines are considered as polygon; polygons are clipped based on the window and the portion which is inside the window is kept as it is and the outside portions are clipped. The polygon clipping is required to deal different cases. Usually it clips the four edges in the boundary of the clip rectangle. The clip boundary determines the visible and invisible regions of polygon clipping and it is categorized as four.

- Visible region is wholly inside the clip window - saves endpoint
- Visible exits the clip window - Save the intersection
- Visible region is wholly outside the clip window - nothing to save
- Visible enters the clip window - save endpoint and intersection

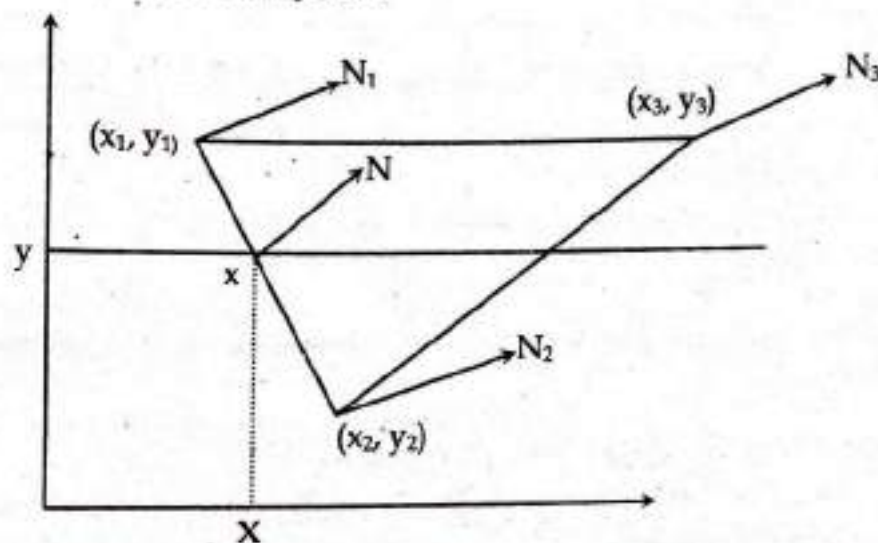


Phong shading

It is the best known shading algorithm, developed by Phong Bui Tuong, is called Phong shading or normal vector interpolation shading. Idea here is to interpolate the normal vectors instead of the light intensity. Then apply the illumination model at each surface point. It provides more accurate calculation of light intensity values and more realistic surface highlights. But it is computationally inefficient.

A more accurate method for rendering a polygon surface is to interpolate normal vector and then apply illumination model to each surface point. This method is called Phong shading or normal vector interpolation method for shading. It displays more realistic highlights and greatly reduces the mach band effect. A polygon surface is rendered with Phong shading by carrying out following calculations.

- Determine the average normal unit vectors at each polygon vertex.
- Linearly interpolate vertex normal over the surface of polygon.
- Apply illumination model along each scan line to calculate the pixel intensities for the surface point.



In figure, N_1, N_2, N_3 are the normal unit vectors at each vertex of polygon surface. For scan-line that intersects an edge, the normal vector N can be obtained by vertically interpolating normal vectors of the vertex on that edge as,

$$N = \frac{y - y_2}{y_1 - y_2} N_1 + \frac{y_1 - y}{y_1 - y_2} N_2$$

Incremental calculations are used to evaluate normal between scan lines and along each individual scan line as in Gouraud shading. Phong shading produces accurate results than the direct interpolation but it requires considerably more calculations.

Algorithm

- Determine the average unit normal vector at each polygon vertex.
- Linearly interpolate the vertex normal over the surface of the polygon.
- Apply an illumination model along each scan line to calculate pixel intensities for the surface points. The intensity value is calculated using the interpolated normal vector.

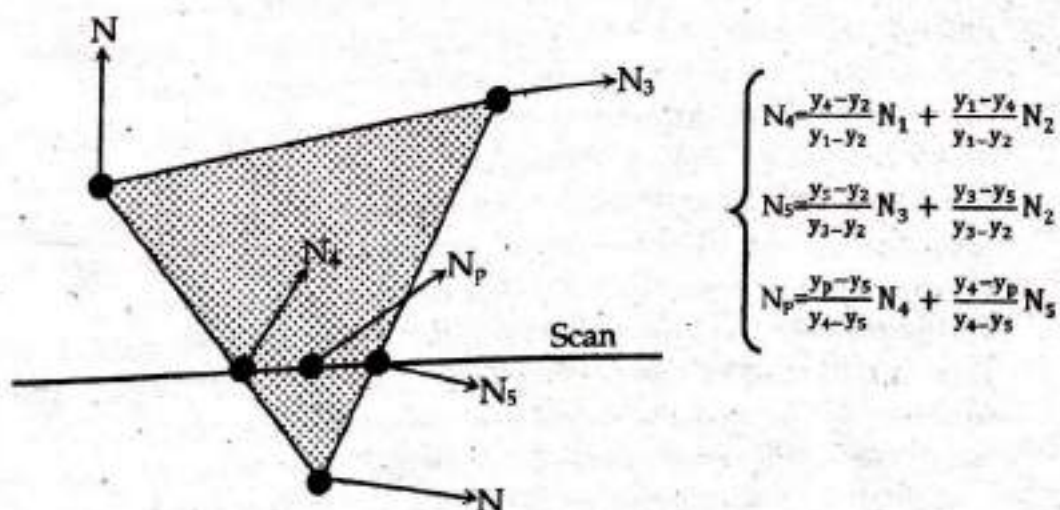


Fig: Phong shading

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

TU QUESTIONS-ANSWERS 2020

Bachelor in Computer Applications
Course Title: Computer Graphics Animation
Code No: CACS305
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.

Group B

[6×5=30]

Attempt any six questions

2. What is computer graphics? Explain different application areas of computer graphics.

Ans: Computer graphics is an art of drawing pictures on computer screens with the help of programming. It involves computations, creation, and manipulation of data. In other words, we can say that computer graphics is a rendering tool for the generation and manipulation of images. Computer Graphics is a field related to the generation of graphics using computers. In short, Computer graphics refer different things in different contexts:

- Pictures, scenes that are generated by a computer.
- Tools used to make such pictures, software and hardware, input/output devices.
- The whole field of study that involves these tools and the pictures they produce.

Application areas of computer graphics

Computer graphics is used today in many different areas of science, engineering, industry, business, education, entertainment, medicine, art and training. All of these are included in the following categories.

1. **User interfaces:** Most applications have user interfaces that rely on desktop windows systems to manage multiple simultaneous activities, and on point-and click facilities to allow users to select menu items, icons and objects on the screen. These activities fall under computer graphics. Typing is necessary only to input text to be stored and manipulated. For example, Word processing, spreadsheet, and desktop-publishing programs are the typical examples where user-interface techniques are implemented.
2. **Plotting:** Plotting 2D and 3D graphs of mathematical, physical, and economic functions use computer graphics extensively. The histograms, bar, and pie charts; the task-scheduling charts are the most commonly used plotting. These all are used to present meaningfully and concisely the trends and patterns of complex data.
3. **Office automation and electronic publishing:** Computer graphics has facilitated the office automation and electronic publishing

which is also popularly known as desktop publishing, giving more power to the organizations to print the meaningful materials Pictures, Drawings, etc.

4. **Computer Aided Drafting and Design:** One of the major uses of computer graphics is to design components and systems of mechanical, electrical, electrochemical, and electronic devices, including structures such as buildings, automobile bodies, airplane and ship hulls, very large scale integrated (VLSI) chips, optical systems and telephone and computer networks. These designs are more frequently used to test the structural, electrical, and thermal properties of the systems.
5. **Scientific and business Visualization:** Generating computer graphics for scientific, engineering, and medical data sets is termed as scientific visualization whereas business visualization is related with the non-scientific data sets such as those obtained in economics. Visualization makes easier to understand the trends and patterns inherent in the huge amount of data sets. It would, otherwise, be almost impossible to analyze those data numerically.
6. **Simulation and modeling:** Simulation is the imitation of the conditions like those, which is encountered in real life. Simulation thus helps to learn or to feel the conditions one might have to face in near future without being in danger at the beginning of the course. For example, astronauts can exercise the feeling of weightlessness in a simulator; similarly a pilot training can be conducted in flight simulator. The military tank simulator, the naval simulator, driving simulator, air traffic control simulator, heavy-duty vehicle simulator, and so on are some of the mostly used simulator in practice. Simulators are also used to optimize the system, for example the vehicle, observing the reactions of the driver during the operation of the simulator.
7. **Entertainment:** Disney movies such as Lion Kings and The Beauty of Beast, and other scientific movies like Jurassic Park, The lost world etc. are the best example of the application of computer graphics in the field of entertainment. Instead of drawing all necessary frames with slightly changing scenes for the production of cartoon-film, only the key frames are sufficient for such cartoon-film where the in between frames are interpolated by the graphics system dramatically decreasing the cost of production while maintaining the quality. Computer and video games such FIFA, Doom, Pools are few to name where graphics is used extensively.
8. **Art and commerce:** Here computer graphics is used to produce pictures that express a message and attract attention such as a new model of a car moving along the ring of the Saturn. These pictures are frequently seen at transportation terminals supermarkets, hotels etc. The slide production for commercial, scientific, or educational presentations is another cost effective use of computer graphics. One of such graphics packages is a PowerPoint.
9. **Cartography:** Cartography is a subject, which deals with the making of maps and charts. Computer graphics is used to produce

both accurate and schematic representations of geographical and other natural phenomena from measurement data. Examples include geographic maps, oceanographic charts, weather maps, contour maps and population-density maps. Surfer is one of such graphics packages, which is extensively used for cartography.

3. How can you draw a circle using mid-point circle algorithm? Explain with the algorithm.

Ans: Midpoint circle Algorithm

In midpoint circle algorithm, we sample at unit intervals and determine the closest pixel position to the specified circle path at each step. For a given radius r , and screen center position (x_c, y_c) we can set up our algorithm to calculate pixel positions around a circle path centered at $(0, 0)$ and then each calculated pixel position (x, y) is moved to its proper position by adding x_c to x and y_c to y .

$$x = x + x_c$$

$$y = y + y_c$$

It uses the symmetric property of circle to reduce computational time. Pixel positions are only calculated within an octant and transfer them through remaining seven octant to get a complete circle.

Here a decision parameter is used to decide the pixel for next plot.

Derivation of p-value for mid-point circle algorithm

To apply the midpoint method, we define a circle function as:

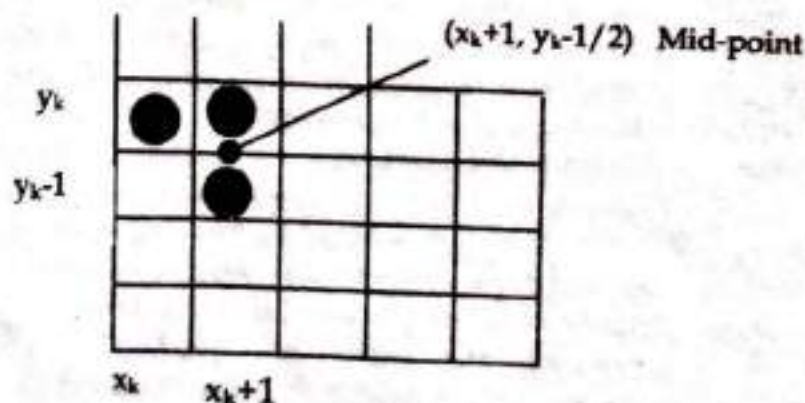
$$f_{\text{circle}} = x^2 + y^2 - r^2$$

To summarize the relative position of point (x, y) by checking sign of f_{circle} function,

$$f_{\text{circle}}(x, y) \begin{cases} < 0, & \text{if } (x, y) \text{ lies inside the circle boundary} \\ = 0, & \text{if } (x, y) \text{ lies on the circle boundary} \\ > 0, & \text{if } (x, y) \text{ lies outside the circle boundary.} \end{cases}$$

The circle function tests are performed for the mid positions between pixels near the circle path at each sampling step. Thus the circle function is decision parameter in mid-point algorithm.

The figure shows the midpoint between the two candidate pixels at sampling position $x_k + 1$, assuming we have just plotted the pixel (x_k, y_k) , we next need to determine whether the pixel at position $(x_k + 1, y_k)$ or $(x_k + 1, y_k - 1)$ is closer to the circle.



Our decision parameter is circle function evaluated at the mid-point

$$\begin{aligned}
 p_k &= f_{\text{circle}}\left(x_k + 1, y_k - \frac{1}{2}\right) \\
 &= (x_k + 1)^2 + \left(y_k - \frac{1}{2}\right)^2 - r^2 = (x_k + 1)^2 + y_k^2 - y_k + \frac{1}{4} - r^2
 \end{aligned}$$

If $p_k < 0$, then mid-point lies inside the circle, so point at y_k is closer to boundary otherwise, $y_k - 1$ closer to choose next pixel position.

Successive decision parameters are obtained by incremental calculation. The decision parameter for next position is calculated by evaluating circle function at sampling position $x_{k+1} + 1$ as,

$$\begin{aligned}
 p_{k+1} &= f_{\text{circle}}\left(x_{k+1} + 1, y_{k+1} - \frac{1}{2}\right)^2 \\
 &= ((x_{k+1} + 1))^2 + \left(y_{k+1} - \frac{1}{2}\right)^2 - r^2 \\
 &= (x_{k+1})^2 + 2x_{k+1} + 1 + (y_{k+1})^2 - (y_{k+1}) + \frac{1}{4} - r^2
 \end{aligned}$$

$$\text{Now, } p_{k+1} - p_k = 2x_{k+1} + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1$$

$$\text{i.e. } p_{k+1} = p_k + 2x_{k+1} + (y_{k+1}^2 - y_k^2) - (y_{k+1} - y_k) + 1$$

Where y_{k+1} is either y_k or $y_k - 1$ depending upon sign of p_k and $x_{k+1} = x_k + 1$

If p_k is negative, $y_{k+1} = y_k$ so we get,

$$p_{k+1} = p_k + 2x_{k+1} + 1$$

If p_k is positive, $y_{k+1} = y_k - 1$ so we get,

$$p_{k+1} = p_k + 2x_{k+1} + 1 - 2y_{k+1}$$

Calculating initial decision parameter:

$$\text{We have } p_k = (x_k + 1)^2 + y_k^2 - y_k + \frac{1}{4} - r^2$$

When $k=0$,

$$p_0 = (x_0 + 1)^2 + y_0^2 - y_0 + \frac{1}{4} - r^2$$

The initial decision parameter is obtained by evaluating the circle function at starting position $(x_0, y_0) = (0, r)$.

$$\text{Hence } p_0 = (0 + 1)^2 + r^2 - r + \frac{1}{4} - r^2 = 1 - r + \frac{1}{4} = \frac{5}{4} - r$$

If p_0 is specified in integer,

$$p_0 = 1 - r.$$

Algorithm for Midpoint circle drawing

1. Start
2. Input radius r and circle Centre (h, k)
3. Initialize $x=0$, $y=r$ and $p_0 = 1 - r$
4. While($x \leq y$)
 - // plot 8 points as
 - Plot pixel($x+h$, $y+k$);
 - Plot pixel($-x+h$, $y+k$);
 - Plot pixel($x+h$, $-y+k$);
 - Plot pixel($-x+h$, $-y+k$);

Plot pixel(y+h, x+k);
 Plot pixel (-y+h, x+k);
 Plot pixel(y+h, -x+k);
 Plot pixel (-y+h, -x+k);

5. If $p_0 < 0$ then

Set, $p_0 = (p_0 + 2x + 1)$

Set, $x = x + 1$

Otherwise,

Set, $p_0 = p_0 + 2x + 1 - 2y$

Set $y = y - 1$

6. End while loop

7. Stop

4. Explain 3D basic geometric transformation with an example.

Ans: The process of moving points in space is called three-dimensional transformations. Methods for geometric transformations in three dimensions are extended from two-dimensional methods by including considerations for the z coordinate. In three-dimensional space, we can now select any spatial orientation for the rotation axis. Some graphics packages handle three-dimensional rotation as a composite of three rotations, one for each of the three Cartesian axes. Represent a 3D transformation by a 3×3 matrix.

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Apply the transformation to a point

$$\begin{aligned} x' &= ax + by + cz \\ y' &= dx + ey + fz \\ z' &= gx + hy + iz \end{aligned} \quad \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \text{Image} = T.M. \times \text{Object}$$

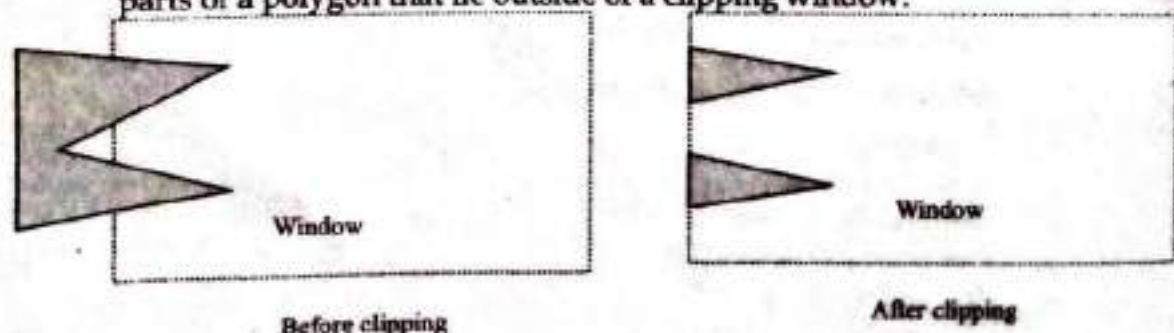
If object and image are represented in row matrix then:

$$\text{Image} = \text{Object} \times T.M.$$

Transformations can be combined as in two-dimension.

5. What is polygon clipping? Explain Sutherland and Hodgman algorithm for polygon clipping.

Ans: A polygon can be defined as a geometric object "consisting of a number of points (called vertices) and an equal number of line segments (called sides or edges)". Polygon clipping is defined as the process of removing those parts of a polygon that lie outside of a clipping window.



Sutherland-Hodgeman Polygon Clipping Algorithm

Sutherland and Hodgeman's polygon-clipping algorithm uses a divide-and-conquer strategy. It clips a given polygon successively against the edges of the given clip window. It starts with the initial set of polygon vertices, first clips the polygon against the left boundary of the window. Then successively against the right boundary, bottom boundary, and finally against the top boundary, as shown in figure

Basic approach:

Clip a polygon against each clip window edge i.e., left, right, bottom, and top.

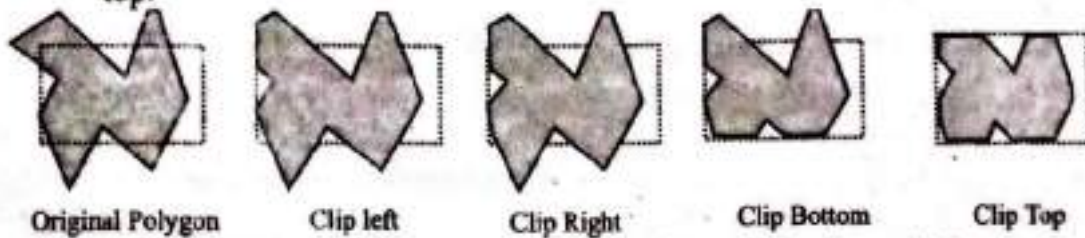
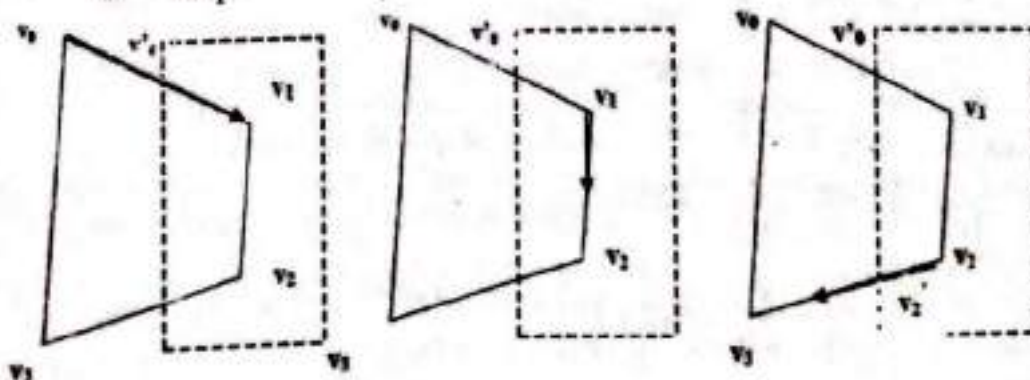


Fig: Clipping a polygon against successive window boundaries

Algorithm

For each clip window boundary perform the following steps:

1. Construct an input vertex list ($v_0, v_1, \dots, v_n$) where $v_0 = v_n$
2. Create empty output vertex list
3. For each pair of adjacent vertices v_i and v_{i+1} perform the following inside-outside test:
 - a. If out-in (v_i is outside the window boundary and v_{i+1} is inside),
 - i. Add intersection point v_i' and v_{i+1} to the output vertex list.
 - b. If in-in (both v_i, v_{i+1} are inside the window boundary),
 - i. add v_{i+1} to output vertex list
 - c. If in-out (v_i is inside the window boundary and v_{i+1} is outside),
 - i. add intersection point v_i' to output vertex list
 - d. If out-out (both v_i, v_{i+1} are outside the window boundary),
 - i. add nothing to output vertex list
4. Stop

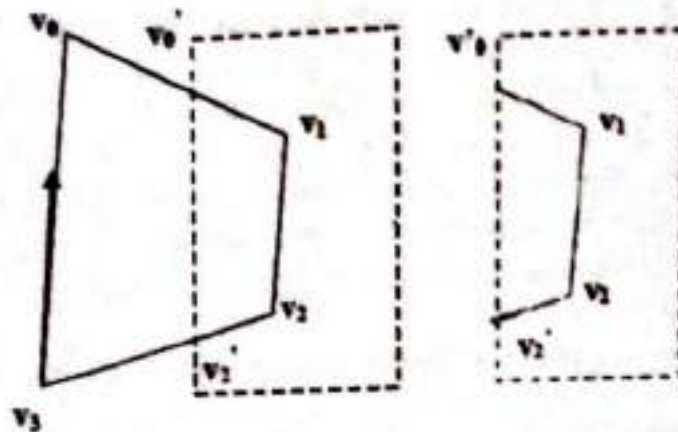


(a) out-in, save v_0', v_1

(b) in-in, save v_2

(c) in-out, save v_2'

Example: Clip the given polygon using Sutherland-Hodgeman Polygon clipping Algorithm.



(d) out-out, save none (e) Clipped polygon

Input vertex list: $\{v_0, v_1, v_2, v_3\}$

Output vertex list: $\{ \}$

For vertices v_0, v_1 , out - in pair so intersection point v_0' and v_1 are added to the output vertex list

Therefore, output vertex list = $\{v_0', v_1\}$

For vertices v_1, v_2 , in-in pair so only v_2 is added to the output vertex list.

Now, output vertex list = $\{v_0', v_1, v_2\}$

For vertices v_2, v_3 , in-out pair so only intersection point v_2' is added to the output vertex list.

Now, output vertex list = $\{v_0', v_1, v_2, v_2'\}$

For vertices v_3, v_0 , out-out pair so nothing is added to the output vertex list.

Finally, output vertex list = $\{v_0', v_1, v_2, v_2'\}$

6. Given a triangle with vertices $A(2,3)$, $B(5,5)$, $C(4,3)$ by rotating 90 degree about the origin and then translating two unit in each direction. Use homogenous transformation matrix to find the new vertices of the triangle.

Solution:

$$\text{Object} = \begin{bmatrix} 2 & 5 & 4 \\ 3 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

Composite Transformation Matrix = $(T(1, 1) \times R(45) \times T(-1, -1))$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -\sqrt{2}+1 \\ 0 & 0 & 1 \end{bmatrix}$$

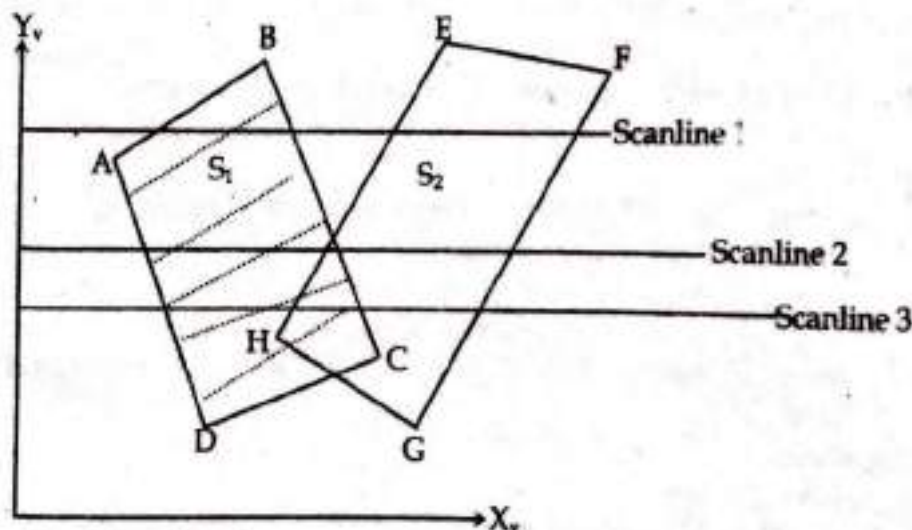
We have,

Image = Transformation Matrix $\times$ Object

$$\text{Image} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & -\sqrt{2}+1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 5 & 4 \\ 3 & 5 & 3 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}}+1 & 1 & \frac{1}{\sqrt{2}}+1 \\ \frac{3}{\sqrt{2}}+1 & \frac{8}{\sqrt{2}}+1 & \frac{5}{\sqrt{2}}+1 \\ 1 & 1 & 1 \end{bmatrix}$$

7. Explain the scan line algorithm for visible surface detection.

Ans: This is image-space method for removing hidden surface which is extension of the scan line polygon filling for polygon interiors. In this method, as each scan line is processed, all polygon surfaces intersecting that line are examined to determine which are visible. Across each scan line, depth calculations are made for each overlapping surface to determine which is nearest to the view plane. When the visible surface has been determined, the intensity value for that position is entered into the image buffer. Instead of filling one surface we deal with multiple surfaces here. As each scan line is processed, all polygon surfaces intersecting that line are examined to determine which are visible. We assume that polygon table contains the co-efficient of the plane equation for each surface as well as vertex edge, surface information, intensity information for the surface, and possibly pointers to the edge table.



Working Procedure

Above figure illustrate the scan line method for locating visible portion of surfaces for pixel position along the line. The active list for scan line one contains information from the edge table for edges AB, BC, EH and FG.

For scan line 1, between edges AB and BC, flag (S_1) = ON and flag (S_2) = OFF. Therefore no depth calculations are necessary and intensity information for surface S_1 is entered from the polygon table into the refresh buffer. Similarly, between edge EH and FG flag (S_1) = OFF and flag (S_2) = ON.

For scan line 2 and 3, the active edge list contains edges AD, EH, BC and FG. Along the scan line 2, from edge AD to edge EH, flag (S_1) = ON and flag (S_2) = OFF.

Between edges EH and BC, Flag (S_1) = ON and flag (S_2) = ON. Therefore depth calculation must be made using the plane coefficient for the two surfaces. For this example the depth of surface S_1 is assumed to be less than

that of S_2 . So intensity for surface S_1 are loaded into the refresh buffer until boundary BC is encountered. Then the flag for the surface S_1 goes off. And intensity for surface S_2 is stored until edge FG is passed.

Characteristics

- It is an image-space method for identifying visible surfaces. It deals with multiple polygons.
- The scan line method solves the hidden surface problem one scan line at a time. Processing of the scan line start from the top to the bottom of the display.
- This method calculates the z-value for only the overlapping surface which is tested by scan line.
- Each scan line is processed; all polygon surfaces intersecting that line are examined to determine which are visible.
- Across each scan line, depth calculations are made for each overlapping surface to determine which surface is nearest to the view plane.
- When the visible surface has been determined, the intensity value for that position is entered into the refresh buffer.
- It requires an edge table, polygon table, active edge list and flag.
- **Edge table contains:** Coordinate end points for each scan line, pointers into the polygon table to identify the surfaces bounded by each line.
- **Polygon table contains:** Coefficients of plane equations for each surfaces, pointers into the edge table, intensity information of the surfaces.
- **Active edge list contains:** Edges that cross the current scan line, sorted in order of increasing x.
- **Flag** is defined for each surface that is set 'ON' or 'OFF' to indicate whether a scan line is inside or outside of the surface. At the left most boundary surface flag is 'ON' and at right most boundary flag is 'OFF'.

Algorithm

1. Start
2. Established data structure.
 - a. Edge table with line endpoints.
 - b. Polygon table with plane coefficients, color and pointer to the edge table.
 - c. Active edge list sorted in increasing order of x.
 - d. A flag for each surface.
3. Repeat for each scan line.
 - a. Update AEL for each scan line y.
 - b. When flag is ON, enter intensity of that surface to refresh buffer.
 - c. When two or more flags are ON, calculate depth and store the intensity of the surface nearest to the view plane.
4. Stop

Advantages

- Any number of overlapping polygon surfaces can be processed with this method.
- Deals with transparent, translucent, and opaque surfaces.

- Can be applied to non-polygonal objects.

Disadvantages

- Depth calculations are performed only when there are overlapping polygons.
- Additional memory buffer is required
- Complex.

8. Explain the architecture of VR system with necessary components.

Ans: Major components of VR system are listed below:

1. PC (Personal Computer)/Console/Smartphone

Virtual reality content, which is the users view inside of a virtual reality headset, is equally important as the headset itself. In order to power these interactive three-dimensional environments, significant computing power is required. This is where PC (Personal Computer), consoles, and smart phones come in. They act as the engine to power the content being produced.

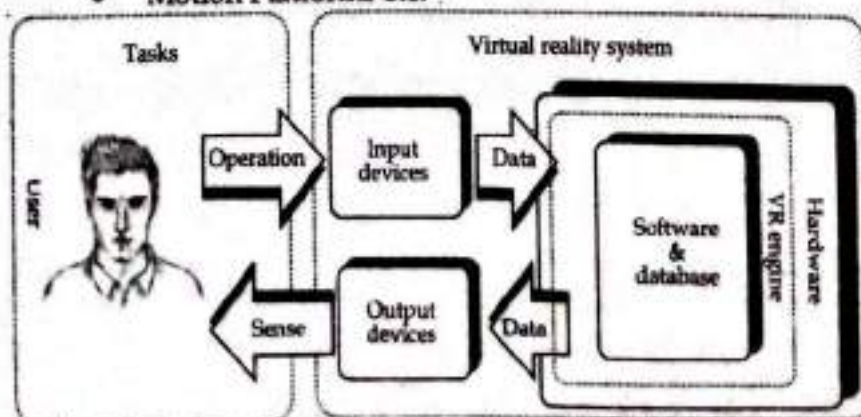
2. Head-Mounted Display

A head-mounted display is a type of device that contains a display mounted in front of a user's eyes. This display usually covers the user's full field of view and displays virtual reality content. Some virtual reality head mounted displays utilize Smartphone displays, including the Google Cardboard and Samsung Gear VR. Head-mounted displays are often also accompanied with a headset to provide for audio stimulation.

3. Input Devices

Input devices are one of the two categories of components that provide users with a sense of immersion (i.e. convincing the human brain to accept an artificial environment as real). They provide users with a more natural way to navigate and interact within a virtual reality environment. Some of the more common forms of virtual reality input devices include:

- Joysticks
- Force Balls/Tracking Balls
- Controller Wands
- Data Gloves
- Track pads
- On-Device Control Buttons
- Motion Trackers/Bodysuits
- Treadmills
- Motion Platforms etc.



Group C

Attempts any two questions

9. Explain the working details of DDA line drawing algorithm. Digitize the line with end points (2, 2) and (10, 5) using Bresenham's line drawing algorithm.

Ans: DDA line drawing Algorithm

The digital differential analyzer (DDA) is a scan conversion line drawing algorithm based on calculating either Δx or Δy from the equation,

$$m = \frac{\Delta y}{\Delta x}$$

We sample the line at unit intervals in one co-ordinate and determine the corresponding integer values nearest to the line path for the other co-ordinates. Here following cases occur:

Case 1: If m is positive ($m > 0$) and $|m| \leq 1$,

Let (x_k, y_k) be the plotted pixel then next pixels going to plot be,

$$\begin{aligned} x_{k+1} &= x_k + \Delta x \\ y_{k+1} &= y_k + \Delta y \end{aligned} \quad \text{Where, } \Delta x \text{ and } \Delta y \text{ are small incremental distance.}$$

$$\text{We have } m = \frac{\Delta y}{\Delta x}$$

$$\Delta y = m \Delta x \quad \dots \dots \dots (1)$$

Also here $|m| \leq 1$, sampling be done along x -axis with $\Delta x = 1$.

Then,

$$\begin{aligned} x_{k+1} &= x_k + 1 \\ y_{k+1} &= y_k + m \end{aligned}$$

Here k takes value from starting point and increase by 1 until final end point. m can be any real value between 0 and 1.

Case 2: If m is positive ($m > 0$) and $|m| > 1$,

Let (x_k, y_k) be the plotted pixel then next pixels going to plot be,

$$\begin{aligned} x_{k+1} &= x_k + \Delta x \\ y_{k+1} &= y_k + \Delta y \end{aligned}$$

Where Δx and Δy are small incremental distance

$$\text{We have } m = \frac{\Delta y}{\Delta x}$$

$$\Delta y = m \Delta x \quad \dots \dots \dots (1)$$

Also here $|m| > 1$, sampling be done along y -axis with $\Delta y = 1$.

$$\text{Then, } \Delta x = \frac{1}{m}$$

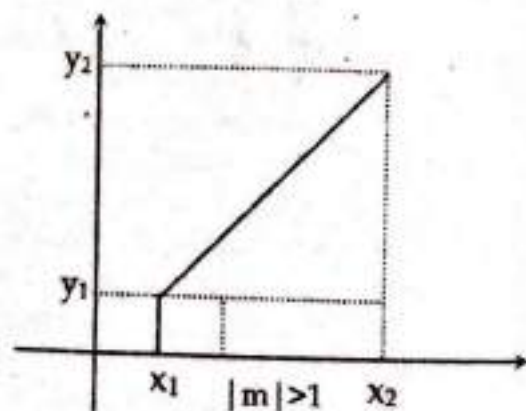
$$x_{k+1} = x_k + \frac{1}{m}$$

$$y_{k+1} = y_k + 1$$

The above equations are under the assumption that the lines are processed from left to right. I.e. left end point is starting.

If the processing is from right to left

Case 3: if $|m| \leq 1$



Let (x_k, y_k) be the plotted pixel then next pixels going to plot be,

$$x_{k+1} = x_k + \Delta x$$

$$y_{k+1} = y_k + \Delta y$$

Where Δx and Δy are small incremental distance

$$\text{We have } m = \frac{\Delta y}{\Delta x}$$

$$\Delta y = m \Delta x \dots \dots \dots (1)$$

Also here $|m| \leq 1$, sampling be done along x-axis with $\Delta x = 1$.

Then

$$x_{k+1} = x_k + 1$$

$$y_{k+1} = y_k + m$$

Case 4: If $|m| > 1$

Let (x_k, y_k) be the plotted pixel then next pixels going to plot be,

$$x_{k+1} = x_k + \Delta x$$

$$y_{k+1} = y_k + \Delta y$$

Where Δx and Δy are small incremental distance

$$\text{We have } m = \frac{\Delta y}{\Delta x}$$

$$\Delta y = m \Delta x \dots \dots \dots (1)$$

Also here $|m| > 1$, sampling be done along y-axis with $\Delta y = 1$.

Then,

$$x_{k+1} = x_k + \frac{1}{m}$$

$$y_{k+1} = y_k + 1$$

$$y_{k+1} = y_k + 1$$

Therefore, in general,

$$\therefore y_{k+1} = y_k \pm m \quad \text{For } |m| < 1 \text{ and}$$

$$x_{k+1} = x_k \pm \frac{1}{m} \quad \text{For } |m| > 1$$

Algorithm for DDA line drawing Algorithm

Step 1: Start

Step 2: Input the line endpoints and store the left endpoint in (x_1, y_1) and right end point in (x_2, y_2) .

Step 3: Calculate the values of dx and dy,

$$dx = x_2 - x_1$$

$$dy = y_2 - y_1$$

Step 4: if(abs(dx) > abs(dy)) then

$$\text{steplength} = \text{abs}(dx)$$

else

$$\text{steplength} = \text{abs}(dy)$$

Step 5: Calculate the values of x-increment and y-increment as,

$$x\text{Increment} = dx / \text{steplength}$$

$$y\text{Increment} = dy / \text{steplength}$$

Step 6: set $x = x_1$ and $y = y_1$;

Step 7: draw the pixel (round(x), round(y));

Step 8: for $k=0$ to steplength do

$$x = x + x\text{Increment}$$

$$y = y + y\text{Increment}$$

Draw the pixel (round(x), round(y));

Step 9: End

Numerical part,

Digitize a line with end points (2, 2) and (10, 5) using BSA.

Solution: Here, $\Delta y = |5-2| = 3$

$$\Delta x = |10-2| = 8$$

Since, $|m| = \Delta y / \Delta x \leq 1$, so we sample at x direction i.e., at each step we simply increment x-coordinate by 1 and find appropriate y-coordinate.

First pixel to plot is (2, 2), which is the starting endpoint.

Now, initial decision parameter, $p_0 = 2\Delta y - \Delta x = 2 \cdot 3 - 8 = -2$

We know that, If $p_k < 0$, then we need to set $p_{k+1} = p_k + 2\Delta y$ and plot pixel (x_k+1, y_k)

And if $p_k \geq 0$, then we need to set $p_{k+1} = p_k + 2\Delta y - 2\Delta x$ then plot pixel (x_k+1, y_k+1)

Here, $p_0 = -2$, we have i.e., $p_k < 0$

So, $p_1 = p_0 + 2\Delta y = -2 + 2 \cdot 3 = 4$

$$p_2 = p_1 + 2\Delta y - 2\Delta x = 4 + 2 \cdot 3 - 2 \cdot 8 = -6$$

$$p_3 = p_2 + 2\Delta y = -6 + 2 \cdot 3 = 0$$

$$p_4 = p_3 + 2\Delta y - 2\Delta x = 0 + 2 \cdot 3 - 2 \cdot 8 = -10$$

$$p_5 = p_4 + 2\Delta y = -10 + 2 \cdot 3 = -4$$

$$p_6 = p_5 + 2\Delta y = -4 + 2 \cdot 3 = 2$$

$$p_7 = p_6 + 2\Delta y - 2\Delta x = 2 + 2 \cdot 3 - 2 \cdot 8 = -8$$

Successive pixel positions and decision parameter calculation is shown in the table below.

K	X	Y	p_k	(x_{k+1}, y_{k+1})
0	2	2	-2 (<0)	(3, 2)
1	3	2	4 (>=0)	(4, 3)
2	4	3	-6 (<0)	(5, 3)
3	5	3	0 (>=0)	(6, 4)
4	6	4	-10 (<0)	(7, 4)
5	7	4	-4 (<0)	(8, 4)
6	8	4	2 (>=0)	(9, 5)
7	9	5	-8 (<0)	(10, 5)

10. Write the different between object space method and image space method. Explain Z buffer algorithm for visible surface detection.

Ans: Different between object space method and image space method

Object-space method is implemented in the physical coordinate system in which objects are described. Object-space methods are generally used in line-display algorithms. Image space method is implemented in the screen coordinate system in which the objects are viewed.

Object space is the 3 dimensional space in which a graphic object is defined. Image space is the projection of the object defined in 3D to two dimensional screen space.

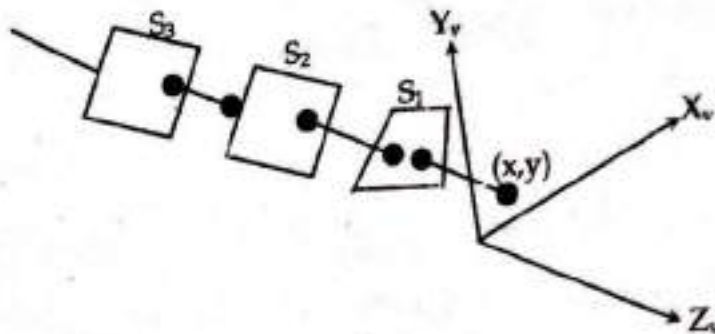
Z buffer algorithm for visible surface detection

Depth Buffer Method is the commonly used image-space method for detecting visible surface. It is also known as z-buffer method. It is an image-space approach. The basic idea is to test the Z-depth of each surface to determine the closest (visible) surface. In this method each surface is processed separately one pixel position at a time across the surface. The depth values for a pixel are compared and the closest (smallest z) surface determines the color to be displayed in the frame buffer. It is applied very

efficiently on surfaces of polygon. Surfaces can be processed in any order. To override the closer polygons from the far ones, two buffers named **frame buffer** and **depth buffer**, are used.

- **Depth buffer** is used to store depth values for (x, y) position, as surfaces are processed ($0 \leq \text{depth} \leq 1$).
- The **frame buffer** is used to store the intensity value of color value at each position (x, y) .

The z -coordinates are usually normalized to the range $[0, 1]$. The 0 value for z -coordinate indicates back clipping plane and 1 value for z -coordinates indicates front clipping plane.



In figure, three surface at varying distance from view plane X_v, Y_v , the projection along (x, y) surface S_1 is closest to the view-plane so surface intensity value of S_1 at (x, y) is saved.

In Z -buffer method, two buffers area are required. A depth buffer is used to store the depth value for each (x, y) position or surface are processed, and a refresh buffer stores the intensity value for each position. Initially all the position in depth buffer is set to 0, and refresh buffer is initialize to background color. Each surface listed in polygon table is processed one scan line at a time, calculating the depth (z -val) for each position (x, y) . The calculated depth is compared to the value previously stored in depth buffer at that position. If calculated depth is greater than stored depth value in depth buffer, new depth value is stored and the surface intensity at that position is determined and placed in refresh buffer.

Algorithm

1. Start
2. Set the buffer values
 - Depthbuffer $(x, y) = 0$
 - Framebuffer $(x, y) = \text{background color}$
3. Process each polygon (One at a time)
 - 3.1. For each projected (x, y) pixel position of a polygon, calculate depth z .
 - 3.1.1. If $Z > \text{depthbuffer}(x, y)$
 - Compute surface color,
 - set depthbuffer $(x, y) = z$,
 - framebuffer $(x, y) = \text{surfacecolor}(x, y)$
4. Stop

After all surfaces are processed, the depth buffer contains the depth value of the visible surface and refresh buffer contains the corresponding

intensity values for that surface. The depth value of the surface position (x, y) is calculated by plane equation of surface.

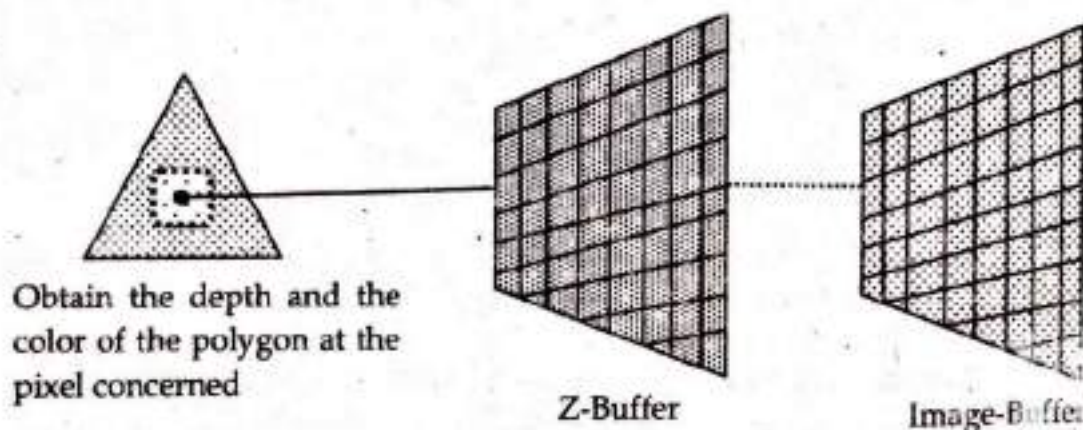
$$Z = \frac{-Ax - By - D}{C}$$

Let Depth Z' at position $(x+1, y)$

$$Z' = \frac{-A(x+1) - By - D}{C}$$

$$\Rightarrow Z' = Z - \frac{A}{C} \dots \dots \dots (1)$$

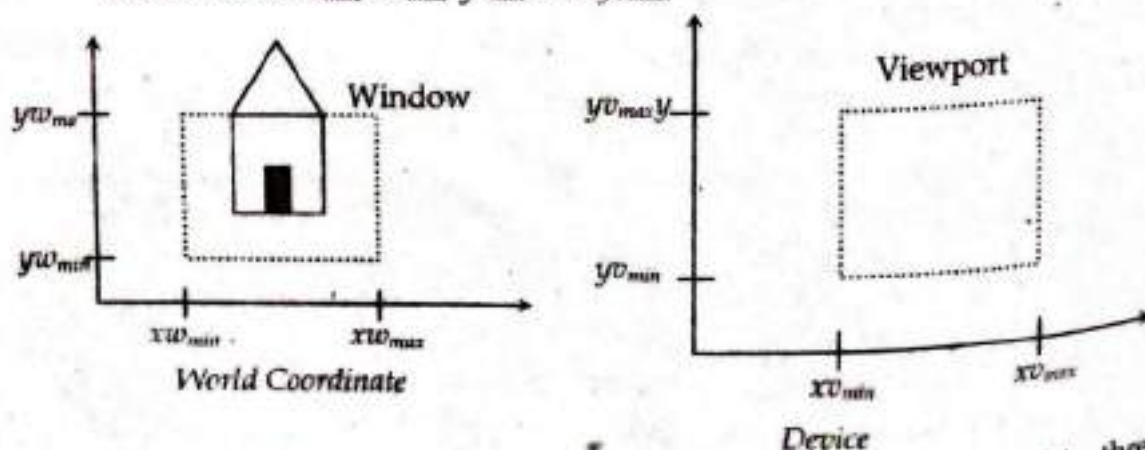
$-\frac{A}{C}$ is constant for each surface so succeeding depth value across a scan line are obtained from preceding values by simple calculation.



11. Derive the formula for window a viewpoint transformation. Given a window bordered by $(0, 0)$ at the lower left and $(4, 4)$ at the upper right. Similarly, a viewport bordered by $(0, 0)$ at the lower left and $(2, 2)$ at the upper right. If a window at position $(2, 4)$ is mapped into the viewport. What will be the position of viewport to maintain same relative placement as in window?

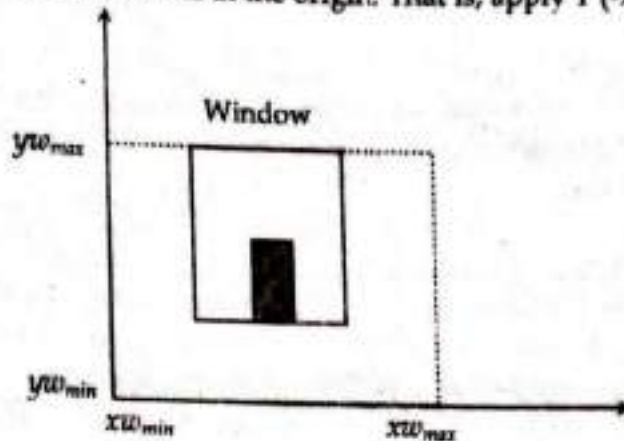
Ans: Window to viewport coordinate transformation

A window can be specified by four world coordinates: xw_{min} , xw_{max} , yw_{min} and yw_{max} . Similarly, a viewport can be described by four device coordinates: xv_{min} , xv_{max} , yv_{min} and yv_{max} .

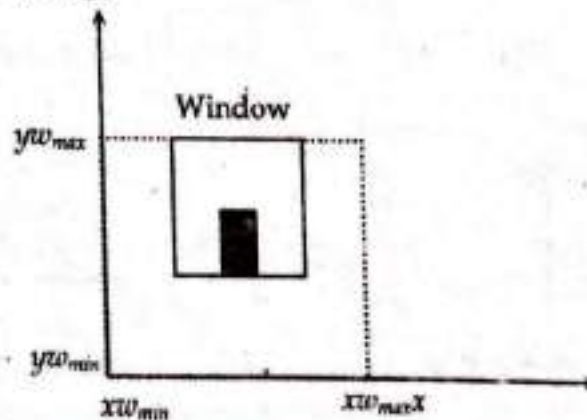


Then, the window-to-viewport transformation can be explained in three steps as below:

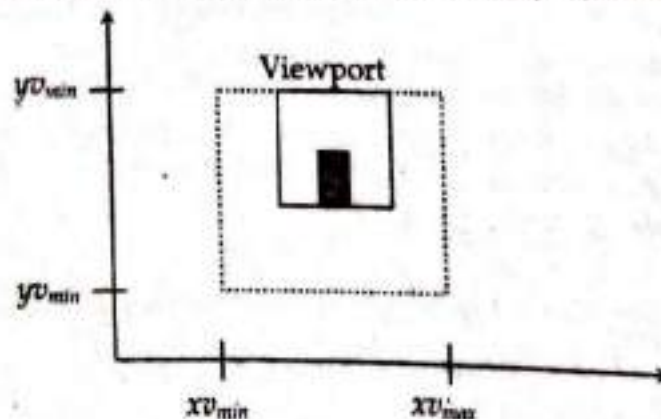
1. Translate object along with window such that the lower left corner of the window is at the origin. That is, apply $T(-X_{wmin}, -Y_{wmin})$.



2. Scale the object and the window such that window has the same dimension as that of a viewport. Simply we can say that we are converting the object into image and window in viewport. That is, apply $S(s_x, s_y)$.



3. Finally, apply another translation to move the viewport to its original position on the screen. That is, apply $T(X_{vmin}, Y_{vmin})$.



Therefore, net transformation (Composite Transformation),

$$T_{WV} = T(xv_{min}, yv_{min}) \cdot S(S_x, S_y) \cdot T(-xw_{min}, -yw_{min}) \dots \dots \dots (i)$$

Here, S_x and S_y are scaling factors, where;

$$S_x = \frac{Xv_{max} - Xv_{min}}{Xw_{max} - Xw_{min}}$$

$$S_y = \frac{Yv_{max} - Yv_{min}}{Yw_{max} - Yw_{min}}$$

Now, writing in matrix form;

$$T_{wv} = \begin{bmatrix} 1 & 0 & x_{vmin} \\ 0 & 1 & y_{vmin} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -y_{wmin} \\ 0 & 1 & -y_{wmin} \\ 0 & 0 & 1 \end{bmatrix}$$

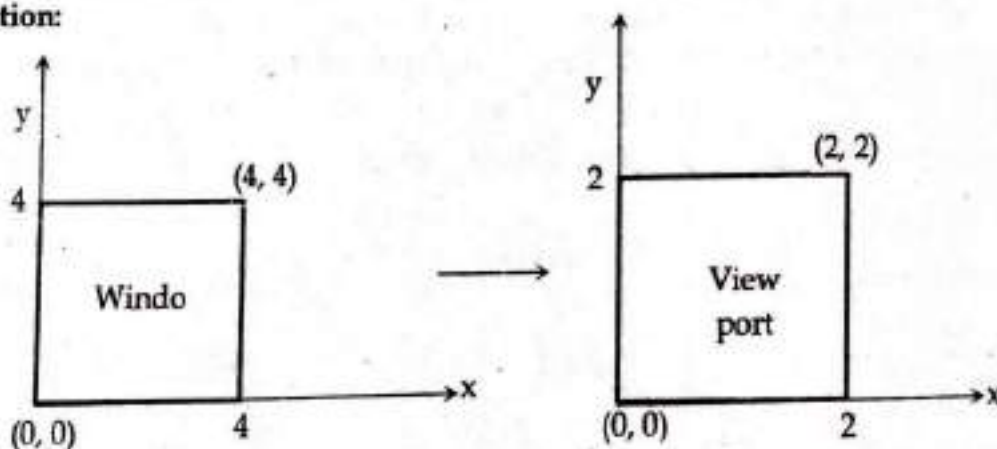
This is required window-to-viewpoint transformation. Let $P(x, y)$ be the world coordinate point that is mapped onto the viewport point $P'(u, v)$, then we must have:

$$P' = T_{wv}.P$$

Numerical part,

Given a window bordered by (0, 0) at the lower left and (4, 4) at the upper right. Similarly, a viewport bordered by (0, 0) at the lower left and (2, 2) at the upper right. If a window at position (2, 4) is mapped into the viewport. What will be the position of viewport to maintain same relative placement as in window?

Solution:



Coordinates for window:

$$X_{wmin} = 0$$

$$Y_{wmin} = 0$$

$$X_{wmax} = 4$$

$$Y_{wmax} = 4$$

Coordinates of view port

$$X_{vmin} = 0$$

$$Y_{vmin} = 0$$

$$X_{vmax} = 2$$

$$Y_{vmax} = 2$$

We know that,

$$S_x = \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}} = \frac{2 - 0}{4 - 0} = 0.5$$

$$\text{And } S_y = \frac{Y_{vmax} - Y_{vmin}}{Y_{wmax} - Y_{wmin}} = \frac{2 - 0}{4 - 0} = 0.5$$

We know that, Composite transformation matrix is given as

$$T_{wv} = T(x_{vmin}, y_{vmin}) \cdot S(S_x, S_y) \cdot T(-x_{wmin}, -y_{wmin})$$

$$T_{wv} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, Image = T_{wv} . Image position

$$= \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

So, relative position of (2, 4) = (1, 2)



MODEL QUESTIONS SETS FOR PRACTICE

MODEL SET 1

Course Title: Computer Graphics and Animation (3 Cr.)

Full Marks: 60

Course Code: CACS305

Pass Marks: 24

Year/Semester: III/V

Time: 3

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10x1=10]

1. Circle (O) the correct answer in the following questions:
 - i. Which of the following is/are side effects of scan conversion?
 - a. Aliasing
 - b. Unequal intensity of diagonal lines
 - c. Over striking in photographic applications
 - d. Local or Global aliasing
 - ii. What is the primary use of clipping in computer graphics?
 - a) Adding graphics
 - b) removing objects and lines
 - c) Zooming
 - d) copying
 - iii. 3D rotation is not same as 2D rotation
 - a) TRUE
 - b) FALSE
 - c) Can be true or false
 - d) cannot say
 - iv. We can change the size of an object using.....
 - a) Shear transformation
 - b) scaling transformation
 - c) Reflection transformation
 - d) parallel transformation
 - v. Which of the following are the true Properties of 3-D Transformation?
 - a) Lines are preserved
 - b) Parallelism is preserved
 - c) Proportional distances are preserved
 - d) all of the above
 - vi. The transformation set is also known as.....
 - a) Translation difference
 - b) Rotation color model
 - c) Color Mapping Functions
 - d) None of these
 - vii. Which of the following is an example of the impact device?
 - a) Laser printer
 - b) Inkjet printer
 - c) Line printer
 - d) none of the above
 - viii. Random scan systems are used for.....
 - a) Color drawing application
 - b) Pixel drawing application
 - c) Line drawing application
 - d) None of the above
 - ix. Shadow mask method is used in.....
 - a) Raster scan system
 - b) Random scan system
 - c) Both (a) & (b)
 - d) None of the above
 - x. Which of the following uses the Beam penetration method?
 - a) Raster scan system
 - b) Random scan system
 - c) Both (a) & (b)
 - d) None of the above

Group B

Attempt any six questions

2. Describe uses of computer graphics in computer field. Define pixel. What is the use of pixel in display devices?

3. How much time is spent scanning across each row of pixels during screen refresh on a raster system with resolution of 1280×1024 and a refresh rate of 60 frames per second?
4. Discuss DDA line drawing algorithm in detail. Also give its advantages and disadvantages.
5. Suppose we have a quadrilateral ABCD with vertices A(3,1), B(-7,-4), C(5,-8) and D(3,0). What is the coordinates of quadrilateral if it is to be doubled in size about origin. Then translate obtained coordinates with (-4, 3).
6. What do you mean by hidden surface removal? Describe any hidden surface removal algorithm with suitable example.
7. Differentiate between diffuse reflection and specular reflection? Also explain Phong illumination model.
8. Describe key components of virtual reality with suitable examples. Write down advantages and disadvantages of virtual reality.

Group C

Attempts any two questions

9. What is Color CRT Display? Explain Beam-penetration and Shadow-mask method.
10. Rasterizing the circle with radius $r=5$ and center = (100, 100) with midpoint circle generation algorithm.
11. How can polygons be clipped? Why is Phong shading also called Normal Vector Interpolation scheme? Explain.

MODEL SET 2

Group A

[10x1=10]

Attempt all the questions

1. Circle (O) the correct answer in the following questions:

- i. Back face removal is an example of.....
 - a) Object space model
 - b) Image space model
 - c) Combination of both
 - d) None of the above
2. Which vertex of the polygon is clipped first in polygon clipping?
 - a) Top right
 - b) bottom right
 - c) Bottom left
 - d) top left
3. Shadow mask method is used in.....
 - a) Raster scan system
 - b) Random scan system
 - c) Both (a) & (b)
 - d) None of the above
4. Which of the following uses the Beam penetration method?
 - a) Raster scan system
 - b) Random scan system
 - c) Both (a) & (b)
 - d) None of the above
5. Which of the following algorithm is used to fill the interior of a polygon?
 - a) Boundary fill algorithm
 - b) Scan line polygon fill algorithm
 - c) Flood fill algorithm
 - d) All of the above
6. _____ refers to simulated motion pictures showing movement of drawn objects.
 - a) Motion
 - b) Animation
 - c) VR
 - d) SMD
7. _____ animation is used to animate things that are smaller than life size.
 - a) Immersive
 - b) Claymotion
 - c) Stop motion
 - d) Augmented
8. The father of animation?

- a) Walt Disney
b) J. Stuart Blackton
c) William Horner
d) J. A. Ferdinand Plateau
9. VR may be useful for students with special needs because _____
a) It allows them to escape from difficult
b) They can attend class from home
c) Teachers can develop personalized lessons for students
d) None of the mentioned
10. Sutherland-Hodgmann algorithm is used for.....
a) Line clipping
b) Point clipping
c) Polygon clipping
d) Hybrid clipping.

- a) Outside, inside b) inside, outside
c) Exact copy, different d) different, an exact copy
- iii. Which of the following techniques is used in Midpoint subdivision algorithm?
a) Binary search b) Bubble sort
c) Linear search d) Sequential search.
- iv. In Cohen-Sutherland algorithm ... region bit code is assigned to each end point of the line.
a) 2 b) 3 c) 4 d) 5
- v. All the hidden surface algorithms employ image space approach except?
a. Back face removal b. Depth buffer method
c. Scan line method d. Depth sort method
- vi. Choose the incorrect statement from the following about the basic ray tracing technique used in image synthesis?
a. In this technique rays are cast from the eye point through every pixel on the screen
b. In this technique, viewing transformation are not supplied to the scene prior to rendering
c. This technique removes hidden surfaces.
d. In this technique rays are cast from the light source to the object in the scene
- vii. Which of the following color models are used for color printing?
a) RGB b) CMY c) CMYK d) CMY and CMYK
- viii. Which of the following are the true Properties of 3-D Transformation?
a) Lines are preserved b) Parallelism is preserved
c) Proportional distances are preserved d) All of the above
- ix. The transformation set is also known as.....
a) Translation difference b) Rotation color model
c) Color Mapping Functions d) None of these
- x. VR may be useful for students with special needs because _____
a) It allows them to escape from difficult
b) They can attend class from home
c) Teachers can develop personalized lessons for students
d) None of the mentioned

Group B

Attempt any six questions

2. Find the new co-ordinates of a unit cube 90 degree rotated about an axis defined by its end points A(2, 1, 0) and B(3, 3, 1).
3. What is the main advantage of using fast Phong shading over shading? Write down advantages and disadvantages of Phong shading.
4. Suppose we have a quadrilateral ABCD with vertices A(3,1), B(-7,-4), C(5,-8) and D(3,0). What is the coordinates of quadrilateral if it is to be doubled in size about origin. Then translate obtained coordinates with (-4, 3).
5. Outline the Z-buffer method. List the advantages and disadvantages of the Z-buffer algorithm.
6. Differentiate between simple reality and mixed reality. Also write down applications of mixed reality with suitable example.
7. Discuss relative advantages and disadvantages of Gourand Shading and Phong Shading.

8. What do you mean by symmetry in octant of a circle? Digitize a circle $(x - 2)^2 + (y - 3)^2 = 25$

Group C

Attempts any two questions

9. Clip a quadrilateral ABCD with coordinates (10, 18) (22, 18) (34, 27) and (10, 37) against a window QRST with coordinate (5, 15) (30, 15) (30, 25) and (5, 25) using Cohen Sutherland algorithm.
10. Explain the concept of Rendering. Explain Gourand shading with their advantages and disadvantages.
11. Given a 25cm x 20cm display operating in 1024 x 768 x 16 color mode which is refreshed 30 times per second, and for which 10% of the refresh cycle is spent in retrace, calculate
- the pixel aspect ratio,
 - the size of the frame buffer, and
 - The required data transfer rate in kilobytes per second.

MODEL SET 4

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
- Which of the following color models are used for color printing?
a) RGB b) CMY c) CMYK d) CMY and CMYK
 - The process of removal of hidden surfaces is termed as.....
a) Clipping b) copying c) Culling d) shorting
 - Which of the following is an example of the impact device?
a) Laser printer b) Inkjet printer
c) Line printer d) none of the above
 - Random scan systems are used for.....
a) Color drawing application b) Pixel drawing application
c) Line drawing application d) None of the above
 - Shadow mask method is used in.....
a) Raster scan system b) Random scan system
c) Both (a) & (b) d) None of the above
 - Which of the following uses the Beam penetration method?
a) Raster scan system b) Random scan system
c) Both (a) & (b) d) None of the above
 - Positive values for the rotation angle Θ define.....
a) Counterclockwise rotations about the end points
b) Counterclockwise translation about the pivot point
c) Counterclockwise rotations about the pivot point
d) Negative direction
 - 3D rotation is not same as 2D rotation
a) TRUE b) FALSE
c) Can be true or false d) cannot say
 - We can change the size of an object using.....
a) Shear transformation b) scaling transformation
c) Reflection transformation d) parallel transformation

- x. What are the three types of VR?
- Immersive, 3D, Non-immersive
 - Immersive, Semi-immersive, Non-immersive
 - Digital, Semi-immersive, Projective
 - 3D, Non-immersive, Digital

Group B

Attempt any six questions

- What do you mean by refresh rate of a display device? Write short note about shadow mask method.
- Consider a RGB raster system is to be designed using 8 inch by 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 8 bits per pixel in the frame buffer. How much storage (in bytes) do we need for the frame buffer?
- Digitize the line with endpoints (1, 6), (6, 10) using DDA line drawing algorithm.
- Consider 256 pixel X 256 scan lines image with 24-bit true color. If 10 minutes video is required to capture, calculate the total memory required? Why intensity assignment is required?
- Why shedding is required in the computer graphics? Explain in detail about constant intensity shading.
- What do you mean by homogeneous coordinates? Explain it with equation and practical application
- Explain the following term with practical applications.
 - 3D Rotation
 - 2D Shear

Group C

Attempts any two questions

- A polygon has four vertices located at A(20, 10) B(60, 10) C(60, 30) D(20, 30). Calculate vertices after applying a transformation matrix to double the size of polygon with point A located on the same place.
- Describe briefly Bresenham's line drawing algorithm. Why we prefer incremental algorithm over DDA?
- What is projection? List out the types of projection. Differentiate parallel and perspective projections.

MODEL SET 5

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x 1 = 10]

- Circle (O) the correct answer in the following questions:
 - How many bit RGB color image is represented by full-color image?
 - 32-bit RGB color image
 - 24-bit RGB color image
 - 16-bit RGB color image
 - 8-bit RGB color image
 - The Cohen-Sutherland algorithm divides the two-dimensional space in how many regions?
 - 4
 - 8
 - 9
 - 23

- iii. Which of the following is a Computer Graphics type?
 - a) Raster and Vector
 - b) Raster and Scalar
 - c) Scalar only
 - d) All of the above
- iv. Which of the following plane is used for 2D transformations?
 - a) Three-dimensional plane
 - b) Two-dimensional plane
 - c) One-dimensional plane
 - d) Four-dimensional Plane
- v. Bitmap is a collection of _____ that describes an image.
 - a) Pixels
 - b) algorithms
 - c) Bits
 - d) colors
- vi. Which of the following is defined as the number of pixels stored in the frame buffer of a graphics system?
 - a) Resolution
 - b) Resolution
 - c) Depth
 - d) none of the mentioned
- vii. Which of the following algorithm is a faster method for calculating pixel positions?
 - a) Parallel line algorithm
 - b) Mid-point algorithm
 - c) DDA line algorithm
 - d) Bresenham's line algorithm
- viii. What does an aspect ratio mean?
 - a) Ratio of vertical points to horizontal points
 - b) Ratio of vertical points to horizontal points and horizontal points to vertical points
 - c) Number of pixels
 - d) Ratio of horizontal points to vertical points
- ix. Which of the following representation represents a three-dimensional object?
 - a) Function
 - b) Point
 - c) Polygon
 - d) Equation
- x. The algorithm which displays line-type attributes by plotting pixel spans is.....
 - a) Raster scan algorithm
 - b) Raster line algorithm
 - c) Random line algorithm
 - d) Random scan algorithm

Group B

Attempt any six questions

2. Explain the visual effect that occurs when during animation of a Gouraud shading polyhedron, the center of a highlight moves from one vertex to another along an edge.
3. Digitize the endpoint (10, 18), (15, 8) using Bresenham's algorithm.
4. Digitize an ellipse with center (20, 20) and x-radius=8 and y-radius=6.
5. Construct the polygon table for an object with six vertexes, eight edge and three surface.
6. A system with 24 bits per pixel and resolution of 1024 by 1024, Calculate the size of frame buffer (in Megabytes).
7. What do you mean by refresh rate of a display device? Write short note about shadow mask method.
8. Consider a RGB raster system is to be designed using 8 inch by 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 8 bits per pixel in the frame buffer. How much storage (in bytes) do we need for the frame buffer?

Group C

Attempts any two questions

9. Transform a square with corners A(2, 2), B(4, 2), C(2, 4), D(4, 4) into a rectangle whose breadth is half of the breadth of square and length is double of it.

10. If a monitor screen has 525 scan lines and an aspect ratio of 3:4 and if each pixel contains 8 bits worth of intensity information, how many bits per second are required to show 30 frames each second?
11. Differentiate between flood fill and boundary fill algorithms.

MODEL SET 6

Candidates are required to give their answer in their own words as far as practicable.

Group- A

[10 x1=10]

Attempt all the questions

1. Circle (O) the correct answer in the following questions:

- i. Choose the incorrect statement from the following about the basic ray tracing technique used in image synthesis?
 - a. In this technique rays are cast from the eye point through every pixel on the screen
 - b. In this technique, viewing transformation are not supplied to the scene prior to rendering
 - c. This technique removes hidden surfaces.
 - d. In this technique rays are cast from the light source to the object in the scene
- ii. The 4-bit code of the bottom-region among the nine regions divided using the Cohen- Sutherland algorithm?
 - a) 0000 b) 0010 c) 0110 d) 0101
- iii. The transformation set is also known as.....
 - a) Translation difference b) Rotation color model
 - c) Color Mapping Functions d) None of these
- iv. If the scaling factors values s_x and s_y are assigned to unequal values then.....
 - a) Uniform rotation is produced b) Uniform scaling is produced
 - c) Differential scaling is produced d) Scaling cannot be done
- v. The best hidden surface removal method used for complex scenes with more than a fewthousand surfaces is?
 - a. depth sorting method b. scan line algorithm
 - c. depth buffer algorithm d. octree method
- vi. All the hidden surface algorithms employ image space approach except?
 - a. Back face removal b. Depth buffer method
 - c. Scan line method d. Depth sort method
- vii. Back face removal is an example of.....
 - a. Object space model b. Image space model
 - c. Combination of both d. None of the above
- viii. Random scan systems are used for.....
 - a) Color drawing application b) Pixel drawing application
 - c) Line drawing application d) None of the above
- ix. The algorithm which displays line-type attributes by plotting pixel spans is.....
 - a) Raster scan algorithm b) Raster line algorithm
 - c) Random line algorithm d) Random scan algorithm

- x. Which of the following algorithm is used to fill the interior of a polygon?
- | | |
|----------------------------|-------------------------------------|
| a) Boundary fill algorithm | b) Scan line polygon fill algorithm |
| c) Flood fill algorithm | d) All of the above |

Group B**Attempt any six questions**

- Clip the polygon ABC against the window PQRS. Coordinates of polygon are A(80,80), B(120,60) and C(75,60) and Coordinates of window are P(50,50), Q(100,50), R(100,100) and S(50,100).
- Compare storage type CRT against refresh type CRT display. List the important properties of phosphor being used in CRTs.
- Describe key components of virtual reality with suitable examples.
- Digitize the line with endpoints (1, 6), (6, 10) using DDA algorithm.
- Explain basic rendering and illumination models. Also describe their advantages and disadvantages.
- Explain Binary Space Partitioning Trees (BSP) with suitable example.
- What is the difference between a local illumination model and a global illumination model?

Group C**Attempts any two questions**

- What are the object space and image space method of hidden surface removal? Describe the back face detection method of hidden surface removal.
- Perform the scaling transformation to the triangle with vertices A(6, 9), B(10, 5), C(4, 3) with scaling factors $S_x = 3$ and $S_y = 2$.
- Explain about parametric cubic curve. Describe the properties of Bezier Curve.



Candidates are required to give their answer in their own words as far as practicable.

Group- A**[10 x1=10]****Attempt all the questions****1. Circle (O) the correct answer in the following questions:**

- All the hidden surface algorithms employ image space approach except?

a. Back face removal	b. Depth buffer method
c. Scan line method	d. Depth sort method
- A line with end point codes 0000 and 0100 is.....

a) Partially invisible	b) completely invisible
c) Completely visible	d) cannot be determined.
- A translation is applied to an object by.....

a) Repositioning it along with straight line path	d) None of the above
b) Repositioning it along with circular path	
c) Both a and b	

- iv. In color raster system, the number of color choices available depends on_____.
 - a) colors in frame buffer
 - b) Amount of storage provided per pixel in frame buffer
 - c) RGB color
 - d) Neither a nor b
- v. Color information can be stored in_____.
 - a) Main memory
 - b) Secondary memory
 - c) Graphics card
 - d) Frame buffer
- vi. Whenever a particular color code is specified in an application program, the corresponding binary value is placed in?
 - a) Color look-up table
 - b) Directly in frame buffer
 - c) a or b
 - d) Video lookup table
- vii. RGB colors on internet applications are called_____.
 - a) Safe colors
 - b) Colors space
 - c) Web colors
 - d) Safe web colors
- viii. Two dimensional color model are_____.
 - a) RGB and CMKY
 - b) RBG and CYMK
 - c) RGB and CMYK
 - d) None
- ix. Color depth can be defined by which can be displayed on a display unit.
 - a) Bits per pixel
 - b) Bytes per pixel
 - c) Megabyte per pixel
 - d) None of these
- x. The types of animation are_____.
 - a) Traditional animation
 - b) Computer animation
 - c) Both a & b
 - d) None of these

Group B

Attempt any six questions

2. Explain the principle of Phong shading algorithm. And also write their advantages and disadvantages over Gouraud shading.
3. Outline the Z-buffer method. List the advantages and disadvantages of the Z-buffer algorithm.
4. Trace DDA algorithm for line with endpoints (1, 2), (5, 6)
5. Find the points on a circle one of its octants with the circle centered at (5, 5) and has a radius of 8 units.
6. What is the fundamental difference in the method of operation of a monochrome CRT and a color CRT?
7. Define pixel. What is the use of pixel in display devices?
8. Rotate the ΔABC by 45° clock wise about origin and scale it by (2, 3) about origin, where A(7,15), B(5,8), and C (10,10).

Group C

Attempts any two questions

9. What is a polygon mesh? Explain the application of polygon mesh with example.
10. Justify that hidden surface removal is required in computer graphics. Explain in detail about depth buffer method.

11. Consider 256 pixels X512 scan lines image with 24-bit true color. If 20 minutes video is required to capture, calculate the total memory required? What is the color intensity model?



Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
 - i. The best hidden surface removal method used for complex scenes with more than a few thousand surfaces is?
 - a. depth sorting method
 - b. scan line algorithm
 - c. depth buffer algorithm
 - d. octree method
 - ii. Sutherland-Hodgeman algorithm is used for.....
 - a) Line clipping
 - b) Point clipping
 - c) Polygon clipping
 - d) Hybrid clipping.
3. Which of the following is not an object-space hidden surface removal algorithm?
 - a. Back Face Culling
 - b. Depth Buffer
 - c. Painter's Algorithm
 - d. All of these.
4. Which of the following techniques is used in Midpoint subdivision algorithm?
 - a) Binary search
 - b) Bubble sort
 - c) Linear search
 - d) Sequential search.
5. Z-buffer algorithm is used for.....
 - a. Frame buffer removal
 - b. Hidden line removal
 - c. Rendering
 - d. Animation.
6. In Cohen-Sutherland algorithm ... region bit code is assigned to each end point of the line.
 - a) 2
 - b) 3
 - c) 4
 - d) 5
7. _____ is the option to create a Circle.
 - a) Shape
 - b) Geometry
 - c) Sphere
 - d) Modify panel
8. HMD stands for?
 - a) Head Mounted Display
 - b) Head Masked Display
 - c) Head Made Display
 - d) Head Mounted Detection
9. We can change the size of an object using.....
 - a) Shear transformation
 - b) scaling transformation
 - c) Reflection transformation
 - d) parallel transformation
10. Which of the following are the true Properties of 3-D Transformation?
 - a) Lines are preserved
 - b) Parallelism is preserved
 - c) Proportional distances are preserved
 - d) All of the above

Group B**Attempt any six questions**

2. What is resolution? What is the impact of resolution in display devices?
3. Rasterizing the line from (10, 5) to (15, 9) using Bresenham's line drawing Algorithm.
4. What do you mean by hidden surface removal? Describe any hidden surface removal algorithm with suitable example.
5. Explain basic rendering and illumination models. Also describe their advantages and disadvantages.
6. How we can apply Virtual Reality for education?
7. Trace BSA for endpoints (20, 30), (30, 22)
8. Represent all the transformations with homogeneous matrix representation. Consider a triangle ABC with vertices A(3, 4), B(7, 4), and C(5, 8). What is the transformation if the triangle is to be doubled in size keeping A(3, 4) fixed? Obtain new vertices of the triangle.

Group C**Attempts any two questions**

9. Develop the Bresenham's line drawing to draw lines of any slope. Compare this with the DDA Algorithm.
10. Given a window bordered by (1, 2) at the lower left and (16, 12) at the upper right, give the screen coordinates of a triangle with vertices (3, 2), (10, 7.5) and (5, 5) when mapped into a viewport with corners (100, 100) and (400, 200). Provide accurate illustrations of the window, viewport, and the untransformed and transformed triangles with your answer.
11. Explain the essential difference between the "Scan-Line" hidden surface removal algorithm and the depth buffer technique.

□□□

Computer Networking New Syllabus

Course Title: Computer Networking (3 Cr.)

Course Code: CACS303

Year/Semester: III/V

Class Load: 6 Hrs. / Week (Theory: 3Hrs.

Tutorial: 1 Hrs. Practical: 2 Hrs.)

Course Description

This course is designed to extend students' knowledge and practice in Graphics hardware, software, and applications. It also provides the knowledge of data structures for graphics, graphics languages, and models for 2D and 3D objects, clipping, hidden surface elimination, depth buffer, raster graphics, shading, and rendering.

Course objectives

The general objectives of this course are to provide theoretical as well as practical knowledge of computer networking to make students capable of implementing, managing and troubleshooting the issues of computer network in their personal as well professional life.

Course Contents

Unit 1: Introduction

6 Hrs

- 1.1 Network as an infrastructure for data communication
- 1.2 Applications of Computer network
- 1.3 Network Architecture
- 1.4 Types of computer Networks
- 1.5 Protocols and Standards
- 1.6 The OSI Reference Model
- 1.7 The TCP/IP Protocol Suite
- 1.8 Comparison between OSI and TCP/IP Reference model
- 1.9 Critiques of OSI and TCP/IP Reference model

Unit 2: The Physical Layer

6 Hrs

- 2.1 Functions of Physical Layer
- 2.2 Data and Signals: Analog and Digital signals, Transmission Impairment, Data Rate Limits, Performance
- 2.3 Data Transmission Media: Guided Media, Unguided Media and Satellites
- 2.4 Bandwidth Utilization: Multiplexing and Spreading
- 2.5 Switching: Circuit switching, Message switching & Packet switching
- 2.6 Telephone, Mobile and Cable network for data Communication

Unit 3: The Data Link Layer

8 Hrs

- 3.1 Functions of Data Link Layer
- 3.2 Data Link Control: Framing, Flow and Error Control
- 3.3 Error Detection and Correction
- 3.4 High-Level Data Link Control (HDLC) & Point-to-Point protocol (PPP)
- 3.5 Channel Allocation Problem
- 3.6 Multiple Access: Random Access (ALOHA, CSMA, CSMA/CD, CSMA/CA), Controlled Access (Reservation, Polling, Token Passing), Channelization (FDMA, TDMA, CDMA)
- 3.7 Wired LAN: Ethernet Standards and FDDI
- 3.8 Wireless LAN: IEEE 802.11 and Bluetooth Standards
- 3.9 Token Bus, Token Ring and Virtual LAN

Unit 4: The Network Layer

8 Hrs

- 4.1 Functions of Network Layer

- 4.2 Virtual circuits and Datagram Subnets
- 4.3 IPv4 Addresses: Address Space, Notations, Classful addressing, Classless addressing, Subnetting and Network Address Translation (NAT)
- 4.4 IPv4 Datagram format and fragmentation
- 4.5 IPv6 Address Structure and advantages over IPv4
- 4.6 Routing Algorithms: Distance Vector Routing, Link State Routing
- 4.7 Internet Control Protocols: ARP, RARP, ICMP
- 4.8 Routing protocols: OSPF, BGP, Unicast, Multicast and Broadcast

7 Hrs

Unit 5: The Transport Layer

- 5.1 Functions of Transport Layer
- 5.2 Elements of Transport Protocols: Addressing, Establishing and Releasing Connection, Flow Control & Buffering, Error Control, Multiplexing & Demultiplexing, Crash Recovery
- 5.3 User Datagram Protocol (UDP): User Datagram, UDP Operations, Uses of UDP, RPC
- 5.4 Principles of Reliable Data Transfer: Building a Reliable Data Transfer Protocol, Pipelined Reliable Data Transfer Protocol, Go Back-N (GBN), Selective Repeat (SR)
- 5.5 Transmission Control Protocol (TCP): TCP Services, TCP Features, TCP Segment Header
- 5.6 Principle of Congestion Control

Unit 6: The Application Layer

5 Hrs

- 6.1 Functions of Application layer
- 6.2 Application Layer Protocols: DNS, DHCP, WWW, HTTP, HTTPS, TELNET, FTP, SMTP, POP, IMAP
- 6.3 Concept of traffic analyzer: MRTG, PRTG, SNMP, Packet tracer, Wireshark.

Unit 7: Network Security

5 Hrs

- 7.1 A Model for Network Security
- 7.2 Principles of cryptography: Symmetric Key and Public Key
- 7.3 Public Key Algorithm- RSA
- 7.4 Digital Signature Algorithm
- 7.5 Communication Security: IPSec, VPN, Firewalls, Wireless Security.

Practical

1. Prepare hardware and software specification for basic computer system.
2. Determine the appropriate placement of networking devices on a network.
3. Identify networking cable standards. Create and test cross-over and straight cables.
4. Configure the IP address of the computer.

5. Create a basic network and share file and folders.
6. Install and configure windows server: Active Directory, User and Group Policy Management.
7. Set the file access permissions and quota in windows server.
8. Configure basic DNS and DHCP services in windows server.
9. Install Linux based OS and practice on basic Linux and networking commands.
10. Configure IP address and subnet in Linux Machine.
11. Install packet tracer and identify the features of packet tracer.
12. Implement the LAN topologies.
13. Demonstrate the use of VLAN.
14. Implement the both static and dynamic router configurations.
15. Install and configure DNS, DHCP, FTP and Web Servers in Linux machine.
16. Capture some packets and analyze the header using Wireshark.
17. Implement the firewall.

Evaluation

Examination Scheme			
Internal Assessment	External Assessment		Total
Theory	Practical	Theory	Practical
20	20 (3 hrs)	60 (3 Hrs)	

1. Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks, 5/e", Prentice Hall, 2011.
2. Behrouz A. Forouzan, "Data Communications and networking" Tata McGraw-Hill.
3. Kurose, Ross, "Computer Networking: A Top-Down Approach", Pearson Education Limited, 2017.
4. Larry L. Peterson and Bruce S. Davie, "Computer Network: A System Approach", Morgan Kaufmann, 5/e, 2012.
5. Matthew Helmke, Andrew Hudson, Paul Hudson "Ubuntu • Unlcashe

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

MODEL QUESTIONS-ANSWERS

Bachelor in Computer Applications
 Course Title: Computer Networking
 Code No: CACS303
 Semester: V

Full Marks: 60
 Pass Marks: 24
 Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.

Group 'A'

Attempt all the questions.

10 × 1 = 10

1. Circle (O) the correct answer in the following questions
 - i. Set of rules that governs data communication is called _____.
 a) Protocol b) standards c) RFCs d) servers
 - ii. Which of this is not a guided media?
 a) Coaxial cable b) twisted pair cable
 c) optical fiber d) wireless LAN
 - iii. The identifier that is used for data transfer in virtual circuit network is called _____.
 a) Global address b) virtual circuit identifier
 c) Network identifier d) IP identifier
 - iv. In classless addressing, there are no classes but addresses are still granted in _____.
 a) IPs b) blocks c) codes d) sizes
 - v. In classful addressing, a large part of available addresses is _____.
 a) Organized b) blocked c) wasted d) communicated
 - vi. Which is not an application layer protocol?
 a) HTTP b) SMTP c) FTP d) TCP
 - vii. Which mode of IPSec should you use to assure the security and confidentiality of data within the same LAN?
 a) AH transport mode b) ESP transport mode
 c) ESP tunnel mode d) AH tunnel mode
 - viii. The entire hostname has a maximum of _____.
 a) 255 characters b) 127 characters
 c) 63 characters d) 31 characters
 - ix. What protocol is not used in the operation of a VPN?
 a) PPTP b) IPSec c) YMUM d) L2TP
 - x. The DHCP server can provide the _____ of the IP addresses.
 a) Dynamic allocation b) automatic allocation
 c) Static allocation d) all of the mentioned

Answer-Key

i. (a)	ii. (c)	iii. (b)	iv. (b)	v. (c)	vi. (d)	vii. (b)	viii. (a)	ix. (c)	x. (d)
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Group B

Attempt any SIX questions.

[6×5=30]

2. Is 192.16.144.64/27 a host, network or broadcast address? In which layer of OSI model do HUB, Switch and Router operate on? [4+1]

Ans: IP address: 192.16.144.64/27

Subnet Mask: 11111111 11111111 11111111 11100000 = 255.255.255.224

Total Subnets = $2^3 = 8$

Valid Subnets (4th octet) = Total subnet mask - Current subnet mask
= 256 - 224 = 32

Networks ID are: 192.16.144.0, 192.16.144.32, 192.16.144.64, 192.16.144.96, 192.16.144.128, 192.16.144.160, 192.16.144.192, 192.16.144.224

Therefore, 192.16.144.64/27 is a network address.

The layers of OSI model where HUB, Switch and Router operate are First Layer (Physical Layer), Second Layer (Data Link Layer) and Third Layer (Network Layer) respectively.

3. Describe the working procedure of Token bus and Token ring. [2.5+2.5]

Ans: Token Bus

The IEEE 802.4 Committee has defined token-bus standards as broadband computer networks, as opposed to Ethernet's baseband transmission technique. Physically, the token bus is a linear or tree-shape cable to which the stations are attached. The topology of the computer network can include groups of workstations connected by long trunk cables. Logically, the stations are organized into a ring. These workstations branch from hubs in a star configuration, so the network has both a bus and star topology. Token bus topology is well suited to groups of users that are separated by some distance. IEEE 802.4 token bus networks are constructed with 75-ohm coaxial cable using a bus topology. The broadband characteristics of the 802.4 standard support transmission over several different channels simultaneously.

Token Passing Mechanism in Token Bus

In token bus network station must have possession of a token before it can transmit on the computer network. When the logical ring is initialized, the highest numbered station may send the first frame. The token and frames of data are passed from one station to another following the numeric sequence of the station addresses. Thus, the token follows a logical ring rather than a physical ring. The last station in numeric order passes the token back to the first station. The token does not follow the physical ordering of workstation attachment to the cable. Station 1 might be at one end of the cable and station 2 might be at the other, with station 3 in the middle. In such a case, there is no collision as only one station possesses a token at any given time. In token bus, each station receives each frame; the station whose address is specified in the frame processes it and the other stations discard the frame. This is depicted in the following diagram:

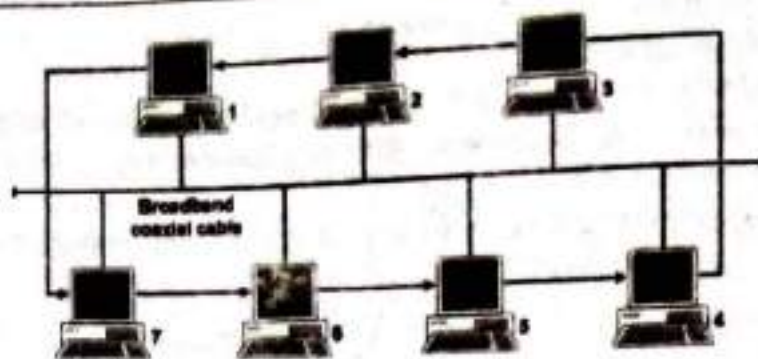


Figure: Token bus network

Token Ring

Token ring is the IEEE 802.5 standard for a token-passing ring in communication networks. A ring consists of a collection of ring interfaces connected by point-to-point lines i.e., ring interface of one station is connected to the ring interfaces of its left station as well as right station. Internally, signals travel around the communication network from one station to the next in a ring.

These point-to-point links can be created with twisted pair, coaxial cable or fiber optics. Each bit arriving at an interface is copied into a 1-bit buffer. In this buffer the bit is checked and may be modified and is then copied out to the ring again. This copying of bit in the buffer introduces a 1-bit delay at each interface.

Token Ring is a LAN protocol defined in the IEEE 802.5 where all stations are connected in a ring and each station can directly hear transmissions only from its immediate neighbor. Permission to transmit is granted by a message (token) that circulates around the ring. A token is a special bit pattern (3 bytes long). There is only one token in the network.

Token Passing Mechanism in Token Ring

If a station has a frame to transmit when it receives a token, it sends the frame and then passes the token to the next station; otherwise it simply passes the token to the next station. Passing the token means receiving the token from the preceding station and transmitting to the successor station. The data flow is unidirectional in the direction of the token passing. In order that tokens are not circulated infinitely, they are removed from the network once their purpose is completed. This is shown in the following diagram -

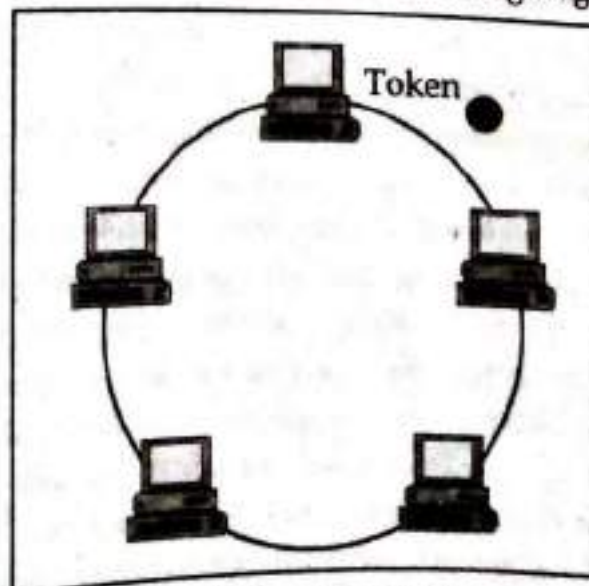


Figure: Token ring network

4. **Why do you think network traffic analysis is carried out? How does IPv6 overcome the disadvantages of IPv4?** [2+3]

Ans: Unlimited information exchange is one of the most significant results of today's advancing computing and information technologies. The world becomes smaller, as people get more connected every day. This evolution has brought many benefits to our society when it comes to information dissemination, international cooperation, business opportunities and more. Unfortunately, these advantages bring with them serious security threats. Information can be disseminated through unsecure avenues because anyone with basic knowledge of computers and internet computing can easily share information online. The various technological innovations have also given birth to a new generation of hackers, whose main objective is to steal and trade valuable information either for money or political purposes. This is why many companies have started to secure their firewalls, update their anti-malware software, and invest in a network security solution. Network traffic analysis helps in warding off different cyber-attacks like:

- Uses Behavioral Patterns to Spot Suspicious Activities
- Helps Pinpoint Invisible Threats
- Minimizes Damage and Profit-loss

IPv6 overcomes the disadvantages of IPv4 as follows:

- **More Efficient Routing:** IPv6 reduces the size of routing tables and makes routing more efficient and hierarchical. IPv6 allows ISPs to aggregate the prefixes of their customers' networks into a single prefix and announce this one prefix to the IPv6 Internet. In addition, in IPv6 networks, fragmentation is handled by the source device, rather than the router, using a protocol for discovery of the path's maximum transmission unit (MTU).
- **More Efficient Packet Processing:** IPv6's simplified packet header makes packet processing more efficient. Compared with IPv4, IPv6 contains no IP-level checksum, so the checksum does not need to be recalculated at every router hop. Getting rid of the IP-level checksum was possible because most link-layer technologies already contain checksum and error-control capabilities. In addition, most transport layers, which handle end-to-end connectivity, have a checksum that enables error detection.
- **Directed Data Flows:** IPv6 supports multicast rather than broadcast. Multicast allows bandwidth-intensive packet flows (like multimedia streams) to be sent to multiple destinations simultaneously, saving network bandwidth. Disinterested hosts no longer must process broadcast packets. In addition, the IPv6 header has a new field, named Flow Label that can identify packets belonging to the same flow.
- **Simplified Network Configuration:** Address auto-configuration (address assignment) is built in to IPv6. A router will send the prefix of the local link in its router advertisements. A host can generate its own IP address by appending its link-layer (MAC) address, converted into Extended Universal Identifier (EUI) 64-bit format, to the 64 bits of the local link prefix.

- **Support for New Services:** By eliminating Network Address Translation (NAT), true end-to-end connectivity at the IP layer is restored, enabling new and valuable services. Peer-to-peer networks are easier to create and maintain, and services such as VoIP and Quality of Service (QoS) become more robust.
- **Security:** IPSec, which provides confidentiality, authentication and data integrity, is baked into in IPv6. Because of their potential to carry malware, IPv4 ICMP packets are often blocked by corporate firewalls, but ICMPv6, the implementation of the Internet Control Message Protocol for IPv6, may be permitted because IPSec can be applied to the ICMPv6 packets.

5. Find Hamming Code for data 01100111. [5]

Ans: Total number of data bits ' m ' = 8

Number of redundant bits r : $2^r \geq m + r + 1$

$$2^r \geq 8 + r + 1$$

$$2^4 \geq 8 + 4 + 1 = 13 \text{ (Satisfied)}$$

Therefore, the value of r is 4 that satisfy the above relation and r_8, r_4, r_2, r_1 are redundant bits.

Total number of bits = $m + r = 8 + 4 = 12$;

0	1	1	0	r_8	0	1	1	r_4	1	r_2	r_1
12	11	10	9	8	7	6	5	4	3	2	1

r_1 : Checking even parity on bits 1, 3, 5, 7, 9, 11

r_1 : 1

r_2 : Checking even parity on bits 2, 3, 6, 7, 10, 11

r_2 : 0

r_4 : Checking even parity on bits 4, 5, 6, 7, 12

r_4 : 1

r_8 : Checking even parity on bits 8, 9, 10, 11, 12

r_8 : 0

Substituting the value of redundant bits, we get,

0	1	1	0	0	0	1	1	1	1	0	1
12	11	10	9	8	7	6	5	4	3	2	1

Therefore, Hamming Code for data 01100111 is 011000111101.

6. Differentiate between TCP and UDP.

Ans: Difference between TCP and UDP are shown in table below:

Characteristics / Description	UDP	TCP
General	Simple, High Speed, Low Functionality	Full-featured, Reliable Data Transfer
Connection Setup	Connectionless	Connection Oriented
Data Interface to Application	Message Based Data is Sent	Stream Based data is sent
Reliability	Unreliable (No ACK)	Reliability (ACK available)
Retransmission	Not Performed	Data is Lost → Retransmission
Flow Control	None	Sliding Window → Flow Control
Overhead	Very Low	Low, Higher than UDP
Transmission Speed	Very High	High, Lower than UDP

Application that uses Protocol	DNS, DHCP, SNMP, (Multimedia Application)	FTP, Telnet, SMTP, DNS, HTTP, etc.
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7. Explain DNS with reference to its hierarchy and records.

Ans: DNS is a global system for translating IP addresses to human-readable domain names. When a user tries to access a web address like "example.com", their web browser or application performs a DNS Query against a DNS server, supplying the hostname. The DNS server takes the hostname and resolves it into a numeric IP address, which then web browser can connect to.

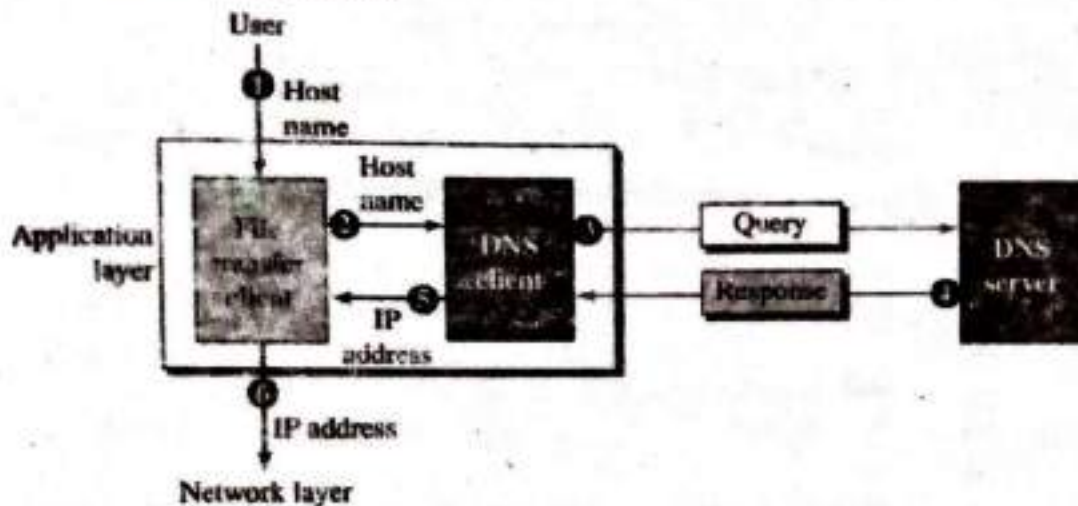


Figure: Purpose of DNS

Let's discuss the concepts and ideas behind the DNS. Above figure shows how TCP/IP uses a DNS client and a DNS server to map a name to an address. A user wants to use a file transfer client to access the corresponding file transfer server running on a remote host. The user knows only the file transfer server name. Such as *example.com*. However, the TCP/IP suite needs the IP address of the file transfer server to make the connection.

The following six steps map the host name to an IP address:

- The user passes the host name to the file transfer client.
- The file transfer client passes the host name to the DNS client.
- Each computer, after being booted, knows the address of one DNS server. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
- The DNS server responds with the IP address of the desired file transfer server.
- The DNS server passes the IP address to the file transfer client.
- The file transfer client now uses the received IP address to access the file transfer server.

Figure below shows an example of how DNS client/server program can support an e-mail program to find the IP address of an e-mail recipient. A user of an e-mail program may know the e-mail address of the recipient; however, the IP protocol needs the IP address. The DNS client program sends a request to a DNS server to map the e-mail address to the corresponding IP address.

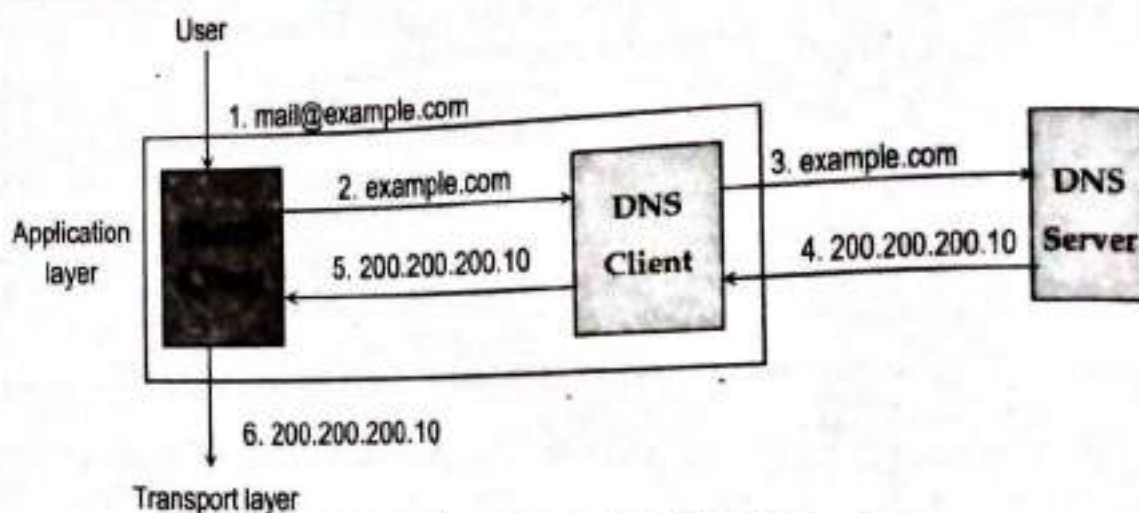


Figure: Example of using DNS service

DNS Hierarchy

The DNS hierarchy is comprised of the following elements:

- i) Root Level
- ii) Top Level Domains
- iii) Second Level Domains
- iv) Sub-Domain
- v) Host

For example, in the figure below, .edu is the top-level domain, berkeley is the second level domain, and .cs is the sub-domain of berkeley. Eos is the host name. A DNS server would store the IP address of the host where its name resides in the tree.

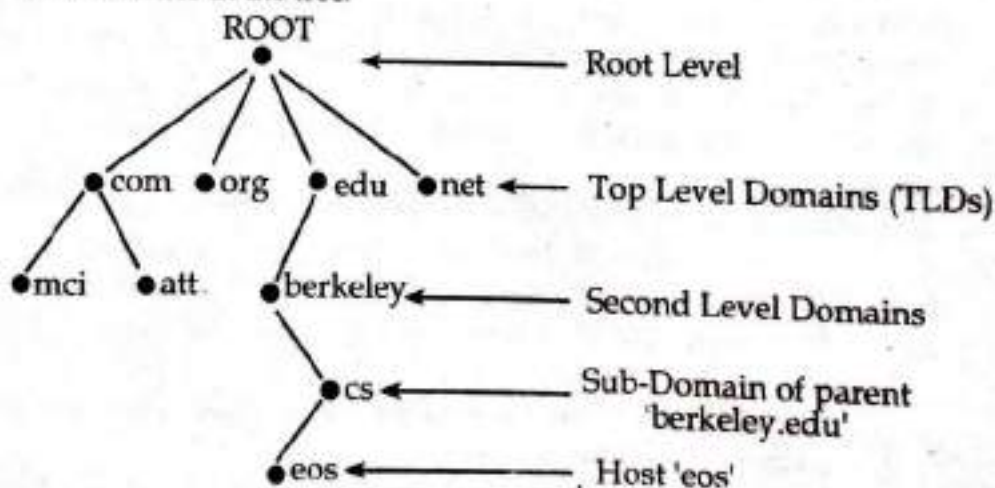


Figure: DNS Hierarchy

Types of DNS Records

Two types of records are used in DNS i.e. question records and resource records.

Question Record

A question record is used by the client to get information from a server. This contains the domain name. The list below describes question record fields.

- **Query name:** This is a variable-length field containing a domain name.
- **Query type:** This is a 16-bit field defining the type of query. Table below shows some of the types commonly used.

Type	Description
A	Address. A 32-bit IPv4 address
PTR	Pointer. It is used to convert an IP address to a domain name.
NS	Name Server. It identifies the authoritative servers for a zone
CNAME	Canonical name. It defines an alias for the official name of a host
SOA	Start of Authority. It marks the beginning of a zone
MX	Mail exchange. It redirects mail to a mail server
AAAA	Address. An IPv6 address
WKS	Well-known services. It defines the network services that a host provides.
AXFR	A request for the transfer of the entire zone.
ANY	A request for all records

- **Query class:** This is a 16-bit field defining the specific protocol using DNS.

Resource Record

Each domain name (each node on the tree) is associated with a record called the resource record. The server database consists of resource records. Resource records are also what is returned by the server to the client.

8. Write short notes on (Any Two): [2.5+2.5]

a. Multiplexing

Ans: Multiplexing is a technique used to combine and send the multiple data streams over a single medium. The process of combining the data streams is known as multiplexing and hardware used for multiplexing is known as a multiplexer. Multiplexing is achieved by using a device called **Multiplexer (MUX)** that combines n input lines to generate a single output line. Multiplexing follows many-to-one, i.e., n input lines and one output line. Demultiplexing is achieved by using a device called **Demultiplexer (DEMUX)** available at the receiving end. DEMUX separates a signal into its component signals (one input and n outputs). Therefore, we can say that demultiplexing follows the one-to-many approach.

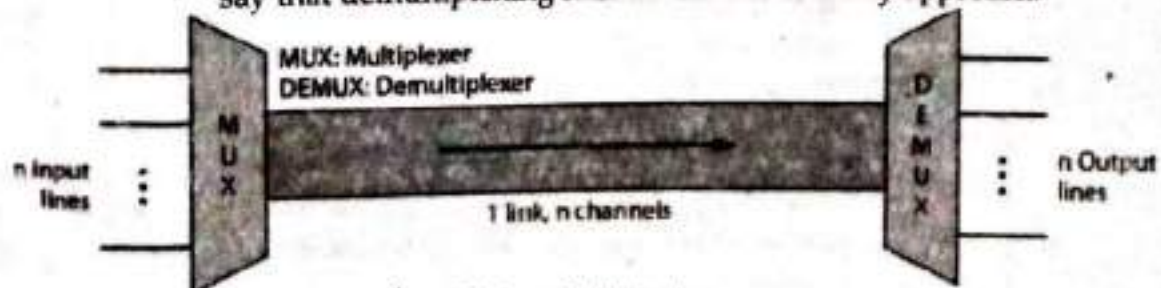


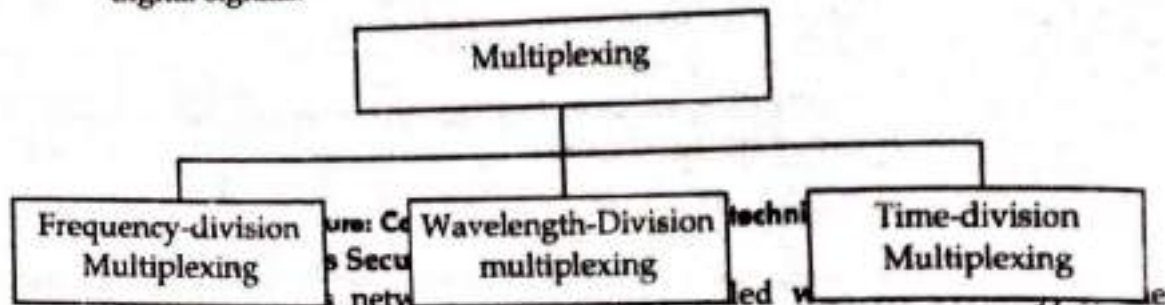
Figure: Multiplexing

Above figure depicts the multiplexing function in its simplest form. There are n inputs to a multiplexer. The multiplexer is connected by a single data link to a demultiplexer. The link is able to carry n separate channels of data. The multiplexer combines (multiplexes) data from the n input lines and transmits over a higher-capacity data link. The demultiplexer accepts the multiplexed data stream, separates (demultiplexes) the data according to channel, and delivers data to the appropriate output lines.

Advantages of Multiplexing:

- More than one signal can be sent over a single medium.
- The bandwidth of medium can be utilized effectively.

There are three basic multiplexing techniques: *frequency-division multiplexing, wavelength-division multiplexing, and time-division multiplexing*. The first two are techniques designed for analog signals, the third, for digital signals.



process of designing, implementing and ensuring security on a wireless computer network. It is a subset of network security that adds protection for a wireless computer network.

Wireless network security primarily protects a wireless network from unauthorized and malicious access attempts. Typically, wireless network security is delivered through wireless devices (usually a wireless router/switch) that encrypts and secures all wireless communication by default. Even if the wireless network security is compromised, the hacker is not able to view the content of the traffic/packet in transit. Moreover, wireless intrusion detection and prevention systems also enable protection of a wireless network by alerting the wireless network administrator in case of a security breach.

The wireless security protocols are:

- Wired Equivalent Policy (WEP):** WEP was developed for wireless networks and approved as a Wi-Fi security standard in September 1999. WEP was supposed to offer the same security level as wired networks, however there are a lot of well-known security issues in WEP, which is also easy to break and hard to configure. Despite all the work that has been done to improve the WEP system it still is a highly vulnerable solution. Systems that rely on this protocol should be either upgraded or replaced in case security upgrade is not possible. WEP was officially abandoned by the Wi-Fi Alliance in 2004.
- Wi-Fi Protected Access (WPA):** This wireless security protocol precedes the WEP. Hence, it is designed to deal with the flaws that are found with the WEP protocol. Most modern WPA applications use a pre-shared key (PSK), most often referred to as WPA Personal, and the Temporal Key Integrity Protocol (TKIP) for encryption. WPA Enterprise uses an authentication server for keys and certificates generation.
- Wi-Fi Protected Access 2 (WPA2):** The WPA2, a successor to WPA, comes with enhanced features and encryption abilities.

For instance, the WPA2 uses Counter Mode Cipher Block Chaining Message Authentication Code Protocol (CCMP) instead of (TKIP). This replacement feature is known to be efficient in encrypting data. Hence, WPA2 is considered the best wireless security protocol. At this time the main vulnerability to a WPA2 system is when the attacker already has access to a secured Wi-Fi network and can gain access to certain keys to perform an attack on other devices on the network.

- iv. **Wi-Fi Protected Access 3 (WPA3):** This one is a recent wireless protocol. It is enhanced in terms of encryption abilities and keeping hackers at bay from both private and public networks. WPA3 will protect against dictionary attacks by implementing a new key exchange protocol. WPA3's expanded encryption for public networks also keeps Wi-Fi users safe from a vulnerability they may not realize exists in the first place. In fact, if anything it might make Wi-Fi users feel too secure.

c. **ICMP**

Ans: Since IP does not have an inbuilt mechanism for sending error and control messages. It depends on Internet Control Message Protocol (ICMP) to provide an error control. ICMP is a protocol used primarily for sending error messages, performing diagnostics, and controlling the flow of data. An example of an error message and of flow control is an ICMP source-quench packet sent by a router to a source host to tell the host to slow down because the router is overloaded. Performing Diagnostics with ICMP: As stated earlier, the ICMP protocol is used for performing diagnostics. An example of using ICMP as a diagnostic tool is with the Ping utility. Ping stands for Packet Internet Groper.

ICMP messages are divided into two broad categories:

- Error-reporting messages
- Query messages.

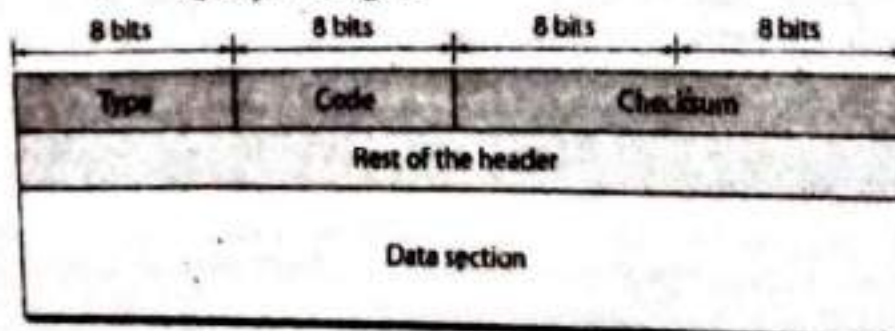


Figure: ICMP message.

The format of an ICMP message is shown above. The 8-bit type code identifies the types of message. Each ICMP message contains three fields that define its purpose and provide a checksum. They are TYPE, CODE, and CHECKSUM fields. The TYPE field identifies the ICMP message, the CODE field provides further information about the associated TYPE field, and the CHECKSUM provides a method for determining the integrity of the message.

Although over twenty ICMP messages have been defined, only a few are used. Table below lists key ICMP messages and the purpose of each.

Table: Examples of ICMP messages with the message number and purpose.

Number	Type	Purpose
0	Echo Reply	Used by the ping problem
3	Destination Unreachable	Datagram could not be delivered
5	Redirect	Host must change a route
8	Echo	Used by the ping program
11	Time Exceeded	TTL expired or fragments timed out
12	Parameter Problem	IP header is incorrect
30	Traceroute	Used by the traceroute program

As the above table illustrates, ICMP contains two message types: messages used to report errors and messages used to obtain information. For example, the *Time Exceeded* and *Destination Unreachable* messages each report an error when a datagram cannot be delivered successfully. A destination address is unreachable if no route exists to the address; a datagram times out if either the TTL count in the header expires or fragments of the datagram do not arrive before the reassembly timer expires. In contrast, the *Echo Request* and *Echo Reply* messages do not correspond to an error. Instead, they are used by the ping application to test connectivity — when it receives an *echo request* message, ICMP software on a host or router sends an *echo reply* that carries the same data as the request. Thus, a ping application sends a request to a remote host, waits for a reply, and either declares that the host is reachable, or after a suitable timeout, declares that the host is unreachable.

Group C

Attempt any TWO questions.

9. Why do we need layered protocol architecture? What are the Functions of 7 layers of OSI reference model? [2×10=20]

Ans: A layered architecture facilitates development in complex environments by grouping specific related functions into separate well-defined layers with clear interfaces. This methodology reduces complexity by breaking the problem space into smaller and simpler components and standardizes interfaces facilitating multi-vendor development and modular component-based engineering. Layered architectures in conjunction with open standards define a common vocabulary necessary for understanding and cooperation in multi-vendor environments and positively results in increased competition and innovation. The need of layered protocol architecture can be summarized as follow: [3+7]

- To Separate Specific Functions in Each Layer.
- Each Layer Should Define a Unique Function.
- To make their Implementation Transparent to Other Components.
- Allows Independent Design and Testing of Each Component.
- Modularization Eases Maintenance and Updating of System.

Functions of 7 layers of OSI model are as follows:

a) Physical Layer:

- Physical characteristics of interfaces and media

- Representation of bits
 - Data Rate or Transmission rate
 - Synchronization of bits
 - Line Configuration
 - Physical Topology
 - Transmission Mode
- b) **Data Link Layer:**
- Enable Node to Node (hop to hop) communication
 - Responsible for transmitting frames from one node to next.
 - Framing
 - Physical addressing
 - Flow Control
 - Error control
 - Access Control [E.g. CSMA/CD]
- c) **Network Layer:**
- Enables Host to Host Communication
 - Responsible for Delivery of Packets
 - Logical Addressing
 - Routing
- d) **Transport Layer:**
- Enables Process to Process Communication.
 - Port Addressing
 - Segmentation and Reassembly.
 - Connection Control
 - Flow Control
 - Error Control
- e) **Session Layer:**
- Control of Dialogues between Applications.
 - Whose Turn is to Transmit?
 - Dialogue Discipline => Half Duplex/ Full Duplex
- f) **Presentation Layer:**
- Data Formats and Coding.
 - Data Compression
 - Encryption
- g) **Application Layer:**
- Responsible for Providing Service to End Users.
 - Mail Transfer Service.
 - File Transfer Service.

10. **Highlight on the importance of routing algorithm. Explain Distance Vector Routing algorithm and compare it with link state routing. [3+3+4]**

Ans: In order to transfer the packets from source to the destination, the network layer must determine the best route through which packets can be transmitted. Whether the network layer provides datagram service or virtual circuit service, the main job of the network layer is to provide the best route. The routing protocol provides this job. The routing protocol is a routing algorithm that provides the best path from the source to the destination. The best path is the path that has the "least-cost path" from source to the destination. Routing is the process of forwarding the packets

from source to the destination but the best route to send the packets is determined by the routing algorithm. A routing algorithm is a set of step-by-step operations used to direct Internet traffic efficiently. When a packet of data leaves its source, there are many different paths it can take to its destination. The routing algorithm is used to determine mathematically the best path to take.

Distance Vector Routing Algorithm: It is a dynamic routing algorithm. A distance-vector routing protocol requires that a router inform its neighbors of topology changes periodically. The term distance vector refers to the fact that the protocol manipulates vectors (arrays) of distances to other nodes in the network. It makes use of Bellman Ford Algorithm for making routing tables. Problems of this algorithm are:

- Count to infinity problem which can be solved by splitting horizon.
- Good news spread fast and bad news spread slowly.
- Persistent looping problem i.e. loop will be there forever.

Distance Vector Routing Algorithm Working:

Step 1) Initialization

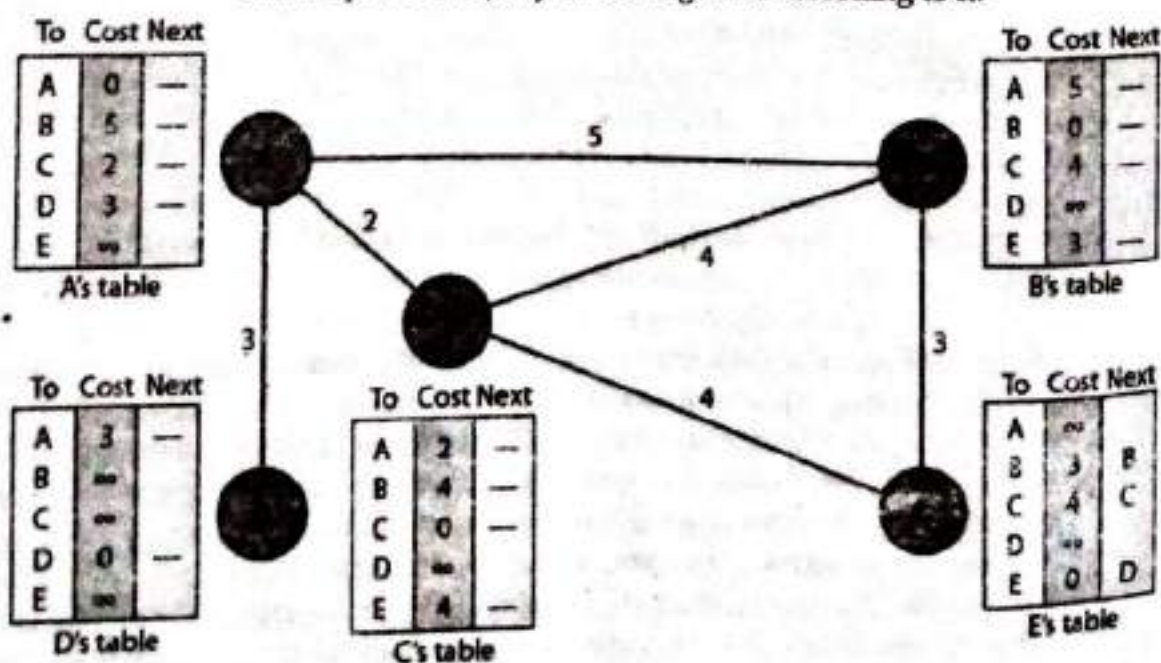
- At the beginning, each node knows the cost of itself and its immediate neighbor. [Node directly connected to it.]
- The distance of any entry that is not a neighbor is marked as infinite (unreachable).

Step 2) Sharing

- Idea is to share the information between neighbors.
- The node C does not know the distance about D, but C's neighbor node A and node B does.
- If node A and B share it routing table with C, node C can also know how to reach node D.

Step 3) Updating in distance vector routing

- When node C get routing tables from A and B, it will compare the distance of both nodes to go to D, whose ever distance is less, node C will update the entry in routing table according to it.

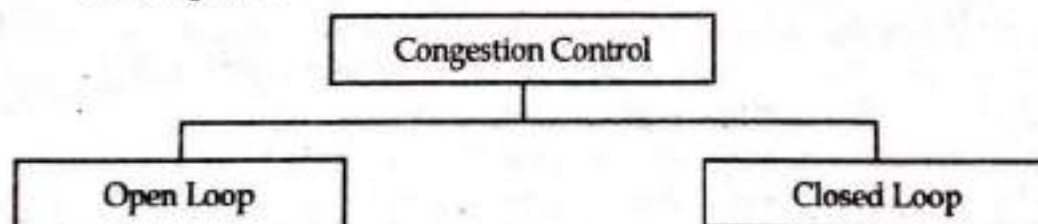


Comparison between Distance Vector Routing and Link State Routing

Distance Vector Routing	Link State Routing
1. Bandwidth required is less due to local sharing, small packets and no flooding.	1. Bandwidth required is more due to flooding and sending of large link state packets.
2. Based on local knowledge since it updates table based on information from neighbors.	2. Based on global knowledge i.e. it has knowledge about entire network
3. Makes use of Bellman Ford algorithm	3. Make use of Dijkstra's algorithm
4. Traffic is less	4. Traffic is more
5. Converges slowly i.e. good news spread fast and bad news spread slowly.	5. Converges faster
6. Count to infinity problem	6. No count to infinity problem
7. Persistent looping problem i.e. loop will there forever.	7. No persistent loops, only transient loops.
8. Practical implementation is RIP and IGRP	8. Practical implementation is OSPF and ISIS

11. Explain various congestion control approaches. [10]

Ans: Congestion control refers to the techniques used to control or prevent congestion. Congestion control techniques can be broadly classified into two categories:



Open Loop Congestion Control

Open loop congestion control policies are applied to prevent congestion before it happens. The congestion control is handled either by the source or the destination.

Policies adopted by open loop congestion control -

- Retransmission Policy:** It is the policy in which retransmission of the packets are taken care. If the sender feels that a sent packet is lost or corrupted, the packet needs to be retransmitted. This transmission may increase the congestion in the network. To prevent congestion, retransmission timers must be designed to prevent congestion and also able to optimize efficiency.
- Window Policy:** The type of window at the sender side may also affect the congestion. Several packets in the Go-back-n window are resent, although some packets may be received successfully at the receiver side. This duplication may increase the congestion in the network and making it worse. Therefore, Selective repeat window should be adopted as it sends the specific packet that may have been lost.
- Acknowledgment Policy:** Since acknowledgement is also the part of the load in network, the acknowledgment policy imposed by the receiver may also affect congestion. Several approaches can be used

to prevent congestion related to acknowledgment. The receiver should send acknowledgement for N packets rather than sending an acknowledgement for a single packet. The receiver should send an acknowledgement only if it has to send a packet or a timer expires.

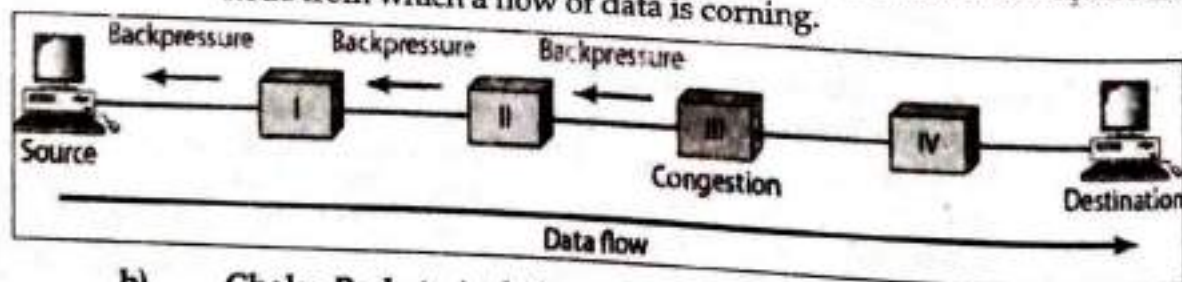
- d) **Discarding Policy:** A good discarding policy adopted by the routers is that the routers may prevent congestion and at the same time partially discards the corrupted or less sensitive package and also able to maintain the quality of a message. In case of audio file transmission, routers can discard less sensitive packets to prevent congestion and also maintain the quality of the audio file.
- e) **Admission Policy:** In admission policy a mechanism should be used to prevent congestion. Switches in a flow should first check the resource requirement of a network flow before transmitting it further. If there is a chance of congestion or there is congestion in the network, router should deny establishing a virtual network connection to prevent further congestion.

All the above policies are adopted to prevent congestion before it happens in the network.

Closed Loop Congestion Control

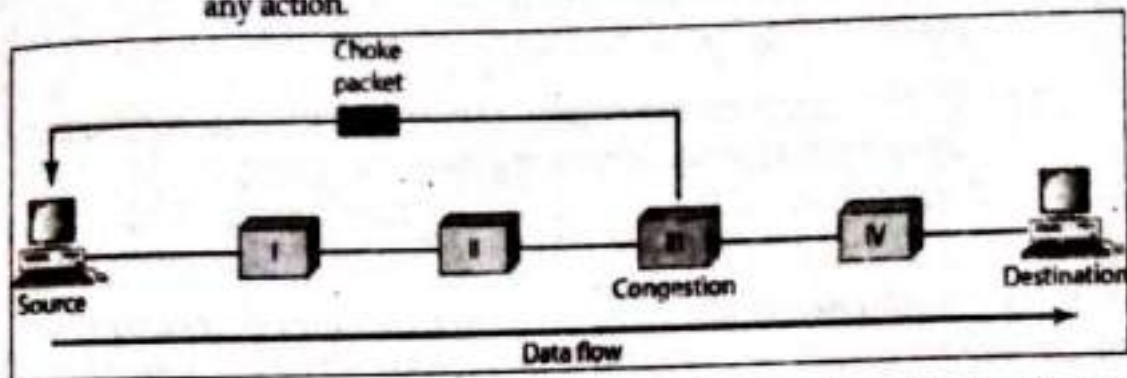
Closed loop congestion control technique is used to treat or alleviate congestion after it happens. Several techniques are used by different protocols; some of them are:

- a) **Back-pressure:** The technique of backpressure refers to a congestion control mechanism in which a congested node stops receiving data from the immediate upstream node or nodes. This may cause the upstream node or nodes to become congested, and they, in turn, reject data from their upstream nodes or nodes. And so on. Backpressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow, to the source. The backpressure technique can be applied only to virtual circuit networks, in which each node knows the upstream node from which a flow of data is coming.



- b) **Choke Packet:** A choke packet is a packet sent by a node to the source to inform it of congestion. Note the difference between the backpressure and choke packet methods. In backpressure, the warning is from one node to its upstream node, although the warning may eventually reach the source station. In the choke packet method, the warning is from the router, which has encountered congestion, to the source station directly. The intermediate nodes through which the packet has traveled are not warned. We have seen an example of this type of control in ICMP. When a router in the Internet is overwhelmed datagrams, it may discard some of them; but it informs the source. Host, using a

source quench ICMP message. The warning message goes directly to the source station; the intermediate routers, and does not take any action.



- c) **Implicit Signaling:** In implicit signaling, there is no communication between the congested node or nodes and the source. The source guesses that there is congestion somewhere in the network from other symptoms. For example, when a source sends several packets and there is no acknowledgment for a while, one assumption is that the network is congested. The delay in receiving an acknowledgment is interpreted as congestion in the network; the source should slow down.
- d) **Explicit Signaling:** The node that experiences congestion can explicitly send a signal to the source or destination. The explicit signaling method, however, is different from the choke packet method. In the choke packet method, a separate packet is used for this purpose; in the explicit signaling method, the signal is included in the packets that carry data. Explicit signaling, as we will see in Frame Relay congestion control, can occur in either the forward or the backward direction.
- **Backward Signaling:** A bit can be set in a packet moving in the direction opposite to the congestion. This bit can warn the source that there is congestion and that it needs to slow down to avoid the discarding of packets.
 - **Forward Signaling:** A bit can be set in a packet moving in the direction of the congestion. This bit can warn the destination that there is congestion. The receiver in this case can use policies, such as slowing down the acknowledgments, to alleviate the congestion.

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

QUESTIONS-ANSWERS TU-2020

Bachelor in Computer Applications
Course Title: Computer Networking
Code No: CACS303
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.

Group B

Attempt any six questions.

[6×5=30]

2. Define protocols. Explain WWW and HTTP protocol.

[1+2+2]

Ans: In information technology, a protocol is the special set of rules that end points in a telecommunication connection use when they communicate. In brief a protocol is defined as a set of rules that governs communication. Protocols exist at several levels in a telecommunication connection. For example, there are protocols for the data interchange at the hardware device level and protocols for data interchange at the application program level. Some examples of network protocols are hyper-text transfer protocol (HTTP), file transfer protocol (FTP), transmission control protocol/ internet protocol (TCP/IP), secure sockets layer etc.

WWW

The World Wide Web (WWW), or the Web, is a repository of information spread all over the world and linked together. The WWW has a unique combination of flexibility, portability, and user-friendly features that distinguish it from other services provided by the Internet. The WWW project was initiated by CERN (European Laboratory for Particle Physics) to create a system to handle distributed resources necessary for scientific research.

Today it is a distributed client-server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called websites, as shown in figure 6.12. Each website holds one or more documents, referred to as web pages. Each web page, however can contain some links to other web pages in the same or other websites. The pages can be retrieved and viewed by using browsers.

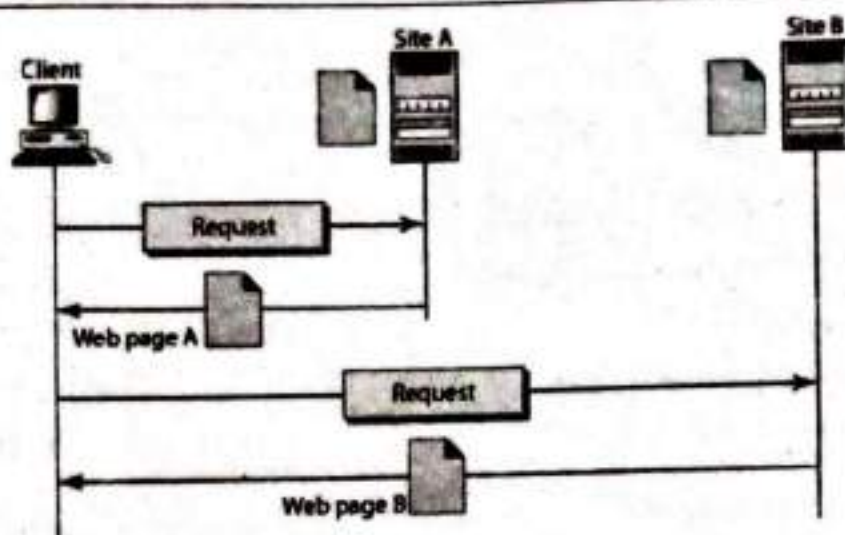


Figure: Architecture of WWW

Let us go through the scenario shown in above figure. The client needs to see some information that it knows belongs to site A. It sends a request through its browser, a program that is designed to fetch Web documents. The request, among other information, includes the address of the site and the Web page, called the URL. The server at site A finds the document and sends it to the client. When the user views the document, s/he finds some references to other documents, including a Web page at site B. The reference has the URL for the new site. The user is also interested in seeing this document. The client sends another request to the new site, and the new page is retrieved.

The above example shows the idea of **hypertext** and **hypermedia**. Hypertext means creating documents that refer to other documents. In a hypertext document, a part of text can be defined as a link to another document. When a hypertext is viewed with a browser, the link can be clicked to retrieve the other document. Hypermedia is a term applied to document that contains links to other textual document or documents containing graphics, video, or audio.

HTTP

The Hypertext Transfer Protocol (HTTP) is a protocol used mainly to access data on the World Wide Web. The protocol transfers data in the form of plain text, hypertext, audio, video, and so on. It is called the Hypertext Transfer Protocol because its efficiency allows its use in a hypertext environment where there are rapid jumps from one document to another.

The HTTP controls the transactions between a web client and a web server. The HTTP protocol transparently makes use of DNS and other Internet protocols to form connections between the web client and the web server, so the user is aware of only the web site's domain name and the name of the document itself. HTTP uses the services of TCP on well-known port 80.

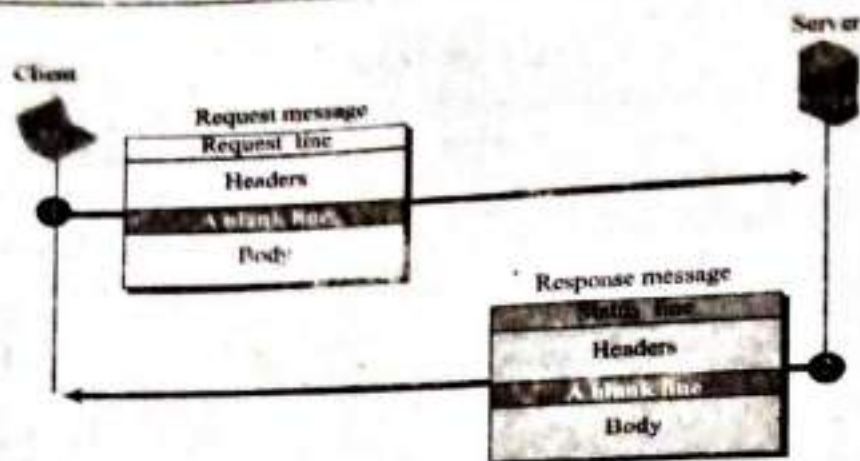


Figure: HTTP transaction

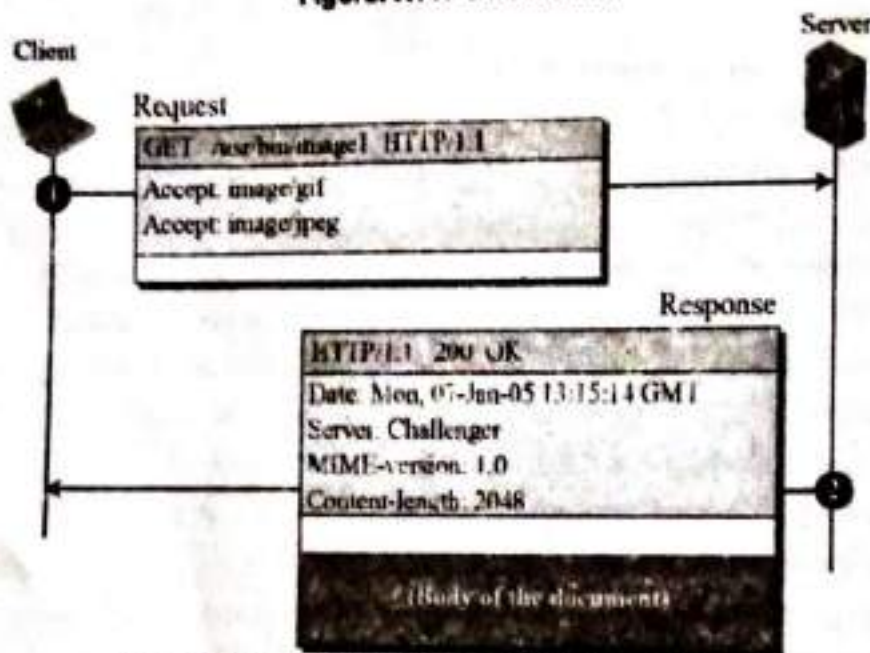


Figure: Request (GET Method) /response example

The above example retrieves a document. We use the GET method to retrieve an image with the path /usr/bin/image1. The request line shows the method (GET), the URL, and the HTTP version (1.1). The header has two lines that show that the client can accept images in the GIF or JPEG format. The request does not have a body. The response message contains the status line and four lines of header. The header lines define the date, server, MIME version, and length of the document. The body of the document follows the header.

3. Define transmission impairment. Explain the causes of impairments.

Ans: When signal travels through the medium, which are not perfect? The imperfection cause signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. Three causes of impairment are attenuation, distortion and noise. [1+4]

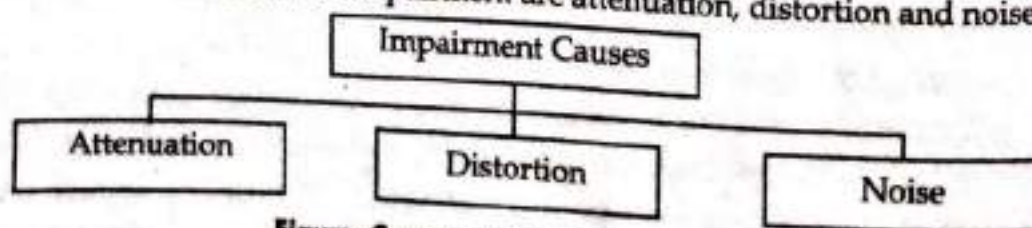


Figure: Causes of Impairment

Attenuation

Attenuation means a loss of energy. For the receiver to interpret the data accurately, the signal must be sufficiently strong. When the signal passes through the medium, it tends to get weaker. As it covers distance, it loses strength. To compensate for this loss, amplifiers are used to amplify the signal. Figure below shows the effect of attenuation and amplification.

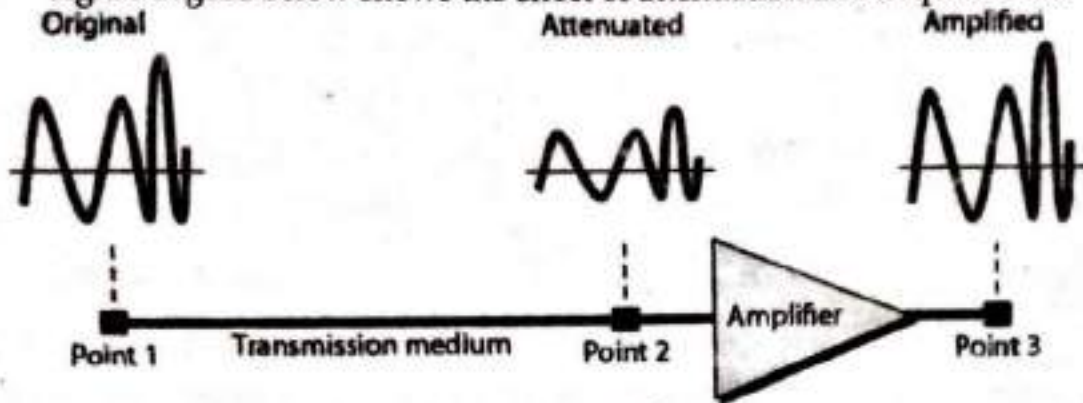


Figure: Attenuation

Distortion

Distortion means that the signal changes its form or shape. Signals are sent over media with pre-defined speed and frequency. If the signal speed and frequency do not match, there are possibilities that signal reaches destination in arbitrary fashion. In digital media, this is very critical that some bits reach earlier than the previously sent ones. In other words, signal components at the receiver have phase different from what they had at the sender. The shape of the composite signal is therefore not the same. Figure below shows the effect of distortion on a composite signal.

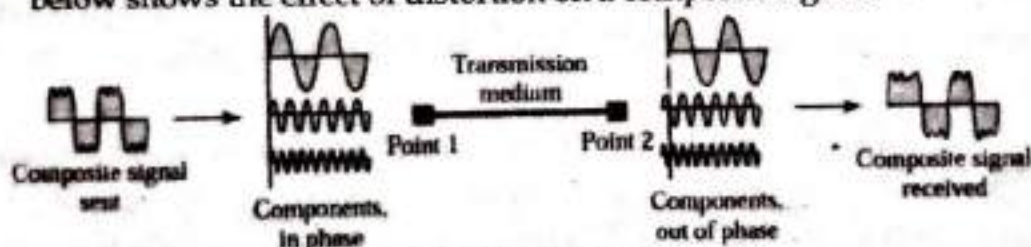


Figure: Distortion

Noise

Random disturbance or fluctuation in analog or digital signal is said to be Noise in signal, which may distort the actual information being carried. Several types of noise, such as thermal noise, intermodulation, crosstalk and impulse noise, may corrupt the signal.

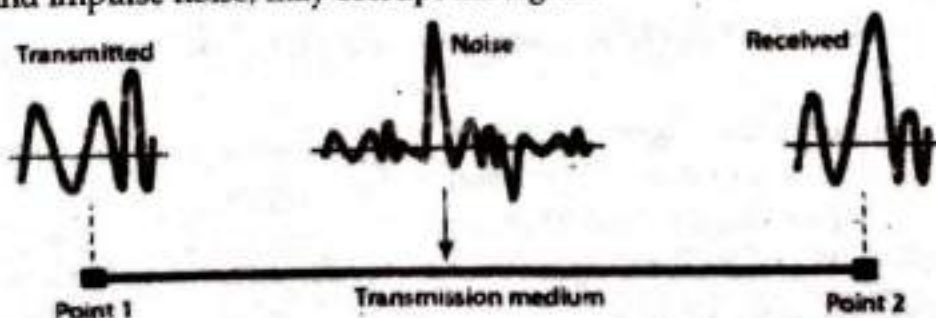


Figure: Noise

- **Thermal Noise**

Heat agitates the electronic conductors of a medium which may introduce noise in the media. Up to a certain level, thermal noise is unavoidable.

- **Intermodulation**
When multiple frequencies share a medium, their interference can cause noise in the medium. Intermodulation noise occurs if two different frequencies are sharing a medium and one of them has excessive strength or the component itself is not functioning properly, then the resultant frequency may not be delivered as expected.
- **Crosstalk**
This sort of noise happens when a foreign signal enters into the media. This is because signal in one medium affects the signal of second medium.
- **Impulse**
This noise is introduced because of irregular disturbances such as lightening, electricity, short-circuit, or faulty components. Digital data is mostly affected by this sort of noise.

3. **Define HDLC. Explain the HDLC frame formats.** [1+4]

Ans: High-level Data Link Control (HDLC) is a bit-oriented protocol for communication over point-to-point and multipoint links. HDLC is a synchronous Data Link layer bit-oriented protocol developed by the International Organization for Standardization (ISO). The current standard for HDLC is ISO 13239. HDLC was developed from the Synchronous Data Link Control (SDLC) standard proposed in the 1970s. HDLC provides both connection-oriented and connectionless service. HDLC uses synchronous serial transmission to provide error free communication between two points.

HDLC defines a Layer 2 framing structure that allows for flow control and error control through the use of acknowledgments. Each frame has the same format, whether it is a data frame or a control frame. When you want to transmit frames over synchronous or asynchronous links, you must remember that those links have no mechanism to mark the beginnings or ends of frames. HDLC uses a frame delimiter, or flag, to mark the beginning and the end of each frame.

To provide the flexibility necessary to support all the options possible in the modes and configurations just described, HDLC defines three types of frames:

1. Information frames (I-frames),
2. Supervisory frames (S-frames), and
3. Unnumbered frames (U-frames).

Each type of frame serves as an envelope for the transmission of a different type of message. I-frames are used to data-link user data and control information relating to user data (piggybacking). S-frames are used only to transport control information. U-frames are reserved for system management. Information carried by U-frames is intended for managing the link itself.

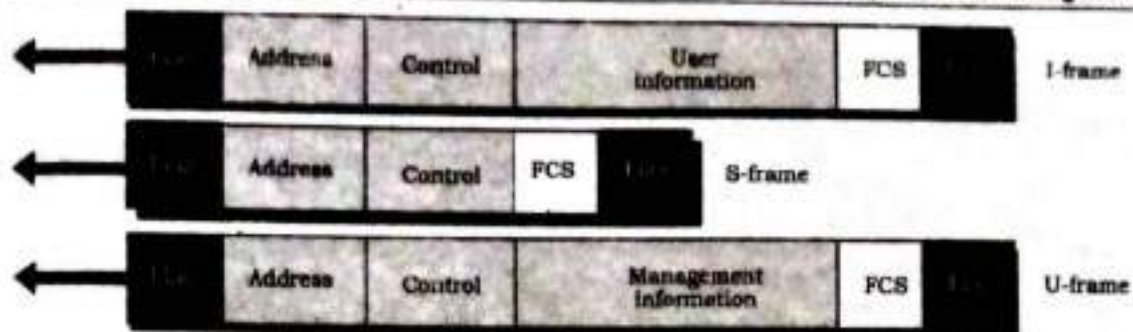


Figure: HDLC frames

Let us now discuss the fields and their use in different frame types.

- **Flag field.** This field contains synchronization pattern 01111110, which identifies both the beginning and the end of a frame.
- **Address field.** This field contains the address of the secondary station. If a primary station created the frame, it contains a to address. If a secondary station creates the frame, it contains a from address. The address field can be one byte or several bytes long, depending on the needs of the network.
- **Control field.** The control field is one or two bytes used for flow and error control. The interpretations of bits are discussed later.

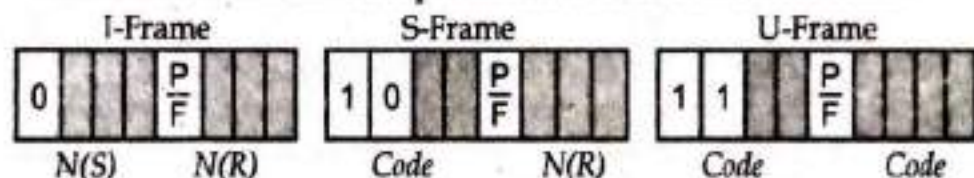


Figure: Control field format for the different frame types

- **Information field.** The information field contains the user's data from the network layer or management information. Its length can vary from one network to another.
 - **FCS field.** The frame check sequence (FCS) is the HDLC error detection field. It can contain either a 2- or 4-byte CRC.
4. **Define IP Address. Specify IPv4 address classes with their address ranges.** [1+4]

Ans: An IP address is a unique address that identifies a device on the internet or a local network. IP stands for "Internet Protocol," which is the set of rules governing the format of data sent via the internet or local network.

IPv4 addresses are 32-bit numbers that are typically displayed in dotted decimal notation. A 32-bit address contains two primary parts: the network prefix and the host number. All hosts within a single network share the same network address. Each host also has an address that uniquely identifies it.

IPv4 classfull addresses

To provide flexibility in the number of addresses distributed to networks of different sizes, 4-octet (32-bit) IP addresses are divided into five different categories or classes: **Class A**, **Class B**, **Class C**, **Class D** and **Class E**. Each address class specifies a different number of bits for its network prefix and host number as shown in following diagram:

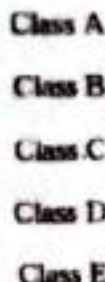


Figure: IPv4 address classes

Class	Leading bit	First Octet IP Range		IP Range		Remark
		Binary	Decimal Dotted	Start	End	
A	0	00000000 - 01111111	0 - 127	0.0.0.0	127.255.255.255	0 and 127 are reserved.
B	10	10000000 - 10111111	128 - 191	128.0.0.0	191.255.255.255	
C	110	11000000 - 11011111	192 - 223	192.0.0.0	223.255.255.255	
D	1110	11100000 - 11101111	224 - 239	224.0.0.0	239.255.255.255	Multicast address
E	1111	11110000 - 11111111	240 - 255	240.0.0.0	255.255.255.255	Reserved for future use

Figure: IPv4 classes and their IP ranges

5. Define Subnetting. Suppose you are given network address: 192.168.10.0 and subnet mask: 255.255.255.240 then calculate total number of subnets and numbers of hosts per subnet. [1+2+2]

Ans: Subnetting is the strategy used to partition a single physical network into more than one smaller logical sub-networks (subnets). Subnets are designed by accepting bits from the IP address's host part and using these bits to assign a number of smaller sub-networks inside the original network. Subnetting allows an organization to add sub-networks without the need to acquire a new network number via the Internet service provider (ISP). Subnetting helps to reduce the network traffic and conceals network complexity. Subnetting is essential when a single network number has to be allocated over numerous segments of a local area network (LAN).
IP address: 192.168.10.0

New Subnet Mask: 255.255.255.240 = 11111111 11111111
 11111111 11110000

Total Subnets = $2^4 = 16$

Total hosts = $2^4 = 16$

Usable hosts = $16 - 2 = 14$

Valid Subnets (4th octet) = 256 - 240 = 16

Starting Address (Network ID)	Usable Host IP Range		Ending Address (Broadcast ID)
	First host	Last host	
192.168.10.0	192.168.10.1	192.168.10.14	192.168.10.15
192.168.10.16	192.168.10.17	192.168.10.30	192.168.10.31
192.168.10.32	192.168.10.33	192.168.10.46	192.168.10.47
192.168.10.48	192.168.10.49	192.168.10.62	192.168.10.63
192.168.10.64	192.168.10.65	192.168.10.78	192.168.10.79
192.168.10.80	192.168.10.81	192.168.10.94	192.168.10.95
192.168.10.96	192.168.10.97	192.168.10.110	192.168.10.111

192.168.10.112	192.168.10.113	192.168.10.126	192.168.10.127
192.168.10.128	192.168.10.129	192.168.10.142	192.168.10.143
192.168.10.144	192.168.10.145	192.168.10.158	192.168.10.159
192.168.10.160	192.168.10.161	192.168.10.174	192.168.10.175
192.168.10.176	192.168.10.177	192.168.10.190	192.168.10.191
192.168.10.192	192.168.10.193	192.168.10.206	192.168.10.207
192.168.10.208	192.168.10.209	192.168.10.222	192.168.10.223
192.168.10.224	192.168.10.225	192.168.10.238	192.168.10.239
192.168.10.240	192.168.10.241	192.168.10.254	192.168.10.255

6. Draw a user Datagram format. Explain UDP operations. [2+3]

Ans: A UDP datagram header contains four fields of two bytes each as shown in figure below.

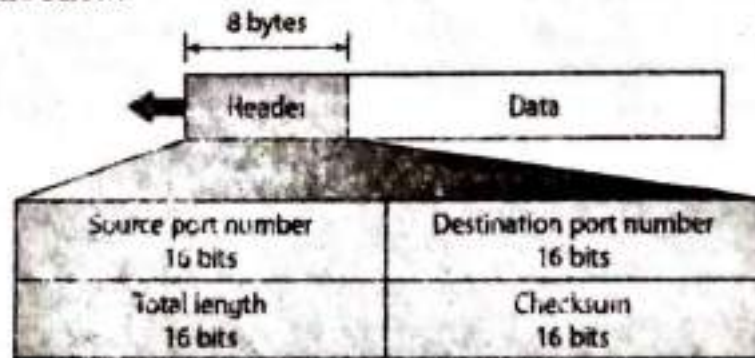


Figure: User Datagram Header Format

- **Source Port:** Source Port is 2 Byte long fields used to identify port number of sources.
- **Destination Port:** It is 2 Byte long fields, used to identify the port of destined packet.
- **Length:** Length is the length of UDP including header and the data. It is 16-bits field.
- **Checksum:** Checksum is 2 Bytes long field. It is the 16-bit one's complement of the one's complement sum of the UDP header, pseudo header of information from the IP header and the data, padded with zero octets at the end (if necessary) to make a multiple of two octets.

UDP operations

UDP (User Datagram Protocol) is an alternative communications protocol to Transmission Control Protocol (TCP) used primarily for establishing low-latency and loss-tolerating connections between applications on the internet.

UDP uses a simple transmission model but does not employ handshaking dialogs for reliability, ordering and data integrity. The protocol assumes that error-checking and correction is not required, thus avoiding processing at the network interface level. UDP is widely used in video conferencing and real-time computer games. The protocol permits individual packets to be dropped and UDP packets to be received in a different order than that in which they were sent, allowing for better performance. UDP network traffic is organized in the form of datagrams, which comprise one message units. The first eight bytes of a datagram contain header information, while the remaining bytes contain message data.

As it is a connectionless protocol, it is not at all reliable protocols when compared to the TCP. It will never offer any sequencing of the data. Hence, the data will arrive at the destination device in the various orders from which it is sent. This will occur in the large networks like the internet, where datagrams take various paths to a destination and also experience the delay in the different router. The UDP is generally the IP with the transport layer port addressing. Sometimes this UDP is also known as the wrapper protocol. [2.5+2.5]

7. Write short notes on (Any Two):

(a) DNS

Ans: DNS is a global system for translating IP addresses to human-readable domain names. When a user tries to access a web address like "example.com", their web browser or application performs a DNS Query against a DNS server, supplying the hostname. The DNS server takes the hostname and resolves it into a numeric IP address, which then web browser can connect to.

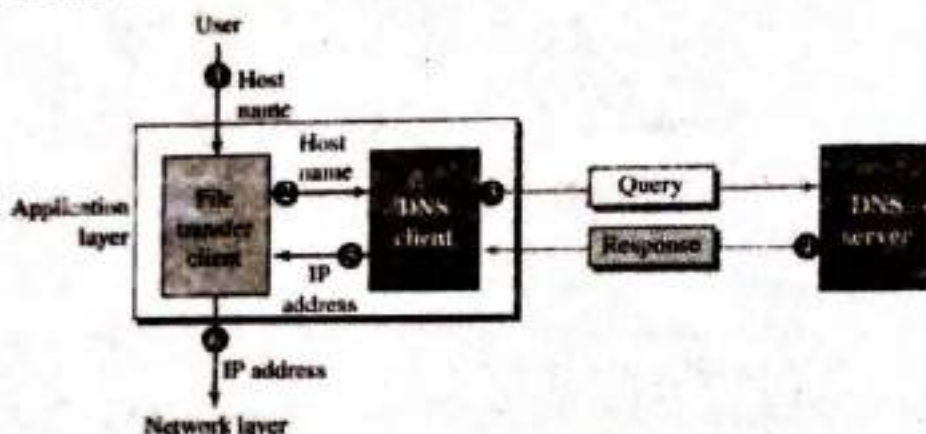


Figure: Purpose of DNS

Let's discuss the concepts and ideas behind the DNS. Above figure shows how TCP/IP uses a DNS client and a DNS server to map a name to an address. A user wants to use a file transfer client to access the corresponding file transfer server running on a remote host. The user knows only the file transfer server name. Such as example.com. However, the TCP/IP suite needs the IP address of the file transfer server to make the connection.

The following six steps map the host name to an IP address:

1. The user passes the host name to the file transfer client.
2. The file transfer client passes the host name to the DNS client.
3. Each computer, after being booted, knows the address of one DNS server. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
4. The DNS server responds with the IP address of the desired file transfer server.
5. The DNS server passes the IP address to the file transfer client.
6. The file transfer client now uses the received IP address to access the file transfer server.

Note that the purpose of accessing the Internet is to make a connection between the file transfer client and server, but before this can happen,

another connection needs to be made between the DNS client and DNS server. In other words, we need at least two connections in this case. The first is for mapping the name to an IP address; the second is for transferring files.

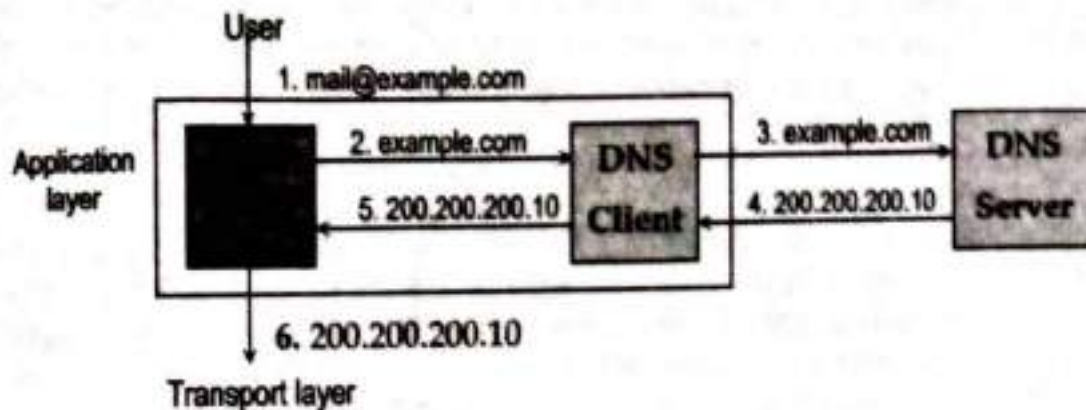


Figure: Example of using DNS service

Above figure shows an example of how DNS client/server program can support an e-mail program to find the IP address of an e-mail recipient. A user of an e-mail program may know the e-mail address of the recipient; however, the IP protocol needs the IP address. The DNS client program sends a request to a DNS server to map the e-mail address to the corresponding IP address.

(b) Public Key Cryptography

Ans: Public Key Cryptography use one key to encrypt data and a different key to decrypt data. One key is public and the other is private. In a public-key encryption system, any person can encrypt a message using the public key of the receiver, and the receiver is the only one that can decrypt it using his private key. Parties exchange secure messages without needing a pre-shared key, as shown in figure below. Asymmetric algorithms are more complex. These algorithms are resource intensive and slower to execute.

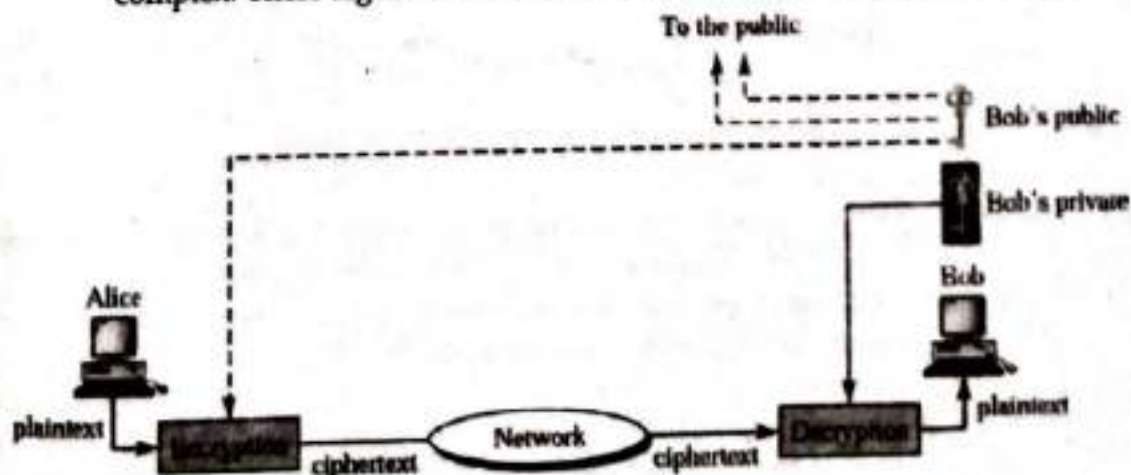


Figure: Public-key cryptography

We have the same situation as the symmetric-key algorithm, with a few exceptions. First, there are two keys instead of one: **public key** and **private key**. To send a secured message to Bob, Alice first encrypts the message using Bob's public key. To decrypt the message, Bob uses his own Private Key. Some algorithm based on public-key cryptography are RSA, DSA, Diffie-Hellman etc.

(c) VPN

Ans: A Virtual Private Network (VPN) provides a secure connection between a sender and a receiver over a public non-secure network such as the Internet. A secure connection is generally associated with private networks. (A private network is a network that is owned, or at least controlled via leased lines, by an organization.) A VPN can transform the characteristics of a public non-secure network into those of a private secure network. VPNs reduce remote access costs by using public network resources. Compared to other solutions, including private networks, a VPN is inexpensive. VPN has two types: *Remote Access VPN* and *Site-to-site VPN*. VPN technology must do the following activities:

- **IP encapsulation:** This involves enclosing TCP/IP data packets within another packet with an IP address of either a firewall or a server that acts as a VPN end-point. This encapsulation of host IP address helps in hiding the host.
- **Encryption:** It is done on the data part of the packet. Just like in SSL, the encryption can be done either in transport mode which encrypts its data at the time of generation or tunnel mode which encrypts and decrypts data during transmission encrypting both data and header.
- **Authentication:** It involves creating an encryption domain which includes authenticating computers and data packets by user for public encryption.

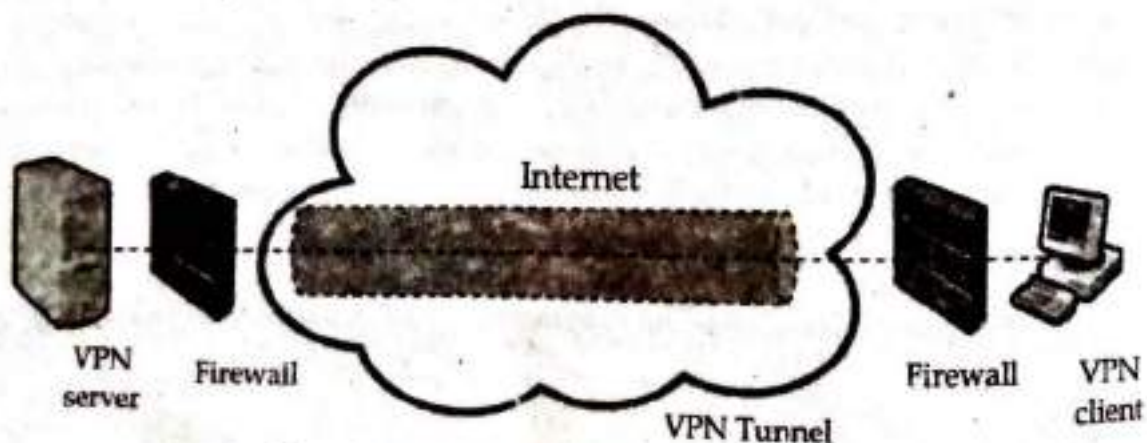


Figure: Virtual Private Network (VPN) Model

Advantages of VPNs

- VPNs are inexpensive; they provide remote and mobile user with access to the corporate network at the price of a local call.
- VPNs also provide the framework for corporate **intranets** and **extranets**. Corporations can exploit the global nature of the Internet and use VPNs to link all branch offices into private networks called intranets. A corporation can make certain sections of its intranet accessible to its vendors and strategic partners by means of the extranet.
- VPN tunnels permit non-routable protocols to be delivered to specific LAN segments in the corporate intranet. In this way, VPNs enable legacy applications to use the intranet.
- VPNs have also contributed significantly to the increased use of private IP addresses. Since applications are tunneled rather than routed across the WAN, companies can assign their own addresses, as long as these addresses are not advertised to the outside world.

Group C

Attempt any TWO questions.

[2×10=20]

8. Critically analyze the OSI reference model.

[10]

Ans: Neither the OSI model and its protocols nor the TCP/IP model and its protocols are perfect. The criticism of the OSI model and its protocols can be summarized as:

(a) **Bad timing**

The competing TCP/IP protocols were already in widespread use by research universities by the time the OSI protocols appeared. While the billion-dollar wave of investment had not yet hit, the academic market was large enough that many vendors had begun cautiously offering TCP/IP products. When OSI came around, they did not want to support a second protocol stack until they were forced to, so there were no initial offerings. With every company waiting for every other company to go first, no company went first and OSI never happened.

(b) **Bad Technology**

The second reason that OSI never caught on is that both the model and the protocols are flawed. The choice of seven layers was more political than technical, and two of the layers (session and presentation) are nearly empty, whereas two other ones (data link and network) are overfull. The OSI model, along with its associated service definitions and protocols, is extraordinarily complex. They are also difficult to implement and inefficient in operation. In addition to being incomprehensible, another problem with OSI is that some functions, such as addressing, flow control, and error control, reappear again and again in each layer.

(c) **Bad Implementations**

Given the enormous complexity of the model and the protocols, it will come as no surprise that the initial implementations were huge, unwieldy, and slow. It did not take long for people to associate "OSI" with "poor quality". Although the products improved in the course of time, the image stuck. In contrast, one of the first implementations of TCP/IP was part of Berkeley UNIX and was quite good. People began using it quickly, which led to a large user community, which led to improvements, which led to an even larger community. Here the spiral was upward instead of downward.

(d) **Bad Politics**

On account of the initial implementation, many people, especially in academia, thought of TCP/IP as part of UNIX, and UNIX in the 1980s in academia was not unlike parenthood and apple pie. OSI, on the other hand, was widely thought to be the creature of the European telecommunication ministries. The very idea of a bunch of government bureaucrats trying to shove a technically inferior standard down the throats of the poor researchers and programmers down in the trenches actually developing computer networks did not aid OSI's cause.

The TCP/IP model and protocols have their problems too. First, the model does not clearly distinguish the concepts of services, interfaces, and protocols. Good software engineering practice requires differentiating between the specification and the implementation, something that OSI does very carefully, but TCP/IP does not. Consequently, the TCP/IP model is not much of a guide for designing new networks using new technologies.

Second, the TCP/IP model is not at all general and is poorly suited to describing any protocol stack other than TCP/IP. Trying to use the TCP/IP model to describe Bluetooth, for example, is completely impossible.

Third, the link layer is not really a layer at all in the normal sense of the term as used in the context of layered protocols. It is an interface (between the network and data link layers). The distinction between an interface and layer is crucial and one should not be sloppy about it.

Fourth, the TCP/IP model does not distinguish between the physical and data link layers. These are completely different. The physical layer has to do with the transmission characteristics of copper wire, fiber optics, and wireless communication. The data link layer's job is to delimit the start and end of frames and get them from one side to the other with the desired degree of reliability. A proper model should include both as separate layers. The TCP/IP model does not do this.

Finally, although the IP and TCP protocols were carefully thought out and well implemented, many of the other protocols were ad hoc, generally produced by a couple of graduate students hacking away until they got tired. The protocol implementations were then distributed free, which resulted in their becoming widely used, deeply entrenched, and thus hard to replace. Some of them are a bit of an embarrassment now.

9. Explain the random-access protocols under the multiple access taxonomy.

Ans:

[10]

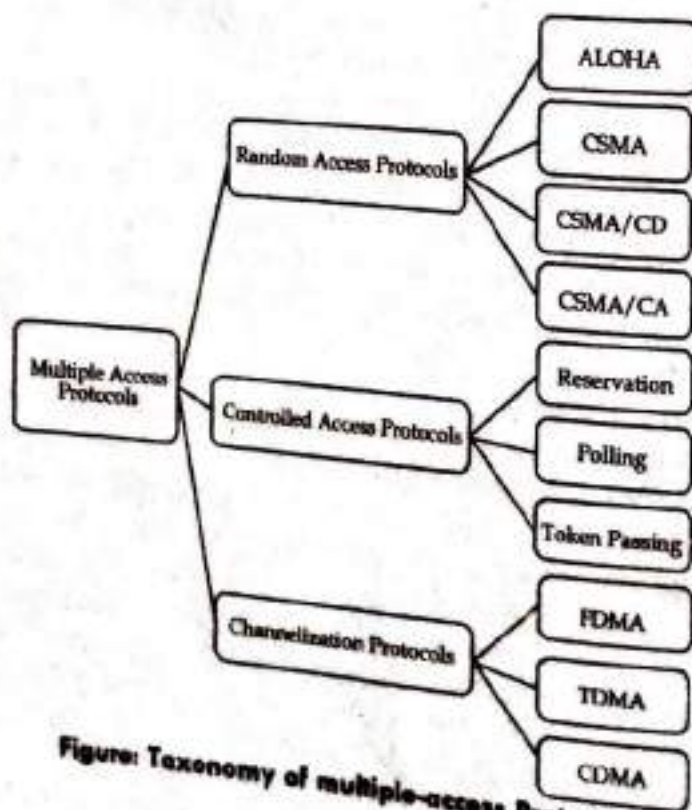


Figure: Taxonomy of multiple-access Protocols

Random Access Protocol

In random access or contention methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send. Any station can send data depending on medium's state (idle or busy). It has two features:

- There is no fixed time for sending data
- There is no fixed sequence of stations sending data

The random-access protocols are further subdivided as:

(a) ALOHA

The Aloha protocol was designed as part of a project at the University of Hawaii. It provided data transmission between computers on several of the Hawaiian Islands involving packet radio networks. Aloha is a multiple access protocol at the data link layer and proposes how multiple terminals access the medium without interference or collision. There are two different versions of ALOHA:

- **Pure ALOHA:**

Pure Aloha is an un-slotted, decentralized, and simple to implement protocol. In pure ALOHA, the stations simply transmit frames whenever they want data to send. It does not check whether channel is busy or not before transmitting. In case, two or more stations transmit simultaneously, collision occurs and frames are destroyed. Whenever any station transmits a frame, it expects the acknowledgement from the receiver. If it is not received within specified time, the station assumes that the frame or acknowledgement has been destroyed. Then, the station waits for a random amount of time and sends the frame again. This randomness helps in avoiding more collisions. This scheme works well in small networks where the load is not much. But in largely loaded networks, this scheme fails poorly. This led to the development of Slotted Aloha.

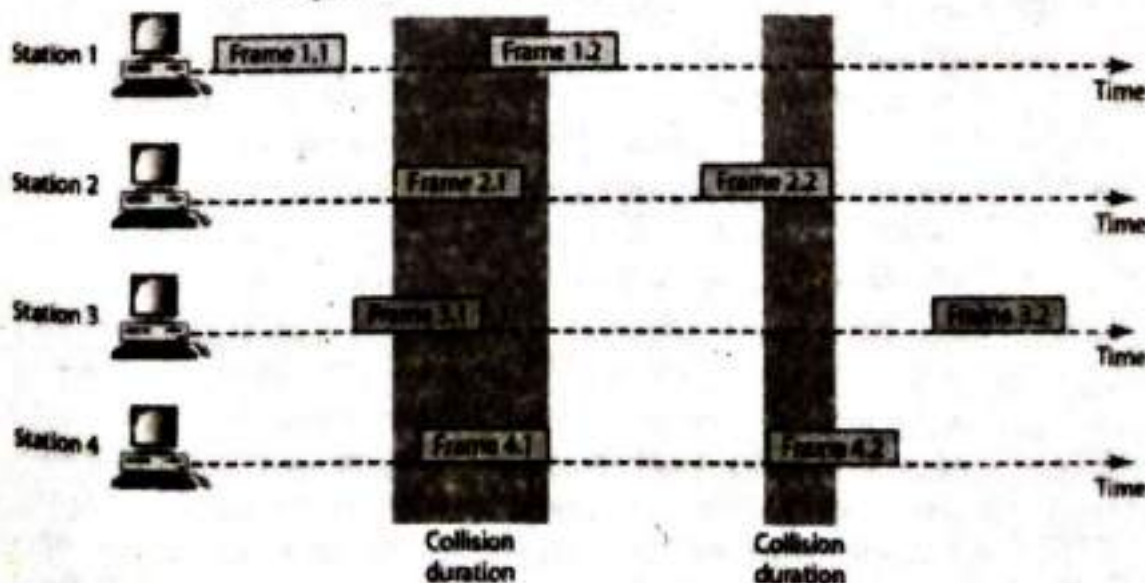


Figure: Frames in Pure ALOHA Network

To assure pure aloha: Its throughput and rate of transmission of frame to be predicted. For that to make some assumption:

- All the frames should be the same length.
- Stations cannot generate frame while transmitting or trying to transmit frame.
- The population of stations attempts to transmit (both new frames and old frames that collided) according to a Poisson distribution.

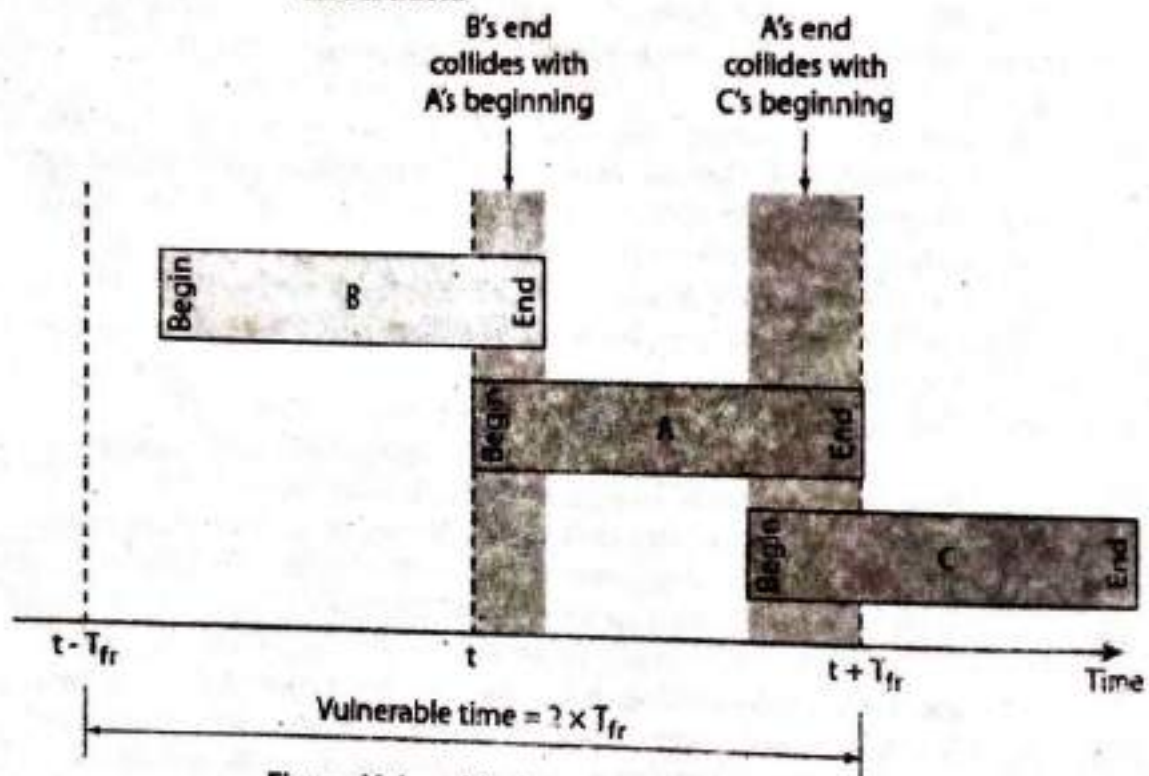


Figure: Vulnerable time for Pure ALOHA Protocol

Pure ALOHA vulnerable Time = $2 \times$ Frame Transmission Time (T_{fr})

Throughput for pure ALOHA (S_{pure}) = $G \times e^{-2G}$

Where G is number of stations wants to transmit in T_{fr} slot.

Maximum throughput (S_{pure})_{max} = 0.184 for $G=0.5$

Which means, in Pure ALOHA, only about 18.4% of the time is used for successful transmissions.

• Slotted ALOHA

This is quite similar to Pure Aloha, differing only in the way transmissions take place. Instead of transmitting right at demand time, the sender waits for some time. In slotted ALOHA, the time of the shared channel is divided into discrete intervals called Slots. The stations are eligible to send a frame only at the beginning of the slot and only one frame per slot is sent. If any station is not able to place the frame onto the channel at the beginning of the slot, it has to wait until the beginning of the next time slot. There is still a possibility of collision if two stations try to send at the beginning of the same time slot. But still the number of collisions that can possibly take place is reduced by a large margin and the performance becomes much well compared to Pure Aloha.

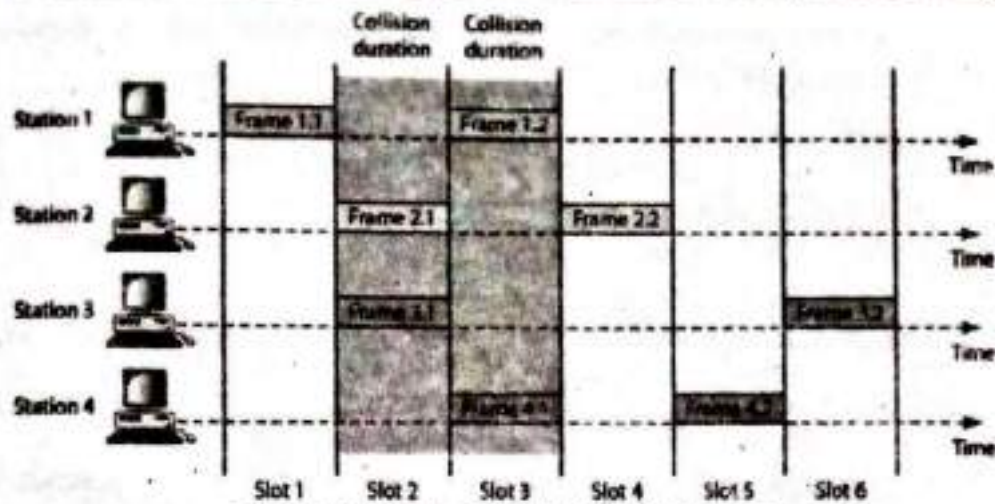


Figure: Frames in Slotted ALOHA Network
A collides with C

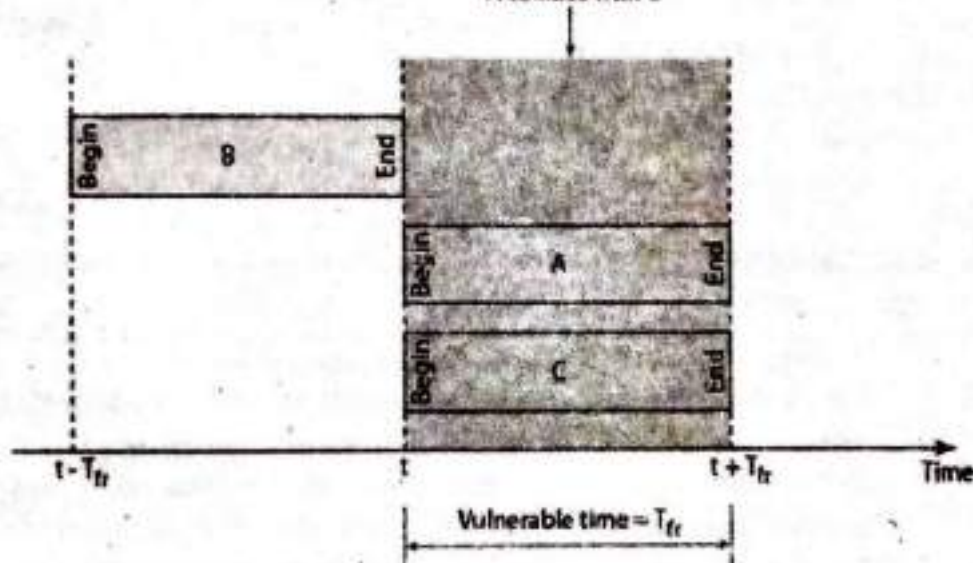


Figure: Vulnerable time for slotted ALOHA protocol

Slotted ALOHA vulnerable Time = Frame Transmission Time (T_f)

Throughput for Slotted ALOHA (S_{slotted}) = $G \times e^{-2G}$

Where G is number of stations wants to transmit in T_f slot.

Maximum throughput (S_{slotted})_{max} = 0.368 for $G=1$

Which means, in slotted ALOHA, about 36.8% of the time is used for successful transmissions.

(b) Carrier Sense Multiple Access (CSMA)

Carrier Sense Multiple Access (CSMA) ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data. If it is idle then it sends data, otherwise it waits till the channel becomes idle. However, there is still chance of collision in CSMA due to propagation delay. For example, if station A wants to send data, it will first sense the medium. If it finds the channel idle, it will start sending data. However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data. This will result in collision of data from station A and B.

CSMA access modes:

1. 1-persistent: The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being

idle and transmits unconditionally (with 1 probability) as soon as the channel gets idle.

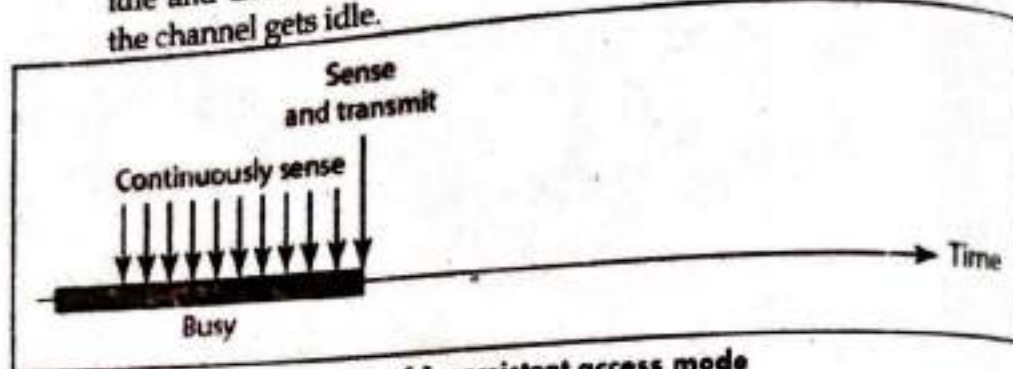


Figure: Behavior of 1-persistent access mode

2. Non-Persistent: The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.

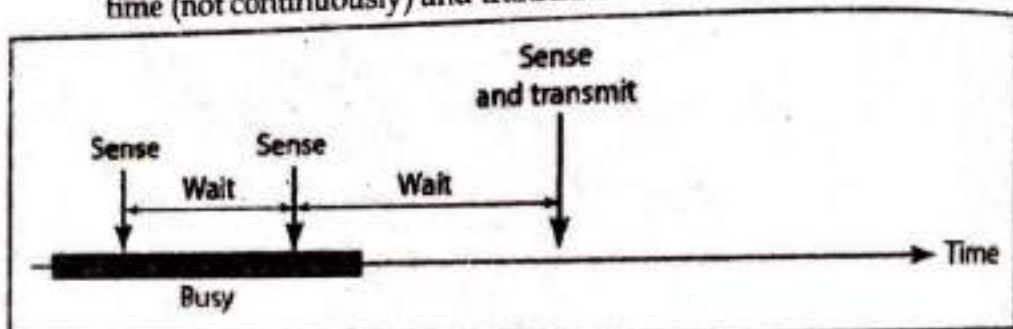


Figure: Behavior of Non-persistent access mode

3. P-persistent: The node senses the medium, if idle it sends the data with p probability. If the data is not transmitted ($(1-p)$ probability) then it waits for some time and checks the medium again, now if it is found idle then it sends with p probability. This repeat continues until the frame is sent. It is used in Wi-Fi and packet radio systems.

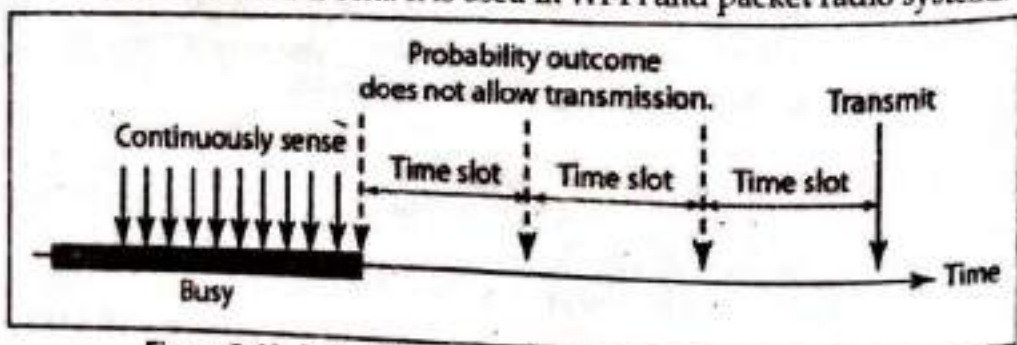


Figure 3.41: Behavior of p-persistent access mode

4. O-persistent: Superiority of nodes is decided beforehand and transmission occurs in that order. If the medium is idle, node waits for its time slot to send data.

(c) Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

The CSMA method does not tell us what to do in case there is a collision. Carrier sense multiple access with collision detection (CSMA/CD) adds on to the CSMA algorithm to deal with collision. In CSMA/CD, the size of a frame must be large enough so that collision can be detected by sender while sending the frame. So, the frame transmission delay must be at least two times the maximum propagation delay.

To better understand CSMA/CD, let us look at the first bits transmitted by the two stations involved in the collision. Although each station continues to send bits in the frame until it detects the collision, we show what happens as the first bits collide. In figure below, stations A and C are involved in the collision.

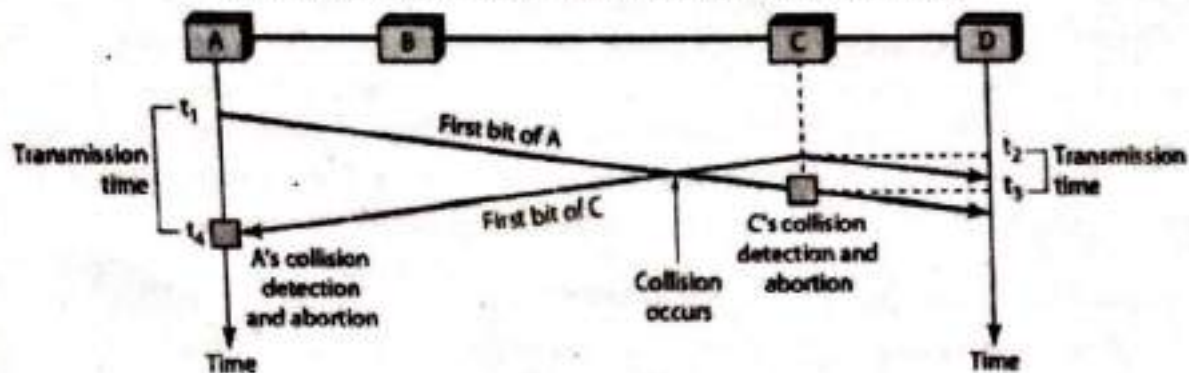


Figure: Collision of the first bit in CSMA/CD

At time t_1 , station A has executed its persistence procedure and starts sending the bits of its frame. At time t_2 , station C has not yet sensed the first bit sent by A. Station C executes its persistence procedure and starts sending the bits in its frame, which propagate both to the left and to the right. The collision occurs sometime after time t_2 . Station C detects a collision at time t_3 when it receives the first bit of A's frame. Station C immediately (or after a short time, but we assume immediately) aborts transmission. Station A detects collision at time t_4 when it receives the first bit of C's frame; it also immediately aborts transmission. Looking at the figure, we see that A transmits for the duration $t_4 - t_1$; C transmits for the duration $t_3 - t_2$.

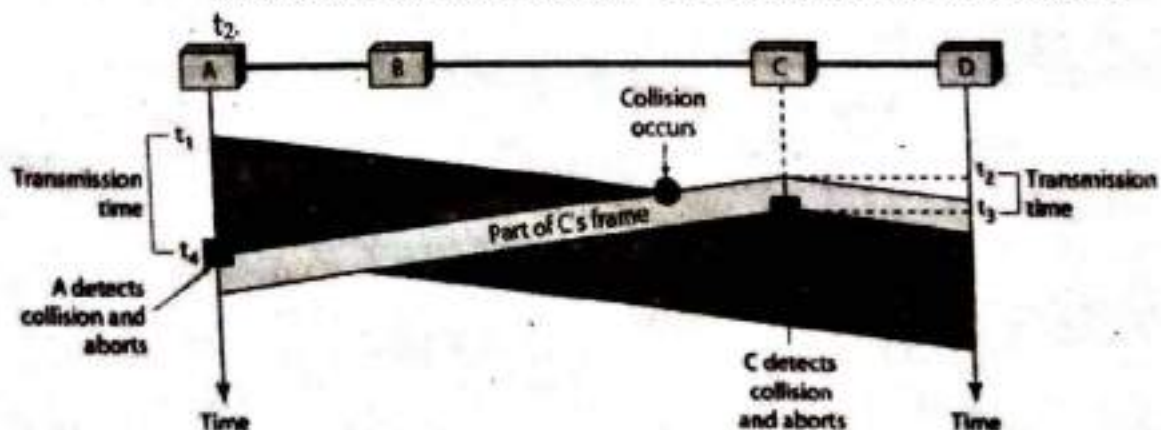


Figure: Collision and abortion in CSMA/CD

(d) **Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)**

Carrier sense multiple access with collision avoidance (CSMA/CA) was invented for wireless networks. The process of collisions detection involves sender receiving acknowledgement signals. If there is just one signal (its own) then the data is successfully sent but if there are two signals (its own and the one with which it has collided) then it means a collision has occurred. To distinguish between these two cases, collision must have a lot of impact on received signal. However, it is not so in wired networks, so CSMA/CA is used in this case.

CSMA/CA avoids collision by:

- Interframe space:** Station waits for medium to become idle and if found idle it does not immediately send data (to avoid collision due to propagation delay) rather it waits for a period of time called Interframe space or IFS. After this time, it again checks the medium for being idle. The IFS duration depends on the priority of station.
- Contention Window** It is the amount of time divided into slots. If the sender is ready to send data, it chooses a random number of slots as wait time which doubles every time medium is not found idle. If the medium is found busy it does not restart the entire process, rather it restarts the timer when the channel is found idle again.

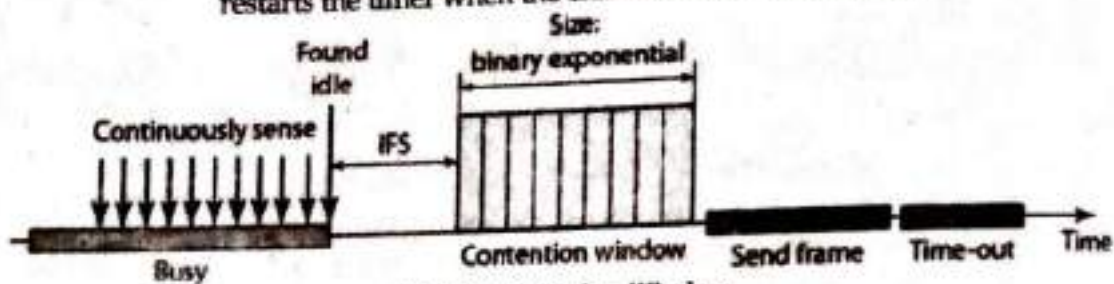


Figure: Contention Window

- Acknowledgement:** Sender re-transmits the data if acknowledgement is not received before time-out.

10. Explain the IPv4 header format in detail. [10]

Ans: Packets in the network (internet) layer are called datagrams. A datagram is a variable-length packet consisting of two parts: **header** and **data**. The header is 20 to 60 bytes in length and contains information essential to routing and delivery. It is customary in TCP/IP to show the header in 4-byte sections. The following figure shows the IP datagram format.

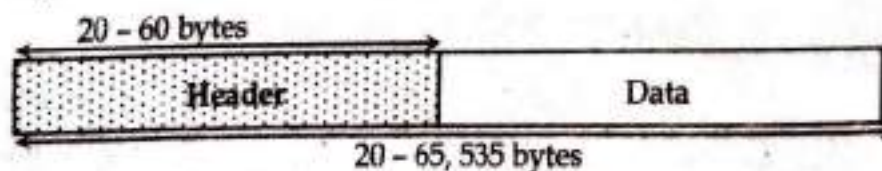


Figure: IP datagram

VER (4 Bits)	HLEN (4 Bits)	TYPE OF SERVICE	TOTAL LENGTH (16 Bits)	
IDENTIFICATION (16 Bits)			FLAGS (3 Bits)	FRAGMENTATION OFFSET (13 Bits)
TIME TO LIVE (8 Bits)		PROTOCOL (8 Bits)	HEADER CHECKSUM (16 Bits)	
SOURCE IP ADDRESS (32 Bits)				
DESTINATION IP ADDRESS (32 Bits)				
OPTIONS + PADDING (0 to 40 bytes)				
DATA (16 Bits)				

Figure: IPv4 header format

Version Number (VER)

These 4 bits specify the IP protocol version of the datagram. By looking at the version number, the router can determine how to interpret the remainder of the IP datagram. Different versions of IP use different datagram formats.

Header Length (HLEN)

Because an IPv4 datagram can contain a variable number of options (which are included in the IPv4 datagram header), these 4 bits are needed to determine where in the IP datagram the data actually begins. Most IP datagrams do not contain options, so the typical IP datagram has a 20-byte header.

Type of Service

The type of service (TOS) bits was included in the IPv4 header to allow different types of IP datagrams (for example, datagrams particularly requiring low delay, high throughput, or reliability) to be distinguished from each other. For example, it might be useful to distinguish real-time datagrams from non-real-time traffic (for example, FTP). The specific level of service to be provided is a policy issue determined by the router's administrator.

Total Length

This is the total length of the IP datagram (header plus data), measured in bytes. Since this field is 16 bits long, the theoretical maximum size of the IP datagram is 65,535 bytes. However, datagrams are rarely larger than 1,500 bytes.

Length of data = total length - header length

Identification

If IP packet is fragmented during the transmission, all the fragments contain same identification number to identify original IP packet they belong to.

Flags

As required by the network resources, if IP Packet is too large to handle, these flags tell if they can be fragmented or not. The first bit is reserved (not used). The second bit is called the do not fragment bit. If its value is 1, the machine must not fragment the datagram. If it cannot pass the datagram through any available physical network, it discards the datagram and sends an ICMP error message to the source host. If its value is 0, the datagram can be fragmented if necessary. The third bit is called the more fragment bit. If its value is 1, it means the datagram is not the last fragment. If its value is 0, it means this is the last or only fragment.

Fragment Offset

This offset tells the exact position of the fragment in the original IP Packet.

Time-to-live

To avoid looping in the network, every packet is sent with some TTL value set, which tells the network how many routers (hops) this packet can cross. At each hop, its value is decremented by one and when the value reaches zero, the packet is discarded.

Protocol

This field is used only when an IP datagram reaches its final destination. The value of this field indicates the specific transport-layer protocol to which the data portion of this IP datagram should be passed. For example,

a value of 6 indicates that the data portion is passed to TCP, while a value of 17 indicates that the data is passed to UDP.

Header Checksum

This field is used to keep checksum value of entire header which is then used to check if the packet is received error-free.

Source Address

This 32-bit field define the address of the Sender (or source) of the packet.

Destination Address

This 32-bit field define the address of the Receiver (or destination) of the packet.

Options

This is optional field, which is used if the value of IHL is greater than 5. These options may contain values for options such as Security, Record Route, Time Stamp, etc.

Data (Payload)

Finally, we come to the last and most important field. In most circumstances, the data field of the IP datagram contains the transport-layer segment (TCP or UDP) to be delivered to the destination. However, the data field can carry other types of data, such as ICMP messages.

□□□



MODEL QUESTIONS SETS FOR PRACTICE

MODEL SET 1

Course Title: Computer Networking

Course No: CACS303

Nature of the Course: Theory + Lab

Semester: V

Full Marks: 60

Pass Marks: 24

Time: 3

Candidates are required to give their answer in their own words as far as practicable.

Group- A

[10x1=10]

Attempt all the questions

1. Circle (O) the correct answer in the following questions:

- i. Computer Network is _____
 - a) Collection of hardware components and computers
 - b) Interconnected by communication channels
 - c) Sharing of resources and information
 - d) all of the above
- ii. Which transmission media provides the highest transmission speed in a network?
 - a) Coaxial cable
 - b) twisted pair cable
 - c) optical fiber
 - d) electrical cable
- iii. Which of the following is the multiple access protocol for channel access control?
 - a) CSMA/CD
 - b) CSMA/CA
 - c) Both CSMA/CD & CSMA/CA
 - d) HDLC
- iv. You have an IP address of 172.16.13.5 with a 255.255.255.128 subnet mask. What is your class of address, subnet address, and broadcast address?
 - a) Class A, Subnet 172.16.13.0, Broadcast address 172.16.13.127
 - b) Class B, Subnet 172.16.13.0, Broadcast address 172.16.13.127
 - c) Class B, Subnet 172.16.13.0, Broadcast address 172.16.13.255
 - d) Class B, Subnet 172.16.0.0, Broadcast address 172.16.255.255
- v. Which of the following is false with respect to TCP?
 - a) Connection-oriented
 - b) Process-to-process
 - c) Transport layer protocol
 - d) Unreliable
- vi. Which is not an application layer protocol?
 - a) HTTP
 - b) SMTP
 - c) FTP
 - d) TCP
- vii. In cryptography, what is cipher?
 - a) algorithm for performing encryption and decryption
 - b) Encrypted message
 - c) Both algorithm for performing encryption and decryption and encrypted message
 - d) Decrypted message
- viii. Communication channel is shared by all the machines on the network in _____.
 - a) Broadcast network
 - b) unicast network
 - c) Multicast network
 - d) any-cast network

- ix. Bluetooth is an example of _____
 a) Local area network
 b) wide area network
 c) Persona area network
 d) virtual private network
- x. A single channel is share by multiple signals by _____
 a) Analog modulation
 b) digital modulation
 c) Phase modulation
 d) multiplexing

Group B

[6×5=30]

Attempt any SIX questions.

2. What do you mean by Internet Protocol Stack?
3. Explain the principle of congestion control.
4. What do you mean by IP datagram fragmentation?
5. Explain Give a detail note on slotted ALOHA protocols.
6. Explain about GBN Sliding window protocol.
7. Explain the term checksum in reference to error detection codes.
8. Write short notes on (ANY TWO):
 a) Bluetooth Standards
 b) Analog vs. Digital Signals
 c) IPSec

[2.5+2.5]

Group C

[2×10=20]

Attempt any TWO questions.

9. What is the significance of layered architecture? Explain the OSI layered architecture with neat sketch.
10. With an example explain the Dynamic routing algorithm used in computer networks.
11. Define DNS. Explain the DNS records and DNS messages.

MODEL SET 2

Candidates are required to give their answer in their own words as far as practicable.

Group- A

[10 x1=10]

Attempt all the questions

1. Circle (O) the correct answer in the following questions:
 i. The structure or format of data is called _____
 a) Syntax b) Semantics c) Struct d) Formatting
- ii. The physical layer provides _____
 a) Mechanical specifications of electrical connectors and cables
 b) Electrical specification of transmission line signal level
 c) Specification for IR over optical fiber
 d) all of the mentioned
- iii. Which of the following tasks is not done by data link layer?
 a) framing b) error control
 c) flow control d) channel coding
- iv. CRC stands for _____
 a) cyclic redundancy check b) code repeat check
 c) code redundancy check d) cyclic repeat check
- v. The network layer is concerned with _____ of data.

- a) bits b) frames c) packets d) bytes
- vi. Which is not a function of network layer?
 a) routing b) inter-networking
 c) congestion control d) error control
- vii. A _____ is a TCP name for a transport service access point.
 a) port b) pipe c) node d) protocol
- viii. In an asymmetric-key cipher, the receiver uses own _____ key to decipher.
 a) private b) public
 c) all of the above d) none of the above
- ix. Virtual terminal protocol is an example of the _____.
 a) transport layer b) presentation layer
 c) application layer d) none of the above
- x. In a network with 26 computers, which topology would require the most extensive cabling?
 a) mesh b) bus c) ring d) star

Group B**[6×5=30]****Attempt any SIX questions.**

2. Describe in details about light wave transmission.
3. Explain the working principle of SMTP.
4. Define data link layer and its services.
5. Explain point-to-point protocol (PPP).
6. Explain the multicasting routine and its applications.
7. What are the responsibilities of Presentation layer and Session layer of OSI Model?
8. Write short notes on (ANY TWO): [2.5+2.5]
 - a. Packet Switching
 - b. DHCP
 - c. Firewalls

Group C**Attempt any TWO questions.****[2×10=20]**

9. Explain the principles of application layer protocols. What do you mean by file transfer?
10. Draw TCP packet header format. Explain 3-way handshaking of Transmission Control Protocol.
11. Define protocol and standard. Explain TCP/IP model with a neat diagram.



Candidates are required to give their answer in their own words as far as practicable.

Group- A**Attempt all the questions****[10 ×1=10]**

1. Circle (O) the correct answer in the following questions:
 - i. In a P2P network, who controls the devices?
 - a) first computer b) last computer
 - c) no device controls d) all devices control

- ii. A computer network consists of _____ number of computers or servers or system.
a) 2 b) 3 c) more than 2 d) all of the above
- iii. The two sub-layers of a data link layer are _____.
a) LLC b) MAC
c) both LLC and MAC d) data layer
- iv. The technique of temporarily delaying outgoing acknowledgements so that they can be hooked onto the next outgoing data frame is called _____.
a) piggybacking b) cyclic redundancy check
c) fletcher's checksum d) parity check
- v. A 4 bytes IP address consists of _____.
a) Only network address b) only host address
c) Network address & host address d) network address & MAC address
- vi. For carrying digital data over long distance using either analog signal or digital signal at approximately spaced points, we must have _____.
a) Amplifiers b) repeater
c) switch d) either amplifier or repeater
- vii. IPSec is designed to provide security at the _____.
a) Transport layer b) network layer
c) Application layer d) session layer
- viii. The packet of information at the application layer is called _____.
a) Message b) segment c) packet d) frame
- ix. When displaying a web page, the application layer uses the _____.
a) FTP protocol b) HTTP protocol
c) SMTP protocol d) TCP protocol
- x. Cryptanalysis is used _____.
a) To find some insecurities in a cryptographic scheme
b) to increase the speed
c) To encrypt the data
d) To make new ciphers

Group B

Attempt any SIX questions.

[6×5=30]

2. What is circuit-switching? How is it different from packet-switching?
3. Discuss different fields of UDP. How is it different from TCP?
4. What is static routing? How is it different from dynamic routing?
5. What is random access protocol? Discuss FDMA in detail.
6. Differentiate between thicknet and thinnet cable.
7. Perform encryption and decryption using the RSA algorithm for $p=5$, $q=7$, $e=7$ and $m=2$.
8. Write short notes on (ANY TWO):
a. SNMP
b. ARP
c. Framing

[2.5+2.5]

Group C

Attempt any TWO questions.

[2×10=20]

9. Why do we need layered protocol architecture? Discuss each layer of TCP/IP architecture along with function of each layer. Compare TCP/IP with OSI mode.
10. Assume a Class B network. Divide this network in 16 different sub-networks. What is new subnet mask?

11. Define multiplexing. What are the benefits of using multiplexing? Discuss Go-Back-N and Selective Repeat protocol in detail.

MODEL SET 4

Candidates are required to give their answer in their own words as far as practicable.

Group- A

[10 x1=10]

Attempt all the questions

1. Circle (O) the correct answer in the following questions:

- i. When collection of various computers seems a single coherent system to its client, then it is called _____.
 - a) Computer network
 - b) distributed system
 - c) Networking system
 - d) mail system
- ii. Who developed standards for the OSI reference model?
 - a) ANSI - American National Standards Institute
 - b) ISO - International Standards Organization
 - c) IEEE - Institute of Electrical and Electronics Engineers
 - d) ACM - Association for Computing Machinery
- iii. An Ethernet jack is _____.
 - a) RJ11
 - b) RJ14
 - c) RJ45
 - d) None
- iv. The size of an IP address in IPv6 is _____.
 - a) 4 bytes
 - b) 128 bits
 - c) 8 bytes
 - d) 100 bits
- v. Which of the following is false with respect to UDP?
 - a) connection-oriented
 - b) low overhead
 - c) Transport layer protocol
 - d) Unreliable
- vi. Two broad categories of congestion control are
 - a) open-loop and closed-loop
 - b) open-control and closed-control
 - c) active control and passive control
 - d) active loop and passive loop
- vii. The communication protocol used by internet is _____.
 - a) HTTP
 - b) WWW
 - c) TCP/IP
 - d) FTP
- viii. Web site is a collection of _____.
 - a) HTML Documents
 - b) Graphic files
 - c) Audio and video files
 - d) all of the above
- ix. The shortest frame in HDLC protocol is usually the _____.
 - a) Information frame
 - b) management frame
 - c) Supervisory frame
 - d) none of the above
- x. Digital signature does not provide _____.
 - a) Nonrepudiation
 - b) privacy
 - c) Authentication
 - d) all of the above

Group B

Attempt any SIX questions.

[6x5=30]

2. What is transmission media? Explain about any three transmission medias in detail.
3. Is 192.16.10.64/26 a host, network or broadcast address? Also write subnet mask and usable hosts of the following network.
4. Explain how a TCP connection can be gracefully terminated?

5. What methods are used so that IPV6 and IPV4 networks are interoperable?
6. Find Hamming Code for data 11010111.
7. Describe how SMTP protocol is used in e-mail applications.
8. Write short notes on (Any Two):
- NAT
 - Virtual LAN
 - Single bit parity

Group C**Attempt any TWO questions.**

9. Explain the reason behind layering the network architecture? Write the functions of all layers of TCP/IP model. [2×10=20]
10. Explain the signature generation and signature verification in digital signature process.
11. Suppose we want to transmit the message 1011 0010 0111 and protect it from errors using the CRC polynomial $x^4 + x^2 + 1$. Use polynomial long division to determine the message that should be transmitted. Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

MODEL SET 5

Candidates are required to give their answer in their own words as far as practicable.

Group- A**Attempt all the questions****[10 x1=10]**

1. Circle (O) the correct answer in the following questions:
- A list of protocols used by a system, one protocol per layer, is called
 - Protocol architecture
 - protocol stack
 - Protocol suite
 - protocol system
 - Which one of the following is an architecture paradigm?
 - peer-to-peer
 - client-server
 - http
 - both peer-to-peer and client-server
 - The first network was called _____.
 - CNNET
 - NSFNET
 - ARPANET
 - ASAPNET
 - The physical layer provides _____.
 - Mechanical specifications of electrical connectors and cables
 - Electrical specification of transmission line signal level
 - Specification for IR over optical fiber
 - all of the mentioned
 - When two or more bits in a data unit has been changed during the transmission, the error is called _____.
 - Random error
 - burst error
 - inverted error
 - double error
 - The network layer protocol for internet is _____.
 - Ethernet
 - internet protocol
 - FTP
 - HTTP
 - ICMP is primarily used for _____.
 - error and diagnostic functions
 - addressing
 - Forwarding
 - routing

- viii. Transmission control protocol _____
 a) is a connection-oriented protocol
 b) Uses a three way handshake to establish a connection
 c) Receives data from applications as a single stream
 d) all of the mentioned
- ix. For separate channels in TDM, it is necessary to use
 a) Time slots
 b) bandpass filters
 c) Differentiation
 d) none of the above
- x. VPN technology uses two simultaneous techniques to guarantee privacy for an organization: _____
 a) SSL; tunneling
 b) IPSec; SSL
 c) IPSec; tunneling
 d) none of the above

Group B**[6×5=30]****Attempt any SIX questions.**

2. What do you mean by Internet Protocol Stack?
3. Define mobile network. Discuss the evolution of mobile network generation from 1G to 5G.
4. Explain POP3 and IMAP4.
5. Explain IPv4 addressing.
6. Explain on ALOHA and Slotted ALOHA protocols.
7. Draw and explain three-way handshake process of TCP.
8. Write short notes on (ANY TWO):
 a) OSPF
 b) Transmission Impairment
 c) Congestion Control

Group C**[2×10=20]****Attempt any TWO questions.**

9. What are the seven layers of OSI model? Explain the structure of TCP Header format.
10. What do you mean by cryptography? Explain the encryption and decryption in cryptosystem.
11. Consider 8 bit 10101010 is transmitted using Hamming distance. Show the actual bits transmitted without any error. If the 6th bit from LSB is inverted during transmission. Show how the error is detected and corrected at the receiver's end. Also, obtain the original data after correction.

**MODEL SET 6**

Candidates are required to give their answer in their own words as far as practicable.

Group- A**[10 x1=10]****Attempt all the questions**

1. Circle (O) the correct answer in the following questions:
 i. Individual pieces of information within a network protocol mean _____
 a) Syntax b) Semantics c) Struct d) Formatting
- ii. A mobile station only communicates with one base station, in a _____
 a) back off b) soft handoff c) hard handoff d) low handoff
- iii. In which one of the following, the slow and fast hopping is used?
 a) GSM b) GPRS c) FHSS d) none of the above

- iv. Which of the following computer networks is built on the top of another network?
 a) Overlay network b) prime network
 c) Chief network d) prior network
- v. The datalink layer is concerned with _____ of data.
 a) Bits b) frames c) packets d) bytes
- vi. Function of network layer.
 a) Routing b) inter-networking
 c) congestion control d) all of the mentioned
- vii. What is the measure (unit) used to represent signaling rate per second?
 a) Baud b) Hz
 c) Bps d) none of the above
- viii. What is the main advantages of UDP?
 a) more overload b) reliable
 c) low overhead d) fast
- ix. Which one of the following protocol deliver/store mail to receiver server?
 a) SMTP b) POP c) IMAP d) HTTP
- x. In the _____ mode, the IPSec header is added between the IP header and the rest of the packet.
 a) Tunnel b) transport c) transition d) none of the above

Group B

Attempt any SIX questions.

[6×5=30]

2. Explain peer-to-peer network model.
3. Your friend cannot communicate with you without protocol, why?
4. Explain how slotted ALOHA improves the performance of system over pure ALOHA.
5. Differentiate between broadband and base band services.
6. Compare and contrast the IPv4 and the IPv6 header files. Do they have any fields in common?
7. What is meant by "domain name"? How is a domain name translated to an equivalent IP address? Explain
8. Write short notes on (ANY TWO):
 a) ICMP
 b) Throughput vs. Good put
 c) Sliding window

Group C

Attempt any TWO questions.

[2×10=20]

9. Define public key cryptography. In RSA system, how private key is generated from given public key? Explain with an example.
10. Explain the various field of the TCP header and the working of the TCP protocol.
11. Explain how does CRC detect the error with multiple bits? Given message is $M(x) = x^7 + x^4 + x^3 + x^2 + 1$ and the generator $G(x) = x^3 + 1$. Show the actual bit string transmitted, suppose the third bit from the left is inverted during the transmission. Show how the error is detected at the receiver's end.

MODEL SET 7

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:

- i. In the layer hierarchy as the data packet moves from the upper to the lower layers, headers are ____
 a) Added b) removed c) rearranged d) modified
- ii. Which of this is not a network edge device?
 a) PC b) smartphone c) servers d) switch
- iii. The device operation at Data Link layer is
 a) Repeater b) router c) bridge d) none of the above
- iv. Which of the following routing algorithms can be used for network layer design?
 a) Shortest path algorithm b) distance vector routing
 c) Link state routing d) all of the above
- v. Application layer offers ____ service.
 a) End to end
 b) Process to process
 c) both end to end and process to process
 d) None of the above
- vi. Which of the following is true with respect to TCP?
 a) Connection-oriented b) Process-to-process
 c) Transport layer protocol d) all of the above
- vii. Pick the odd one out.
 a) File transfer b) file download
 c) Interactive games d) e-mail
- viii. Which of the following is a fundamental principle of wireless communication?
 a) Electromagnetic waves b) microwaves
 c) Both a) and b) d) none of the above
- ix. In cryptography, what is cipher?
 a) Encrypted message
 b) Algorithm for performing encryption and decryption
 c) Both algorithm for performing encryption and decryption and encrypted message
 d) Decrypted message
- x. A proxy firewall filters at ____
 a) Physical layer b) data link layer
 c) Network layer d) application layer

Group B

Attempt any SIX questions.

[6x5=30]

2. Explain client server system. How is it different from peer-to-peer system?
3. Explain the final delivery of email to the end user using POP3.
4. Assume a class A network and divide it into four subnets. What is the value of new subnet mask?

5. Discuss CRC as an error detection mechanism.
6. What is congestion control? Why do we need it?
7. Differentiate between circuit switching, packet switching and message switching.
8. Write short notes on (ANY TWO): [2.5+2.5]
 - a) IEEE 802.3
 - b) Nyquist bit Rate
 - c) IPSec tunnel modes

Group C

Attempt any TWO questions.

9. Define Protocol. Why do we need layered protocol architecture? Discuss each layer of TCP/IP protocol architecture in detail [2×10=20]
10. Define transmission media. Differentiate between guided and unguided transmission media. Discuss each guided transmission media in detail.
11. What is routing? Discuss link state routing algorithm in detail.

MODEL SET 8

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
 - i. Which transmission media is easy to install inside a city to create a network quickly?
 - a) Cable media
 - b) fiber optic
 - c) Radio frequency devices
 - d) all of the above
 - ii. Which one of the following is an architecture paradigm?
 - a) peer-to-peer
 - b) client-server
 - c) Http
 - d) both peer-to-peer and client-server
 - iii. Which is the network that can cover an entire city?
 - a) LAN
 - b) MAN
 - c) WAN
 - d) all of the above
 - iv. Which is the layer that converts raw bits to frames and frames to raw bits in the OSI model?
 - a) Application layer
 - b) transport layer
 - c) Network layer
 - d) data link layer
 - v. When two or more bits in a data unit has been changed during the transmission, the error is called _____.
 - a) Random error
 - b) burst error
 - c) inverted error
 - d) double error
 - vi. In reality, Internet Protocol recognizes only _____.
 - a) An IP address
 - b) a location of the host
 - c) a postal mail address
 - d) none of the above

- vii. FTP is used for _____
 a) Uploading files only b) downloading files only
 c) Both (A) and (B) d) none of the above
- viii. Which of the network allow different speed links?
 a) Message-switched networks b) packet-switched networks
 c) circuit-switched networks d) none of the mentioned
- ix. Three security goals are _____
 a) Confidentiality, cryptography, and nonrepudiation
 b) Confidentiality, encryption, and decryption
 c) Confidentiality, integrity, and availability
 d) None of the above
- x. In _____ cryptography, the same key is used in both directions.
 a) symmetric-key b) asymmetric-key
 c) public-key d) none of the above

Group B**Attempt any SIX questions.****[6×5=30]**

2. Discuss FTP in detail.
3. What is Subnetting? Assume a class C network and divide it into four subnets. What is the value of new subnet mask?
4. How does the system correct error after error detection?
5. What is transmission media? Discuss wireless transmission media in detail.
6. Suppose you have got 50 PCs to set up in a LAN, which cable, device, protocol, you prefer and why justify with reason.
7. What are sliding window protocol? Explain one -bit sliding window protocol with an appropriate diagram.
8. Write short notes on (ANY TWO): [2.5+2.5]
 - a) Single bit parity
 - b) Leaky bucket
 - c) Cable Network

Group C**Attempt any TWO questions.****[2×10=20]**

9. Discuss the relationship between transport layer and network layer. Discuss TCP as a transport layer protocol along with its segment structure.
10. Design a general model for network security. Explain.
11. Why do we need routing algorithm? Discuss distance vector routing algorithm in detail.

Dot Net Technology

Course Title: DotNet Technology (3 Cr.)

Course Code: CACS302

Year/Semester: III/5th

Class Load: 6 Hrs. I Week (Theory: 3Hrs. Practical: 3 Hrs.)

Course Description:

This course covers different concepts of .NET framework. It also covers basic to advanced features of C# language including language basics, creating types and inheritance, delegates, events, lambda expressions, LINQ, working with databases, and developing web applications using ASP.NET.

Course Objectives:

The primary objective of this course is to provide concepts of .NET framework and different concepts of C# programming language and make students familiar with their uses and applications.

Course Details

Unit 1: Introducing C# and the .NET Framework (7 Hrs.)

Object Orientation; Type Safety; Memory Management; Platform Support; C# and CLR; CLR and .NET Framework; Other Frameworks; Framework Overview; .NET Standard 2.0; applied Technologies

Unit 2: The C# Language Basics (12 Hrs.)

Writing Console and GUI Applications; Identifiers and Keywords; Writing Comments; Data Types; Expressions and Operators; Strings and Characters; Arrays; Variables and Parameters; Statements (Declaration, Expression, Selection, Iteration, and Jump Statements); Namespaces

Unit 3: Creating Types in C# (12 Hrs.)

Classes; Constructors and Destructors; this Reference; Properties; Indexers; Static Constructors and Classes; Finalizers; Dynamic Binding; Operator Overloading; Inheritance; Abstract Classes and Methods; base Keyword; Overloading; Object Type; Structs; Access Modifiers; Interfaces; Enums; Generics

Unit 4: Advanced C# (14 Hrs.)

Delegates; Events; Lambda Expressions; Exception Handling; Introduction of LINQ; Working with Databases; Writing Web Applications using ASP.NET

Laboratory Work: The laboratory work includes writing console and/or GUI programs in C#

- To implement basic language features
- To create classes and objects and to implement different object-oriented features
- To implement inheritance

- To implement advanced features like delegates, event handling, lambda expressions, exception handling
- To implement LINQ and database applications

Text Books:

1. C# 7.0 in a Nutshell (7th Edition), the Definitive Reference, Joseph Albahari & Ben Albahari, O'Reilly.
2. Microsoft Visual C# Step by Step (9th Edition), John Sharp, Pearson Education.

Reference Books:

1. C# 7.0 All-in-One for Dummies (1st Edition), John Paul Mueller, Bill Sempf, Chuck Sphar, John Wiley & Sons, Inc.
2. Professional C# 7 and .NET Core 2.0 (7th Edition), Christian Nagel, John Wiley & Sons, Inc.

Teaching Methods:

The teaching faculties are expected to create environment where students can update and upgrade themselves with the current scenario of computing and information technology with the help of topics listed in the syllabus. The general teaching pedagogy that can be followed by teaching faculties for this course includes class lectures, laboratory activity, group discussions, case studies, guest lectures, research work, project work, assignments (Theoretical and Practical), and written and verbal examinations.

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

MODEL QUESTIONS-ANSWERS

Bachelor Level / fifth-semester / Science
Computer Science and Information Technology (CACS302)
(DotNet Technology)

Full marks: 60

Pass marks: 24

Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.

Group 'A'

Attempt all the questions.

10 × 1 = 10

1. Circle (O) the correct answer in the following questions
- i. .NET Framework is an internal _____ component that supports the execution of applications created by using various programming languages.
 - a. Internet
 - b. Windows
 - c. Hardware
 - d. Language
- ii. What is CLR?
 - a. Common Language Runtime
 - b. Compiler Language Runtime
 - c. Compiler Library Runtime
 - d. Common Library Runtime
- iii. Which Is the Following Is Not a Component of the CLR?
 - a. Class loader
 - b. Garbage collector
 - c. .NET Framework
 - d. JIT Compiler
- iv. Which of the following provides automatic memory management and resolves the issue of memory leaks and invalid memory references?
 - a. Security engine
 - b. Garbage collector
 - c. JIT compiler
 - d. Debugger
- v. Which is the String method used to compare two strings with each other?
 - a. CompareTo()
 - b. Compare()
 - c. Copy()
 - d. ConCat()
- vi. Which of the following options is not a Bitwise Operator in C#?
 - a) &, |
 - b) ^, ~
 - c) <<, >>
 - d) +=, -=
- vii. Arrange the operators in the increasing order as defined in C#. !=, ?, &, ++, &&
 - a) ? : <&& < != <& < ++
 - b) ? : <&& < != < ++ <&
 - c) ? : <&& < < != < ++
 - d) ? : <&& < != <& < ++
- viii. Which method has the same name as that of its class?
 - a) delete
 - b) class
 - c) constructor
 - d) none of the mentioned
- ix. Which operator among the following signifies the destructor operator?
 - a) ::
 - b) :
 - c) ~
 - d) &
- x. Operator used to free the memory when memory is allocated?
 - a) new
 - b) free
 - c) delete
 - d) none of the mentioned

Answer-Key

i. (b)	ii. (a)	iii. (c)	iv. (b)	v. (a)	vi. (d)	vii. (d)	viii. (c)	ix. (d)	x. (c)
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Group B

[6x5=30]

Attempt any SIX questions.

1. What is C#? Explain different features of C# language.

Ans: C# is a simple, modern, object oriented language derived from C++ and Java. It aims to combine the high productivity of Visual Basic and the raw power of C++. Simply C# is a modern, type safe programming language, object oriented language that enables programmers too quickly and easily build solutions for the Microsoft .NET platform.

Features of c#

1. Simple

- Pointers are missing in C#.
- Unsafe operations such as direct memory manipulation are not allowed.
- In C# there is no usage of "::" or "->" operators.
- Since it's on .NET, it inherits the features of automatic memory management and garbage collection.
- Varying ranges of the primitive types like Integer, Floats etc.
- Integer values of 0 and 1 are no longer accepted as Boolean values. Boolean values are pure true or false values in C# so no more errors of "=" operator and "==" operator.
- "==" is used for comparison operation and "=" is used for assignment operation.

2. Modern

- C# has been based according to the current trend and is very powerful and simple for building interoperable, scalable, robust applications.
- C# includes built in support to turn any component into a web service that can be invoked over the internet from any application running on any platform.

3. Object oriented

- C# supports Data Encapsulation, inheritance, polymorphism, interfaces.
- (int, float, double) are not objects in java but C# has introduces structures(structs) which enable the primitive types to become objects.

4. Type safe

- In C# we cannot perform unsafe casts like convert double to a Boolean.
- Value types (primitive types) are initialized to zeros and reference types (objects and classes) are initialized to null by the compiler automatically.
- Arrays are zero base indexed and are bound checked.
- Overflow of types can be checked.

5. Interoperability

- C# includes native support for the COM and windows based applications.
- Allowing restricted use of native pointers.
- C# allows the users to use pointers as unsafe code blocks to manipulate your old code.

- Components from VB NET and other managed code languages and directly be used in C#.
- 6. **Scalable and updateable**
- .NET has introduced assemblies which are self-describing by means of their manifest. manifest establishes the assembly identity, version, culture and digital signature etc. Assemblies need not to be register anywhere.
- To scale our application we delete the old files and updating them with new ones. No registering of dynamic linking library.
- Updating software components is an error prone task. Revisions made to the code. Can affect the existing program C# support versioning in the language. Native support for interfaces and method overriding enable complex frame works to be developed and evolved over time.

2. **What are control statements? Explain different types of control statements with examples in detail.**

Ans: Control statements help with the control flow of the program. But, do we mean by control flow. Control flow is the order in which the statements execute. For example, the program's flow is usually from top to bottom (Sequential flow), but what if we want a particular block of code to only be executed when it fulfills a specific condition or wants to run a block of the code a specified number of times. These statements will ensure a smooth and intentional flow of a program.

C# program control statements can be put into the following categories: selection, iteration, and jump.

- **Selection statements** allow your program to choose different paths of execution based upon the outcome of an expression or the state of a variable.
- **Iteration statements** enable program execution to repeat one or more statements (that is, iteration statements form loops).
- **Jump statements** allow your program to execute in a nonlinear fashion.

Selection Statements

C# supports two selection statements: if and switch. These statements allow you to control the flow of your program's execution based upon conditions known only during run time.

The if statement is C#'s conditional branch statement. It can be used to route program execution through two different paths. Here is the general form of the if statement:

```
if (condition) statement1;
else statement2;
```

The if works like this: If the condition is true, then statement1 is executed. Otherwise, statement2 (if it exists) is executed. In no case will both statements be executed. For example, consider the following:

```
using System;
class Program
{
    static void Main(string[] args)
    {
```

```

int a = 10, b = 20;
if (a > b) //false
    Console.WriteLine(a + " is greatest!");
else
    Console.WriteLine(b + " is greatest!");
}
}

```

Switch statement

The switch statement is C#'s multi-way branch statement. It provides an easy way to dispatch execution to different parts of your code based on the value of an expression. As such, it often provides a better alternative than a large series of if-else-if statements.

Here is a simple example that uses a switch statement:

```

using System;
class Program
{
    static void Main(string[] args)
    {
        int n = 2;
        switch (n)
        {
            case 1:
                Console.WriteLine("This is case 1");
                break;
            case 2:
                Console.WriteLine("This is case 2");
                break;
            case 3:
                Console.WriteLine("This is case 3");
                break;
            default:
                Console.WriteLine("Invalid value");
                break;
        }
    }
}

```

Iteration Statements

C#'s iteration statements are for, while, do-while and for-each loop. These statements create what we commonly call loops. As you probably know, a loop repeatedly executes the same set of instructions until a termination condition is met. A loop statement allows us to execute a statement or group of statements multiple times.

for loop

A for loop is a repetition control structure that allows you to efficiently write a loop that needs to be executed a specific number of times. A for loop is useful when you know how many times a task is to be repeated.

The syntax of a for loop is -

```

for(initialization; Boolean_expression; update) {
// Statements
}

```

Following is an example code of the for loop in Java.

```

using System;
class Program
{
static void Main(string[] args)
{
//find sum of numbers from 1 to 10
inti, sum = 0;
for (i = 1; i <= 10; i++)
sum = sum + i;
Console.WriteLine("Sum = {0}",sum);
}
}

```

Jump Statements

The C# jump statements are **break**, **continue**, **goto**, **return**, and **throw**. These statements transfer control to another part of your program.

Using break

In C#, the **break** statement has three uses. First, as you have seen, it terminates a statement sequence in a switch statement. Second, it can be used to exit a loop. Third, it can be used as a "civilized" form of **goto**.

Using break to Exit a Loop

By using **break**, you can force immediate termination of a loop, bypassing the conditional expression and any remaining code in the body of the loop. When a **break** statement is encountered inside a loop, the loop is terminated and program control resumes at the next statement following the loop. Here is a simple example:

```

using System;
class Program
{
static void Main(string[] args)
{
inti;
for (i = 1; i <= 10; i++)
{
Console.WriteLine("{0}\t",i);
if (i == 5)
break;
}
Console.WriteLine("\nLoop Terminated !");
}
}

```

3. Write a program in C# to overload unary operators.

Ans: // C# program to illustrate the unary operator overloading

```

using System;
namespace Calculator
{
class Calculator

```

```

    public int number1, number2;
    public Calculator(int num1, int num2)
    {
        number1 = num1;
        number2 = num2;
    }
    public static Calculator operator -(Calculator c1)
    {
        c1.number1 = -c1.number1;
        c1.number2 = -c1.number2;
        return c1;
    }
    public void Print()
    {
        Console.WriteLine("Number1 = " + number1);
        Console.WriteLine("Number2 = " + number2);
    }
}
class EntryPoint
{
    static void Main(String []args)
    {
        Calculator calc = new Calculator(15, -25);
        calc = -calc;
        calc.Print();
    }
}

```

4. What do you mean by constructor? Explain different types of constructors in detail along with programs.

Ans: A constructor in C# is a block of code similar to a method that's called when an instance of an object is created. Here are the key differences between a constructor and a method:

- A constructor doesn't have a return type.
- The name of the constructor must be the same as the name of the class.
- Unlike methods, constructors are not considered members of a class.
- A constructor is called automatically when a new instance of an object is created.

All classes have constructors, whether you define one or not, because C# automatically provides a default constructor that initializes all member variables to zero. However, once you define your own constructor, the default constructor is no longer used. In C#, constructors can be divided into 5 types

- Default Constructor
- Parameterized Constructor
- Copy Constructor

- Static Constructor
- Private Constructor

Default Constructor in C#

A constructor without any parameters is called a default constructor; in other words, this type of constructor does not take parameters. The drawback of a default constructor is that every instance of the class will be initialized to the same values and it is not possible to initialize each instance of the class with different values. The default constructor initializes:

Example

using System;

namespace DefaultConstructor

{

class addition

{

int a, b;

public addition() //default constructor

{

a = 100;

b = 175;

}

public static void Main()

{

additionobj = new addition(); //an object is created , constructor is called

Console.WriteLine(obj.a);

Console.WriteLine(obj.b);

Console.Read();

}

}

}

Parameterized Constructor

A constructor with at least one parameter is called a parameterized constructor. The advantage of a parameterized constructor is that you can initialize each instance of the class with a different value.

using System;

namespace Constructor

{

class paraconstructor

{

public int a, b;

public paraconstructor(int x, int y)

{

a = x;

b = y;

}

}

class MainClass

{

```

static void Main()
{
    paraconstructor v = new paraconstructor(100, 175);
    Console.WriteLine("\t");
    Console.WriteLine("value of a=" + v.a);
    Console.WriteLine("value of b=" + v.b);
    Console.Read();
}
}
}

```

Copy Constructor

The constructor which creates an object by copying variables from another object is called a copy constructor. The purpose of a copy constructor is to initialize a new instance to the values of an existing instance.

Syntax

```

public employee(employee emp)
{
    name=emp.name;
    age=emp.age;
}

```

The copy constructor is invoked by instantiating an object of type employee and bypassing it the object to be copied. Example
 employee emp1=new employee (emp2);

Static Constructor

When a constructor is created using a static keyword, it will be invoked only once for all of the instances of the class and it is invoked during the creation of the first instance of the class or the first reference to a static member in the class. A static constructor is used to initialize static fields of the class and to write the code that needs to be executed only once.

Syntax

```

class employee
{
    // Static constructor
    static employee(){}
}

```

Private Constructor

When a constructor is created with a private specifier, it is not possible for other classes to derive from this class, neither is it possible to create an instance of this class. They are usually used in classes that contain static members only. Some key points of a private constructor are:

```

using System;
namespace defaultConstructor
{
    public class Counter
    {
        private Counter() //private constructor declaration
        {

```

```

    public static int currentview;
    public static int visitedCount()
    {
        return ++currentview;
    }
}
class viewCountedetails
{
    static void Main()
    {
        // Counter aCounter = new Counter(); // Error
        Console.WriteLine();
        Counter.currentview = 500;
        Counter.visitedCount();
        Console.WriteLine("Now the view count is: {0}", Counter.currentview);
        Console.ReadLine();
    }
}

```

5. What do you mean by generics? Write a C# program to create generic class and generic methods.

Ans: Generic means the general form, not specific. In C#, generic means not specific to a particular data type. C# allows you to define generic classes, interfaces, abstract classes, fields, methods, static methods, properties, events, delegates, and operators using the type parameter and without the specific data type. A type parameter is a placeholder for a particular type specified when creating an instance of the generic type. A generic type is declared by specifying a type parameter in an angle brackets after a type name, e.g. TypeName<T> where T is a type parameter.

/ C# program to show working of user defined Generic classes using System;

```

public class GFG<T> {
    private T data;
    public T value
    {
        get
        {
            return this.data;
        }
        set
        {
            this.data = value;
        }
    }
}

```

```
// Driver class
class Test {
    static void Main(string[] args)
    {
        GFG<string> name = new GFG<string>();
        name.value = "BCA 5th";
        GFG<float> version = new GFG<float>();
        version.value = 5.0F;
        Console.WriteLine(name.value);
        Console.WriteLine(version.value);
    }
}

// C# program to show multiple type parameters in Generics
using System;
public class GFG {
    // Generics method
    public void Display<TypeOfValue>(string msg, TypeOfValue value) {
        Console.WriteLine("{0}:{1}", msg, value);
    }
}

// Driver class
public class Example {
    public static int Main()
    {
        GFG p = new GFG();
        p.Display<int>("Integer", 122);
        p.Display<char>("Character", 'H');
        p.Display<double>("Decimal", 255.67);
        return 0;
    }
}
```

6. Differentiate ADO and ADO.NET with example.

Ans: The Major Difference between ADO and ADO.NET is that ADO is a COM-based Library and ADO.net is a CLR based Library.

ADO	ADO.Net
It is a COM (Component Object Modelling) based Library.	It is a CLR (Common Language Runtime) based Library.
ADO works in the connected mode to access data.	ADO.Net works in the disconnected mode to access data.
Locking features is available	Locking features is not available
Data is stored in Binary Format	Data is stores in XML
XML integration is not possible	XML integration is possible
It uses Record Set to store the data from data source	It uses Dataset to store the data from data source

Using classic ADO, you can obtain information from one table or set of tables through join. You cannot fetch records from multiple tables independently	Dataset object of ADO.Net includes collection of Data Table wherein each Data Table will contain records fetched from a particular table. Hence multiple table records are maintained independently
In ADO, we can create only Client side cursor.	In ADO.Net, we can create both Client & Server side cursor
Derives information about data implicitly at run time, based on metadata that is often expensive to obtain.	Leverages known metadata at design time in order to provide better run-time performance and more consistent run-time behavior.
Firewall might prevent execution of Classic ADO	ADO.Net has firewall proof and its execution will never be interrupted
we cannot send multiple transaction using a single connection instance	we can send multiple transaction using a single connection instance

6. What do you mean by anonymous method? Explain with example.

Ans: an anonymous method is a method without a name. Anonymous methods in C# can be defined using the delegate keyword and can be assigned to a variable of delegate type.

Example:

```
public delegate void Print(int value);
static void Main(string[] args)
{
    Print print = delegate(int val)
    {
        Console.WriteLine("Inside Anonymous method. Value: {0}", val);
    };

    print(100);
}
```

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

TU QUESTIONS-ANSWERS 2020

Bachelor Level / fifth-semester / Science

Computer Science and Information Technology (CACS302)

(DotNet Technology)

Full marks: 60

Pass marks: 24

Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.

Group 'A'

Attempt all the questions.

10 × 1 = 10

1. Circle (O) the correct answer in the following questions

- i. .NET Framework is an internalcomponent that supports the execution of applications created by using various programming languages.
 a. Internet b. Windows c. Hardware d. Language
- ii. What is CLR?
 a. Common language Runtime b. Compiler Language Runtime
 c. Compiler Library Runtime d. Common Library Runtime
- iii. Which of the following is not a component of the CLR?
 a. Security engine b. Garbage collector
 c. .NET Framework d. JIT compiler
- iv. Which of the following provides automatic memory management and resolve the issue of memory leaks and invalid memory references?
 a. Security engine b. Garbage collector
 c. JIT Compiler d. Debugger
- v. What is CTS?
 a. Common type specification b. common type safe
 c. compiler type structure d. common type system
- vi. What is CLS?
 a. compiler library specification b. common library specification
 c. compiler language specification d. common language specification
- vii. Which of the following namespace provides the classes that allow you to debug your application step by step?
 a. System b. System Diagnostics
 c. system object d. system security
- viii. What is GAC?
 a. Garbage collector b. Global assembly collector
 c. Global access cache d. Global assembly cache
- ix. What is COM?
 a. Component object model b. common object model
 c. computer oriented model d. common oriented model
- x. code that targets the common language runtime is known as....
 a. Unmanaged code b. Distributed code
 c. Managed code d. Native code

Answer-Key

i. (b)	ii. (a)	iii. (c)	iv. (b)	v. (d)	vi. (d)	vii. (b)	viii. (d)	ix. (a)	x. (c)
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Group B

Attempt any SIX questions.

[6 × 5=30]

2. Differentiate Object Oriented Programming and Object Based Programming. Explain some of the major features of C# language. [2+3]

Ans: Object-oriented programming (OOP) is a programming paradigm based on the concept of "objects", which may contain data, in the form of fields, often known as attributes; and code, in the form of procedures, often known as methods. For example, a person is an object which has certain properties such as height, gender, age, etc. It also has certain methods such as move, talk, and so on.

Object-Oriented Languages (OOP) follow all the concepts of OOPs whereas, Object-based languages don't follow all the concepts of OOPs like inheritance and polymorphism. Object-oriented languages do not have the inbuilt objects whereas Object-based languages have the inbuilt objects, for example, JavaScript has window object.

Examples for Object Oriented Languages include Java, C# whereas Object-based languages include VB etc.

Here are the significant difference between Object-oriented Programming Language and Object-based Programming Language:

Object-oriented Programming Language	Object-based Programming Language
All the characteristics and features of object-oriented programming are supported.	All characteristics and features of object-oriented programming, such as inheritance and polymorphism are not supported.
These types of programming languages don't have a built-in object. Example: C++.	These types of programming languages have built-in objects. Example: JavaScript has a window object.
Java is an example of object-oriented programming language which supports creating and inheriting (which is reusing of code) one class from another.	VB is another example of object-based language as you can create and use classes and objects, but inheriting classes is not supported.

Major features of C# language

The major features of C# from an object-oriented perspective are:

a. **Unified type system**

The fundamental building block in C# is an encapsulated unit of data and functions called a type. C# has a unified type system, where all types ultimately share a common base type. This means that all types, whether they represent business objects or are primitive types such as numbers, share the same basic functionality. For example, an instance of any type can be converted to a string by calling its ToString method.

b. **Classes and interfaces**

In a traditional object-oriented paradigm, the only kind of type is a class. In C#, there are several other kinds of types, one of which is an interface. An interface is like a class, except that it only describes members. The implementation for those members comes from types that implement the interface. Interfaces are particularly useful in scenarios where multiple inheritance is required (unlike

languages such as C++ and Eiffel, C# does not support multiple inheritance of classes).

c. **Properties, methods, and events**

In the pure object-oriented paradigm, all functions are methods. In C#, methods are only one kind of function member, which also includes properties and events. Properties are function members that encapsulate a piece of an object's state, such as a button's colour or a label's text. Events are function members that simplify acting on object state changes.

While C# is primarily an object-oriented language, it also borrows from the functional programming paradigm. Specifically:

a. **Functions can be treated as values**

Through the use of delegates, C# allows functions to be passed as values to and from other functions.

b. **C# supports patterns for purity**

Core to functional programming is avoiding the use of variables whose values change, in favour of declarative patterns. C# has key features to help with those patterns, including the ability to write unnamed functions on the fly that "capture" variables (lambda expressions), and the ability to perform list or reactive programming via query expressions. C# also makes it easy to define read-only fields and properties for writing immutable (read-only) types.

3. **Explain overview of Microsoft .NET framework and its components in detail.** [5]

Ans: DotNet is the framework for which we develop applications. It sits in between our application programs and operating system. It provides an object oriented environment. It ensures safe execution of the code by performing required runtime validations. For example, it is never possible to access an element of an array outside the boundary. Similarly, it is not possible to a program to write into another programs area, etc. The runtime validations performed by .NET makes the entire environment robust.

It is a programming infrastructure created by Microsoft for building, deploying, and running applications and services that use .NET technologies, such as desktop applications and Web services. The .NET Framework consists of:

- the Common Language Runtime
- the Framework Class Library

Common Language Runtime (CLR)

CLR is the basic and Virtual Machine component of the .NET Framework. It is the run-time environment in the .NET Framework that runs the codes and helps in making the development process easier by providing the various services such as remoting, thread management, type-safety, memory management, robustness etc. Basically, it is responsible for managing the execution of .NET programs regardless of any .NET programming language. It also helps in the management of code, as code that targets the runtime is known as the Managed Code and code doesn't target to runtime is known as Unmanaged code.

Framework Class Library (FCL)

The class library is a comprehensive, object-oriented collection of reusable types that you can use to develop applications ranging from traditional command-line or graphical user interface (GUI) applications to applications based on the latest innovations provided by ASP.NET, such as Web Forms and XML Web services.

It is the collection of reusable, object-oriented class libraries and methods etc. that can be integrated with CLR. Also called the Assemblies. It is just like the header files in C/C++ and packages in the java. Installing .NET framework basically is the installation of CLR and FCL into the system. The core libraries are sometimes collectively called the Base Class Library (BCL). The entire framework is called the Framework Class Library (FCL). .Net framework provides multiple advantages to the programmers in comparison to the advantages provided by other platforms. Microsoft has integrated various modern as well as existing technologies of application development in .NET framework. These technologies are highly efficient for modern as well as future business application. There are following components of .NET framework:

1. .NET Class Library
2. Common Language runtime
3. Dynamic Language runtime
4. Application domains
5. .Net Framework Security
6. Cross Language interoperability
7. Side by side execution
8. Common Type System

1. **.NET Class Library**

.NET framework contains multiple classes that are readily available for developers. The classes in the FCL (framework class library) are grouped under multiple namespaces.

2. **Common Language Runtime**

CLR provides interoperability between different language, like C#, VB, Visual C++, by providing a common environment for the execution of code written in of these languages.

3. **Dynamic Language runtime**

DLR provides to execute dynamic languages on .NET Framework by adding some special services to the CLR.

4. **Application domains**

It is used to isolate the process of different applications and can be defined by .NET framework.

5. **.NET Framework Security**

.NET framework provides multiple tools that can be used by developers to protect the resources and code from unauthorized users.

6. **Cross Language interoperability**

Object or compiled code of one language can be used in other .NET compatible language.

7. **Side by side execution**

In the same application we can use multiple versions of CLR simultaneously.

8. Common Type System

CTS is used to maintain data integrity across the code written in different .NET compliant languages. CTS also used to prevent data loss when a type in one language transfers data to its equivalent type in other language.

4. What do you mean by property in C# language? How it is different from method? Compare automatic property with other types of property with suitable example. [1+1+3]

Ans: Property in C# is a member of a class that provides a flexible mechanism for classes to expose private fields. Internally, C# properties are special methods called accessors. A C# property have two accessors, get property accessor and set property accessor. A get accessor returns a property value, and a set accessor assigns a new value. The value keyword represents the value of a property.

Properties look like fields from the outside, but internally they contain logic, like methods do. Properties are named members of classes, structures, and interfaces. Member variables or methods in a class or structures are called **Fields**. Properties are an extension of fields and are accessed using the same syntax. They use **accessors(get and set)** through which the values of the private fields can be read, written or manipulated. Usually, inside a class, we declare a data field as private and will provide a set of public SET and GET methods to access the data fields. This is a good programming practice since the data fields are not directly accessible outside the class. We must use the set/get methods to access the data fields.

An example, which uses a set of set/get methods, is shown below.

using System;

class MyClass

```
{
    private int x;
    public void SetX(int i)
    {
        x = i
    }
    public int GetX()
    {
        return int GetX()
    }
}
```

class MyClient

```
{
    public static void Main()
    {
        MyClass mc = new MyClass();
        mc.SetX(10);
        int xVal = mc.GetX()
        Console.WriteLine(xVal);
    }
}
```

But C# provides a built in mechanism called properties to do the above. In C#, properties are defined using the property declaration syntax. The general form of declaring a property is as follows.

<access_modifier><return_type><property_name>

```
{
    get
    {
    }
    set
    {
    }
}
```

For example, the above program can be modified with a property X as follows.

```
using System;
class MyClass
{
    private int x;
    public int X
    {
        get
        {
            return x;
        }
        set
        {
            x = value;
        }
    }
}
class MyClient
{
    public static void Main()
    {
        MyClass mc = new MyClass();
        mc.X = 10;
        int xVal = mc.X;
        Console.WriteLine(xVal); // Displays 10
    }
}
```

Compare automatic property with other types of property

Automatic Properties

The most common implementation for a property is a getter and/or setter that simply reads and writes to a private field of the same type as the property. An automatic property declaration instructs the compiler to provide this implementation. We can improve the first example in this section by declaring CurrentPrice as an automatic property.

```
{
```

```

...
public decimal CurrentPrice { get; set; }
}
Example of Automatic Property
class Chk
{
    public int a { get; set; }
    public int b { get; set; }
    public int sum
    {
        get
        {
            return a + b;
        }
    }
}
class Test
{
    static void Main()
    {
        Chk obj = new Chk();
        obj.a = 10;
        obj.b = 5;
        Console.WriteLine("Sum of "+obj.a+" and "+obj.b+" = "+obj.sum);
        Console.ReadKey();
    }
}

```

Output:

Sum of 10 and 5 = 15

5. Define constructor. Explain different types of constructors used in C# with example. [1+4]

Ans: A constructor in C# is a block of code similar to a method that's called when an instance of an object is created. Here are the key differences between a constructor and a method:

- A constructor doesn't have a return type.
- The name of the constructor must be the same as the name of the class.
- Unlike methods, constructors are not considered members of a class.
- A constructor is called automatically when a new instance of an object is created.

All classes have constructors, whether you define one or not, because C# automatically provides a default constructor that initializes all member variables to zero. However, once you define your own constructor, the default constructor is no longer used. In C#, constructors can be divided into 5 types

- Default Constructor
- Parameterized Constructor
- Copy Constructor

- Static Constructor
- Private Constructor

Default Constructor in C#

A constructor without any parameters is called a default constructor, in other words, this type of constructor does not take parameters. The drawback of a default constructor is that every instance of the class will be initialized to the same values and it is not possible to initialize each instance of the class with different values. The default constructor initializes:

Example

using System;

namespace DefaultConstructor

```
{
    class addition
    {
        int a, b;
        public addition() //default constructor
        {
            a = 100;
            b = 175;
        }
    }
}
```

public static void Main()

```
{
    additionobj = new addition(); //an object is created , constructor is called
    Console.WriteLine(obj.a);
    Console.WriteLine(obj.b);
    Console.Read();
}
```

Parameterized Constructor

A constructor with at least one parameter is called a parameterized constructor. The advantage of a parameterized constructor is that you can initialize each instance of the class with a different value.

using System;

namespace Constructor

```
{
    class paraconstructor
    {
        public int a, b;
        public paraconstructor(int x, int y)
        {
            a = x;
            b = y;
        }
    }
}

class MainClass
```

```

static void Main()
{
    paraconstructor v = new paraconstructor(100, 175);
    Console.WriteLine("\t");
    Console.WriteLine("value of a=" + v.a);
    Console.WriteLine("value of b=" + v.b);
    Console.Read();
}
}

```

Copy Constructor

The constructor which creates an object by copying variables from another object is called a copy constructor. The purpose of a copy constructor is to initialize a new instance to the values of an existing instance.

Syntax

```

public employee(employee emp)
{
    name=emp.name;
    age=emp.age;
}

```

The copy constructor is invoked by instantiating an object of type employee and bypassing it the object to be copied. Example
 employee emp1=new employee (emp2);

Static Constructor

When a constructor is created using a static keyword, it will be invoked only once for all of the instances of the class and it is invoked during the creation of the first instance of the class or the first reference to a static member in the class. A static constructor is used to initialize static fields of the class and to write the code that needs to be executed only once.

Syntax

```

class employee
{
    // Static constructor
    static employee(){}
}

```

Private Constructor

When a constructor is created with a private specifier, it is not possible for other classes to derive from this class, neither is it possible to create an instance of this class. They are usually used in classes that contain static members only. Some key points of a private constructor are:

using System;

namespace defaultConstructor

```

{
    public class Counter
    {
        private Counter() //private constructor declaration
        {
        }
    }
}

```

```

    public static int currentview;
    public static int visitedCount()
    {
        return ++currentview;
    }
}
class viewCountedetails
{
    static void Main()
    {
        // Counter aCounter = new Counter(); // Error
        Console.WriteLine();
        Counter.currentview = 500;
        Counter.visitedCount();
        Console.WriteLine("Now the view count is: {0}", Counter.currentview);
        Console.ReadLine();
    }
}

```

6. Define inheritance. Write a C# program to demonstrate multilevel and multiple inheritance. [1+2+2]

Ans: The process by which one class acquires the properties (data members) and functionalities (methods) of another class is called **inheritance**. The aim of inheritance is to provide the reusability of code so that a class has to write only the unique features and rest of the common properties and functionalities can be extended from the another class.

The idea behind inheritance in C# is that you can create new classes that are built upon existing classes. When you inherit from an existing class, you can reuse methods and fields of the parent class. Moreover, you can add new methods and fields in your current class also.

The syntax of C# Inheritance

```
class Subclass-name : Superclass-name
```

```
{
    //methods and fields
}
```

The : indicates that you are making a new class that derives from an existing class. The meaning of "extends" is to increase the functionality.

In the terminology of C#, a class which is inherited is called a parent or superclass, and the new class is called child or subclass.

Multilevel Inheritance

Multilevel inheritance refers to a child and parent class relationship where a class extends the child class. For example, class C extends class B and class B extends class A.

Syntax of Multilevel Inheritance:

```
class A //base class
```

```
{
    //data members & methods
}
```

```

class B : A //derived class
    //also base class of C
    {
        //data members & methods
    }

```

```

class C : B //derived class
    {
        //data members & methods
    }

```

Example:

using System;

class A

```

    {
        public int a, b, c;
        public void ReadData(int a, int b)
        {
            this.a = a;
            this.b = b;
        }
        public void Display()
        {
            Console.WriteLine("Value of a is: "+a);
            Console.WriteLine("Value of b is: " + b);
        }
    }

```

class B : A

```

    {
        public void Add()
        {
            base.c = base.a + base.b;
            Console.WriteLine("Sum="+base.c);
        }
    }

```

class C : B

```

    {
        public void Sub()
        {
            base.c = base.a - base.b;
            Console.WriteLine("Difference=" + base.c);
        }
    }

```

class Level

```

    {
        static void Main()
        {
            C obj = new C();
            obj.ReadData(20,5);
        }
    }

```

```

obj.Display();
obj.Add();
obj.Sub();
Console.ReadLine();

```

Multiple Inheritance

When one class extends more than one classes then this is called multiple inheritance. For example: Class C extends class A and B then this type of inheritance is known as multiple inheritance. C# doesn't allow multiple inheritance. We can use interfaces instead of classes to achieve the same purpose.

Syntax of Multiple Inheritance:

```

interface A
{
    //methods (only signature)
}

class B
{
    //data members & methods
}

class C : B,A
{
    //data members & methods
}

```

Example:

```
using System;
```

```
interface A
```

```

{
    //doesn't contain data members
    //method only contain signature
    int CalculateArea();
    int CalculatePerimeter();
}

```

```
class B
```

```

{
    public int l, b;
    public void ReadData(int l, int b)
    {
        this.l = l;
        this.b = b;
    }
}

```

```
class C : A, B
```

```

{
    public int CalculateArea()
    {

```

```

        ReadData(10,5);
        int area=l*b;
        return area;
    }
    public int CalculatePerimeter()
    {
        ReadData(15, 10);
        int peri = 2*(l+b);
        return peri;
    }
}

// driver class
class Inter
{
    static void Main(string[] args)
    {
        C obj = new C();
        console.WriteLine("Area of Rectangle=" + obj.CalculateArea());
        Console.WriteLine("Perimeter of Rectangle=" +
            obj.CalculatePerimeter());
        Console.ReadKey();
    }
}

```

7. What is generics? List different types of generic classes. Explain delegate with example. [1+1+3]

Ans: Generics in C# and .NET procedure many of the benefits of strongly-typed collections as well as provide a higher quality of and a performance boost for code.

Generics are very similar to C++ templates but having a slight difference in such a way that the source code of C++ templates is required when a template is instantiated with a specific type and .NET Generics are not limited to classes only. In fact, they can also be implemented with Interfaces, Delegates and Methods.

Generic Classes

The Generic class can be defined by putting the <T> sign after the class name. It isn't mandatory to put the "T" word in the Generic type definition. You can use any word in the TestClass<> class declaration.

```
public class TestClass<T> { }
```

The **System.Collection.Generic** namespace also defines a number of classes that implement many of these key interfaces. The following table describes the core class types of this namespace.

Generic class	Description
Collection<T>	The basis for a generic collection Comparer compares two generic objects for equality
Dictionary<TKey, TValue>	A generic collection of name/value pairs
List<T>	A dynamically resizable list of Items
Queue<T>	A generic implementation of a first-in, first-out (FIFO) list
Stack<T>	A generic implementation of a last-in, first-out (LIFO) list

C# - Delegates

The delegate is a reference type data type that defines the method signature. We can define variables of delegate, just like other data type, that can refer to any method with the same signature as the delegate. There are three steps involved while working with delegates:

- Declare a delegate
- Set a target method
- Invoke a delegate

A delegate can be declared using the delegate keyword followed by a function signature, as shown below.

Delegate Syntax

[Access modifier] delegate [return type] [delegate name]([parameters])

Complete Example:

```
using System;
public delegate void MyDelegate(string msg);
public class Program
{
    public static void Main()
    {
        MyDelegatedel = ClassA.MethodA;
        del("Hello World");
        del = ClassB.MethodB;
        del("Hello World");
        del = (string msg) => Console.WriteLine("Called lambda
expression: " + msg);
        del("Hello World");
    }
}
public class ClassA
{
    public static void MethodA(string message)
    {
        Console.WriteLine("Called ClassA.MethodA() with parameter: " + message);
    }
}
public class ClassB
{
    public static void MethodB(string message)
    {
        Console.WriteLine("Called ClassB.MethodB() with parameter: " + message);
    }
}
```

Input / Output

Called ClassA.MethodA() with parameter: Hello World

Called ClassB.MethodB() with parameter: Hello World

Called lambda expression: Hello World

8. What do you mean by lambda expression? Explain different types of lambda expression used in C# with example. [1+4]

Ans: Lambda expressions in C# are used like anonymous functions, with the difference that in Lambda expressions you don't need to specify the type of

the value that you input thus making it more flexible to use. The ' $\Rightarrow$ ' is the lambda operator which is used in all lambda expressions. The Lambda expression is divided into two parts, the left side is the input and the right is the expression.

The Lambda Expressions can be of two types:

1. Expression Lambda: Consists of the input and the expression.

Syntax:

input $\Rightarrow$ expression;

2. Statement Lambda: Consists of the input and a set of statements to be executed. It can be used along with delegates.

Syntax:

input $\Rightarrow$ { statements };

Basic example of lambda expression:

```
class LambdaTest
{
    static int test1()  $\Rightarrow$  5;
    static int test2(int x)  $\Rightarrow$  x + 10;
    static void Main(string[] args)
    {
        int x = test1();
        int res = test2(x);
        Console.WriteLine("Result is: "+res);
    }
}
```

Group C

Attempt any TWO questions.

[2x10=20]

9. a) Write a program to create user registration form in one ASP.NET web page and display filled data in another page. [7]

Ans:

```
<body>
<form id="form">
<div>
<table class="auto-style1">
<tr>
<td>
<asp:Label ID="Label1" Text="User Name"></asp:Label>
</td>
<td>
<asp:TextBox ID="username" required="true"></asp:TextBox></td>
</tr>
<tr>
<td>
<asp:Label ID="Label6" Text="Email ID"></asp:Label>
</td>
<td>
<asp:TextBox ID="EmailID" TextMode="Email"></asp:TextBox></td>
</tr>
<tr>
<td>
<asp:Label ID="Label2" Text="Password"></asp:Label></td>
```

```

<td>
<asp:TextBoxID="TextBox2"
TextMode="Password"></asp:TextBox></td>
</tr>
<tr>
<td>
<asp:Label ID="Label3" Text="Confirm Password"></asp:Label></td>
<td>
<asp:TextBoxID="TextBox3" TextMode="Password"></asp:TextBox></td>
</tr>
<tr>
<td>
<asp:Label ID="Label4" Text="Gender"></asp:Label></td>
<td>
<asp:RadioButton ID="RadioButton1" GroupName="gender" Text="Male" />
<asp:RadioButton ID="RadioButton2" GroupName="gender"
Text="Female" /></td>
</tr>
<tr>
<td>
<asp:Label ID="Label5" Text="Select Course"></asp:Label></td>
<td>
<asp:CheckBox ID="CheckBox1" />
<asp:CheckBox ID="CheckBox2" />
<asp:CheckBox ID="CheckBox3" />
</td>
</tr>
<tr>
<td>
<asp:Button ID="Button1" Text="Register" CssClass="btn btn-primary"
OnClick="Button1_Click"/>
</td>
</tr>
</table>
<asp:Label ID="message"Font-Size="Medium"
ForeColor="Red"></asp:Label>
</div>
</form>
<table class="auto-style1">
<tr>
<td class="auto-style2"><asp:Label ID="ShowUserNameLabel"
></asp:Label></td>
<td>
<asp:Label ID="ShowUserName" ></asp:Label></td>
</tr>
</table>

```

```

<td class="auto-style2"><asp:Label
ID="ShowEmailIDLabel"
></asp:Label></td>
<td>
<asp:Label ID="ShowEmail" ></asp:Label></td>
</tr>
<tr>
<td class="auto-style3"><asp:Label
ID="ShowGenderLabel"
></asp:Label></td>
<td class="auto-style4">
<asp:Label ID="ShowGender" ></asp:Label></td>
</tr>
<tr>
<td class="auto-style2"><asp:Label
ID="ShowCourseLabel"
></asp:Label></td>
<td>
<asp:Label ID="ShowCourses" ></asp:Label></td>
</tr>
</table>
</body>
</html>

```

b) Write a program for handling exception in ASP.NET. [3]

Ans: An exception is a problem that arises during the execution of a program. A C# exception is a response to an exceptional circumstance that arises while a program is running, such as an attempt to divide by zero. Exceptions provide a way to transfer control from one part of a program to another. C# exception handling is built upon four keywords: **try**, **catch**, **finally**, and **throw**.

- **try** - A try block identifies a block of code for which particular exceptions is activated. It is followed by one or more catch blocks.
- **catch** - A program catches an exception with an exception handler at the place in a program where you want to handle the problem. The catch keyword indicates the catching of an exception.
- **finally** - The finally block is used to execute a given set of statements, whether an exception is thrown or not thrown. For example, if you open a file, it must be closed whether an exception is raised or not.
- **throw** - A program throws an exception when a problem shows up. This is done using a throw keyword.

Example:

```

using System;
class HandledException
{
    public static void Main()
    {
        int x = 0;
        int intTemp = 0;
        try
        {
            intTemp = 100/x;

```

```

        Console.WriteLine("Not executed line");
    }
    catch
    {
        Console.WriteLine("oException");
    }
    Console.WriteLine("Result is {0}", intTemp);
}
}

```

10. a) How virtual method is used to achieve dynamic binding in C#? Explain with the help of suitable program. [1+4]

Ans: Polymorphism, Late Binding, Method overriding etc. Whereas in static polymorphism we overload a function; in dynamic polymorphism we override a base class function using virtual or override keyword. Method overriding means having two or more methods with the same name, same signature but with different implementation.

Example

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace RuntimePolymorphism
{
    class Program
    {
        static void Main(string[] args)
        {
            Chocolate ch = new Chocolate();
            ch.flavor();
            Console.ReadKey();
        }
    }
    class IceCream
    {
        public IceCream()
        {
            Console.WriteLine("Class : Icecream");
        }
        public virtual void flavor()
        {
            Console.WriteLine("IceCream Type : Vanilla");
        }
    }
    class Chocolate : IceCream
    {
        public Chocolate()
        {
            Console.WriteLine("Class : Chocolate");
        }
    }
}

```

```

    }
    public override void flavor()
    {
        Console.WriteLine("IceCream Type : Chocolate");
    }
}

```

- b) Define operator overloading. Write a C# program to overload binary operator. [1+4]

Ans: Operator overloading gives the ability to use the same operator to do various operations. It enables to make user-defined implementations of various operations where one or both of the operands are of a user-defined class. Only the predefined set of C# operators can be overloaded. To make operations on a user-defined data type is not as simple as the operations on a built-in data type. To use operators with user-defined data types, they need to be overloaded according to a programmer's requirement. An operator can be overloaded by defining a function to it. The function of the operator is declared by using the operator keyword.

Syntax:

```

    accessSpecifier className operator Operator_symbol
(parameters)
{
    // Code
}

```

C# program to illustrate the Binary Operator Overloading
using System;

namespace BinaryOverload

```

{
    class Calculator
    {
        public int number = 0;
        // no-argument constructor
        public Calculator() {}
        // parameterized constructor
        public Calculator(int n)
        {
            number = n;
        }
        // Overloading of Binary "+" operator
        public static Calculator operator + (Calculator Calc1, Calculator Calc2)
        {
            Calculator Calc3 = new Calculator(0);
            Calc3.number = Calc2.number + Calc1.number;
            return Calc3;
        }
        // function to display result
        public void display()
        {
            Console.WriteLine("{0}", number);
        }
    }
}

```

```

    }
    class CalNum
    {

```

```

        // Driver Code
        static void Main(string[] args)
    {

```

```

        Calculator num1 = new Calculator(200);
        Calculator num2 = new Calculator(40);
        Calculator num3 = new Calculator();
        num3 = num1 + num2;

```

```

        num1.display(); // Displays 200
        num2.display(); // Displays 40
        num3.display(); // Displays 240
    }
}

```

11. a) What is LINQ? Write a program to select employees whose salary is greater than 20000 and whose address is Kathmandu using LINQ. [1+4]

Ans: LINQ (Language Integrated Query) is uniform query syntax in C# to retrieve data from different sources and formats. It is integrated in C#, thereby eliminating the mismatch between programming languages and databases, as well as providing a single querying interface for different types of data sources.

For example, SQL is a Structured Query Language used to save and retrieve data from a database. In the same way, LINQ is a structured query syntax built in C# to retrieve data from different types of data sources such as collections, ADO.Net DataSet, XML Docs, web service and MS SQL Server and other databases.

Program part,

```
using System;
```

```
using System.Collections.Generic;
```

```
using System.Linq;
```

```
public class Employee
```

```
{
    int ID;
```

```
    string Name;
```

```
    string Address;
```

```
    int Salary;
```

```
    public override string ToString()
    {

```

```
        return ID + " " + Name + " " + Address + " " + Salary;
    }

```

```
    static void Main(string[] args)
    {

```

```
        List<Employee> employees = new List<Employee>()
        {

```

```
            new Employee {ID=101, Name="Sumit", Address="Palpa", Salary=40000},
        }
    }
}

```

```

new Employee (ID=102, Name="Kiran" ,Address="Kathmandu",
Salary=6000),
new Employee (ID=103, Name="Suman" , Address ="Kanchanpur",
Salary=7000),
new Employee (ID=104, Name="Raman" , Address ="Kathmandu",
Salary=90000),
);
IEnumerable<Employee> Query =
from emp in employees
where emp.Salary> 20000 AND Address="Kathmandu"
select emp;
Console.WriteLine("ID Name Address Salary");
Console.WriteLine("=====");
foreach (Employee s in Query)
{
    Console.WriteLine(s.ToString());
}
Console.WriteLine("=====");
}

```

b) Write a C# program to show insert and select operation in database. [5]

Ans:

```

using MySql.Data.MySqlClient;
using System;
using System.Data;
namespace DatabaseTest
{
    class Program
    {
        MySqlConnection conn;
        MySqlCommand command;
        void CreateConnection()
        {
            string constr = "SERVER=localhost;
DATABASE=dbtest;
UID=root; PASSWORD=";
conn = new MySqlConnection(constr);
conn.Open();
}
        void SelectRecords(string sql)
        {
            command = new MySqlCommand(sql, conn);
            MySqlDataAdapter adapter =
            new MySqlDataAdapter(command);
            DataTable dt = new DataTable();
            adapter.Fill(dt);
            if (dt.Rows.Count != 0)
            {
                Console.WriteLine("Sid\t Name\t
Address");
            }
        }
    }
}

```

```

        for (inti = 0; i<dt.Rows.Count;i++)
        {
            stringsid =
            dt.Rows[i]["sid"].ToString();
            string name =
            dt.Rows[i]["name"].ToString();
            string address =
            dt.Rows[i]["address"].ToString();
            Console.WriteLine(sid+"\t"+name
            +"\t"+address);
        }
    }
}

static void Main(string[] args)
{
    Programobj = new Program();
    try
    {
        obj.CreateConnection();
        x: Console.WriteLine("1.Insert\t2.Select");
        Console.WriteLine("Enter your choice: ");
        int n = Convert.ToInt32(Console.ReadLine());
        stringsql="",nm="",add="";
        int id=0;
        switch (n)
        {
            case 1:
                Console.WriteLine("Enter Name of Student: ");
                nm = Console.ReadLine();
                Console.WriteLine("Enter Address of Student: ");
                add = Console.ReadLine();
                sql = "INSERT INTO
                tblStudent(name,address)VALUES('"+nm+"','"+add+"')";
                obj.InsertUpdateDelete(sql);
                break;
            case 2:
                sql = "SELECT * FROM tblStudent";
                obj.SelectRecords(sql);
                break;
            default:
                Console.WriteLine("Wrong Choice");
                break;
        }
        goto x;
    }
}
}

```



MODEL QUESTIONS SETS FOR PRACTICE

MODEL SET 1

Bachelor Level / fifth-semester / Science
Computer Science and Information Technology (CACS302)
(DotNet Technology)

Full marks: 60

Pass marks: 24

Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.

Group 'A'

Attempt all the questions.

10 × 1 = 10

1. Circle (O) the correct answer in the following questions

- i. What is CLS?
 - a. Compiler Library Specification
 - b. Common Library Specification
 - c. Compiler Language Specification
 - d. Common Language Specification
- ii. What is DLL?
 - a. Dynamic-Link Language
 - b. Direct Link Language
 - c. Dynamic-Link Library
 - d. Direct Link Library
- iii. Which Of The Following Namespace Provides The Classes That Allow You To Debug Your Application Step By Step?
 - a. System
 - b. System.Diagnostics
 - c. System.Object
 - d. System.Security
- iv. What Is GAC?
 - a. Garbage collector
 - b. Global assembly collector
 - c. Global access cache
 - d. Global assembly cache
- v. What Is COM?
 - a. Component object model
 - b. Common object model
 - c. Computer oriented model
 - d. Common oriented model
- vi. Which of the following components of the .NET framework provide an extensible set of classes that can be used by any .NET compliant programming language?
 - a. .NET class libraries
 - b. Common Language Runtime
 - c. Common Language Infrastructure
 - d. Component Object Model
- vii. Which of the following statements correctly define .NET Framework?
 - a. It is an environment for developing, building, deploying and executing Desktop Applications, Web Applications and Web Services.
 - b. It is an environment for developing, building, deploying and executing only Web Applications.
 - c. It is an environment for developing, building, deploying and executing Distributed Applications.
 - d. It is an environment for developing, building, deploying and executing Web Services.
- viii. Which of the following constitutes the .NET Framework?
 - a. ASP.NET Applications
 - b. CLR
 - c. Framework Class Library
 - d. None of above

- ix. Code that targets the Common Language Runtime is known as
 a. Unmanaged code b. Distributed code
 c. Managed Code d. Native Code
- x. Which of the following statements is correct about Managed Code?
 a. Managed code is the code that is compiled by the JIT compilers.
 b. Managed code is the code where resources are Garbage Collected.
 c. Managed code is the code that runs on top of Windows.
 d. Managed code is the code that is written to target the services of the CLR.

Group B

[6x5=30]

Attempt any SIX questions.

- What are different LINQ operators? Explain.
- Explain indexers with the help of suitable example.
- What do you mean by static class? Explain with suitable program.
- Write a C# program to find product of any two matrices.
- What are other popular frameworks related to .NET? Explain.
- Explain different .NET implementations in brief.
- Differentiate OOP and POP in detail.

Group C

[2x10=20]

Attempt any TWO questions.

- Differentiate single and multilevel inheritance with example.
 - Differentiate error and exception with example. Write a suitable program to demonstrate exception handling.
- Write a program to show the use of auto-increment and auto-decrement operators.
 - Explain DataReader, DataAdapter, Dataset and DataTable with example.
- What do you mean by generics? Write a C# program to create generic class and generic methods.
 - Write a C# program to find factorial of any number.



Group A

Attempt all the questions

[10x1=10]

- Circle (O) the correct answer in the following questions:
- Code snippet for having a named instance of SQL Server would be.....
 - "Server=localhost\sql express"
 - "Server=local\sql express"
 - "Server=host\sql express"
 - "Ser=localhost\sql express"
- Syntax for closing and opening the connection in ADO.net is.....
 - sqlConn.Open() and sqlConn.close()
 - sqlConn.open() and sqlConn.Close()
 - sqlConn.Open() and sqlConn.Close()
 - none of the mentioned
- _____ object is used to fill a DataSet/DataTable with query results in ADO.net.
 - DataReader
 - Dataset
 - DataAdapter
 - DataTables

- iv. ASP stands for.....
 - a) Active Server Pages
 - b) Access Server Pages
 - c) Active Server Platform
 - d) Active Server Programming
- v. Web.config file is used.....
 - a) Configures the time that the server-side codebehind module is called
 - b) To store the global information and variable definitions for the application
 - c) To configure the web server
 - d) To configure the web browser
- vi. Difference between Response.Write() and Response.Output.Write().
 - a) Response.Output.Write() allows you to buffer output
 - b) Response.Output.Write() allows you to write formatted output
 - c) Response.Output.Write() allows you to flush output
 - d) Response.Output.Write() allows you to stream output
- vii. What class does the ASP.NET Web Form class inherit from by default?
 - a) System.Web.UI.Page
 - b) System.Web.UI.Form
 - c) System.Web.UI.Page
 - d) System.Web.Form
- viii. Attribute must be set on a validator control for the validation to work.....
 - a) ControlToValidate
 - b) ControlToBind
 - c) ValidateControl
 - d) Validate
- ix. The DataReader object is an alternative to the.....
 - a) DataSet
 - b) DataAdapter
 - c) Both A and B
 - d) None of the above
- x. The file extension an ASP.NET web form is.....
 - a) .docx
 - b) .aspx
 - c) .jpeg
 - d) none of the above

Group B

Attempt any SIX questions.

[6x5=30]

1. Write a suitable program to demonstrate LINQ query.
2. What do you mean by properties in C#? Explain with example.
3. Explain automatic property in C# with example.
4. Write a C# program to add any two matrices.
5. Explain different backend technologies associated with .NET framework.
6. Explain the scope of .NET technology.
7. What is OOP? Explain features of OOP.

Group C

Attempt any TWO questions.

[2x10=20]

1.
 - (a) Create student registration form in ASP.NET with proper form validations.
 - (b) What are different validation controls used in ASP.NET? Explain with example.
2. Write a program for entering college general information in one ASP page and display it in another page.
3.
 - (a) Differentiate class and struct. Write a C# program using struct.
 - (b) Explain different access supported by C# with example.

MODEL SET 3

Candidates are required to give their answer in their own words as far as practicable.
Group- A

[10 x1=10]

Attempt all the questions

1. Circle (O) the correct answer in the following questions:

- i. Correct method to define + operator is?
 - a) public sample operator +(int a, int b)
 - b) public abstract operator +(int a, int b)
 - c) public static sample operator +(int a, int b)
 - d) public abstract sample operator +(int a, int b)
- ii. Which keyword is used to declare a base class method while performing overriding of base class methods?
 - a) this
 - b) virtual
 - c) Override
 - d) extend
- iii. The process of defining a method in a subclass having same name & type signature as a method in its superclass is known as?
 - a) Method overloading
 - b) Method overriding
 - c) Method hiding
 - d) None of the mentioned
- iv. Which of the given modifiers can be used to prevent Method overriding?
 - a) Static
 - b) Constant
 - c) Sealed
 - d) final
- v. Select the correct statement from the following?
 - a) Static methods can be a virtual method
 - b) Abstract methods can be a virtual method
 - c) When overriding a method, the names and type signatures of the override method must be the same as the virtual method that is being overridden
 - d) We can override virtual as well as no virtual methods
- vi. Which of the following cannot be used to declare a class as a virtual?
 - a) Methods
 - b) Properties
 - c) Events
 - d) Fields
- vii. Where the properties can be declared?
 - a) Class
 - b) struct
 - c) Interface
 - d) All of the mentioned
- viii. Select the modifiers which can be used with the properties?
 - a) Private
 - b) Public
 - c) Protected Internal
 - d) All of the mentioned
- ix. Select the correct statement about properties of read and write in C#.NET?
 - a) A property can simultaneously be read or write only
 - b) A property cannot be either read only or write only
 - c) A write only property will only have get accessor
 - d) A read only property will only have get accessor
- x. What is meant by the term generics?
 - a) parameterized types
 - b) class
 - c) structure
 - d) interface

Group B

[6x5=30]

Attempt any SIX questions.

1. What are different exception classes? Explain.
2. What do you mean by LINQ? Differentiate LINQ and SQL with example.
3. What do you mean by static constructor? How it is different than other types of constructors? Explain with example.

4. Explain the use of this keyword in C# with example.
5. Write a C# program to sort name of 10 persons in alphabetical order.
6. Differentiate parameters pass by value and by reference with example.
7. Explain operator precedence with example.

Group C

Attempt any TWO questions.

[2x10=20]

1. Write a program ASP.NET to find simple interest. Use text fields for inputs and label for output. Your program should display the result when the user presses a button.
2. (a) what is polymorphism? Explain compile time and runtime polymorphism with program in detail.
(b) Write a program to show the use of break, continue and return.
3. Write a program using ADO.NET to extract name of those students who lives in Palpa district, assuming that the student table has four attributes (ID, name, district, and age).



MODEL SET 4

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
 - i. Which of the following is the correct way of implementing an interface addition by class maths?
 - a) classmaths : addition {} b) class maths implements addition {}
 - c) class maths imports addition {} d) none of the mentioned
 - ii. Access specifiers which can be used for an interface are?
 - a) Public b) Protected
 - c) Private d) All of the mentioned
 - iii. The number of levels of inheritance are?
 - a) 5 b) 4 c) 3 d) 2
 - iv. Select the class visibility modifiers among the following:
 - a) Private, protected, public, internal
 - b) Private, protected, public, internal, protected internal
 - c) Private, protected, public
 - d) All of the mentioned
 - v. In Inheritance concept, which of the following members of base class are accessible to derived class members?
 - a) static b) protected c) private d) shared
 - vi. A class member declared protected becomes member of subclass of which type?
 - a) public member b) private member
 - c) protected member d) static member
 - vii. Which form of inheritance is not supported directly by C# .NET?
 - a) Multiple inheritance b) Multilevel inheritance
 - c) Single inheritance d) Hierarchical inheritance
 - viii. If no access modifier for a class is specified, then class accessibility is defined as?
 - a) public b) protected c) Private d) internal

- ix. The process of defining two or more methods within the same class that have same name but different parameters list?
- | | |
|-----------------------|--------------------------|
| a) Method overloading | b) Method overriding |
| c) Encapsulation | d) None of the mentioned |
- x. Which of these can be overloaded?
- | | |
|--------------------------------|--------------------------|
| a) Constructors | b) Methods |
| c) Both Constructors & Methods | d) None of the mentioned |

Group B

Attempt any SIX questions.

[6x5=30]

1. What are different validation controls used in ASP.NET? Explain with example.
2. Explain exception handling in ASP page with suitable program.
3. Write a C# program to demonstrate group by operation in LINQ.
4. Differentiate constructor and destructor with example.
5. What is namespace? Explain the use of namespace in C# program with example.
6. What are other popular frameworks related to .NET? Explain.
7. Explain different .NET implementations in brief.

Group C

Attempt any TWO questions.

[2x10=20]

1. Write a program ASP.NET to find simple interest. Use text fields for inputs and label for output. Your program should display the result when the user presses a button.
2. Write a program using ADO.NET to extract name and age of those students who lives in Banepa city, assuming that the student table has four attributes (ID, name, city, and age).
3. (a) What do you mean by operators? Explain different operators used in C# with examples.
(b) What is multiple inheritance? How it is achieved in C#? Explain with suitable example.

MODEL SET B

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

(10 x1=10)

1. Circle (O) the correct answer in the following questions:
 - i. The modifier used to define a class which does not have objects of its own but acts as a base class for its subclass is?

a) Sealed	b) Static	c) New	d) Abstract
-----------	-----------	--------	-------------
 - ii. Choose the correct statements among the following:

a) An abstract method does not have implementation
b) An abstract method can take either static or virtual modifiers
c) An abstract method can be declared only in abstract class
d) All of the mentioned
 - iii. Which of the following modifiers is used when an abstract method is redefined by a derived class?

a) Overloads	b) Override	c) Base	d) Virtual
--------------	-------------	---------	------------

- iv. Which among the following cannot be used as a datatype for an enum in C#.NET?
a) short
b) double
c) int
d) all of the mentioned
- v. Which operator among the following signifies the destructor operator?
a) ::
b) :
c) ~
d) &
- vi. Operator used to free the memory when memory is allocated?
a) new
b) free
c) delete
d) none of the mentioned
- vii. Select wrong statement about destructor in C#?
a) A class can have one destructor only
b) Destructors cannot be inherited or overloaded
c) Destructors can have modifiers or parameters
d) All of the mentioned
- viii. What is the return type of destructor?
a) int
b) void
c) float
d) none of the mentioned
- ix. How many values does a function return?
a) 0
b) 2
c) 1
d) any number of values
- x. The capability of an object in Csharp to take number of different forms and hence display behaviour as according is known as _____
a) Encapsulation
b) Polymorphism
c) Abstraction
d) None of the mentioned

Group B

Attempt any SIX questions.

[6x5=30]

1. Explain any five form control elements with example.
2. Create student registration form in ASP.NET with proper form validations.
3. Write a C# program to demonstrate order by operation in LINQ.
4. Write a C# program to find maximum and minimum number from array.
5. What is ASP.NET? Explain different elements of ASP.NET with example.
6. Explain any three operators with example.
7. Write a C# program to generate multiplication table from 1 to 10.

Group C

Attempt any TWO questions.

[2x10=20]

1. Write a program for entering college general information in one ASP page and display it in another page.
2.
 - (a) Write a C# program to sort 'n' numbers in ascending order.
 - (b) What do you mean by .NET framework? Explain the overview of .NET framework in detail.
3.
 - (a) What is LINQ? Write a program to select employees whose salary is greater than 10000 and whose address is Palpa using LINQ.
 - (b) Write a C# program to show update and delete operation in database.

MODEL SET 6

Candidates are required to give their answer in their own words as far as practicable.

Group- A**[10 x1=10]**

Attempt all the questions

1. Circle (O) the correct answer in the following questions:

- i. Which of the following constitutes the .NET Framework?
 - a. ASP.NET Applications
 - b. CLR
 - c. Framework Class Library
 - d. None of above
- ii. Code that targets the Common Language Runtime is known as
 - a. Unmanaged code
 - b. Distributed code
 - c. Managed Code
 - d. Native Code
- iii. What is the return type of constructors?
 - a) int
 - b) float
 - c) void
 - d) none of the mentioned
- iv. Which method has the same name as that of its class?
 - a) delete
 - b) class
 - c) constructor
 - d) none of the mentioned
- v. In a connection string _____ represents name of the database
 - a) Data Source
 - b) Initial Catalog
 - c) Catalog Initial
 - d) Database
- vi. What are the Command Object Methods?
 - a) ExecuteNonQuery
 - b) ExecuteReader
 - c) ExecuteScalar
 - d) All of the above.
- vii. Which ado.net class provide disconnected environment?
 - a) DataReader
 - b) DataSet
 - c) Command
 - d) None of the above.
- viii. Which database is the ADO.NET SqlConnection object designed for?
 - a) Access
 - b) Microsoft SQL Server
 - c) MySQL
 - d) Oracle
- ix. The first step of configuring a DataAdapter is to select:
 - a) An adapter object.
 - b) A connection object.
 - c) A database object.
 - d) A dataset object.
- x. To check the status of the connection, _____ property is useful
 - a) ConnectionStatus
 - b) Status
 - c) State
 - d) ConnectionState

Group B

Attempt any SIX questions.

[6x5=30]

1. Write a C# program to demonstrate complete CRUD operation of a Student.
2. What is ASP.NET? Explain different elements of ASP.NET with example.
3. Write a C# program to demonstrate aggregate functions in LINQ.
4. Define constructor. Explain parameterized constructors used in C# with example.
5. Define inheritance. Write C# program to demonstrate hybrid inheritance.
6. What do you mean by method overriding? Explain the use of virtual function for method overriding.

7. What is operator overloading? What are different operators that can be overloaded in C#? Explain.

Group C

Attempt any TWO questions.

[2x10=20]

1. (a) Write a C# program to sort given elements in descending order.
(b) What is polymorphism? Explain compile time and runtime polymorphism with program in detail.
2. (a) Differentiate ADO and ADO.NET with example.
(b) Explain the process of creating connection in ADO.NET.
3. (a) What is generics? List different types of generic classes. Explain delegate with example.
(b) What do you mean by lambda expression? Explain different types of lambda expression used in C# with example.

MODEL SET 7

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
 - i. Which of these clauses will be executed even if no exceptions are found?
a) throws b) finally c) throw d) catch
 - ii. Which of these exceptions will occur if we try to access the index of an array beyond its length?
a) ArithmeticException b) ArrayException
c) ArrayArgumentOutOfRangeException d) IndexOutOfRangeException
 - iii. To implement delegates, the necessary condition is?
a) class declaration b) inheritance
c) runtime polymorphism d) exceptions
 - iv. To generate a simple notification for an object in runtime, the programming construct to be used for implementing this idea?
a) namespace b) interface
c) delegate d) attribute
 - v. Choose the incorrect statement among the following about the delegate?
a) delegates are of reference types b) delegates are object oriented
c) delegates are type safe d) none of the mentioned
 - vi. Which of the following is an incorrect statement about delegate?
a) a single delegate can invoke more than one method
b) delegates could be shared
c) delegates are type safe wrappers for function pointers
d) delegate is a value type
 - vii. Which of these keywords must be used to monitor exceptions?
a) try b) finally c) throw d) catch
 - viii. Which of these keywords is used to manually throw an exception?
a) try b) finally c) throw d) catch

- xi. Assume 2 columns named as Product and Category how can be both sorted out based on first by category and then by product name?
- varsortedProds = `_db.Products.OrderBy(c =>c.Category)`
 - varsortedProds = `_db.Products.OrderBy(c =>c.Category) + ThenBy(n =>n.Name)`
 - varsortedProds = `_db.Products.OrderBy(c =>c.Category) . ThenBy(n =>n.Name)`
 - all of the mentioned
- x. Choose the wrong statement about the LINQ?
- The main concept behind the linq is query
 - linq makes use of foreach loop to execute the query
 - It is not required that linq should make use of IEnumerable interface
 - None of the mentioned

Group B

Attempt any SIX questions.

[6x5=30]

- Explain DataReader, DataAdapter, Dataset and Datatable with example.
- Write a program to show insert and delete operation in database.
- Write a C# program to demonstrate join, concat and union operation in LINQ.
- What are control statements? Explain different types of control statements with examples in detail.
- Explain the benefits of using C# language.
- What is CLR? Explain its features.
- What is an array? Differentiate rectangular and jagged array with example.

Group C

Attempt any TWO questions.

[2x10=20]

- Write a C# program to find sum of 20 numbers in an array.
 - Create a simple class named Person that contains basic information like name, age, gender, etc. Your class should also contain functions/methods for storing and displaying data.
- Define event. How can we declare and invoke event? Explain with the help of suitable program.
 - What do you mean by anonymous method? Explain with example.
- What do you mean by static constructor? How it is different than other types of constructors? Explain with example.

MODEL SET 8

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

- Circle (O) the correct answer in the following questions:
 - Which among the following is NOT an exception?
 - Stack Overflow
 - Arithmetic Overflow or underflow
 - Incorrect Arithmetic Expression
 - All of the mentioned

- ii. Which among the following is considered as .NET Exception class?
 - a) Exception
 - b) StackUnderflow Exception
 - c) File bound Exception
 - d) All of the mentioned
- iii. Which of the following is the object oriented way to handle run time errors?
 - a) Error codes
 - b) HRESULT
 - c) OnError
 - d) Exceptions
- iv. Choose the namespace in which the interface IEnumerable is declared?
 - a) System.Collections
 - b) System.Collections.Generic
 - c) Both System.Collections & System.Collections.Generic
 - d) None of the mentioned
- v. Can we use linq to query against a DataTable?
 - a) Yes
 - b) No
 - c) Either Yes or No
 - d) none of the mentioned
- vi. Select the correct statement about an Exception?
 - a) It occurs during loading of program
 - b) It occurs during Just-In-Time compilation
 - c) It occurs at run time
 - d) All of the mentioned
- vii. Which of these keywords is not a part of exception handling?
 - a) try
 - b) finally
 - c) thrown
 - d) catch
- viii. What is the return type of destructor?
 - a) int
 - b) void
 - c) float
 - d) none of the mentioned
- ix. How many values does a function return?
 - a) 0
 - b) 2
 - c) 1
 - d) any number of values
- x. The capability of an object in Csharp to take number of different forms and hence display behaviour as according is known as _____.
 - a) Encapsulation
 - b) Polymorphism
 - c) Abstraction
 - d) None of the mentioned

Group B

Attempt any SIX questions.

[6x5=30]

1. Differentiate ADO and ADO.NET with example.
2. Explain the process of creating connection in ADO.NET.
3. Write a C# program to demonstrate where condition in LINQ.
4. What do you mean by operators? Explain different operators used in C# with examples.
5. Differentiate parameters pass by value and by reference with example.
6. Explain operator precedence with example.
7. What do you mean by interface? Explain use of interface in C# with example.

Group C

Attempt any TWO questions.

[2x10=20]

1. Write a program using ADO.NET to extract name of those students who lives in Kathmandu district, assuming that the student table has four attributes (ID, name, district, and age).
2. (a) What is multiple inheritance? How it is achieved in C#? Explain with suitable example.
3. (b) What do you mean by interface? Explain use of interface in C# with example.
- (a) Write a C# program to sort name of 10 persons in alphabetical order.
- (b) Write a C# program to add any two matrices.

MIS and E-Business

Course Title: MIS and E-Business

Course Code: CACS301

Year/Semester: 111/V

Class Load: Hrs./Week (Theory: 3Hrs, Practical 2Hrs.)

Course Description

This course provides the fundamental knowledge of Management Information System and E-Business and its associated infrastructure, security and protection issues, payment system and ultimately live implementation of any web application in Web server.

Objectives:

The general objectives of this course is to know fundamental knowledge of MIS/E-Business and implement the web application in Web Server.

Course Details

Unit 1: Introduction to E-Commerce

(7 LH)

Definitions and Concepts; Defining E-Business; Pure Versus Partial EC; EC Framework, Classification of EC; Benefits of E-Commerce; Electronic Markets; Role of Internet and Web in E-Commerce; The Limitations and Barriers of EC; Social Networks and Social Network Services; M-Commerce: Concept, Scope, Attributes, Benefits; Location-based 1-commerce, I-Commerce Infrastructure, Location-Based Services and Applications.

Unit 2: The Network Infrastructure for E-Commerce

8 LH

Introduction to Information Superhighway (I-Way), Components of the I-Way, Internet as a network infrastructure. Wireless Application Protocol: Wireless Application Protocol (WAP); Architecture of WAP; Working of WAP; Wireless Technologies: ADSL, WiMAX, WLAN, WMAN Wi-Fi, UMTS (3G), LTE (4G), (5G NR). Security Issues related to Wireless Communications.

Unit 3: Introduction to Management Information System

8 LH

Data, information, computer based information system (CBIS), Information System Resources, Management information system, Transaction processing (TPS) system, decision support system (DSS), and executive information system (EIS), SCM, CRMS and international/Systems: Introduction, Supply Chain Management Systems, Customer Relationships Management Systems. Enterprise systems and Challenges of Enterprise Systems Implementations- Managing the implementation, International Information Systems-Outsourcing and off-shoring.

Unit 4: E-Commerce Security and Fraud Issues and Protections

10 LH

Basic EC Security Terminology, The Threats, Attacks and Attackers, EC Security Requirements: Confidentiality, Integrity, Availability, Authentication, Authorization and Non repudiation. Technical attack: Viruses, Worms, Trojan Horses, Heart bleed, Distributed

Denial of server, Cryptoblocker, Page hijacking, Botnets, Malvertising, ransom ware, sniffing Non-Technical malware attack: Social Phishing, Pharming, Identity Theft and Identify Fraud, Spam Attacks; EC defense Strategy: access control (Authorization and Authentication, Biometric Systems), encryption and PKI (Symmetric Key Encryption, Asymmetric Key Encryption, Certificate Authority (CA), Secure Socket Layer (SSL). Securing e-commerce networks: Firewalls, Virtual Private Networks, Intrusion Detection Systems (IDS), intrusion prevention System (IPS)

Unit 5: E-payment systems

6 LH

Online payment cards (credit cards, charge cards, debit cards, smart cards), processing cards in online, credit card payment procedure, e-micropayments, e-checking and its processing in online. Automated clearing house (ACH) network, mobile payments (Digital wallet), mobile payment participants and issues, international payments, emerging EC payment systems and issues: crypto currency, virtual currency. A case study of emerging trends in online payment system in Nepal.

Unit 6: Launching a Successful EC Website

10LH

Planning Online Businesses (Business Plan, The cost-benefit, risk elements of a business case, funding a New Online Business, EC Model selection), The process of building a website, basic hierarchical website structure, website hosting and obtaining a domain name (cPanel management, upload EC site to the web Server), web content creation and management: (Web content, Categories and Types of Content, Content Management and Maintenance, Catalog Content and its Management), Website design criteria, Site Map and Navigation, Web page layout grid, Colors and Graphics, Website Usability, Performance, Website Promotion, implementation of Payments system, Website Promotion, Search Engine Optimization (SEO).

Laboratory Works:

32LH

Laboratory works should be done covering all the topics listed above and a project work should be carried out by individually implementing a fully functioning e-commerce web application along with payment system.

Reference Books:

1. Efraim Turban, D. K. (2018). *Electronic Commerce: A Managerial and Social Networks Perspective*. New York: Springer
2. Chaffey, D. (2009). *E-Business and E-Commerce Management: Strategy, Implementation and Practice (4th Edition)*. Harlow: Prentice Hall.
3. Kalakota, A. B. (1996). *Frontiers of Electronic Commerce*. Pearson.
4. Kenneth C. Laudon, C.G. (2014), *E-Commerce: business.technology.society*. New York: Pearson Education Limited.
5. Kenneth C. Laudon, J.P. (2014). *Management Information System: MANAGING THE DIGITAL FIRM*. New York Pearson Education Limited.

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

MODEL QUESTIONS-ANSWERS

Bachelor Level / fifth-semester / Science
Computer Science and Information Technology (CACS301)

Full marks: 60
Pass marks: 24
Time: 3 hours

Group A

Attempt all the questions.

10 × 1 = 10

1. Circle (O) the correct answer in the following questions.

- i. Management information systems (MIS)
 - a. create and share documents that support day-today office activities
 - b. process business transactions (e.g., time cards, payments, orders, etc.)
 - c. capture and reproduce the knowledge of an expert problem solver
 - d. use the transaction data to produce information needed by managers to run the business.
- ii. Which of the following describes e-commerce?
 - a. Doing business electronically
 - b. Doing business
 - c. Sale of goods
 - d. All of the above
- iii. Which is a function of E-commerce.....
 - a. marketing
 - b. Advertising
 - c. warehousing
 - d. all of the above
- iv. Any successful e-commerce application requires
 - a. I-way
 - b. Highway network
 - c. carry goods from point to point
 - d. None of the above
- v. Electronic Data Interchange is necessary in
 - a. B2C e-Commerce
 - b. C2C e-Commerce
 - c. B2B e-Commerce
 - d. Commerce using internet
- vi. TCP/IP is capable of connecting multiple independent networks through
 - a. routers
 - b. switches
 - c. hubs
 - d. Cables
- vii. The internet, cable TV, computer networks are applications of _____.
 - a. Information Super highway
 - b. ARPANET
 - c. ESNET
 - d. EBONE
- viii. EFT stands for.....
 - a. Electronic fund transmission
 - b. Electronic foreign transmission
 - c. Electronic feature transfer
 - d. Electronic feature transmit
- ix. SSL stands for.....
 - A. Socket secure layer
 - B. Secondary segment layer
 - C. Secure sockets layer
 - D. Secure system layer
- x. _____ is a unique call number that provides the exact location of every data in the universal network.
 - a. URL
 - b. Http
 - c. www
 - d. none

Answer-Key

i. (d)	ii. (a)	iii. (d)	iv. (a)	v. (c)	vi. (a)	vii. (a)	viii. (a)	ix. (c)	x. (a)
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Group B

[6x5=30]

Attempts any SIX questions.

1. What is SEO? Explain the importance of SEO for the growth of a website.

Ans: SEO stands for "search engine optimization." In simple terms, it means the process of improving your site to increase its visibility when people search for products or services related to your business in Google, Bing, and other search engines. The better visibility your pages have in search results, the more likely you are to garner attention and attract prospective and existing customers to your business.

Search Engine Optimization (SEO) is the process of increasing the quantity and consistency of traffic to your website through organic search engine performance. In layman's words, it is the method of enhancing the website's exposure for related searches. The higher your pages' exposure in search results, the more likely you are to gain interest and attract new and current customers to your business.

Search Engine Optimization (SEO) is the process of getting a website to the top of a search engine's rankings to increase the amount of traffic it receives. Businesses and individuals use SEO to increase the popularity of their websites and content to increase traffic and, as a result, revenue. Companies also employ SEO experts to introduce certain techniques to increase organic traffic. Organic traffic is traffic that comes to a website automatically rather than because of commercial search campaigns like pay-per-click ads (PPC).

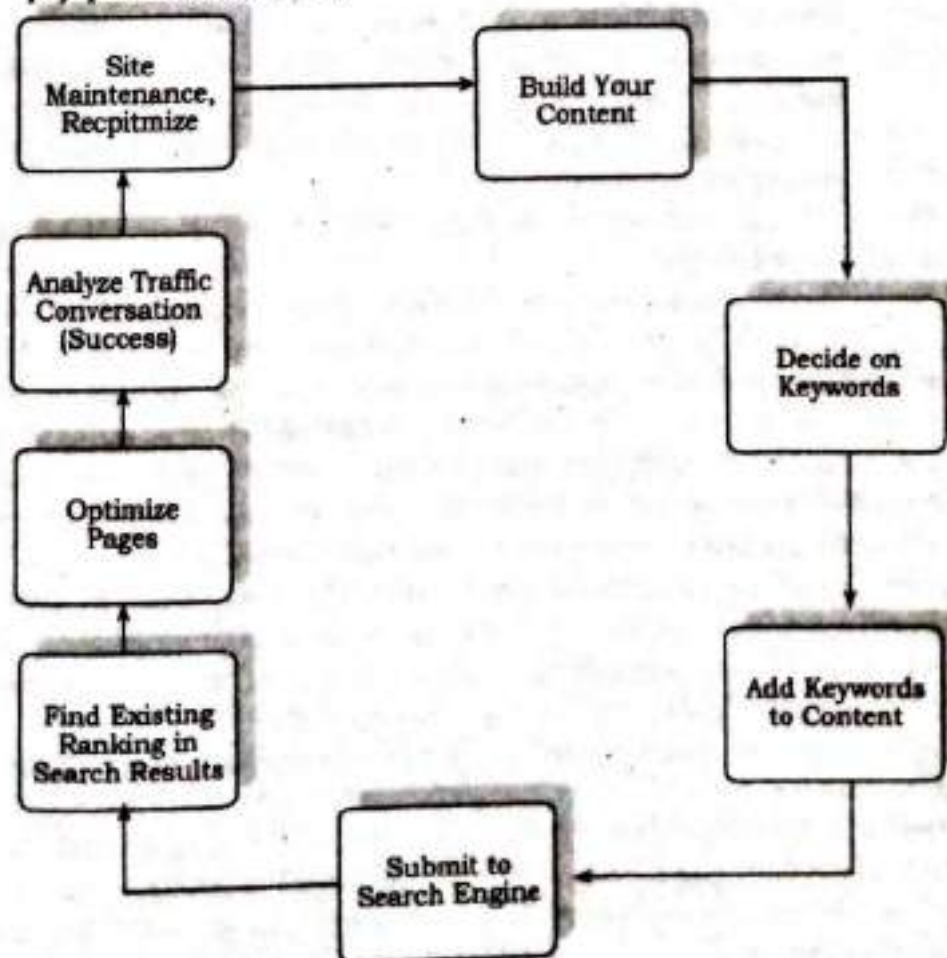


Figure: SEO Cycle

There are two kinds of SEO: on-page and off-page. On-page SEO consists of anything you can control on your own website to improve your rankings. This refers to things like site speed, keyword presence, header text, and so on. On the other hand, off-page SEO consists of anything done on other websites that may influence your rankings. Generally, this only refers to links.

Why is SEO Important?

- SEO is Google's way of determining which sites deserve to rank highly for each query entered into its search engine. Without modern SEO, it would be extremely simple to manipulate the search results so that the site with the most links or the most pages—all easily generated by software—consistently ranked #1.
- SEO is important because it keeps the search results fair. It reduces the ability to manipulate these results as much as possible, so that the sites appearing for each search are there because they deserve to be there. Hard work and a website that appeals to visitors correlate with high search engine rankings, so if your site meets these criteria, you'll have a better chance at showing up in the results.
- Users trust search engines, and achieving a top spot in search engine rankings signals to searchers that your site is a credible source. The higher you rank in results pages, the more clicks and traffic your site will generate. SEO also improves user experience, making it more likely for customers to become repeat buyers.
- SEO is cost-effective. Competitive industries can spend large amounts of money on paid website traffic. Of course, you can still purchase ad space, but if you're working with a tight budget, SEO is a great way to drive qualified traffic to your site without directly paying for it.

2. What is virtual currency? Explain the various types of virtual currencies popular in the market.

Ans: Virtual currency is a form of unregulated digital currency that can only be used online. It is only stored and transacted by designated devices, smartphone or device applications, or specialized digital wallets, and all transactions are conducted over secure, dedicated networks. Virtual money is a subset of the digital currency category, which also contains crypto currencies that operate on the block chain network. Virtual currency is money that is stored on the block chain network but is not managed by a central bank. Since digital currency is literally money issued by a bank in digital form, virtual currency is not the same as digital currencies. Since virtual money is uncontrolled, it experiences wild price swings and the only real factor driving trade is consumer interest.

There are two major types of virtual currencies – **centralized and decentralized.**

There is a central administrator or registry for a centralized virtual currency. The issuer of a virtual currency is usually the central administrator of that currency. In a managed currency market, the function is equivalent to that of a central bank. Ripple (XRP) is an example of a centralized virtual currency.

There is no central administrator or registry for a decentralized currency. Instead, the transactions of a shared virtual currency can be authenticated by a distributed framework. Decentralized currencies are based on block chain networks, such as Bitcoin, Litecoin, and Ethereum. A block chain network combines encryption with a catalog of documents known as blocks. As a transaction is requested, it is broadcast across the network with several nodes. Once the network has authenticated the transaction, a permanent and irreversible block containing the transaction data is attached to the existing block chain. Then the transaction is finished and properly registered.

When compared to a centralized virtual currency scheme, the decentralized peer-to-peer network eliminates the need for a central administrator and, as a result, it eliminates the risk of a centralized security breach. Decentralization also allows for greater transparency between parties and lower transaction costs due to the lack of intermediaries. On the other hand, the lack of a centralized jurisdiction, raises regulatory questions. The decentralized setup can be used for money laundering and other criminal transactions.

Virtual currencies popular in the market are listed below;

1. Bitcoin (BTC)

As the harbinger of the crypto currency era, Bitcoin is still the coin people generally reference when they talk about digital currency. Its mysterious creator — allegedly Satoshi Nakamoto — debuted the currency in 2009 and it's been on a roller-coaster ride since then. However, it wasn't until 2017 that the cryptocurrency broke into popular consciousness.

2. Ethereum (ETH)

Ethereum — the name for the crypto currency platform — is the second name you're most likely to recognize in the crypto space. The system allows you to use ether (the currency) to perform a number of functions, but the smart contract aspect of Ethereum helps make it a popular currency.

3. Binance Coin (BNB)

Binance Coin is the crypto currency issued by Binance, among the largest crypto exchanges in the world. While originally created as a token to pay for discounted trades, Binance Coin can now be used for payments as well as purchasing various goods and services.

4. Cardano (ADA)

Cardano is the crypto currency platform behind ada, the name of the currency. Created by the co-founder of Ethereum, Cardano also uses smart contracts, enabling identity management.

5. Solana (SOL)

Launched in March 2020, Solana is a newer crypto currency and it touts its speed at completing transactions and the overall robustness of its "web-scale" platform. The issuance of the currency, called SOL, is capped at 480 million coins.

6. Tether (USDT)

Tether's price is anchored at \$1 per coin. That's because it is what's called a **stablecoin**. Stablecoins are tied to the value of a specific asset, in Tether's case, the U.S. Dollar. Tether often acts as a

medium when traders move from one crypto currency to another. Rather than move back to dollars, they use Tether. However, some people are concerned that Tether isn't safely backed by dollars held in reserve but instead uses a short-term form of unsecured debt.

7. **XRP (XRP)**

Formerly known as Ripple and created in 2012, XRP offers a way to pay in many different real-world currencies. Ripple can be useful in cross-border transactions and uses a trust-less mechanism to facilitate payments.

8. **Polkadot (DOT)**

Launched in May 2020, Polkadot is a digital currency that connects the technology of block chain from many different crypto currencies. A co-founder of Ethereum is one of Polkadot's inventors, and some industry watchers believe Polkadot is looking to dethrone Ethereum.

9. **Dogecoin (DOGE)**

Originally created as a joke after the run-up in Bitcoin, Dogecoin takes its name from an internet meme featuring a Shiba Inu dog. Unlike many digital currencies limiting the number of coins in existence, Dogecoin has unlimited issuance. It can be used for payments or sending money.

10. **USD Coin (USDC)**

Like Tether, USD Coin is a stablecoin pegged to the dollar, meaning that its value should not fluctuate. The currency's founders say that it's backed by fully reserved assets or those with "equivalent fair value" and those assets are held in accounts with regulated U.S. institutions.

11. **Shiba Inu (SHIB)**

Shiba Inu is the latest dog-based meme token to see its price soar in a very short amount of time. It is an Ethereum-based altcoin that was anonymously created in August 2020 featuring a Shiba Inu dog as its mascot. Shiba Inu is widely considered to be an alternative to Dogecoin.

12. **Terra (LUNA)**

Using its currency Luna, Terra is a platform that helps backstop a range of stablecoins based on real currencies such as the dollar or euro. Terra helps stabilize the price of stable coins through various technical means, and it also supports smart contracts.

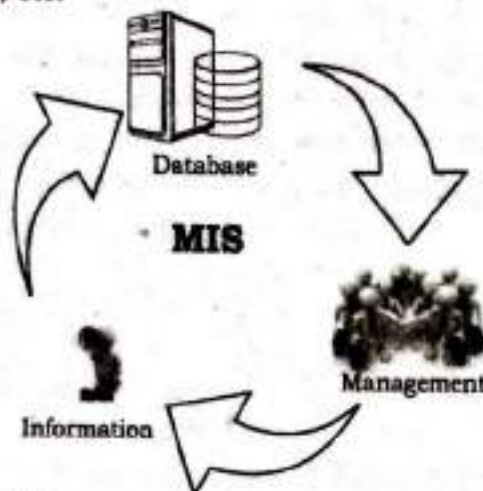
3. **What is MIS? Describe benefits of MIS.**

Ans: Management Information Systems (MIS) is the use of information technology, people, and business processes to record, store and process data to produce information that decision makers can use to make day to day decisions. The full form of MIS is Management Information Systems. The purpose of MIS is to extract data from varied sources and derive insights that drive business growth. MIS professionals help firms realize maximum benefit from investment in personnel, equipment, and business processes. MIS is a people-oriented field with an emphasis on service through technology.

Components of MIS

The major components of a typical MIS long-form (Management Information System) are:

- People - people who use the information system
- Data - the data that the information system records
- Business Procedures - procedures put in place on how to record, store and analyze data
- Hardware - these include servers, workstations, networking equipment, printers, etc.
- Software - these are programs used to handle the data. These include programs such as spreadsheet programs, database software, etc.



Benefits from MIS

MIS plays the role of information generation, communication, problem identification and helps in the process of decision making. The MIS, therefore, plays a vital role in the management, administration and operations of an organization. The benefits accruable from an effective MIS are summarized thus:

- The system is expected to fulfill the informational needs of an individual, a group of individuals, the management functionaries (managers and the top management).
- MIS satisfies the diverse needs through a variety of systems such as Query Systems, Analysis Systems, Modeling Systems, and Decision Support Systems. The MIS helps in strategic planning, management control, operational control and transaction processing.
- MIS helps the clerical personnel in transaction processing and address their queries on data pertaining to the transaction, the status of a particular record and references on a variety of documents.
- The MIS helps the junior management personnel by providing them with the operational data for planning, scheduling and control, and helps them further in decision making at the operations level to correct an offcontrol situation.
- The MIS helps the middle management in short-term planning, target setting and controlling the business operations. It is supported by the use of the management tools of planning and control.
- The MIS assist the top management in goal setting, strategic planning, and evolving the business plans and their implementations.

4. What is an information system? Why are information systems so essential in business today?

Ans: Information system, an integrated set of components for collecting, storing, and processing data and for providing information, knowledge, and digital products. Business firms and other organizations rely on information systems to carry out and manage their operations, interact with their customers and suppliers, and compete in the marketplace. Information systems are used to run inter-organizational supply chains and electronic markets. For instance, corporations use information systems to process financial accounts, to manage their human resources, and to reach their potential customers with online promotions. Many major companies are built entirely around information systems. These include eBay, a largely auction marketplace; Amazon, an expanding electronic mall and provider of cloud computing services; Alibaba, a business-to-business e-marketplace; and Google, a search engine company that derives most of its revenue from keyword advertising on Internet searches. Governments deploy information systems to provide services cost-effectively to citizens. Digital goods—such as electronic books, video products, and software—and online services, such as gaming and social networking, are delivered with information systems. Individuals rely on information systems, generally Internet-based, for conducting much of their personal lives: for socializing, study, shopping, banking, and entertainment.

Following are the benefits of information system

New Products and Services

Any business striving to enhance and to give a strong hold on the future has to instill a well-organized Business Information System. An IS can help in analyzing independent processes and enables organized work activities. Hence an information system entitles the companies to understand how the company generates, develops and sells the services or products.

Information Storage

Keeping a log of activities is important for all the organizations, to understand the reason for the problems and so to provide solution to the same. Business Information System makes it simple to store operational data, revision histories, communication records and documents. The storing of data manually involves a lot of time and money. A sophisticated Information system stores the information in the database which simplifies the process of finding the data easily.

Simplified Decision Making

Business Information System, eases the process of decision making and simplifies the process of delivering the required information and hence assists in taking better decisions instantly.

Behavioral Change

Business Information System can be effectively implemented to help communication better between the employers and the employees. Information Systems work better as it stores documents and files in folders that can be accessed and shared by the employees. This ensures to oversee the flow of information between the management and the lower-level employees. This also allows the front-line employees to be a part of the decision making process and hence feel motivated and committed towards doing a task.

5. Explain the advantages to society because of e-commerce?

Ans: It helps reduce the cost of products because more and more people can use the product. It helps to enable the rural areas to access the service and products because they can use any product what all others are using in town or cities. It helps the government deliver public services such as healthcare, education, and social service at minimal cost and better. Most importantly, no need to travel anywhere; the products will be delivered to the defined address and benefits the customer, organization, and environment.

6. **Define E-Check. Write the limitations & advantages of E-Check.**

Ans: E-Check is a digital version of a paper check and is also known as an electronic check, online check, internet check, and direct debit. E-Checks use the Automated Clearing House (ACH) to direct debit from a customer's checking account into a merchant's business bank account, with the help of a payments processor.

Advantage of E-Check

Faster Processing

Faster processing times provide a key advantage for business owners. Paper checks must go through numerous steps before the money moves from the customer's account to the merchant's, which can take several days. An electronic check often processes in half that time, which means the business gets its money faster. This allows businesses to more easily manage their bills and creates a more stable financial situation for the business.

Fee and Labor Reduction

Businesses that employ electronic checks spend less money on check processing fees, which lets them devote more financial resources to core operations. Electronic checks also require less hands-on labor by employees and management, which allows the business to either reduce its overall labor force or devote that employee time to customer service, inventory management and other mission critical efforts. It also reduces the need to raise product or service costs to offset the labor costs and fees associated with paper checks.

Customer Payment Options

Some customers do not possess a debit or credit card. This limits purchasing options, especially from online vendors. Business that accept electronic checks provide you with access to goods or services that might otherwise remain unavailable to you. For example, if you want to start a website, you need to buy a domain name and purchase web hosting services. If domain registrars and hosting services only accept credit or debit card payments and you can only provide a check, you cannot start your website. If they accept electronic checks, however, you get the chance to start your website without needing to get a credit or debit card.

Disadvantage of E-Check

Fraud Potential

As computers process electronic checks, hackers can potentially get access to your banking information. Some fraudulent businesses also offer electronic checks as a means to get you to hand them your banking information. The Federal Trade Commission suggests you not provide electronic check information to businesses you do not know and trust, whether online or over the phone. Legitimate merchants typically provide you with transparent information about how they process electronic checks.

Errors and Reduced Float

The computer-driven nature of electronic checks also makes them subject to computer errors. For example, a glitch in the processing might lead to a double withdrawal on your account or an incorrect withdrawal amount. Electronic checks also limit the amount of "float," the time between writing a check and when the business cashes it. If you write a check to cover your cable bill with the expectation that the check will not be cashed for a week, but the cable company performs an electronic check conversion three days later, you can find your account overdrawn.

7. Define Credit Card. Write the working mechanism of Credit Cards.

Ans: A credit card is a thin rectangular piece of plastic or metal issued by a bank or financial services company that allows cardholders to borrow funds with which to pay for goods and services with merchants that accept cards for payment. Credit cards impose the condition that cardholders pay back the borrowed money, plus any applicable interest, as well as any additional agreed-upon charges, either in full by the billing date or over time.

How Credit Cards Work

Credit cards can be used to make purchases online or in stores and pay bills. When you use a credit card for either one, your card details are sent to the merchant's bank. The bank then gets authorization from the credit card network to process the transaction. Your card issuer then has to verify your information and either approve or decline the transaction.

If the transaction is approved, the payment is made to the merchant and your card's available credit is reduced by the transaction amount. At the end of your billing cycle, your card issuer will send you a statement showing all the transactions for that month, your previous balance and new balance, your minimum payment due and your due date.

The grace period is the period of time between the date of a purchase on your card and the due date listed on your statement. During this period if you pay your bill in full by the due date, no interest charges accrue.

But if you carry a balance month to month, your card issuer can charge you interest. Your credit card's annual percentage rate or APR reflects the cost of carrying a balance on an annualized basis. Your APR includes both your interest rate and other costs, such as an annual fee if your card has one.

Most credit cards have a variable APR that's tied to the Prime Rate. This means your card's APR can change over time, though the CARD Act of 2009 sets strict guidelines on when credit card companies can and can't raise your rate.

Group C

Attempts any TWO questions.

1. Define social networks. Describe the capabilities of social networking services. [2 × 10=20]

Ans: The term social networking refers to the use of internet-based social media sites to stay connected with friends, family, colleagues, customers, or clients. Social networking can have a social purpose, a business purpose, or both, through sites like Facebook, Twitter, LinkedIn, and Instagram. Social networking is also a significant base for marketers seeking to engage customers. Facebook remains the largest and most popular social network. A Social Networking service is an online platform developed for people to build social networks with other people who share similar personal interests.

career goals, activities, or even real-life relationships. It is an internet-based service that requires the user to create a profile of his own. Such profiles contain their personal information along with the areas of interest. However, every person may choose as to what must be included in the profile. Most people display stuff that makes them impressionable rather than stuff they actually like and admire. Users can also choose whom they wish to connect with and also manage their privacy settings as to how much they wish to reveal about their activities and to which sort of audience. Of-course, each SNS platform has its own methodology of functioning and operations, but its sole purpose is to build a network. People all around the world can share photos, videos, audios, posts, ideas, events on Social Media Sites. A Social Networking Service (SNS) works on the following major basis:

- Internet-based applications
- User-generated content (UGC) forms the lifeline of the SNS mechanism
- Service-specific profiles are created by users
- Social networks are developed online by connecting user's profiles with those of other individuals or groups.

Earlier, SNS had a major purpose of social connections among people. However, due to the advent of technology, ample developments have taken place in the way SNS is utilized by various classes of people.

- People use SNS to connect to their loved ones online, in and around their lives. Be it friends, relatives, colleagues, family members. It has also enabled the development of new connections without having any prior information about the users at the other end.
- Employers use Social media sites to fetch information about their potential candidates. Also, special groups are formed to share knowledge-related content either specific to their organization or in general.
- Businesses are searching for potential customers, and Social Media is the best place they can fetch data about the interest of the consumers. Statistically, businesses have been ahead in analyzing the content on SNS to expand their business horizons and similarly use such data to gain more customers.
- Investigative agencies use the data on SNS to find signals of crime or information of activities of the criminals or suspects.
- Educators use SNS to spread knowledge and connect to their students.

2. **Explain wireless security issues. Comparatively explain 2G, 3G, 4G and 5G Technology to provide wireless services.**

Ans: Wireless local area networks (WLANs) transmit and receive data using radio waves rather than wires. This lack of a physical barrier makes WLANs vulnerable to unlawful interception, eavesdropping, hacking and a range of other cyber security issues.

Wireless network security issues and threats

The three most common WLAN security threats include:

- denial of service attacks - where the intruder floods the network with messages affecting the availability of the network resources
- spoofing and session hijacking - where the attacker gains access to network data and resources by assuming the identity of a valid user

- eavesdropping - where unauthorized third parties intercept the data being transmitted over the secure network

Second Generation (2G)

2G refers to the second generation of mobile networks based on GSM. The radio signals used by the 1G network were analog, while 2G networks were digital. 2G capabilities were achieved by allowing multiple users on a single channel via multiplexing. During 2G, cellular phones were used for data along with voice. Some of the key features of 2G were:

- Data speeds of up to 64 kbps
- Use of digital signals instead of analog
- Enabled services such as SMS and MMS (Multimedia Message)
- Provided better quality voice calls
- It used a bandwidth of 30 to 200 KHz

Third Generation (3G)

The 3G standard utilizes Universal Mobile Telecommunications System (UMTS) as its core network architecture. 3G network combines aspects of the 2G network with new technologies and protocols to deliver a significantly faster data rate. By using packet switching, the original technology was improved to allow speeds up to 14 Mbps. It used Wide Band Wireless Network that increased clarity. It operates at a range of 2100 MHz and has a bandwidth of 15-20 MHz. Some of the main features of 3G are:

- Speed of up to 2 Mbps
- Increased bandwidth and data transfer rates
- Send/receive large email messages
- Large capacities and broadband capabilities

Fourth Generation (4G)

The main difference between 3G and 4G is the data rate. There is also a huge difference between 3G and 4G technology. The key technologies that have made 4G possible are MIMO (Multiple Input Multiple Output) and OFDM (Orthogonal Frequency Division Multiplexing). The most important 4G standards are WiMAX and LTE. While 4G LTE is a major improvement over 3G speeds, it is technically not 4G. What is the difference between 4G and LTE?

Even after it was widely available, many networks were not up to the required speed of 4G. 4G LTE is a "fourth generation long term evolution", capable of delivering a very fast and secure internet connection. Basically, 4G is the predetermined standard for mobile network connections. 4G LTE is the term given to the path which has to be followed to achieve those predefined standards. Some of the features of 4G LTE are:

- Support interactive multimedia, voice, and video.
- High speed, high capacity and low cost per bit (Speeds of up to 20 Mbps or more.)
- Global and scalable mobile networks.
- Ad hoc and multi-hop networks.

Fifth Generation (5G)

5G networks operate on rarely used radio millimeter bands in the 30 GHz to 300 GHz range. Testing of 5G range in mmWave has produced results approximately 500 meters from the tower. Using small cells, the

deployment of 5G with millimeter wave based carriers can improve overall coverage area. Combined with beam forming, small cells can deliver extremely fast coverage with low latency.

Low latency is one of 5G's most important features. 5G uses a scalable orthogonal frequency-division multiplexing (OFDM) framework. 5G benefits greatly from this and can have latency as low as one millisecond with realistic estimates to be around 1 - 10 seconds. 5G is estimated to be 60 to 120 times faster than the average 4G latency.

Active antenna 5G encapsulated with 5G massive MIMO is used for providing better connections and enhanced user experience. Big 5G array antennas are deployed to gain additional beam forming information and knock out propagation challenges that are experienced at mmWave frequency ranges.

Further, 5G networks clubbed with network slicing architecture enables telecom operators to offer on-demand tailored connectivity to their users that is adhered to Service Level Agreement (SLA). Such customized network capabilities comprise latency, data speed, latency, reliability, quality, services, and security.

3. **What is disaster recovery planning? Explain the step involved in disaster recovery planning with figure.**

Ans: A disaster recovery plan (DRP) portrays how an organization will deal with potential disasters. It comprises of the precautions so that the effects of such disaster will be reduced and the business will be up quickly with comparatively minimal loss and/or disruption of productivity. With the proper disaster recovery plan, the recovery process will get simpler. Organizations with disaster recovery plan can sustain in case of catastrophes with minimal or no loss of data, infrastructure and business revenue then recover from it in maximum speed. The principal objective of any disaster recovery plan is to help the organization maintain its business continuity in minimal damage and data loss. The main focus of business continuity plan is on the procedures to be implemented to restore the mission critical business operations after the disaster, which helps business on maintaining the faith and confidence of their customers and other business entities. The terms 'disaster' and 'recovery' are nowadays being replaced by 'continuity' and 'resumption' so the disaster recovery plan is also being referred as business continuity and resumption plan.

In the Nepalese context, neither the government nor the most of the private sector has any specific plans for disaster prevention and data recovery. The government has set up Government Integrated Data Centre (GIDC) to store data. But, as of yet, it does not have a disaster recovery or backup center and very few ministries have stored their data in the center. However, the security measures implemented by banks are comparatively better as they are more sensitive than the government sector. For example, Everest Bank, a leading private sector bank in Nepal has implemented the managed disaster recovery solution. Appnomic System (a leading infrastructure management services company in India that provides Application Performance Management and Remote Infrastructure Management to effectively manage IT assets of large and medium scale enterprises) has developed an efficient, affordable and end-to-end managed disaster recovery solution to

ensure higher levels of availability of mission critical applications in the event of a sudden disaster in the primary datacenter. This solution is delivered from a remote management center in Bangalore, India. Similarly, most of the commercial banks in Nepal use Finacle banking software because of its effective disaster recovery system, which includes an effective data storage system and online data replication system for an alternative backup. As the basic disaster recovery system, Nepalese banks store the replica of their electronic records in backup sites in different parts of the country far from their headquarters.

Developing a Disaster Recovery Plan

Developing a disaster recovery plan follows various steps and slight change in the steps may also be needed depending upon the nature of the business firm. The steps involved in developing a disaster recovery plan mentioned by Gupta (1998) are shown in figure below.

First of all before developing DRP the business firms analyze the impact of disaster in the organizational environment, which is referred as business impact analysis or business resumption plan. The business impact analysis need to be made as detailed as possible so that no critical elements will not be missed during the disaster. Likewise, the business impact analysis also finds the alternative solutions to make the business up, which may also suggest going to manual working environment for the period until the information system resume its regular operation.

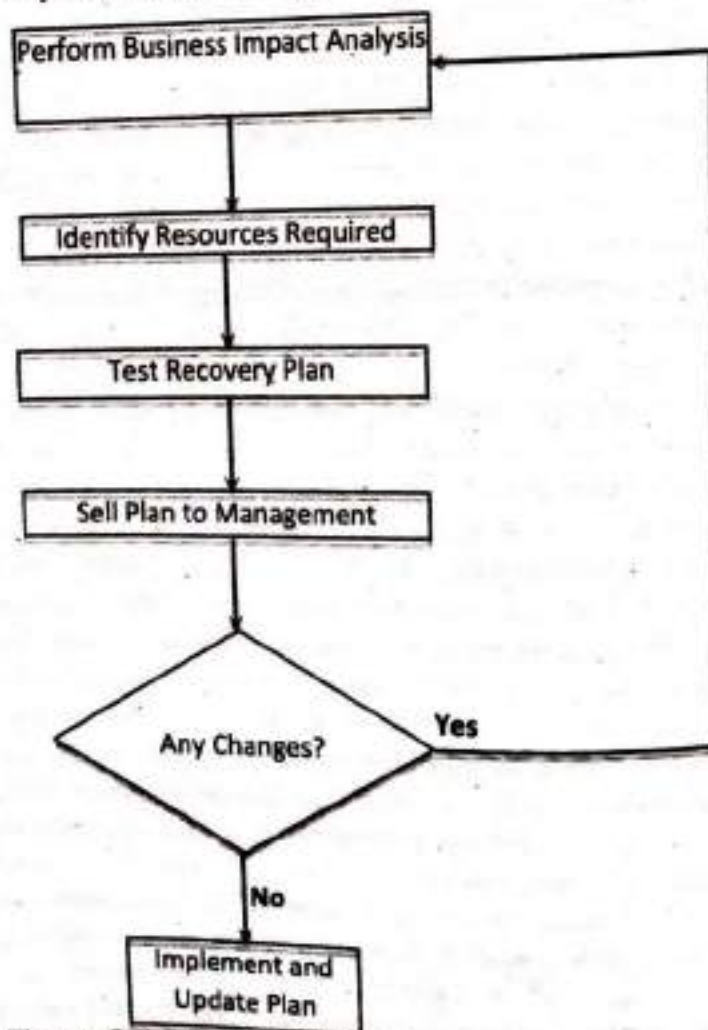


Figure: Steps on developing disaster recovery plan

Gupta (1998) cite three step in developing the business resumption plan:

- i. Take inventory of all corporate assets, mission critical business processes, and resources that are essential to operating the business and prioritize them.
- ii. Identify the situations that are indicated as disasters, the individuals responsible for declaring the disaster and the processes of declaring disasters.
- iii. Develop the course of action that each employee in the organization must take to make company operational after a disaster strikes.

The second step is to identify the resources required to recover from the disaster, which include money, time, personnel, and facilities and that should be specifically dedicated to disaster recovery. The third step is to simulate the recovery plan to ensure that it works as planned and required critical elements are not missed. The steps of recovery plans needs to be performed numerous times and the complete plan also need to be practiced several times so that all employees stay informed on the procedures to follow if a disaster strikes. Cooperation and long-term commitment from the executive level of the organization is extremely important in the success of disaster recovery planning. Proper execution of DRP is only possible with the group effort of executives, IS manager and employees.

The Disaster Recovery Plan must need to be updated to include new organization assets and new risks factors, which can increase the vulnerability of the information system.

After the complete DRP is prepared, the organization has to identify the alternatives to implement it. The possible alternatives can be listed as follows:

- Use the service of a disaster service bureau, which provides temporary disaster recovery services such as immediate access to time-sharing environment, access to utilities and temporary staffs.
- Manage the reciprocal arrangements with other companies, which use similar equipment and applications. With these arrangements, one company can use other's resources in case of emergency.
- Outsource the disaster recovery operations to commercial vendors who provide recovery services including backup system and communication network. This may cost high in the case of larger organizations, which need sophisticated infrastructure.
- Build a backup site with the same technologies and applications as in the central along with backup facilities for program and data. This alternative is best suited for large companies due to high cost. Nevertheless, the companies who can't build their own backup site can lease the backup facilities at a safe location.

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

TU QUESTIONS-ANSWERS 2020

Bachelor Level / fifth-semester / Science
Computer Science and Information Technology (CACS301)

Full marks: 60

Pass marks: 24

Time: 3 hours

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group B

[6x5=30]

Attempts any SIX questions.

1. How properties like ubiquity, richness and information density make e-commerce better than traditional commerce?

Ans: Ubiquity: Internet/Web technology is everywhere, at work, home, and elsewhere, and anytime, providing a ubiquitous market space, a marketplace removed from a temporal and geographical location. The technology reaches across national boundaries.

Richness: Information richness refers to the complexity and content of a message. Internet technology allows for rich video, audio, and text messages to be delivered to large numbers of people. The technology works through interaction with the user.

Information density: Information density is the total amount and quality of information available to all market participants. Internet technology reduces information costs and raises quality of information, enabling price transparency (the ease for consumers of finding a variety of prices) and cost transparency (the ability of consumers to determine the actual costs of products). Information density allows merchants to engage in price discrimination (selling goods to targeted groups at different prices).

2. Differentiate B2C, B2B and C2C e-commerce with examples.

Ans: B2B: Business to Business

This type of websites for business is suitable for the companies that sell products or services to another company, which is an intermediate buyer who then sells the product to the final customer. They help other companies establish a solid foundation for the long-term commercial interrelations between the companies. B2B websites may come in various types. One of them helps to receive information from partners, the other creates accounts for payment for the products or services and establish contracts. An example of B2B web-platform would be a website selling vehicle's components that some auto manufacturer will purchase in order to produce his own product. For instance, among the most well-known B2B websites is the Alibaba B2B Marketplace.

B2C: Business to Consumer

B2C websites for business are the most popular. Web platforms that follow the B2C model are suitable for companies which sell products or services directly to a customer online. One of the largest B2C websites is Amazon.

A customer can view products on the page, choose a product and order it. The Business to Consumer model doesn't require a middleman and reduces the cost of the goods for the ultimate consumers. B2C sites aim to make easy for shoppers to buy products and enjoy this process. Example: A sports store selling Adidas and Nike shoes together

C2C: Consumer to Consumer

Consumer to Consumer website serves as a mediator between the clients and gives an opportunity to sell or purchase goods directly. Through C2C web-service consumers can sell their assets like cars, or rent a room by publishing their information on the website. One customer may buy a product of another consumer by viewing the description on the website.

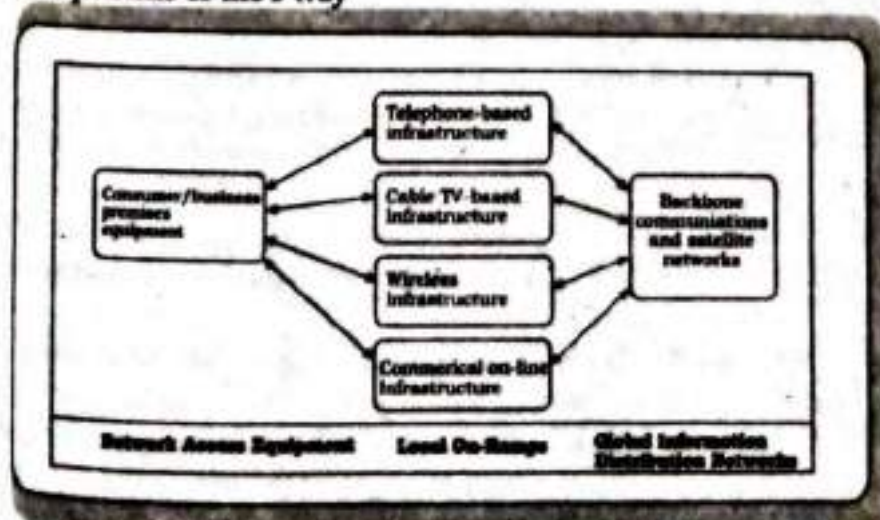
C2C manages utilized things or products. This web innovation encourages us to sell our utilized properties or resources like bicycle, vehicle, house, any electronic things or kitchen things by utilizing this model.

1. What is the role of information superhighway in e-commerce? List the components of information superhighway. [2+3]

Ans: An information superhighway is a telecommunications infrastructure or system (as of television, telephony, or computer networks) used for widespread and usually rapid access to information.

The information superhighways revolution is bringing people of different backgrounds from around the world into a global information superhighway. The Internet provides a worldwide platform connecting thousands of networks around the world. There is a variety of information available on the Internet for users. It has been considered as a forum for users to share worldwide information resources. The resources are so vast that several of us really cannot grasp or understand the I-ways fully. It has become a 'global information library' which allows the users to participate in the group conversation, look for any information, and so on. It can often be considered as a hybrid environment of postal services, citizen's band radio, libraries, and neighborhood community centers where we can spend time with our friends.

Components of the I-way



It consists of various components which may be broadly categorized as:

1. **Consumer access equipment:** - which is at the consumer end and enables the consumer to access the network. It consists of the hardware such as computers, modems, routers, switches for

2. computer networks, set-top boxes for television networks and software platforms such as browsers and operating systems.
 3. **Access road or media:** - provide the communication backbone for the transmission of data and information. The access providers can also be differentiated into four categories: telecom based cable TV, based, wireless-based, or computer-based online systems.
 4. **Global information distribution networks:** providing the infrastructure for connecting across the countries and continents. They include such networks as the long-distance telephone lines, satellite networks, and the internet.
4. What is international information system? Describe the concepts of outsourcing and offshoring in the system. [1+4]

Ans: An international information systems consists of the basic information systems required by organizations to coordinate worldwide trade and other activities.

Outsourcing

Outsourcing involves handing off internal functions to a third party. This may take the form of selling physical plant to a retailer in order to repurchase products or services or moving a whole company segment to a third-party and then repurchasing the operation. The basic philosophy is to delegate transactional tasks to professionals, allowing a company to concentrate on its core competencies. This ideal has served as the foundation for decades of commerce. Outsourcing has resulted in tremendous prosperity for the global economy.

Outsourcing is a corporate process in which a firm hires a third party to provide tasks and produce products that were formerly done in-house by the company's own employees and staff. Outsourcing is a cost-cutting strategy used by businesses to reduce costs. As a result, it has the potential to impact a wide variety of employment, from customer service to engineering to the back office.

There are many reasons why a business may choose to outsource a particular task, job, or a process. For example, some of the recognized benefits of outsourcing include:

- **Improved focus on core business activities:** outsourcing can free up your business to focus on its strengths, allowing your staff to concentrate on their main tasks and on the future strategy
- **Increased efficiency** - choosing an outsourcing company that specializes in the process or service you want them to carry out for you can help you achieve a more productive, efficient service, often of greater quality
- **Controlled costs** - cost-savings achieved by outsourcing can help you release capital for investment in other areas of your business
- **Increased reach** - outsourcing can give you access to capabilities and facilities otherwise not accessible or affordable
- **Greater competitive advantage:** outsourcing can help you leverage knowledge and skills along with your complete supply chain

Outsourcing involves handing over direct control over a business function or process to a third party. As such, it comes with certain risks. For example, when outsourcing, you may experience problems with:

- Service delivery - which may fall behind time or below expectation
- Confidentiality and security - which may be at risk
- Lack of flexibility - contract could prove too rigid to accommodate change
- Management difficulties - changes at the outsourcing company could lead to friction
- Instability - the outsourcing company could go out of business

Offshoring

Offshoring, unlike outsourcing, is mainly a geographical operation. Goods are expensive in the West since the labor used to manufacture and sell them is expensive. Offshoring takes advantage of cost differences by relocating manufacturers from expensive nations to cheaper markets in order to resell the goods at a substantial discount in the West and gain maximum profit. Consumer items such as clothes and appliances have become more affordable due to decades of productive offshoring, which has aided technological advancements.

Offshoring is the practice of relocating a business's operations unit such as production or services to a distant country where labor or materials are cheaper. The company's goal here is not global retailing; rather, it is to reduce the cost of production and other supporting services. The corporation will benefit from the creation of this separate business unit for other tasks because it will be able to focus more on its core business operations.

Advantages of offshoring

- **Concentrate on Core Business:** When the company offshore it's other services, it can lay more focus on its core functions.
- **Cost Reduction:** The most crucial reason or benefit of offshoring is to cut down labor cost and other operating expenses.
- **Cheap and Skilled Workforce:** It is an opportunity to get competent and cost-efficient labor available in a developing nation.
- **Complete Assistance:** The offshore team holds expertise in its field, providing a relevant solution to every problem related to the production or services offshored.
- **Better Control:** The Company can ensure proper management and regulation of all its operations if it opts for offshoring.
- **Streamlines Process:** It assures that a dedicated team is working on the offshore production or services to complete the assignment efficiently and effectively.
- **24/7 Operations:** It's another advantage is that the company can continue a 24/7 service which is otherwise not possible in the domestic business unit.
- **Tax and Other Benefits:** Many developing countries provide various types of incentives like tax holidays, to attract companies for foreign direct investment.
- **Risk Mitigation:** When the business operations are offshored to multiple countries, the company has a lower risk of failure or bankruptcy.

Disadvantages of Offshoring

- **Language Barrier:** When the company enters into a developing country for relocating its business operations; it finds a huge gap between the local language of the area and its native language.
- **Communication Barrier:** Due to the difference in language and understanding, the organization fails to establish effective communication with the offshoring team.
- **Long Distance Locations:** Usually the offshoring units are located in geographically distant locations which make it difficult for the management to visit these subsidiaries often.
- **Ethical Issues:** If the organization pays dissatisfactory remuneration to the offshoring team or in any case if they feel that the facilities are improper, it leads to ethical problems and spreading of negative brand image.
- **Socio-Cultural Problems:** When the company operates in a different country, it needs to familiarize with the cultural and social environment of that place; otherwise, it may face specific socio-cultural issues.
- **Geographical or Political Unrest:** The geographical or political problems like riots, protest, shutdowns, political instability, etc. may prevail in the country where the company relocates its unit; ultimately affecting the business.
- **Security and Safety Issues:** The Company's offshoring unit mostly faces a threat to lockout and closure if the natives find it to be unethical or harmful for their nation.

5. Define availability, authentication and authorization. How they can be enforced in e-commerce security? [3+2]

Ans: **Availability**

The information must be available when the authorized person or system requests it. The computing systems used to store and process the information, the security controls used to protect it, and the communication channels used to access it must be functioning correctly. The system that refers itself as the 'highly available' aim to remain available at all the time. Such system must avoid service disruptions due to power outages, hardware failures, and system upgrades. Those systems must also prevent the security threats like denial-of-services, which creates the flood of request causing the system to shut down and unavailable for some time.

Authentication

Authenticity is the important issue of information security, which insures the data; transactions, communications or electronic documents are genuine. Authenticity must also validate that both parties involved in the communication are who they claim to be.

Authorization

Authorization is the process of giving someone permission to do or have something. In multi-user computer systems, a system administrator defines for the system which users are allowed access to the system and what privileges of use (such as access to which file directories, hours of access, amount of allocated storage space, and so forth). Assuming that someone has logged in to a computer operating system or application, the

system or application may want to identify what resources the user can be given during this session. Thus, authorization is sometimes seen as both the preliminary setting up of permissions by a system administrator and the actual checking of the permission values that have been set up when a user is getting access.

6. What is e-checking? Describe its working mechanism. [2+3]

Ans: An electronic check, or e-check, is another form of payment used to make payments on the Internet. It is intended to perform the same function as a conventional paper check. As the check is in electronic form, processing can be done in fewer steps than traditional checks. An e-check is a part of the electronic banking field and part of a subcategory of transactions referred to as electronic fund transfers (EFT). It also provides more security features than traditional paper checks, such as authentication, public key cryptography, digital signatures, and encryption.

Working mechanism

There are a few steps involved to process an electronic check:

- **Request authorization:** Before proceeding with the payment, the company must first obtain the customer's permission. This can be accomplished by an online payment form, a signed order form, or a registered phone call.
- **Payment set-up:** The Company enters the payment details into the online payment processing program after authorization. If it is a recurring payment, this information also includes the details of the recurring schedule.
- **Finalize and submit:** Once information is properly entered into the payment software, the business clicks "Save" or "Submit" to start the ACH transaction process.
- **Deposit funds:** The payment is automatically withdrawn from the customer's bank account, the online software sends a payment receipt to the customer, and the payment itself is deposited into the business' bank account. Funds are typically deposited into the merchant's bank account three to five business days after the transaction is initiated.

7. Describe the website design criteria that one should show while designing an e-commerce website.

Ans: The following table describes the important criteria to be followed while designing the website.

Navigation	Is it easy for visitors to find their way around the site?
Consistency	Are design elements, especially look, terminology, icons, and user's feeling, consistent across all pages? Does everything match?
Response time	How long does it take to access the site? Does the site comply with efficient rules (e.g., the 12-second rule?)
Appearance	Is the site aesthetically pleasing? Are the colors visually pleasing? Does the site's look express the company's desired image? Is the content easy to read, easy to navigate, and easy to understand?
Quality Assurance	Are the site's calculators, navigation links, visitor registration processes, search tools etc. in working order?
Availability	Are all broken links fixed promptly? Is the site always available to access?

Interactivity	Does the site encourage the visitor to play an active role in learning about the business's products or services? Are all appropriate contact details available on the website for visitors to submit feedback and ask questions? Is the information (contact names, emails, addresses, telephone number etc.) current?
Content	How much multimedia is on the site? How current and relevant is the content? Is it easy to read? Informative?
Usability	How easy is it to use the site? How easy is it to learn to use the site?
Security	Is the customer's information protected? Does the customer feel safe accessing and using the site and, for example, submitting credit card information?
Scalability	Does the site design provide an easy format for future updates? Will the site's growth and increased usage protect the initial investment in site construction?

Group C

[2×10=20]

Attempts any TWO questions.

8. What do you mean by Wireless Application Protocol (WAP)? How it works? Discuss the layered architecture of WAP. [2+2+6]

Ans: Wireless Application Protocol (WAP)

Wireless Application Protocol (WAP) is a specification of various communication protocols, which standardize the way wireless devices communicates on a network. It is a technical standard for accessing information over a mobile wireless network. The mobile device can be used for accessing various services and information available on the internet such as email, web browsing, instant messaging, play games, view weather forecast and traffic updates etc. A WAP browser installed on the mobile device use this protocol for browsing the pages, which are generally created with the wireless markup language (WML), which is defined as XML 1.0 application. WAP is based upon the existing concept of the World Wide Web (WWW), and the backend functioning also remains like WWW but WAP uses the markup language WML to access the WAP services. These days, almost all modern mobile devices support modern browsers which fully supports HTML, so they do not need to use WML. The WAP provides very small user interface, the information access speed is slow with WAP and it is less secured protocol.

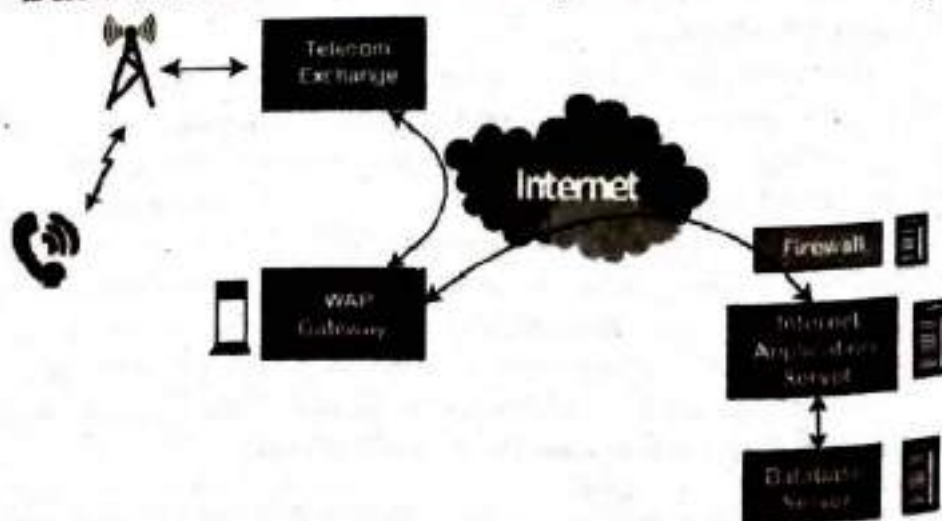
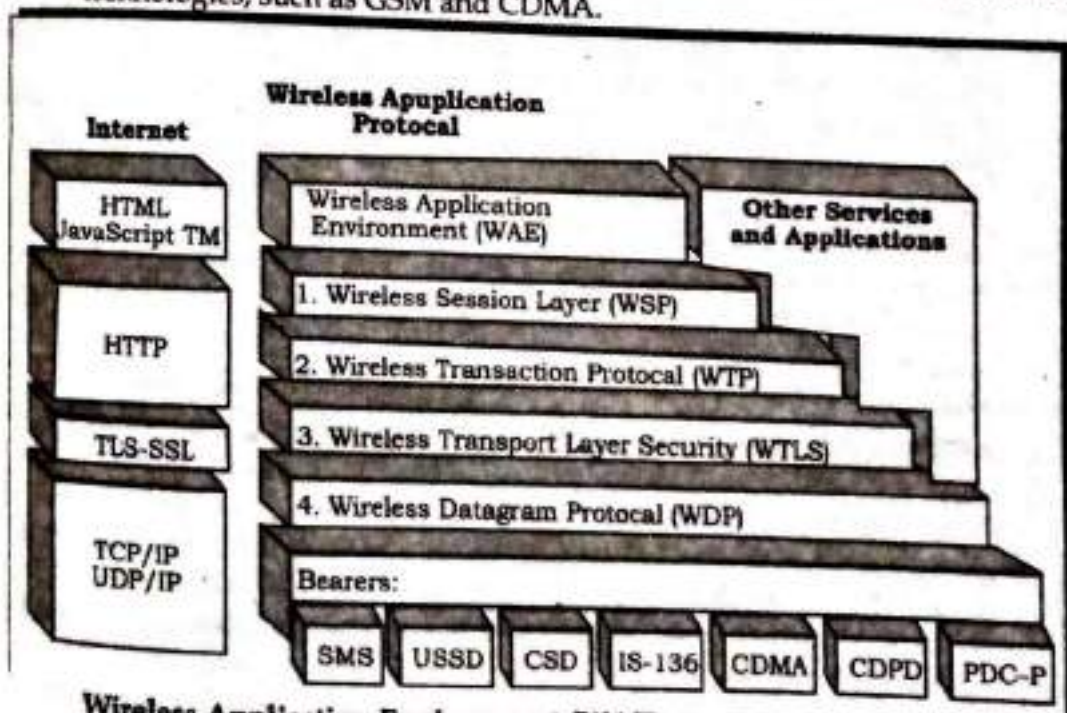


Figure: WAP

The working mechanism of WAP model is like that of client-server model but WAP model uses WAP gateway as a mediator between the client and server. The client opens the browser and enters the website address it wants to view, the mobile device sends the URL encoded request via a network to a WAP gateway using WAP protocol. Such request is called as encoding request. The WAP gateway translates that encoding request into an HTTP URL request and sends it to the server over the Internet. When the request reaches a specified Web server, the server processes the request and sends the response back to the WAP gateway. After receiving the server's response, the WAP gateway processes and sends the webpage to the WAP mobile device as a WML file which is processed and displayed by the browser.

Architecture of WAP

WAP architecture provides a scalable and extensible environment for application development for mobile devices, which is the layered design of protocol stack. Each layer is accessible by layers above. That protocol suite allows the interoperability of WAP equipment such as mobile phones and WAP software such as WAP enabled browsers and different network technologies, such as GSM and CDMA.



Wireless Application Environment (WAE)

The WAE layer contains mobile device specifications, and the content development programming languages such as WML, and WML Script. So, it is important layer to the developers.

Wireless Session Protocol (WSP)

The WSP layer is responsible for managing fast connection suspensions and reconnections.

Wireless Transaction Protocol (WTP)

The WTP layer runs on top of a datagram service, such as User Datagram Protocol (UDP) and is part of the standard suite of TCP/IP protocols used to provide a simplified protocol suitable for low bandwidth wireless stations. This layer manages transaction support for requests and responses to the server.

Wireless Transport Layer Security (WTLS)

The WTLS layer integrates security features that are based upon the Transport Layer Security (TLS) protocol standard. WTLS is responsible for managing privacy, authentication services and data integrity check through public key cryptography.

Wireless Datagram Protocol (WDP)

WDP is the bottom-most protocol in the suite that presents consistent data format to higher layers of WAP protocol stack there by offering the advantage of carrier independence to application developers.

9. How important web contents are for e-commerce systems? Discuss their types. Justify, with examples, how web contents can be used to promote cross-selling, up-selling and promotions in e-commerce. [2+2+6]

Ans: Web content refers to the text, photographs, audio, and video that appear on a Web page. Since it exposes the brand and its products to consumers, creating and managing content is vital to website growth. Content creation is the process of identifying a new topic to write about, deciding which form you want the content to take, formalizing your strategy, and then actually generating it. Most content development processes need several rounds of edits before the content is ready for publication. Content on the website can take many different forms – blog posts, videos, eBooks, Tweets, infographics, and advertisements etc. are a few.

Types of content

Providing content to Ecommerce sites may be a complex job due to the variety and quantity of the contributing sources. In addition, content may include foreign languages and must be updated frequently. Content may include all types of media. Finally, content may involve security, quality, privacy protection, and permission issues. Frequently changing content such as weather updates, share market updates is referred to as dynamic Web content. On the other hand, the content such as about page or contact address page are usually constant and may updated infrequently, such pages are called static Web content. Dynamic content is created with programming logic and data stored on the database, but static content is directly written on the standard HTML files. Dynamic material that is up to date draws new and returning users and keeps them for longer period on the site. As a result, dynamic Web material helps to increase customer satisfaction.

Content on the website should include more than just information about the product itself which is referred to as the primary content. A website also should include secondary content that offers marketing opportunities. Some examples of secondary content are:

- * **Cross-selling:** Using content for cross-selling means offering complementary or related products and services to increase sales. Amazon.com offers people who read personalized recommendations such as "Customers who bought this product also bought..." and items which are "Frequently bought together." Accessories, add-on products, extended warranties, and gift wrapping are examples of cross-selling opportunities that companies can offer to buyers on the product pages or in the purchase process. Another example of cross-selling is if you buy a car online, you may be offered insurance and financing.

- **Up-selling:** Up-selling material entails promoting a more expensive version of a product in order to increase revenue and benefit. Amazon.com suggests "great buy" book combinations that is an offer to buy two related books for slightly more than the price of one. Amazon also offers the used copies of a book at a cheaper price than the new book. Up-selling activities typically include offering products with a different design, color, fabric, or size.
 - **Promotions:** Secondary contents such as coupon, rebate, discount, or special service can increase the sales and also help on improving customer service. E-commerce sites may offer free shipping if the purchase amount is greater than \$500. Similarly, customers get discount offers during the festive period.
 - **Comments:** Additional contents on the website such as product reviews, testimonials, recommendations, or 'how to use the product' attracts the visitors of the website. For instance, Amazon.com book pages always have editorial and customer reviews of the book. The "look inside this book" feature helps website users to sample certain book excerpts and decide whether the content is suitable or not.
10. **Discuss in detail the security defends strategies that you can implement while securing e-commerce systems from security attacks and threats.**
- Ans: Security issues in ecommerce is not something online businesses can neglect. In fact it should be a priority for most online stores so their customers are able to enjoy a smooth and safe shopping experience. Your ecommerce security lets your customers protect themselves from cyber-attacks and fraud. The better your security protocols are, the better your brand will uphold its reputation and earn the trust of the customers. Major security defends strategies while securing e-commerce systems from security attacks and threats are listed below:
1. **Switch to HTTPS**
Using outdated HTTP protocols makes you vulnerable to attacks. I strongly recommend that you switch to HTTPS which displays the trustee green lock sign that says "secured" next to the URL bar on your customer's computer. HTTPS protocols not only protect the sensitive information users submit, but their user data as well. Another benefit you get from upgrading to HTTPS is higher ranking on Google's search page since Google considers HTTPS as a ranking factor.
 2. **Secure Your Servers and Admin Panels**
Most ecommerce platforms come with default passwords that are ridiculously easy to guess. And if you don't change them you are exposing yourself to preventable hacks. Use complex password and usernames and change them frequently.
 3. **Payment Gateway Security**
While it may make processing payments more convenient, having credit card numbers stored on your database is a liability. It's nothing less than an open invitation for hackers where you put your brand's reputation and your customer's sensitive information on the line. If you fall victim to a security breach, and hackers get

their hands on credit card data, all you can do is to say goodbye to your business because the heavy fines will force you into bankruptcy.

In order to save your business from this terrible fate, you should never store credit card information on your servers and ensure your payment gateways security is not at risk. Additionally, you can use third-party payment processing systems to carry out the process off-site. Popular ecommerce payment processing options include PayPal, Stripe, Skrill, and Wordplay.

4. **Antivirus and Anti-Malware Software**

Hackers can use stolen credit card information to place orders from anywhere in the world. An antivirus or an anti-fraud software can help you with this serious ecommerce issue. They use sophisticated algorithms to flag any malicious transactions to help you can take further action. They provide a fraud risk score which can help proprietors determine if a certain transaction is legitimate.

5. **Use Firewalls**

Another effective ecommerce recommendation is to use firewall software and plugins that are pocket-friendly yet effective. They keep untrusted networks at bay and regulate traffic that enters and leaves your site. It offers selective permeability and only allows trusted traffic in. They also protect against cyber threats such as SQL injections and cross-site scripting.

6. **Secure your website with SSL certificates**

Secure sockets layer (SSL) certificates are files that link a key to transactions on different paths on a network. These certificates are associated with credit card details and transactions to regular queries. SSL certificates encrypt data to protect it from interception in between different destinations. The information you send from your end to the server is secure.

If you want to conduct any type of business on your site, you require SSL certificates, so that every process that takes place on your site is secure. Besides, it provides you with a certificate of ownership so hackers can't use your site as a counterfeit for phishing.

7. **Employ Multi-Layer Security**

You can fortify your security by using various layers of security. You can use a wide-spread Content Delivery Network or CDN to protect your site against DDoS attacks and malevolent incoming traffic. They do so by utilizing machine learning to filter out the malicious traffic from regular traffic.

You can also use two-factor authentication to squeeze in an additional layer of security. Two-factor authorization requires a standard username and password combination as well as an extra code that is sent as an email to the user or as an SMS to their provided phone number. This ensures that only the user can access the service even if their username and password are at risk.

8. **Ecommerce Security Plugins**

Security plugins are a simple way to enforce security protection on your website. They provide protection against bad bots, SQLi, XSS,

code injections and hundreds of other severe attacks. One of the most secure, easy to implement, feature rich security plugin is Astra. It helps automatically secure your site and virtually patch software by preventing malicious requests from ever reaching your website.

9. **Backup Your Data**

Data loss due to hardware malfunction or cyber-attacks is not uncommon. And if you don't backup your data regularly, you are at the risk of losing it for good. You should do it yourself and not trust anyone else to do it for you. Employ automatic backup service so that even if you forget to do it manually, all your data will be backed up automatically.

You can go one step further and make a copy of the backup, so you will have a contingency plan available if you lose your original backup. Another option is to choose a managed ecommerce web hosting service that automatically creates backups for you, like Cloudways.

10. **Stay Updated**

The importance of regularly updating Word Press core, security tools, and plugins can be stressful, however, install security updates and patches as soon as they release because hackers can use bots that identify which websites use outdated software. That makes outdated software a serious liability.

11. **Opt for a Solid Ecommerce Platform**

It is important that you choose a secure ecommerce platform that regularly updates itself and offers top-notch security. Ecommerce platform tools safeguard you against common threats and frequently provide you with updates. PrestaShop, Magento and WooCommerce are some popular choices.

12. **Train Your Staff Better**

Your staff should be aware of laws and policies pertaining to the protection of user information. They should not share login credentials, and you should review the personnel who have access to sensitive customer information.

Once your employee tenders their resignation, expunge their details and revoke all their access to keep them from committing a cybercrime against your business.

Strong passwords require a good combination of characters, symbols, and numbers that are near-impossible to brute-force or guess. You can also keep users away from creating profiles with weak passwords. You can also adopt the two-factor authentication system in case they are using weak passwords. Or if the user submitted information is sensitive and susceptible to hacking.

Give these approaches due consideration because some customers might consider them a hassle and might just leave your website altogether. Do ensure that you aren't making your customers jump through unnecessary hoops.

Or you can bypass this whole process and simply let them sign up via Facebook or Google which offer world-class cyber security.



MODEL QUESTIONS SETS FOR PRACTICE

MODEL SET 1

Bachelor Level / fifth-semester / Science
Computer Science and Information Technology (CACS301)
Year/Semester: III/V

Full marks: 60

Pass marks: 24

Time: 3

Candidates are required to give their answers in their own words as far as practicable.

Group A

Attempt all the questions

[10x1=10]

1. Circle (O) the correct answer in the following questions:

- i. In an information system which one is not a technology driver for an information system?
 - a. knowledge asset management
 - b. networks and the Internet
 - c. object technologies
 - d. enterprise applications
- ii. Who have concerned with tactical (short-term) operational problems and decision making?
 - a. supervisors
 - b. mobile managers
 - c. middle managers
 - d. executive managers
- iii. Management information systems (MIS)
 - a. Development and share documents that support day-to-day office activities.
 - b. process business transactions (e.g., time cards, payments, orders, recapture and reproduce the knowledge of an expert problem solver
 - c. use the transaction data to produce information needed by managers to run the business
 - d. none of the above
- iv. Project Management ensures that.....
 - a. system is developed at minimum cost
 - b. project's risk is assessed
 - c. project's feasibility is assessed
 - d. both A and B
- v. Abbreviate EDI.....
 - a. Electronic data interface
 - b. Electronic data type interface
 - c. Electronic device interface
 - d. Electronic data integer
- vi. Users pay in advance for the privilege of getting information or service.....
 - a. debit
 - b. credit
 - c. real-time
 - d. none
- vii. Which one of the following is not one of the major types of e-commerce?
 - a. C2B
 - b. B2C
 - c. B2B
 - d. C2C
- viii. The best products to sell in B2C e-commerce are.....
 - a. Small products
 - b. Specialty products
 - c. Digital products
 - d. Fresh products
- ix. LANs can operate over.....
 - a. Fiber lines
 - b. microwaves
 - c. TI lines
 - d. Co-axial cables

- x. A company that provides private computer network support services to other companies is called.....
- Backbone network
 - Private Network
 - public network
 - Value-added network

Group B**Attempts any SIX questions.****[6x5=30]**

- What is business plan? Explain the steps to carry while preparing a business plan.
- What is bitcoin? Explain the advantages and disadvantages of using bitcoin for payment.
- Portray the security and system reliability problems created by employees.
- Software defects affect system reliability and security. Explain.
- Differentiate between data and information giving suitable examples. Explain the process of converting data into information.
- What do you mean by E-Payment System? List the different types of E-Payment System.
- Define E-Check. Write the limitations & advantages of E-Check.

Group C**Attempts any TWO questions.****[2x10=20]**

- If you are supposed to develop a website for a football team, what management, organization, and technologies issues might you encounter?
- Define complementary assets and describe their relationship to information technology.
- What is Internet & the WWW? How have they changed the role played by information systems in organizations?

MODEL SET 2**Group A****Attempt all the questions****[10x1=10]**

- Circle (O) the correct answer in the following questions:
 - Which of the following is not suitable for a B2C transaction?
 - Clothes
 - Flowers
 - Airline reservation
 - None
 - _____ e-commerce transaction has the advantage of eliminating middlemen.
 - B2G
 - B2C
 - B2B
 - B2E
 - Which among the following is an example of C2C?
 - e-Bay
 - Amazon.com
 - Rentalic.com
 - All of these
 - Which of the following is part of the four main types for e-commerce?
 - b2b
 - b2c
 - c2b
 - all of the above
 - Which segment do eBay, Amazon.com belong?
 - b2bs
 - b2cs
 - c2bs
 - c2cs
 - Which segment is eBay an example?
 - b2b
 - c2b
 - c2cd
 - none of the above
 - The best products to sell in B2C e-commerce are.....
 - small products
 - digital products
 - specialty products
 - fresh products

- viii. The solution for all business needs is.....
 a. EDI
 b. ERP
 c. SCM
 d. none of the above
- ix. All of the following are techniques B2C e-commerce companies use to attract customers, except:
 a. registering with search engines
 b. viral marketing
 c. online ads
 d. virtual marketing
10. What is the percentage of customers who visit a Web site and actually buy something called?
 a. affiliate programs
 b. click-through
 c. spam
 d. conversion rate

Group B

[6x5=30]

Attempts any SIX questions.

1. Explain the importance of color and graphics in a website.
2. What is blockchain? Explain its importance.
3. What do you mean by computer crime? Provide examples of crime in which computers are targets and two examples in which computers are used as instruments of crime.
4. What are identity theft and phishing? Explain why identity theft is such a big problem today.
5. Define a system and explain the system components.
6. Describe 5G technology along with its advantages.
7. Write the working mechanisms of E-Check with the help of neat diagram.

Group C

Attempts any TWO questions.

[2x10=20]

1. What is the status of e-commerce in Nepal? Describe at least five different e-commerce sites available in Nepal briefly.
2. What types of electronic payment system are required in e-commerce? Explain.
3. Discuss the application of e-commerce in service sector. Differentiate between B2B and B2c. Explain the major activities of B2C.



Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
 i. What are plastic cards the size of a credit card that contains an embedded chip on which digital information can be stored?
 a. customer relationship management systems cards
 b. e-government identity cards
 c. fedex cards
 d. smart cards
- ii. Which form of e-commerce currently accounts for about 97% of all e-commerce revenues?
 a. b2b
 b. b2c
 c. c2b
 d. c2c
- iii. _____describes e-commerce.
 a. Doing business offline
 b. Doing business electronically
 c. Surfing online
 d. Both A & B

- iv. Social networking is.....
- Group-centric
 - Individual-centric
 - Organization-centric
 - Society-centric
- v. A key role of Management Information Systems is.....
- To develop and share documents that support day-to-day organizational activities.
 - To process business information
 - To materialize the business transaction data and produce insightful information which assists managers in decision making
 - None of the above
- vi. Which one of the following is not a prerequisite for a good MIS?
- Database
 - Support from Staff
 - Control and maintenance of MIS
 - MIS executives
- vii. Which of the following is not an objective of MIS?
- Supports in decision-making
 - Provides insightful information
 - Assist management people
 - Recruit people for system
- viii. Which one of the following is not included in a Transaction Processing System?
- Input
 - Process
 - Output
 - Feedback
- ix. Which one is a true for a good ERP?
- Flexible ERP
 - Modularity
 - Extend within the company
 - all of these
- x. The most important reason for failure of MIS is.....
- Users involvement form initial phase of development
 - Noninvolvement of end-user
 - Documentation of specification
 - All of these

Group B

Attempts any SIX questions.

[6x5=30]

1. Explain various techniques that can be used to promote the website.
2. What is cryptocurrency? Explain any five cryptocurrencies currently available.
3. What is malware? Distinguish among a virus, a worm, and a Trojan horse.
4. Explain how hackers create security problems and damage the information systems.
5. What is an information system? Why are information systems so essential in business today?
6. How can WiMAX technology revolutionize the Internet access mechanism?
7. What are the components of information superhighway? Explain each of the components in details.

Group C

Attempts any TWO questions.

[2x10=20]

1. What is outsourcing? Explain its advantages and disadvantages.
2. Explain the architectural framework for electronic commerce with practical example.
3. What do you mean by information security? What are the security issues to be taken into account while designing a security system for e-commerce? Explain.

MODEL SET 4

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x 1 = 10]

1. Circle (O) the correct answer in the following questions:
 - i. The dimension of e-commerce that enables commerce across national boundaries is called.....
 - a) Interactivity b) Global reach c) Richness d) Ubiquity
 - ii. Which of the following describes e-commerce?
 - a) Doing business electronically b) Doing business
 - c) Sale of goods d) All of the above
 - iii. E-commerce involves.....
 - a) Marketing b) Adverting
 - c) Warehousing d) All of the above
 - iv. Which is not a function of E-commerce
 - a) Marketing b) Advverrtising
 - c) Warhousing d) None of the above
 - v. What are ballpoint pens purchases by a clothing company?
 - a) Direct materials b) Indirect materials
 - c) EDI d) None of the above
 - vi. What is another name for materials which cannot be linked to a specific product or process?
 - a) Direct materials b) Indirect materials
 - c) EDI d) None of the above
 - vii. Most individuals are familiar with which form of e-commerce?
 - a) B2B b) B2C c) C2B d) C2C
 - viii. The best products to sell in B2C e-commerce are:
 - a) Small products b) Digital products
 - c) Specialty products d) Fresh products
 - ix. The solution for all business needs is.....
 - a) EDI b) ERP c) SCMd) None of the above
 - x. An industry is less attractive for suppliers when the rivalry among existing competitors is:
 - a) High b) Low c) More d) Less

Group B

Attempts any SIX questions.

[6x5=30]

1. What is SEO? Explain the importance of SEO for the growth of a website.
2. What is virtual currency? Explain the various types of virtual currencies popular in the market.
3. List and explain the most important tools and technologies for safeguarding information resources?
4. What are the most common threats against contemporary information systems? Describe.
5. How are information systems transforming business and what is their relationship in globalization?

6. Explain ADSL technology and its advantages traditional dialup connection.
7. Describe the framework of m-commerce applications. Explain the major categories of m-commerce applications.

Group C

Attempt any TWO questions.

[2x10=20]

1. What is e-payment? Describe the working mechanism of online credit card transaction.
2. Differentiate between electronic cheque payment and credit card payment. Explain the cheque transaction protocol used in e-commerce.
3. Why is security issues important in e-commerce? What are the security issues to be taken into account while designing a security system for e-commerce? Explain.

MODEL SET 5

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
 - i. Which of the following is not related to security mechanism
 - a. encryption
 - b. decryption
 - c. e-cash
 - d. all the above
 - ii. Which of the following statements accurately reflect the impact of technology?
 - a. Technology has caused buyer power to increase
 - b. Technology has lessened the entry barriers for many industries
 - c. Technology has increased the threat of substitute products and services
 - d. all of the above
 - iii. A business cannot be all things to all people. Instead, a business must:
 - a. Identify target customers
 - b. Identify the value of products/services as perceived by customers
 - c. all of the above
 - d. None of the above
 - iv. A business competing in a commodity like environment must focus on which of the following?
 - a. Price
 - b. Ease / speed of delivery
 - c. Ease of ordering
 - d. all of the above
 - v. Amazon.com is well-known for which e-commerce marketing technique?
 - a. Banner ads
 - b. Pop-up ads
 - c. Affiliate programs
 - d. viral marketing
 - vi. What type of ad appears under a web page?
 - a. pop-under ad
 - b. pop-up ad
 - c. banner ad
 - d. discount ad
 - vii. An electronic check is one form of what?
 - a. e-commerce
 - b. online banking
 - c. e-cash
 - d. check

- viii. What is the name for direct computer-to-computer transfer of transaction information contained in standard business documents?
 a. internet commerce b. e-commerce
 c. transaction information transfer d. electronic data interchange
- ix. Which of the following is used in B2B to pay for purchases?
 a. e-commerce
 b. financial electronic data interchange
 c. electronic data exchange d. electronic checks
- x. Secure Sockets Layers does which of the following?
 a. creates a secure, private connection to a web server
 b. encrypts information
 c. sends information over the internet
 d. all of the above

Group B

Attempts any SIX questions.

[6x5=30]

1. Explain the parameters to consider while integrating the payment system on a website.
2. Describe the trends that is emerging in the online payment sector of Nepal.
3. What are the components of an organizational framework for security and control?
4. Describe the capabilities of digital firm. Why are digital firms so powerful?
5. Explain wireless security issues. Explain the architecture of WAP.
6. Define Smart Cards. How Smart Cards works? Explain
7. What is SSL? Explain the working mechanism of SSL in e-commerce web site.

Group C

Attempts any TWO questions.

[2x10=20]

1. What is e-payment? Discuss how credit card transactions, electronic checks and stored value payment system work. Mention their pros and cons.
2. What is Search Engine Optimization (SEO)? How on page SEO can be done?
3. Consider a company is planning to establish a B2B e-commerce system. Now describe in detail about the possible types of B2B Business Models the company can adapt.



Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
 i. When a transaction is processed online, how can the merchant verify the customer's identity?
 a. use secure sockets layers
 b. use secure electronic transactions
 c. use electronic data interchange
 d. use financial electronic data interchange

2. What is a model of a proposed product, service, or system?
 - a. Prototyping
 - b. Prototype
 - c. Proof-of-concept prototype
 - d. Selling prototype
- iii. Which factor represents how well your system can adapt to increased demands?
 - a. Availability
 - b. Accessibility
 - c. Reliability
 - d. Scalability
- iv. What defines all of the steps or business rules, from beginning to end, required for a process to run correctly?
 - a. Workflow
 - b. EIP
 - c. Workflow system
 - d. SLA
- v. Which is a function of ERP?
 - a. Warehousing
 - b. sales
 - c. scheduling
 - d. all of the above
- vi. What software detects and removes or quarantines computer viruses?
 - a. Backup
 - b. Anti-virus
 - c. Firewall
 - d. Biometrics
- vii. Which is the most important component of IT?
 - a. Information
 - b. People
 - c. Information technology
 - d. Computers
- viii. What is an electronic representation of cash?
 - a. Digital cash
 - b. Electronic cash
 - c. E-cash
 - d. All of the above
- ix. The E-payment mechanism widely used in B2B is
 - a. Credit card
 - b. Debit card
 - c. Master card
 - d. E-check
- x. Digital signature is a
 - a. Digital id, send as an attachment to a web page/e-mail/message
 - b. Is used for verifying the attachments send using web
 - c. Both a and b
 - d. None of these

Group B

Attempts any SIX questions.

[6x5=30]

1. What do you mean by website usability? Explain the usability factors.
2. Explain various cards which can be used for online payment.
3. Describe how security and control provide value for businesses.
4. Describe the challenges and opportunities of globalization in a flatten world.
5. Comparatively explain 2G, 3G, 4G and 5G Technology to provide wireless services.
6. Explain working mechanism of Secure Sockets Layer (SSL), describe the mechanisms of securing e-commerce networks.
7. What is cryptocurrency? Explain any five cryptocurrencies currently available.

Group C

Attempts any TWO questions.

[2x10=20]

1. What is MIS auditing? How it plays vital role in securing information system.
2. Differentiate pure e-commerce from partial e-commerce with an example. Describe what features make e-commerce better than traditional commerce.
3. What are the security threats in e-commerce? How client server and data transactions securities are maintained in e-commerce?

MODEL SET 7

Candidates are required to give their answer in their own words as far as practicable.

Group- A

[10 x1=10]

Attempt all the questions

1. Circle (O) the correct answer in the following questions:

- i. The smart card which use antenna for reading and writing data
 - a) Contact smart card
 - b) Contactless smart card
 - c) Proximity smart card
 - d) Both B and C
- ii. Which e-government arena is being used when governmental agencies send out and accept bids for work?
 - a) G2G
 - b) G2B
 - c) G2C
 - d) G2G
- iii. Which one is not an offline payment mode?
 - a) Cash on delivery
 - b) Cash before delivery
 - c) Demand drafts
 - d) e-cheque
- iv. OTP stands for.....
 - a) On Time Password
 - b) On Time processing
 - c) One time processor
 - d) None
- v. Which will harm computer resources.....
 - a) Worm
 - b) Virus
 - c) Trojan horse
 - d) All the above
- vi. The viruses spreads through
 - a) Mails
 - b) Pen drives and cds
 - c) Websites
 - d) All the above
- vii. Which one is not a threat for E-commerce?
 - a) Trojan horse
 - b) Viruses
 - c) Worm
 - d) None
- viii. Which of the following is not a common approach to phishing?
 - a. Send an official-looking e-mail asking for confidential information
 - b. Ask people to fill out and return a fake letter using postal mail
 - c. Ask people to click a link in an e-mail to open a submission form on a web page
 - d. All of the above are uncommon approaches to phishing
- ix. Which item can a password be used to protect?
 - a. Network
 - b. File
 - c. Folderd. All of the above
- x. All of the following are considered biometrics, except:
 - a. Fingerprint
 - b. Retina
 - c. Password
 - d. Voice

Group B

Attempts any SIX questions.

[6x5=30]

1. Describe the online processing of the credit card.
2. What is echeck? Explain the processing of eCheck.
3. List and explain the basic principles of security.
4. 'Information systems are so essential for running and managing a business today'. Explain.
5. What is information system literacy? How does it differ from computer literacy?
6. What is WAP? Explain the working mechanism of WAP.

7. How does one protect against spam, pop-ups, phishing, spyware, advertising, and ransomware?

Group C

Attempts any TWO questions.

1. Explain the purpose of risk assessment. How it is conducted for information systems? [2x10=20]
2. What is the information superhighway? Discuss its components
3. How preserving confidentiality and integrity ensures security in e-commerce? Discuss various mechanisms used to ensure data and message security in e-commerce systems.

MODEL SET B

Candidates are required to give their answer in their own words as far as practicable.

Group- A

Attempt all the questions

1. Circle (O) the correct answer in the following questions: [10 x1=10]
 - i. Which of the following describes e-commerce?
 - a. Doing business electronically
 - b. Doing business
 - c. Sale of goods
 - d. All of the above
 - ii. Which of the following is part of the four main types for e-commerce?
 - a. B2B
 - b. B2C
 - c. C2Bd. All of the above
 - iii. Which segment do eBay, Amazon.com belong?
 - a. B2Bs
 - b. B2Cs
 - c. C2Bs
 - d. C2Cs
 - iv. Which type of e-commerce focuses on consumers dealing with each other?
 - a. B2B
 - b. B2C
 - c. C2B
 - d. C2C
 - v. Which segment is eBay an example?
 - a. B2B
 - b. C2B
 - c. C2Cd. None of the above
 - vi. Which type deals with auction?
 - a. B2B
 - b. B2C
 - c. C2B
 - d. C2C
 - vii. In which website Global Easy Buy is facilitated?
 - a. Ebay.com
 - b. Amazon.com
 - c. Yepme.com
 - d. None of these
 - viii. The best products to sell in B2C e-commerce are:
 - a. Small products
 - b. Digital products
 - c. Specialty products
 - d. Fresh products
 - ix. Which products are people most likely to be more uncomfortable buying on the Internet?
 - a. Books
 - b. Furniture
 - c. Movies
 - d. All of the above
 - x. Which products are people most likely to be comfortable buying on the Internet?
 - a. Books
 - b. PCs
 - c. CDs
 - d. All of the above

Group B

[6x5=30]

Attempts any SIX questions.

1. What is e-micropayments? Explain with the examples.
2. Explain various digital wallets popular in Nepal.
3. Why are information systems becoming more vulnerable these days? Explain with examples.
4. Information systems are more than computer. Discuss it with reference to organization, management, and technology dimensions of IS.
5. What is digital signature? Explain the requirements of digital signature in secure transaction.
6. Difference between E-commerce and Traditional Commerce?
7. How does e-commerce work?

Group C

Attempts any TWO questions.

[2x10=20]

1. How security and control are interrelated? Explain the requirement of government regulation on securing the information system.
2. What is e-payment? Explain the functions of e-payment system. Why is orientation and standardization required for e-payment business system?
3. Explain the e-commerce architecture and its components in detail with diagram.

□□□

Introduction to Management

Course Title: Introduction to Management (3 cr.) Course code: CAMG 304

Year/Semester: III N

Class load: 3 Hrs./Week (Theory:3Hrs)

Course Objectives

This course aims to impart the basic management knowledge, and skills to the students so as to enhance their managerial capabilities and enable them to apply in the practical field.

Course Contents

Unit 1: Introduction

Management: concepts, meaning and functions. Types of managers. Managerial roles and skills. Organization and management. Changing perspectives of organization. **LH4**

Unit 2: Perspectives in Management

Classical Perspective: scientific management, administrative management and bureaucracy. Behavioral Perspective: Hawthorne studies, human relations movement, and emergence of organizational behavior. Quantitative Perspective: management science and operations management. Integrating perspectives: systems and contingency perspectives. Emerging management issues and challenges. **LH7**

Unit 3: Planning and decision making

Concept of planning. Levels of Planning: Strategic, Tactical and operational. Steps in Planning. Tools for planning. Decision Making: meaning, types and process. Decision making conditions- certainty, risk and uncertainty. **LH7**

Unit 4: Organizing

Concept of organizing, process and principles of organizing. Organization Architecture: vertical differentiation- tall versus flat hierarchies, horizontal differentiation - functional structure, multidivisional structure, geographic structure, and matrix structure. Outline authority and staff authority. Delegation of authority. Centralization, Decentralization and Devolution: meaning, reasons, advantages and disadvantages. Staffing: concept and importance. **LH9**

Unit 5: Leading and communication

Concept and qualities of leadership. Transformational and transactional leadership, Leadership Styles: autocratic, democratic, and participative. Concept of managerial ethics. Motivation: concept, importance, and techniques. Communication: meaning, process, and networks. Types of communication, Barriers to effective communication. **LH7**

Unit 6: Controlling and total quality management

Concept, purpose, Process and types of controls. Essentials of effective control systems. Control tools and techniques. Quality: Concept and importance. Total Quality Management: concept, components, principles, tools and techniques. Emerging issues in quality management. **LH5**

Unit 7: Organizational Change and Development

Concept and nature, forces, paradigm shifts and areas (structure, technology, business process and behaviors) of organizational change. Resistance to change. Overcoming resistance to change. Concept of Organizational Development **LH5**

Unit 8: Technology, Organization and Management

Concept of technology, approaches to technology and organization, social networking, use of technology in people management. **LH4**

References

Charles W.L. Hill and Steven L. McShane, Principles of Management, Tata Mc-Graw-Hill Company, New Delhi.

Griffin, Ricky W., Management. AITBS Publishers and Distributors, New Delhi. Hitt, M.A.,

J.S. Black and Porter, L.W., Management, Pearson Education, New Delhi Laurie J. M. Management and organizational Behaviour, Pearson, New Delhi

Evaluation:

Internal Assessment: 40 marks External Assessment: 60 marks Total: 100 marks Teaching methods: The major teaching methods include the case analysis, project work, term paper, assignments, and quiz. The instructor decides the learning strategies based on the nature of session/class.

TRIBHUVAN UNIVERSITY
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

**MODEL QUESTIONS-ANSWERS/
TU QUESTIONS-ANSWERS 2020**

Bachelor in Computer Applications
Course Title: Introduction to Management
Code No. CAMG 304
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 Hours

Group B

Attempt any Six Question

1. What is management? Explain the functions of Management.

Ans: Management gets the jobs done through and with people by using the process of **planning, organizing, staffing, leading and controlling** to achieve goals efficiently and effectively in a **dynamic environment**. Management collects all the resources and utilizes it for the achievement of organizational objectives.

According to Marry Parker Follet:

Management is the art of getting things done through people.

According to Henry Fayol:

To manage is to forecast and plan, to organize, to command, to coordinate and to control.

Management = Manage (Arrangement) + Men (People or HR in the organization) + T (Tactfully)

So, management is the tactful arrangement of people in the organization

In conclusion, Management is the art of getting things done through and with people by using the process of Planning, organizing, staffing, leading and controlling. Management is the essence of each and every organization. It coordinates all the efforts to achieve the organizational objectives.

Functions of Management (POSLOC)

- i. **Planning:** Planning is deciding the future course of action at present. It involves setting goals and course of actions to achieve them. It is Answering in advance about what to do, how to do it, when to do, it and who is to do it. It is time bound.
- ii. **Organizing:** It is determining how to group activities and resources. It establishes structure. It involves following:
 - Defining task required to achieve goals.
 - Grouping the task in department.
 - Designing a structure
 - Establishing authority and responsibility for each position.
 - Coordinating task to achieve harmony of efforts.
- iii. **Staffing:** Staffing is putting the right person in right place. It is defining what should a position need to do and qualification need for that position. It involves:
 - Human resource planning

- Job analysis (job description and specification)
 - Acquiring, Development, Utilization and maintenance of People.
 - iv. **Leading:** Leading is **influencing or motivating** people to perform task in the organization. It is directing the activities of people. It involves:
 - Leadership
 - Motivation
 - Communication
 - v. **Controlling:** It consists of making correction in the organization after comparing the standard and actual performance. It involves:
 - Setting standard
 - Measuring actual performance
 - Finding and analyzing deviations
 - Corrective actions.
2. Define scientific management. Explain the major contribution made by F.W. Taylor.

Ans: Scientific Management Theory:

- This theory focuses on job. It is concerned with improving the performance of individual workers.
- Propounded by F.W. Taylor. He is known as father of scientific management.
- The principles of scientific management are described in his book entitled the 'Principles of scientific management' which was published in 1911
- Productivity and efficiency are the goals of scientific management theory. It states that productivity can be improved endlessly.
- Scientific management theory is an attitude and philosophy, which discards the traditional method of hit and miss, rule of thumb, trial and error of managing work and workers.

Principles of Scientific Management:

- a) **One best way:** There is one best method of performing each job. Time and motion study should be used to find the best method of performing each element of the job. There should be no place for guesswork or rule of thumb. Scientific method should be used to define the "one best way".
- b) **Specialization:** There should be division of work for specialization. Worker should be assigned with one small unit of job in order to get the benefit of specialization.
- c) **Worker Education and Development:** Properly trained and developed workers produce the best. Compensation systems should be based on 'earn more by performing more'. Differential piece rate system should be used.
- d) **Management-Labour Harmony:** Management and labour has common interest in increasing productivity. There should be harmony in their relations.

Contribution of scientific management theory

- a) It promoted mass production.
- b) It facilitated job design through specialization and standardization of work.
- c) It gave importance to proper selection, training and compensation of workers for improving efficiency.

- d) It introduced rational method to solve management problem. It replaced guesswork.
- e) It laid the foundation of modern management theory.

Limitations of Scientific Management Theory:

- a) It is mechanistic
- b) Neglects human aspects
- c) There is no one best method of doing job in all work situations.
- d) Work become monotonous
- e) It is mainly applicable to labour intensive work.

3. Describe the formal and informal groups with suitable examples.

Ans:

- Group is the collection of two or more than two people
- Group has common goals/objectives
- Group shares common identity.
- They interact and are interdependent to each other.

Types of Groups:

- i. **Formal Groups:** Formal groups are the collection of people depending on the formal organizational structure. They are established under legal or formal organizational authority. It has work-orientation.
 - a) **Command Group:** Group of subordinates who need to report to the same supervisor in the scalar chain are called command Groups. They are represented on organizational charts. It is permanent in nature. Example Departmental head and his immediate officer.
 - b) **Task Group:** It consists of people working together to complete a particular task. It involves cross-command relationship. Members may not necessarily report to the same supervisor. Example Special Task force and Ad-hoc committee are the example of task group.
 - All command groups are also task group. But all task group are not command group.
- ii. **Informal Group:** Informal Groups are the group which are created naturally in the work environment. It is neither formally structured nor organizationally determined. They are based on friendship. They have common interest. The membership is voluntary.
 - a) **Interest Group:** It is people working together to attain a specific objective. Members have common interests to achieve such objective. Employees having common concerns regarding work schedules, sports, cafeteria etc.
 - b) **Friendship Group:** It is social alliance. Friendship groups are composed of employees who enjoy each other in group. People speaking same language or coming from same geographical area, similar age group. It may be extended outside the organization.

4. What is planning? Discuss the step involved in planning process.

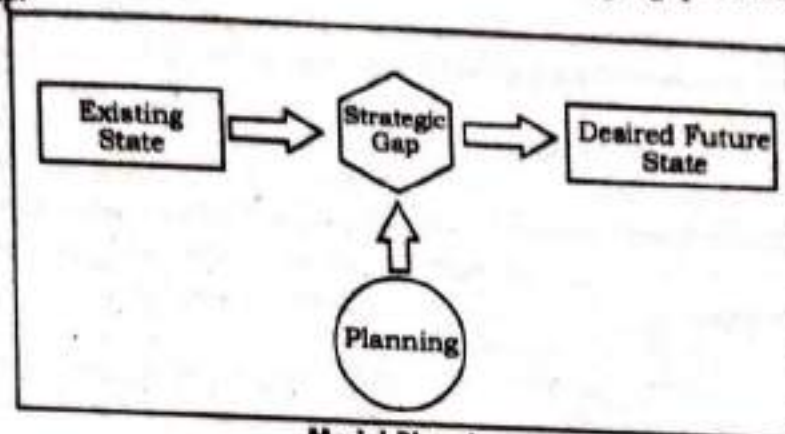
Ans: Concept of Planning:

- Planning is the predetermination of future course of action.
- It is the process of setting goals and choosing the actions to achieve those goals.
- It is the blueprint of action.

- It is deciding in advance about all WH questions (like what to do, how to do, when to do it and who is to do it.)
- Planning selects future course of action from the alternatives.
- It serves as guidance for allocating resources. It also minimizes the risk and uncertainty.
- It also serves as the standard for the effective control mechanism

Definition:

Planning is deciding in advance about what to do, how to do it, when to do it and who is to do it. It provides the ends to be achieved. -S.P. Robbins
Planning takes organizations to where it wants to be. The difference between existing state and the desired state is called 'strategic gap'. Planning bridges this gap.



Model Planning

Planning Process/Process of planning/steps of planning:

The planning should be systematic. So, the planning process involves series of activities or steps as follows:

- Analyze Opportunities:** This is based on SWOT analysis (Strengths, Weaknesses, Opportunities and threats). Environment Scanning is done to generate information. Strengths and weaknesses are the outcome of internal environments and opportunities and threats are the outcome of external environment. Internal environment involves organizational goals, culture, resources and structure; whereas external environment are political, economic and socio-cultural environment and Technology. In some extent, internal environment is controllable to the management, while it has to be adjusted itself with changing external environment.
- Setting Goals:** Goals are what organizations want to achieve in future. Achievement of a goal is the destination. They provide direction to the plan. They are set for the entire organization. The goal should be SMART.
- Develop Premises:** Premises are the key assumptions. It is the assumption of the future on the basis of which the plan is formulated. Forecasting is important for premises. Premises can be about sales, products, customer need, price, cost, technology, tax rates, tax exemption etc.
- Determine and Evaluate Alternatives:** Management has to identify all alternatives for achieving goals/objectives efficiently. Alternatives are evaluated in terms of contribution of goals achievement. Their feasibility is assessed. Quantitative and qualitative tools are used for

evaluation. Evaluation is done on the basis of cost, quality and time. Computer can be a great help.

- v. **Select a Course of Action:** After the evaluation of alternatives best alternative of course of action is selected. Cost, profits and judgment are important for selecting best course of action. Best course of action will satisfy the need of the organization.
- vi. **Formulate Action Plans:** Action plan for each activity is prepared. It is the details of best course of action consisting of activities, responsibilities to be assigned, timelines and method. The integrated action plan become programme.
- vii. **Prepare Budget:** Budget is the estimation of income and expenditure. Budget is a required to prepare to implement the plan. Budget is prepared for each activity to find out how much will it cost.

5. **What are the barriers of effective communication in any organizations? Explain**

Ans: Concept of Communication:

- Communication is the process of Transferring/Transmitting information on common interest from one person to another.
- It is the exchange of facts, opinions, ideas, suggestion and other information from one person to another.

Barriers to Effective Communication:

Barriers to communication are the factors which prevent effective communication. Those are the obstacles that creates problems in transmission. They are as follows:

- a. **Physical Barriers:** Physical barriers includes distance, noise and physical arrangement.
 - **Distance:** It is physical distance between sender and receiver. For example physical distance of Taplejung and Bajhang.
 - **Noise:** It is the unwanted sounds or disturbance. It minimizes the effective transmission of messages. It distract the sender and receiver.
 - **Physical arrangement:** It includes the layout of office. For example People of same department working in different floor (for example top floor and ground floor). This obstruct the communication
- b. **Psychological Barriers:**
 - **Selective Perception:** selective attention, distortion and retention are selective perception
 - **Filtering:** The sender filter the information which is favourable to receiver
 - **Lack of trust:**
 - **Emotions:** hate, anger, fear love etc also makes the communication less effective
 - **Work pressure:** work pressure also creates adverse for the effective communication.
- c. **Organizational Barriers:**
 - **Poor Planning:** Poor planning of The message design, encoding, channel selection creates disturbed communication.
 - **Structure Complexity:** Complex organizational structure and unnecessary hierarchy serve as barriers to effective communication

- **Influence:** Influence by supervisor or higher authority also affects communication
- **Organizational Distance**
- **Bad Timing:** lack of time for the communication also disturb the communication.
- **Technological problem:** disturbance on email, internet, electricity, telephone also cause disturbance.
- d. **Semantic Barriers:** It is related about the language related problems associated with communication.
 - **Language:** The meaning intended by the sender can be different from the meaning understood by receiver due to the differences in language.
 - **Jargon:** Unfamiliar Jargon or technical language disrupts communication.

6. Why is motivation important to achieve organizational goals?

Ans:

- Motivation is inducement for better performance toward goal Achievement.
- It is the act of stimulating, inspiring or energizing employee for higher performance.

Importance of Motivation:

The productivity depends upon the capability of employee(ability) and willingness. The capability of the employee can be increased through the training and development. Capable employee might not generate the higher productivity unless they have the willingness to work. The willingness of the employee can be established or increased through the motivation. So the motivation is important to achieve organizational goals which can be explained through the following points:

- i. **Understanding Behaviour:** Behaviour of the employees can be understand through the motivation. Different motivational theories can be applied to understand the behaviour of the employees.
- ii. **Develops Positive Attitude:** Different monetary and non-monetary techniques are used to motivate employees. It will create positive attitude to the employee as they assume themselves integral part of the organization.
- iii. **Creates Willingness:** Increase in salary, promotion, job rotation, employee etc increases the satisfaction of the employees. Satisfied employees are always willing to work with efficient and effective manner. Finally, Willingness of the employee can be increase through the motivation.
- iv. **Optimal Use of Resources:** Motivated employees use available resources efficiently. The waste of resources is discarded. It also saves the cost
- v. **Improves Productivity:** Motivated employees work willingly to increase productivity. Similarly, different performance-based incentives are used to motivated employees that also improves the productivity.
- vi. **Employee Retention:** Satisfied employees don't want to switch the organization. Long term service of the employee creates strategic advantages to the achievement of organizational goals.

- vii. **Managing Change:** Satisfied employees manages the change required for the organization.
- viii. **Employee Commitment:** Motivation is important for employee-commitment. The committed employees are devoted to the organization of higher performance.
7. **Define delegation of Authority. Explain major features of delegation of Authority.**

Ans: Delegation of Authority:

- Delegation of Authority is the process of transferring authority to the subordinates.
- It is the means for getting the jobs done through other people.
- Superior assign a portion of their workload to subordinates through authority delegation.
- Decision making assigns authority and responsibility along with accountability as well.
- Delegation pushes decision making downward.
- Delegation of Authority makes the job done quick with minimum use of resources. It also saves time.

Major Feature of Authority Delegation:

- a. **Balance of Authority Responsibility:** Authority and responsibility should be equal or balanced. Responsibility without authority and Authority without responsibility is discarded.
- b. **Absoluteness of responsibility:** Responsibility can't be delegated. In other words, Delegation of Authority establishes the responsibility to the subordinates as well but the final responsibility is always retained with the superior.
- c. Legitimate subordinates to perform specific tasks autonomously.
- d. **Restoration of delegated authority:** Authority delegation can be revoked by the superior at any time. Superior can increase, decrease or revoke the authority at any time if required.
- e. **Scalar Chain:** Authority should be delegated through the hierarchy to positions.
- f. **Management by Exception:** Only Exceptional matters should be referred to higher levels for decision making.
- g. **Acceptance of Subordinates willingly**
- h. **Performance monitoring:** The Authority delegation can be made more effective through the timely monitoring the performance.

Group-C

Attempt any Two Questions.

8. **Planning is head and controlling is tail. Justify.**

2×10=20

Ans: Planning is head and controlling is tail. To justify this statement first we have go through the basic definition of Planning and controlling.

Planning:
Planning is the predetermination of future course of action. It is the process of setting goals and choosing the actions to achieve those goals. It is the blueprint of action. It is deciding in advance about all WH questions (like what to do, how to do, when to do it and who is to do it.). Planning selects future course of action from the alternatives. It serves as guidance for allocating resources. It also minimize the risk and uncertainty. It also serves as the standard or yardstick for the effective control mechanism.

Controlling: Controlling is the process and measuring actual performance and comparing it with planned performance and taking corrective actions if there is deviation found in actual and planned performance. Planned performance is also known standard. Control is managerial process. It ensures that actual activities conform to planned activities.

Justification:

- Planning, organizing, staffing, leading and controlling are the process/functions of management. This statement explains that the planning comes at first and controlling comes at last.
- In the process of controlling, planning serves as standard or yardstick for the effective control. Controlling is always related with planning. This statement also justifies that the planning comes at first and controlling comes at last. For example the organization plans to make profit of 5 crores for 1 years period of time. The 5 crores profit plan is serves as standard or yardstick. If the actual profit is deviated from 5 crores then the controlling is required. Therefore, planning always comes at first and controlling comes at last. So we can conclude that the planning is head and controlling is tail.

9. Define leadership. Explain different style of leadership with suitable example.

Ans: Concept of Leadership:

- Leadership is guiding and influencing people to achieve goals willingly and enthusiastically in a given situation.
- It is about inspiring people.
- In other words, Leadership is the ability (or Power) to inspire and influence Subordinates towards achievement of organizational objectives.
- Leader, follower and situation are the major components of leadership.
- Leader: People who Command (direct or lead) subordinates.
- Follower: People who follow the Command (direct or lead)
- Situation: It is the condition

Leadership Styles: Leadership style deals with the way leaders influence the followers. It can be boss-centered, subordinate-centered or relations-centered. Leadership styles can be:

- Autocratic, Democratic, Free Rein Styles:
- Managerial Grid Style
- Four Systems of Management Style
- Situational Style

1. **Autocratic, Democratic, Free Rein Styles:**

Autocratic: If the Power and decision making is centralized then it is called autocratic leadership style. Leaders make all the decisions. They order what to do and how to do. Subordinates are not consulted. For example, the General manager decides about the operation of machinery, units to be produced, worker in production division can be considered as Autocratic leadership style.

Democratic: If the Power and Decision making is decentralized then it is called Democratic leadership style. Discussion, consultation and participation is encouraged. Leaders consult with subordinates about decisions. Subordinates willingly cooperate with the leader. Authority is

delegated to subordinates. For example, the General manager decides about the operation of machinery, units to be produced, worker in production division can be considered as Democratic leadership style if the General manager consult with production manager/ officer or involves the production manager/officer in decision making process.

Free Rein (Laissez Fair): if the Leader Grants Authority and responsibility to a group then it is called Free Rein (Laissez Fair) leadership style. The role of leader is to provide advice and direction as requested by the subordinates. Power and Decision making is entrusted to the subordinates. For example, the General Manager delegates the authority to a team to develop the Enterprise Resource Planning (ERP) software. In this example, the team members are fully autonomous to work to develop the ERP software. They only consult with general manager for Exceptional cases only. They have full authority to decide under the given parameters.

2. Managerial Grid Style (Blake and Mouton):

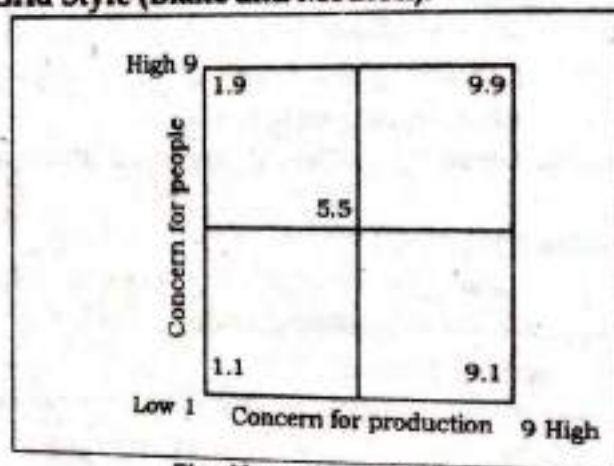


Fig: Managerial Grid

- 1.1 Manager (Impoverished Style or poor leadership): leadership style that has very little concern for production or people.
 - 9.9 Manager (Team Builder Style): leadership style that has High concern for people and production. This is ideal leadership style.
 - 5.5 Manager (Middle of Roader Style): leadership style that has Balanced concern for production and people.
 - 9.1 Manager (Task Manager Style): leadership style that has high concern for production and little concern for people.
 - 1.9 Manager (Country Club): leadership style that has high concern for people, little concern for production.
- ## 3. Four Systems of Management Style

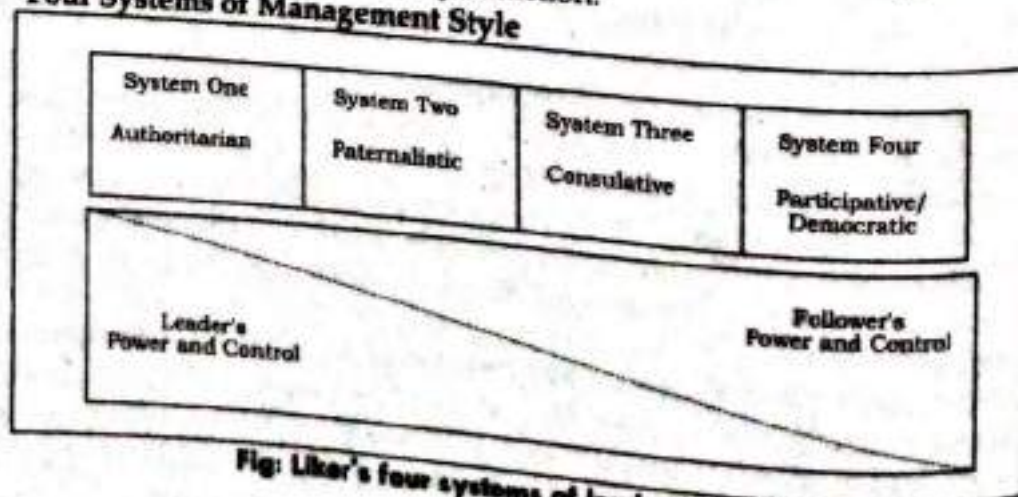


Fig: Likier's four systems of leadership style

- I. System One: Exploitive-Authoritative; Autocratic leadership. It is leader-centered. Leader has power and control.
- II. System Two: Benevolent-Authoritative; Patronizing leadership. It is leader-centered too.
- III. System Three: Consultative; subordinates consulted. It is Group centered. Followers have power and control.
- IV. System Four: Participative; democratic leadership. It is leader and follower centered. Followers have power and control.

According to this approach, system four managers are successful leaders.

4. Situational (Contingency) Style:

- There is no one style which is best for all situations.
- It states that leadership style is situational.
- Structure of task, leader-led relations and position power of leader are the situation variables that affect leadership style.

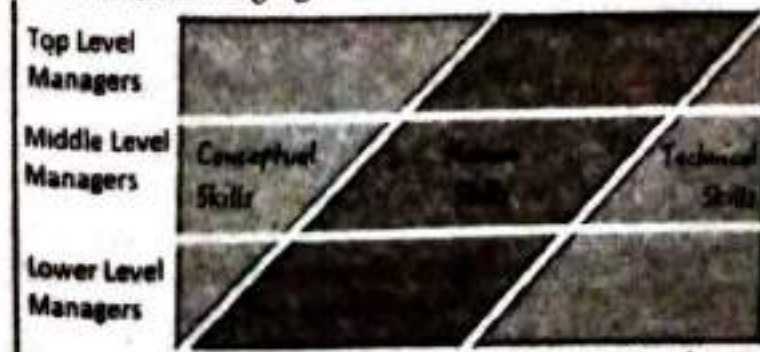
10. What are managerial skills? What skills are essential for a successful manager?

Ans: Skill is the ability or proficiency to perform a particular task.

1. **Technical Skills:** Technical skill is the ability to use equipment, techniques or procedures of specific work. They should know how to perform specific task assigned to the subordinate. Computer skills, accounting skills, quality control skills are some examples. Such skills are needed for the supervision of subordinates.
2. **Human Skills:** Human Skill is the ability to work with subordinate. Human skills refers to the ability to work with, understand and motivate employee. Communication skills, negotiation skills, leadership skills, motivation skills, conflict management skills are some examples.
3. **Conceptual Skills:** Conceptual skill is the mental ability to analyse and diagnose complex situation or problems. It involves seeing the organization as a whole and understanding how its parts will affect the whole. It is ability to analyse the environment and set the strategy. It reflect ability to organize information and judge relationships.

Skills are essential for a successful manager?

Generally, more the skills there will be more chances of being the successful manager. However, The skills required for each of the managers can be shown in the following figures.



Top level Managers: This is the highest level of management. It is responsible for gaining overall objectives of the organization. They coordinate among the different departments and units of the organization performing complex and varied nature of job. Board Chairman, Chief Executive Officer (CEO), Managing Director (MD), General Manager (GM), President are the example of Top level Manager.

Basic Functions:

- To define the overall objectives of the organization.
- To prepare strategic plans and policies of the organization.
- To Set up organizational Structure
- To Appoint departmental managers
- To coordinate review and control the performance of all departments.
- To represent the organization.

To perform the above task the top level managers require mostly the conceptual skills. They have to set the overall objectives of the organization the need to have the problem solving skills and idea generating skills which is ultimately related with the conceptual skills. Similarly, they also have to deal with the people inside and outside the organization the also need to have Human skills. Furthermore, top level managers require technical skills too but they need not to have expert knowledge on operating machinery and equipment.

Middle Level Managers: This level of management consists of **functional heads (Department managers or Division head or plant superintendent)** of different departments and is responsible for preparing and implementing departmental plans. Human resource manager, Production Manager, Marketing Manager, Procurement Manager are middle level manager.

Basic Functions:

- To play the role of mediator between top-level and first line management.
- To implement Strategies, plans and policies laid down by top level management.
- Prepare departmental plans and strategies on the basis of guidance and information from top level
- To Supervise and control the activities of lower level managers.

To perform the above task, middle level manager needs the conceptual, human, technical skills equally because they are mediator between top-level and lower level managers.

Lower Level Managers: This is the lowest level management and is directly involved in implementation of plans. This level is known as first line or operating level of management. It is directly involved in the actual operation of production, marketing, financing, accounting etc. Supervisors, foreman, sales officer, accounts officers and other operational heads are the example of lower level managers.

Basic Function:

- To make day-to-day plans and to implement plans formulated by middle level management.
- Assign responsibilities and duties to the employees.
- To supervise the activities of subordinates.
- To maintain close and harmonious relation among all the employees.
- To submit the progress report to the middle level management.
- To create better environment for work

To perform the above task lower level-managers require the technical skills the most. They need to deal with the workers so they require the human skills. Lower-level managers are ok with less conceptual skills because they are not directly concerned with idea generation and setting mission, vision, objectives and strategy of the organization.

TU QUESTIONS-ANSWERS 2018

Bachelor in Computer Applications
 Course Title: Introduction to Management
 Code No. CAMG 304
 Semester: V

Full Marks: 60
 Pass Marks: 24
 Time: 3 Hours

Group B

Attempts any SIX question.

6×5=30

2. Define Management? What Kinds of skills required for successful manager?

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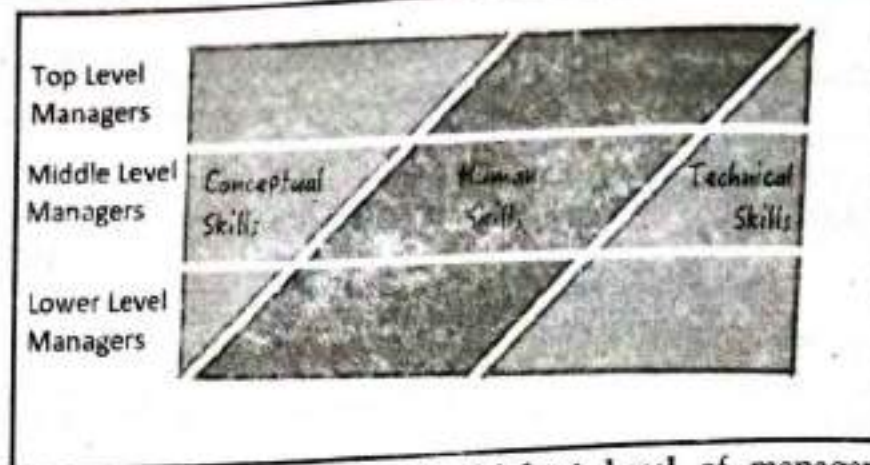
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To perform the above task the top level managers require mostly the conceptual skills. They have to set the overall objectives of the organization the need to have the problem solving skills and idea generating skills which is ultimately related with the conceptual skills. Similarly, they also have to deal with the people inside and outside the organization the also need to have Human skills. Furthermore, top level managers require technical skills too but they need not to have expert knowledge on operating machinery and equipment.

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- To Supervise and control the activities of lower level managers.

To perform the above task, middle level manager needs the conceptual, human, technical skills equally because they are mediator between top-level and lower level managers.

Lower Level Managers: This is the lowest level management and is directly involved in implementation of plans. This level is known as first line or

operating level of management. It is directly involved in the actual operation of production, marketing, financing, accounting etc. Supervisors, foreman, sales officer, accounts officers and other operational heads are the example of lower level managers.

Basic Function:

- To make day-to-day plans and to implement plans formulated by middle level management.
- Assign responsibilities and duties to the employees.
- To supervise the activities of subordinates.
- To maintain close and harmonious relation among all the employees.
- To submit the progress report to the middle level management.
- To create better environment for work

To perform the above task lower level managers require the technical skills the most. They need to deal with the workers so they require the human skills. Lower level managers are ok with less conceptual skills because they are not directly concerned with idea generation and setting mission, vision, objectives and strategy of the organization.

3. Describe the major characteristics of contingency theory. Why there is "no one best way" to manage.

Ans: Contingency Theory/ Situation Theory (Approach): This theory is developed in 1960 by Tom Burns and GM Stalker, Paul Lawrence and Jay Lorsch. It is the most recent development in Management theory. This theory recognizes the situational nature of management.

This Theory states that managerial practice depends on situational factors. There is no one best way in the world of management. Each situation is unique and demands unique managerial action. Result differ because situations differ. Managers should be situation sensitive in their behaviours. Effective managerial behaviour in one situation cannot be generalized to other situations. Different situations call for different management approaches. The effect of cause depends on situations.

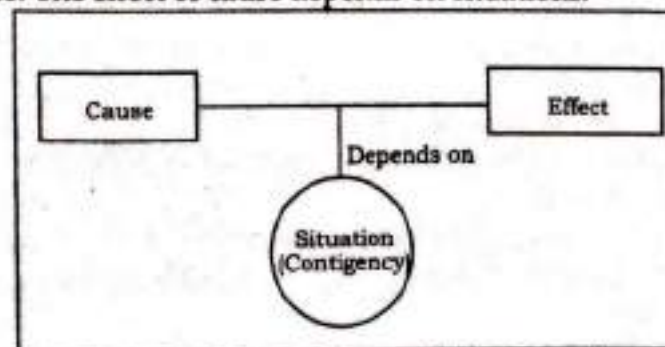


Fig: Contingency Theory of Management

- Prevailing contingency variables determine management approach and practice. They are

- Organization size
- Technology
- Uncertainty
- Individual differences
- Types of work being done.

Features of Contingency Theory:

- i. Management is externally situational: the conditions of the situation will determine which techniques and control system should be designed to fit the particular situation.

- ii. Management is entirely situational.
- iii. There is no best way of doing anything.
- iv. One needs to adapt himself to the circumstances.
- v. It is a kind of "if" "then" approach.
- vi. It is a practically suited.
- vii. Management policies and procedures should respond to environment.
- viii. Managers should understand that there is no best way of managing. It dispels the universal validity of principles.

Application of Contingency Theory of Management are:

- a) Analyze and understand the situation case-by case
- b) Examine the validity and applicability of management approaches to the situations at hand.
- c) Match the management approach to the situation. Identify choice to specific situations.
- d) Implement the choice.

There is "no one best way" to manage

The main theme of situational theory is that managerial practice depends on situational factors. There is no one best way in the world of management. This theory states that leadership styles, organization structure, job design, motivational approaches, control systems and management of change and conflict should be situation specific.

For example, there are two types of employees in the organization. One type of employee are active, self-motivated and always wants to work for the betterment of the organization. Another type of employees is lazy, always wants to skip the office job and they are not self-motivated. Other management theories advocate for the one best way in all the situation. But Situation/contingency has different opinion. For the above condition, manager should deal with the situation. For that first type of employees, managers are should motivate with the better compensation, salary, bonus, incentive, autonomy and challenging work. But for the second type of employees, managers should use better control techniques and use the disciplinary action like decrease in pay, demotions and firing from the organization if required.

So there is no one best way of management.

4. What is planning? Explain the different level of planning.

Ans: Concept of Planning:

- Planning is the predetermination of future course of action.
- It is the process of setting goals and choosing the actions to achieve those goals.
- It is the blueprint of action.
- It is deciding in advance about all WH questions (like what to do, how to do, when to do it and who is to do it.)
- Planning selects future course of action from the alternatives.
- It serves as guidance for allocating resources. It also minimize the risk and uncertainty.
- It also serves as the standard for the effective control mechanism

Definition:

Planning is deciding in advance about what to do, how to do it, when to do it and who is to do it. It provides the ends to be achieved. -S.P. Robbins
Planning takes organizations to where it wants to be. The difference between existing state and the desired state is called 'strategic gap'. Planning bridges this gap.

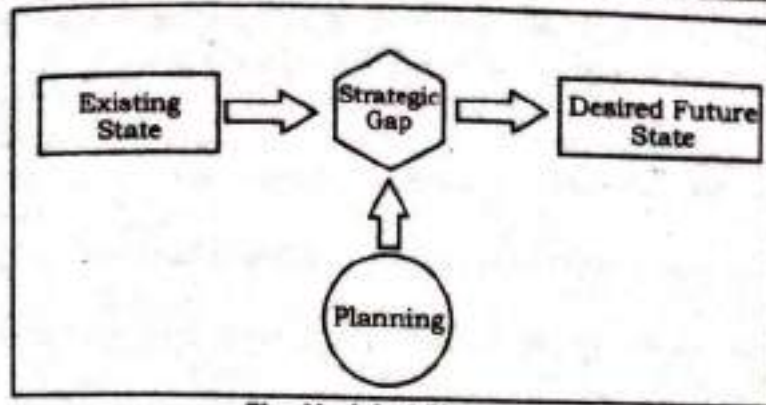


Fig: Model of Planning

Level of planning:

- a) **Corporate Plan (Strategic Plan):** It is organization wide. It establishes overall corporate mission, goals and strategies.
 - **Mission:** Mission is the vision. It represents to the philosophy and ideology of the organizations. It is a statement of its fundamental unique purpose for setting a business apart from other similar firms, and for identifying the unique scope of the business operation.
 - **Goals:** Goals are results to be achieved. They represent the end point of planning.
 - **Strategy:** Strategy or strategic goal is the long term objective of the organization. It refers to long-term broad action plan in key result areas. Strategy provides **scope and direction**. It provides competitive advantage.

Corporate plans are long-term plans for 5 years or more prepared by top level management. They are based on the SWOT analysis. It provide long-term direction and scope to the organization.

- b) **Tactical Plan(Divisional/Departmental Plan):**

- It specifies how overall goals are to be achieved. It defines goals for implementing corporate plan. It identifies priorities of the divisional activities. It is prepared for each division or department.
- The tactical plan allocates tasks and resources. It is prepared for divisional activities such as production, marketing, finance, human resources. It is prepared by middle level managers. The time duration is 1 to 5 years.

- c) **Operational Plan(Unit/Functional Plan):** They are unit level plans set for lower level managers. In other words, Operational plans are sub-division of tactical goals. It is concerned with day to day activities. They are specific in terms of results. They are measurable. They are supported by action plans. It identifies the best mix of resources to be allowed to the unit. Operational plans should be consistent with tactical plans.

5. **Define organizational structure. What are the differences between centralization and decentralization?**

Ans: **Concept of Organizational Structure:**

- Organizational structure is the process of creating functional structure of organization required to implement plans and decisions for attaining organizational goals.
- It is a process of determining and dividing the tasks, creating job positions and their link, assigning job responsibilities to the right position and allocating authority to make decisions.
- It creates shape of responsibility-authority relation.

- It ensures implementation of plans and strategies, utilization of organizational resources to achieve organizational goals.

Definition:

"Organizing is the process of defining and grouping activities of the enterprise and establishing authority relationship among them".

-Theo Haimann

Differences between centralization and decentralization:

Centralization:

- Centralization is the process of systematically retaining power and authority in the hands of higher level managers.
- In centralization, all the decisions are made by the higher level managers.
- Top Managers make all the decisions. Subordinates simply carry them out.

Decentralization:

- Decentralization is the result of delegation of Authority.
- It is transfer of decision making authority downward.
- Every organization practices some degree of decentralization.
- It is autonomy in decision making
- Decentralization pushes down decision making to managers who are closest to the action.

Comparison Chart

BASIS FOR COMPARISON	CENTRALIZATION	DECENTRALIZATION
Meaning	The retention of powers and authority with respect to planning and decisions, with the top management, is known as Centralization.	The dissemination of authority, responsibility and accountability to the various management levels, is known as Decentralization.
Involves	Systematic and consistent reservation of authority	Systematic dispersal of authority.
Communication Flow	Vertical	Open and Free
Decision Making	Slow	Comparatively faster
Advantage	Proper coordination and Leadership	Sharing of burden and responsibility
Power of decision making	Lies with the top management.	Multiple persons have the power of decision making.
Implemented when	Inadequate control over the organization	Considerable control over the organization
Best suited for	Small sized organization	Large sized organization

6. Define communication and explain the various steps in the communication process.

Ans: Communication:

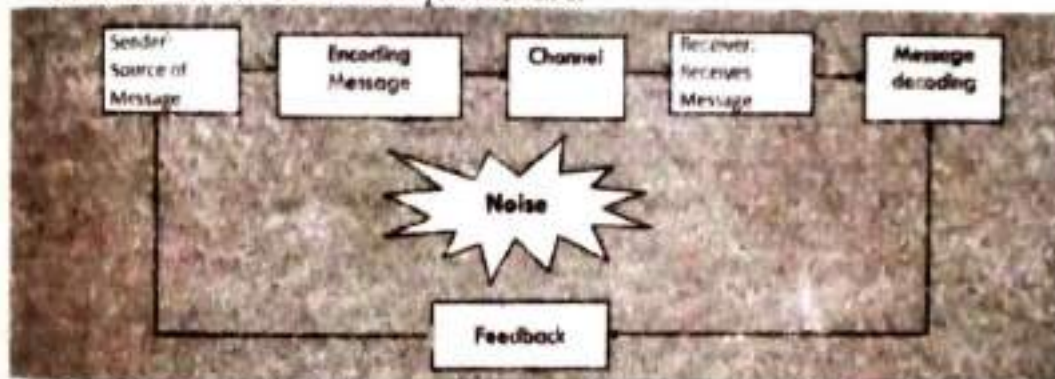
- The word 'Communication' is derived from the Latin word 'Communis' which means 'common'. Literally, communication means to inform, to tell or to spread information and message to all the concerned people so that it will be common to them.
- Communication is the process of Transferring/Transmitting information on common interest from one person to another.
- It is the exchange of facts, opinions, ideas, suggestion and other information from one person to another.

"Communication is the process of transmitting information from one person to another" -R.W. Griffin

Characteristics/Features of Communication:

- i. Minimum two person
- ii. Two way process
- iii. Pervasive function
- iv. Complete and rational process
- v. Continuous function
- vi. Oral or Written
- vii. Formal or informal
- viii. Coding and decoding
- ix. Channel based
- x. Noise

Process of Communication: Communication process consists of steps between the sender and the receiver that result in transference and understanding of meaning. Communication is a two-way process. The elements of communication process are:



- i. **Sender:** Sender is the source of information. It can be an individual, group or organization. Sender has willingness to communicate, has a need to do or is under obligation to pass information to other person, department or organization. Depending upon the nature of job responsibility, sender creates information to flow.
- ii. **Encoding Message:** Message is what is to be communicated. It can be ideas, information and emotions. Encoding is converting a message to understandable form. Message is encoded either in the form of words, symbols, sign or picture.
- iii. **Channel or Medium:** It is the medium for message transmission. It is pathway. Channel can be
 - Audio-Visual(TV, radio)
 - Print Media (Newspaper, magazine)
 - Mechanical (Telephone, fax, e-mail, computer network)
 - Personal (face to face contacts) etc.
- iv. **Receiver (Receives Message):** Receiver is a person who receive and understands the message of the sender. Communication does not take place if it is not received by the receiver. It does not serve any purpose if it has reached the receiver but is not understood correctly.
- v. **Decoding:** It is a process whereby the receiver interprets encoded message into meaningful information. The effectiveness of communication depends on how effectively receiver has decoded the message. Effective decoding is the condition in which receiver understands the message as the theme and meaning of sender.

- vi. **Feedback:** It is the response or reaction of the communication process. It is directed towards the communicator to facilitate future communication. It represents the return flow of communication as the impact of initial communication. It is a process of communicating back to sender regarding how effectively the message is understood by the receiver.
- vii. **Noise:** Noise is the disturbance in course of communication. It is indispensable in communication process. There can be through various ways as mechanical defects, noise of machinery, distance between sender and receiver, network problem, environment etc.

7. **What is leadership? Explain the different style of leadership.**

Ans: **Concept of Leadership:**

- Leadership is guiding and influencing people to achieve goals willingly and enthusiastically in a given situation.
- It is about inspiring people.
- In other words, Leadership is the ability (or Power) to inspire and influence Subordinates towards achievement of organizational objectives.
- Leader, follower and situation are the major components of leadership.
- Leader: People who Command (direct or lead) subordinates.
- Follower: People who follow the Command (direct or lead)
- Situation: It is the condition

"Leadership is the ability to influence a group toward the achievement of goals"-Stephen Robbins

Features of Leadership:

- i. Influence Process
- ii. Unidirectional Authority: Leaders possess authority to influence followers. But followers lack authority to direct the leader. The authority is one way from leader to the follower.
- iii. Goal directed:
- iv. Continuous Process
- v. Three Element: Leader, Follower, Situation
- vi. Motivational Function

Leadership Styles: Leadership style deals with the way leaders influence the followers. It can be boss-centered, subordinate-centered or relations-centered. Leadership styles can be:

- Autocratic, Democratic, Free Rein Styles;
- Managerial Grid Style
- Four Systems of Management Style
- Situational Style

i. **Autocratic, Democratic, Free Rein Styles:**

Autocratic: if the Power and decision making is centralized then it is called autocratic leadership style. Leaders make all the decisions. They order what to do and how to do. Subordinates are not consulted. For example, the General manager decides about the operation of machinery, units to be produced, worker in production division can be considered as Autocratic leadership style.

Democratic: if the Power and Decision making is decentralized then it is called Democratic leadership style. Discussion, consultation and participation is encouraged. Leaders consult with subordinates about decisions. Subordinates willingly cooperate with the leader. Authority is delegated to subordinates. For example, the General manager decides about

the operation of machinery, units to be produced, worker in production division can be considered as Democratic leadership style if the General manager consult with production manager/ officer or involves the production manager/officer in decision making process.

Free Rein (Laissez Fair): if the Leader Grants Authority and responsibility to a group then it is called Free Rein(Laissez Fair) leadership style. The role of leader is to provide advice and direction as requested by the subordinates. Power and Decision making is entrusted to the subordinates. For example, the General Manager delegates the authority to a team to develop the Enterprise Resource Planning (ERP) software. In this example, the team members are fully autonomous to work to develop the ERP software. They only consult with general manager for Exceptional cases only. They have full authority to decide under the given parameters.

ii. **Managerial Grid Style (Blake and Mouton):**

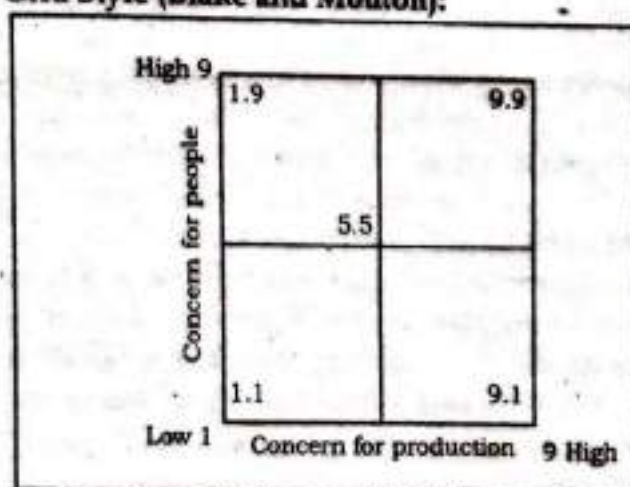


Fig: Managerial Grid

- 1.1 Manager (Impoverished Style or poor leadership): leadership style that has very little concern for production or people.
- 9.9 Manager (Team Builder Style): leadership style that has High concern for people and production. This is ideal leadership style.
- 5.5 Manager (Middle of Roader Style): leadership style that has Balanced concern for production and people.
- 9.1 Manager (Task Manager Style): leadership style that has high concern for production and little concern for people.
- 1.9 Manager (Country Club): leadership style that has high concern for people, little concern for production.

iii. **Four Systems of Management Style**

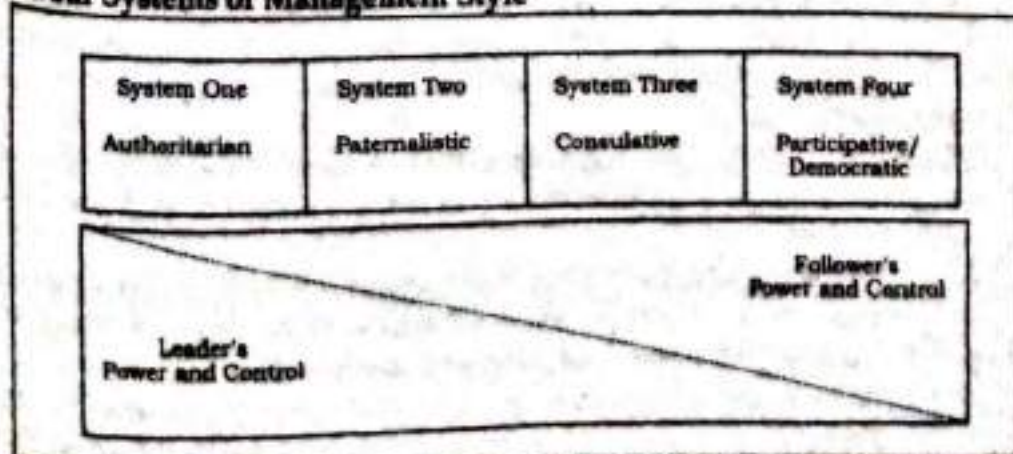


Fig: Likier's four systems of Leadership style

- iv. System One: Exploitive-Authoritative; Autocratic leadership. It is leader centered. Leader has power and control.
 - v. System Two: Benevolent-Authoritative; Patronizing leadership. It leader centered too.
 - vi. System Three: Consultative; subordinates consulted. It is Group centered. Followers have power and control.
 - vii. System Four: Participative; democratic leadership. It is leader and follower centered. Followers have power and control.
- According to this approach, system four managers are successful leaders.
- viii. **Situational (Contingency) Style:**

- There is no one style which is best for all situations.
- It states that leadership style is situational.
- Structure of task, leader-led relations and position power of leader are the situation variables that affect leadership style.

8. **What are the major techniques of control? Explain.**

Ans: Controlling is the process and measuring actual performance and comparing it with planned performance and taking corrective actions if there is deviation found in actual and planned performance. The major Tools and Techniques of control can be explained as follows:

i. **Budgetary Control:**

A budget is a statement of future expenditures and incomes for planning and controlling financial resources. It is generally for one year. It focuses on pre control. Budget can be variable or fixed. It is a tool for annual plan control. It plans and control the acquisition and use of resources. It finds and corrects deviations:

Types of Budget:

- **Operating Budget:** They consist of plans for the use of raw materials, goods and services. They specify quantity and costs. Cost Centre, Revenue Centre, Profit Centre, Invest Centre are the types of operating budgets.
- **Financial Budget:** They consist of plans for mobilizing and spending financial resources. They ensure that funds will be available when required. Cash budget, capital expenditure budget and master budgets are the example.
- **Programme Budget:** Budget based on the goals and priority of each programme is called programmed. The programme consists of different types of project
- **Zero Based Budget:** It requires each unit to take a fresh look at activities and priorities each year. It starts from a clean slate. Managers must justify each expenditure from a zero base.

ii. **Financial Control:**

Financial controls are accounting-Based. They provide an insight into organization's performance during or after the operation. They can be of following types:

- **Financial statements:** They Provide a profile of financial circumstances. They track monetary value of transactions. Balance sheet, Income statement and cash flow statement are the example.
- **Ratio analysis:** Ratios are used for efficiency and profitability control. They show relationships between figures for evaluating financial

performance. Liquidity ratio debt ratio, Profitability ratio, Operating ratio are the examples

- Break-even Analysis: This analysis identifies break-even point where revenue equals cost. Costs are classified into variable and fixed. Profit is possible only when sales exceed the break even point. It is also known as cost-volume-profit analysis
 - Auditing: It is comprehensive, systematic, independent and periodic examination of truth and fairness of financial statements. It examines compliance, efficiency and performance. It can be internal, external or management audit.
- iii. **Reports/Visits:**
- Regular: They are related to production, marketing, finance and human resource performance.
 - Special: They deal with complex problems, such as impact of tax change, make or buy decisions.
 - Visits: They are done for personal observation through walking around to various units. Problems are solved on the spot.
- iv. **PERT/CPM Network Techniques:** These are modern quantitative techniques of control
- PERT (Programme Evaluation and Review Technique): A PERT chart is a Project management tool that provides a graphical representation of a project's timeline. The Program Evaluation Review Technique (PERT) breaks down the individual tasks of a project for analysis.
 - CPM Network: The critical path method (CPM) is a step-by-step project management technique for process planning that defines critical and non-critical tasks with the goal of preventing time-frame problems and process bottlenecks. The CPM is ideally suited to projects consisting of numerous activities that interact in a complex manner.

Group C

Attempts any TWO questions.

[2x10=20]

9. Decision making is very important managerial function. With the statement; discuss the meaning and process of decision making. Also explain the types of decision making.

Ans: Managerial function consists of planning, organizing, staffing, leading, controlling. All the managerial functions are done through the decision making. Formulating strategic planning can be taken as example. Strategic planning is the match between the strength and opportunities after accessing the internal and external environment. For that management develops different alternatives (Strategies) matching strength and opportunities. Among them management selects the best strategy among the different strategies. So Decision making is very important managerial function.

Meaning of Decision Making:

- Choosing the best alternative among the alternatives is call decision (or decision making).
- Manager always performs through the decision making.
- It is a means to achieve ends.
- It is a path selected for moving the organization from existing state to desired state.
- All the organizational problems are solved from the decision-making process.

Definition: Decision making is selection of a course of action from among the alternatives. -Koontz and Weihrich

Nature/Feature/Characteristics of Decision Making:

- i. Goal Oriented
- ii. Dynamic
- iii. Pervasive
- iv. Continuous
- v. Problem Solving
- vi. Choice
- vii. Long term Commitment

Decision Making Process:

- i. **Recognize and define the problem:** The first step is clearly defining a problem with causes. A problem is the gap between the existing situation and the desired situation. The decision-making process is entirely focused to solve the problem.
- ii. **Identifying appropriate alternatives:** Alternatives are possible solution of the problems. Managers should identify the alternatives to solve the problems. Alternatives are developed on the basis of information collection and data analysis.
- iii. **Evaluate the alternatives:** Each alternative should be evaluated in terms of costs, time, quality, and legal constraints, human and other resources. The alternatives are ranked according to their benefit cost ratios.
- iv. **Choose the best alternative:** Most feasible alternative is selected. Sometimes more than one alternative are also selected. Experience, experimentation, research & Analysis and judgment of decision maker are important for choosing the best alternatives. The best alternative is expected to solve the problems.
- v. **Implement the selected alternative:** Implementation is putting a decision into action. It should be accepted by people willingly. All the required resources are gathered to implement the alternatives.
- vi. **Evaluate and follow-up:** This step is essential to evaluate the decision effectiveness in solving the problem. Its progress is monitored and necessary adjustments are made to ensure successful implementation of decision.

Types of Decisions:

- i. **On the basis of frequency of recurrence:**
 - a. **Programed Decisions:** Programed decisions are routine and repetitive in nature. If a particular situation occurs, managers will often, develop a routine procedure for handling it. Decision makers know in advance what decisions he/she has to take in particular situations. In practice, more than 90% of managerial decisions are of this type. Middle and lower-level managers deal with such decisions. Example: payroll procedures, recruiting regular worker in the factory.
 - b. **Non-Programmed decisions:** Non-Programmed decisions are relatively unstructured and occur less frequently. They deal with complex, novel and non-routine problems. Each problem is new and different from those experienced in the past. Managers apply judgment, intuition and creativity to solve problems. There are no standard operating procedures for it. Top level managers deal with non-programmed decisions.
- ii. **On the basis of nature of the decisions:**

- a. **Strategic Decision:** Those decisions which are made by top level management for long term purpose are called strategic decisions. Long term goals, objectives, strategic plans are strategic decisions.
 - b. **Tactical Decision:** Tactical decisions are the decisions made by department level managers for their respective departments. Such decision translate strategic decision into departmental actions.
 - c. **Operational Decision:** Decisions taken to solve the problems regarding day to day operation are called operational decision. Authority is delegated to lower level management to make such decisions.
- iii. **On the basis of Participants involved in decision making process:**
- a. **Individual Decisions:** The decision which is made by a single person is called individual decision. They are generally prevalent in small organizations.
 - b. **Group Decisions:** The decision which is made by a group of persons is called Group decision. Under this method, group members participate in discussion and share their opinion, ideas and views to reach in decision.
- iv. **Other decisions:**
- Organizational and personal decisions
 - Policy and operational decisions
 - Financial and non-financial decisions
 - Short term and long term decisions
10. **Quality improvement has no barriers and limitations. On the light of this statement, explain the concept of Total Quality Management and explain the different tools of TQM.**

Ans: Quality is perception of excellence by customers to satisfy needs. Quality improvement is continuous process. Quality consists of Conformance (Meeting requirement), Aesthetics (Look, taste, feel, smell), Reliability, Durability, Serviceability, Perception and Performance. Lets take the most influence charter of product as quality as conformance i.e. meeting the requirement of customer. Right now, Samsung is a big name in **phone companies** and has emerged as one of the **top mobile brands** in the world. But before a decade ago the nokia has ruled the market. People's requirement shifted from the nokia's features to android version. So people found the Samsung, apple or other phone appropriate or qualitative for the customers. Nokia couldn't improve the quality so it got failure in the market. In order to survive in the market, the business organization should always look after its quality continuously.

Concept of Total Quality Management:

- Total Quality management is a management philosophy of continuously improving product quality through everyone's commitment and involvement to satisfy customers' need.
- Quality is the perception towards effectiveness and utility of any goods and services to satisfy the needs. As the expectation of the people may change continuously, the same level of quality cannot satisfy all the customers. So, the quality of the product must be improved continuously. It is believed that there is no limited scope for quality improvement. In this ground, management should also focus its efforts for the gradual and continuous improvement of quality. This concept of gradual and continuous improvement in product quality is developed as a management philosophy which is known as total quality management.

"Total Quality management is a strategic commitment by top management to change its whole approach to business to make quality a guiding factor in everything it does." -R.W. Griffin

Different Tools of TQM:

- i. **Right for the first time:** Employees ensure quality while they work. They do the right things first time. The result is zero defect. Basic concept is "you don't have second chance to make first impression".
- ii. **Benchmarking:** Benchmarking is the process of comparing and learning from best practices of other organizations that produce superior performance. They do exceptionally high quality things. Such a comparison may be in terms of various dimensions as general practices, services, product design, business process and/or administrative processes. It can be done internally or externally. Internal benchmarking compares the performance of specific unit, department, process, employees, etc. with the best performer within the organization whereas external benchmarking is the comparison with best performer of industry or similar business. Benchmarking tries to recognize the need to change and determine what direction should be taken by observing those who have changed successfully. Benchmarking is successfully used as a process of introspection and search for the best practice in total quality management. Benchmarking is referred to as the process by which an organization measures their products, services, and practices against its most difficult competitors, or those organizations recognized as leaders in the same industry.
- iii. **Outsourcing:** It is subcontracting services and operations to outside firm or person who can perform them cheaper and better. This system of outsourcing improves efficiency, reduces cost, enhances productivity, and improves quality. It helps top management to focus the attention only on core activities and strategic issues. Outsourcing of operations in sectors that could provide a competitive advantage to small, medium and large enterprises is a completely new method that has emerged in recent years. The need to develop information and communication technologies in order to increase performance within a competitive environment has led to the development of various IT and computer departments within firms.
- iv. **ISO 9001:** They are set of quality standards created by International Organization for Standardization (ISO). Organizations obtain certification from ISO. ISO Standards cover product testing, employee training, record keeping, supplier relations and revise policies and procedures. The ISO 9000 family contains following standards:
 - ISO 9001:2015: Quality management systems-Requirements
 - ISO 9000:2015: Quality management systems- Fundamentals and vocabulary(definitions)
 - ISO 9004:2009: Quality management systems-Managing for the sustained success of an organization (continuous improvement)
 - ISO 9011:2011: Guidelines for auditing management systems
- v. **Statistical Quality Control:** It includes a set of specific statistical tools that can be used to monitor quality. This method extremely uses the quality. It is also one of the major tools used for quality assurance.

Benefits of Statistical Quality Control

 - It identifies the bottle necks and trouble spots.

- It provide same ans of detect in gerrorat inspection. It leads to more uniform quality of production.
 - It helps increase the customer satisfaction through uniform goods and services. SQC helps reduce inspection costs.
 - It assists in reducing number of rejects and saves the cost of material.
 - It promotes the understanding and appreciation of quality control.
- vi. **Value Added Analysis:** It is comprehensive evaluation of all work activities, material flows and paperwork to determine the value they add for customers.
- vii. **Just-in-time Inventory Management (JIT):** Inventories are received just-in-time to be used up by production. JIT is the management philosophy that believes in the right product at the right or just in time of demand. JIT has two understandings as 1. customers cannot wait for the long time. So, they should get the product at the time of demand, 2. over inventory of the finished goods to serve the customers can increase the material handling and inventory cost. Thus, JIT is the technique that assures the material, man and machinery at the ready position at the optimal level so that inventory cost remains low and at the same time product delivery becomes effective. Just in Time (JIT) is pioneered by Toyota in the 1950s and adopted successfully all over the world. The philosophy demands eliminating waste and improving product quality to maximize returns on investment. The basic JIT principles are to make only what is needed, when needed, and in the amount needed.
- viii. **Speed:** Speed is the time needed to get the activities accomplished. In other words, Speed is the time required to the organization to get something accomplished. It can be emphasized in any area including developing, producing, promoting and distributing goods and services. The organization can get competitive advantages over others only when it is better, smarter, and faster than their competitors. Thus, speed of operation is one of the major aspects in managing total quality of products. TQM increases speed for developing, making and distributing products. Speed become a part of organization culture.
- ix. **Training:** employees are provided continuous training in quality matters. Quality circles serve as training grounds for TQM.
- II. **Explain the concept of change in organization. Why do employees resist change in organization? And suggest various measures to overcome such resistivity to change.**

Ans: **Concept of Change:**

- Change is a shift from one state to another state.
- Organization change is any substantive modification to some part of the organization.
- Organization change includes changing organization's Structure, technology and process, and business model to gain competitive advantage.

Change is inevitable. In order to be successful in the rapidly changing competitive world and to adapt to the environmental changes, organizations plan change.

Nature (or Features) of Organizational Change:

- a. Shift from one state to another
- b. Change is continuous
- c. Change is inevitable
- d. Reasons for success and failure of business

Forces of Change: Internal and External

Resistance to Change:

a. Individual Sources of resistance

- i. **Habit:** Learning new skills and knowledge for new job, new technology or the new reporting system become source of resistance to change. Because of habit of doing easy jobs, individual resists organizational change.
- ii. **Security:** Employees feel threatened of physical security, social security and job security with changes in the organization.
- iii. **Economic Factors:** People think that if they cannot perform well in the changing technology their pay may be low. So they resist the change.
- iv. **Fear of Unknown:**
- v. **Selective information processing:**

b. Organizational Sources of Resistance:

- i. **Structural Inertia:** Organizations have built-in mechanisms to produce stability. Formal policies, rules, procedures and job descriptions favor stability. Change is resisted because of this structural inertia.
- ii. **Group Inertia:** Groups in organizations have norms that promote group conformity. Change threatens group norms and conformity. This group inertia causes resistance to change.
- iii. **Limited Focus:** Organizations consist of interdependent departments. Changes in one department affect other departments. Change in one department may be resisted by other departments.
- iv. **Threat to Resource Allocation:** If change threatens established practices of resources allocation, it may be resisted by departments getting lesser resources.
- v. **Threat to power relationship:** If change threatens long-established power relationships within organization. It may be resisted. Participative decision making and self-managed teams are often seen as threats by supervisors.
- vi. **Conflicts:** Interdepartmental conflicts can also resist change.
- vii. **Organizational culture:** It refers to a system of shared meaning held by the members of an organization. Change may be resisted if organizational culture is not conducive to change.

Overcoming Resistance to change:

- a. **Communicating Change:** Effective Communication clears misunderstanding and provides logic for change. Resistance to change may subside when the need for change is properly understood.
- b. **Participation:** Employee involvement in planning and implementation of change reduces resistance to change.
- c. **Facilitation and support:** Supportive efforts and facilitation through counseling, therapy, new skills training reduce resistance to change.
- d. **Negotiation:** Negotiations with powerful resisters of change through reward package can reduce resistance to change.
- e. **Co-Option:** The leaders resisting change are given a key role in change decision through membership in the committee planning change.
- f. **Coercion:** Force is applied to overcome resistance to change. Job transfer, demotion are used.
- g. **Rewarding Constructive behaviour**
- h. **Implementing change fairly**
- i. **Selecting people who accept change**

3. "Planning is the blueprint about the future course of action" Explain this statement. Also explain about the Tools for planning.
4. What is organizing? Explain the process of organizing.
5. Explain the concept and qualities of leadership.
6. What is communication? State and explain characteristics of communication.
7. State the various features of controlling.
8. Explain the concept of technology. Why technology is useful for the business?

Group C

[2x10=20]

Attempts any TWO questions.

9. Explain the emerging concepts of management regarding workforce diversity, knowledge management, outsourcing and learning organizations.
10. What is business environment? Explain the impacts of business environment on operating the business.
11. "Centralization and Decentralization is opposite to each other." Do you agree with this statement. Also mention the factor that are responsible for Decentralization.

MODEL SET 2

Candidates are required to give their answers in their own words as far as practicable.

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:
 - i. Who is the father of Scientific Management?
 - a) Henery Fayol b) Peter Drucker c) Dr Alfred Marshall d) FW Taylor
 - ii. To manage is to forecast and plan, to organize, to command, to co-ordinate and to control. This definition was given by _____.
 - a) Peter Drucker b) Henry Fayol c) Louis Allan d) Terry
 - iii. Father of Administrative management _____.
 - a) Mary Parkett. b) Lillian Gilbert. c) Henry Fayol. d) Elton Mayo.
 - iv. Espirit de corps means _____.
 - a) union is strength b) service is our motto.
 - c) buyer beware d) product is our strength.
 - v. Every subordinate should receive orders from and be accountable to only one superior is _____.
 - a) Unity of direction b) Unity of command
 - c) Centralization d) Scalar chain.
 - vi. The chain of command from the highest authority to the lowest level in the organization is _____.
 - a) Unity of direction b) Unity of command.
 - c) Centralization d) Scalar chain
 - vii. Study of the movements of both the workers and the machine to eliminate wasteful _____ movement is _____.
 - a) fatigue study b) time study
 - c) motion study d) work-study

- viii. The assumptions about future derived from forecasting and used in planning are known as _____
- a) planning premises b) freehold premises
c) business premises d) corporate premises
- ix. A chronological sequence of steps to be undertaken to enforce a policy and to attain an objective is known as _____
- a) objective b) strategies c) procedures d) rules
- x. _____ is the harmonizing or synchronizing of individual efforts with the purpose of achieving group goals.
- a) coordination b) control
c) decision making d) delegation

Group B

Attempts any SIX question.

6×5=30

2. What are managerial roles? Explain.
3. What are the planning Methods? Explain
4. What is strategic planning. What are the steps of strategic planning?
5. What is SWOT Stands for? Explain in detail about SWOT analysis. Explain its importance.
6. Explain the transactional and Transformational leadership.
7. Explain the approaches to technology and organization.
8. One best way is recommended by FW Taylor in Scientific Management theory. What is one best way and how can be one best way can be achieved?

Group C

Attempts any TWO questions.

[2×10=20]

9. Explain Theory X and Y. Also explain the relevance of this theory in current scenario of the business organization.
10. Organizing focuses attention on the structure and process of allocation of jobs so that common objectives can be achieved. In the light of the above statement, explain the principles of organization that would help in achieving organizational objectives.
11. What is communication? Explain the various steps of communication.

MODEL SET 3

Candidates are required to give their answers in their own words as far as practicable.

Attempt all the questions

[10 ×1=10]

1. Circle (O) the correct answer in the following questions:

Management is an organ, organs can be described and defined only through their functions. This definition was given by _____

- a) Peter F. Drucker. b) Terry
c) Louis Allan d) Henry Fayol.

- ii. Management is what a manager does _____
 a) Peter F. Drucker
 b) Terry
 c) Louis Allan
 d) Henry Fayol
- iii. Allotment of work to each worker on the basis of the capacity of an average worker functioning in the normal working condition is _____
 a) social task planning
 b) scientific task planning
 c) not a planning
 d) scientific organizing
- iv. Planning lays down the overall objective, strategies and policies for the total enterprise is _____
 a) corporate planning
 b) divisions planning
 c) unit planning
 d) department planning.
- v. The process whereby a manager shares his work and authority with his subordinates is _____
 a) Decentralisation
 b) Responsibility
 c) Delegation
 d) decision making
- vi. Formal authority flows from upwards to downwards in _____
 a) formal organization.
 b) informal organization.
 c) business or organization
 d) strategic organization.
- vii. Leadership behaviour is influenced by certain qualities of a person is ____
 a) Followers theory
 b) Trait theory.
 c) Situational theory
 d) Managerial grid.
- viii. All decision-making power is centralized in the leader is under ____
 a) autocratic style
 b) liberal leader
 c) democratic leader
 d) institutional leader
- ix. Control exercised while the activity is in progress is _____
 a) concurrent control
 b) feedforward control
 c) feedback control
 d) preserving control.
- x. The Lewin's Process consist of.....
 a) Unfreezing
 b) Moving stage
 c) Refreezing stage
 d) all of Above

Group B

Attempts any SIX question.

6×5=30

2. How the organization and Management is interlinked? Explain.
3. What are the decision making conditions or situations?
4. What are the emerging concept in organizing structure?
5. What is managerial ethics. Why it is required for long term success of the business?
6. Define scientific management. Explain the major contribution made by F.W. Taylor.
7. What is social networking. Explain the uses of social networking for making business successful.
8. Write about the communication structure.

Group C

Attempts any TWO questions.

9. What is Decision theory? Explain its contributions and limitations.

[2×10=20]

10. An organization's structure is a means to help management achieve its objectives. If you are appointed as a manager of a construction company, what form of organizational structure and would you recommend? Explain with reasons.
11. What is authority? Explain the line and staff authority.



MODEL SET 4

Candidates are required to give their answers in their own words as far as practicable.

Attempt all the questions

[10 x 1 = 10]

1. Circle (O) the correct answer in the following questions:
 - i. To manage is to forecast and plan, to organize, to compound, to co-ordinate and to control. This definition was given by _____.
 - a) Peter F. Drucker.
 - b) Henry Fayol
 - c) Louis Allan
 - d) Terry
 - ii. Luther Gullik classifies the function of management as _____.
 - a) POSDCORB
 - b) POSTCARD
 - c) POSDCORB
 - d) POSDORBC
 - iii. Father of Administrative management _____.
 - a) Mary Parkett
 - b) Lillian Gilbert
 - c) Henry Fayol
 - d) Elton Mayo
 - iv. Pioneer of Human Relations is _____.
 - a) Mary Parkett
 - b) Lillian Gilbert
 - c) Henry Fayol
 - d) Elton Mayo
 - v. The first and foremost function of management is _____.
 - a) planning
 - b) organizing
 - c) controlling
 - d) coordination
 - vi. Strategic planning is _____.
 - a) long term planning
 - b) medium term planning.
 - c) short term planning
 - d) annual planning.
 - vii. The selection of best alternative from many alternatives is known as _____.
 - a) selection
 - b) decision-making
 - c) organizing
 - d) budgeting
 - viii. The decision which is taken within the purview of the policy of the organization is _____.
 - a) programmed decision
 - b) non-programmed decision
 - c) major decision
 - d) minor decision
 - ix. _____ deals with appointing people and placing them at the appropriate jobs.
 - a) Human resources
 - b) Recruitment.
 - c) Staffing
 - d) Placement.
 - x. Management is the art and science of decision making and leadership _____.
 - a) Harold Koontz.
 - b) Donald J. Clough.
 - c) Louis Allan
 - d) Terry.

Group B

Attempts any SIX question.

6×5=30

2. What is management? Explain the functions of Management.
3. Explain the 14 principles of management given by Henry Fayol.
4. Explain the different approaches of organizing.
5. What is motivation? What are the importances of motivation?
6. Explain the concept of organizational development. State and explain different characteristics of OD.
7. Explain the uses of Technology in people management.

Group C

Attempts any TWO questions.

[2×10=20]

8. Explain about the changing perspective of organization.
9. What is behavioural perspective of management. Explain with different studies or theories.
10. Control is a management function that focuses on the process of monitoring activities to ensure that they are being accomplished as planned. As a manager of an organization, what types of control system would you recommend to achieve planned result?

MODEL SET 5

Candidates are required to give their answers in their own words as far as practicable.

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:

- i. Henry Fayol laid down _____
a) 12 principles b) 13 principles c) 14 principles d) 15 principles
- ii. Control system of an organization has no influence over the _____
a) internal environment b) external environment
c) customers d) government
- iii. The chain of command from the highest authority to the lowest level in the organization is _____.
a) Unity of direction b) Unity of command.
c) Centralization d) Scalar chain.
- iv. A study relating to the fixing of the working hours with rest periods to recoup the energy while performing in a job is called _____.
a) fatigue study b) time study c) motion study d) work-study
- v. Goals, aims, purposes, missions and target is also known as _____.
a) objective b) strategies c) policies d) rules
- vi. Contingent plan to meet environmental pressures is _____.
a) objective b) strategies c) policies d) rules
- vii. Finding ways to reduce is a key responsibility of management.
a) Dissatisfaction b) Uncertainty
c) Stress d) None of the above
- viii. The and control systems should be altered to support the strategic human resource function.

- a) Appointment
- c) Job allotment

- b) Reward
- d) None of the above

10. In which step of the planning process pros and cons of each alternative is examined.
- a) Developing Premises
 - b) Setting up Objective
 - c) Evaluating alternatives course of action
 - d) Selecting an alternative premises
11. Which of the following is not a process of controlling?
- a) Analyzing deviations
 - b) Integrates employees efforts
 - c) Taking corrective measures
 - d) Setting performance standards

Group B

Attempts any SIX question.

6×5=30

2. What is management science theory? Explain the contribution and limitations of management science theory.
3. What is multidivisional structure and geographic structure?
4. How can you motivate the employees if you are the chief executive officer of a business organization. Explain with the help of techniques of motivation.
5. State the advantages and limitations of vertical communication.
6. What is organizational change? Explain the various characteristics of organizational change.
7. Explain the concept of organizational intervention in organization development.
8. Define Management? What Kinds of skills required for successful manager?

Group C

Attempts any TWO questions.

[2×10=20]

9. State and explain the emerging challenges of management
10. What is organizational architecture? Explain in details.
11. Decision making is very important managerial function. With the statement; discuss the meaning and process of decision making. Also explain the types of decision making.

MODEL SET 6

Candidates are required to give their answers in their own words as far as practicable.

Attempt all the questions

[10 × 1 = 10]

1. Circle (O) the correct answer in the following questions:

i. Management is _____

- a) an art
- b) a science
- c) an art and a science
- d) an art not science

ii. The Book "General and Industrial Management" was written by _____

- iii. Every subordinate should receive orders from and be accountable to only one superior is _____.
- a) Harold Koontz b) Terry c) Louis Allan d) Henry Fayol
- iv. A statement of expected results expressed in numerical terms for a definite period of time in the future is known as _____.
- a) Unity of direction b) Unity of command
c) Centralization d) Scalar chain
- v. What is the full form of PERT:
- a) Performance evaluation report technique
b) Personnel enquiry retrieving technique
c) Programme evaluation and review technique
d) Programme Evaluation and Renew Technique
- vi. Assembling of facilities and personnel for collecting, processing, information that is required by managers is called
- a) PERT b) CPM
c) Ratio analysis d) Management information Technique
- vii. Which of the following is not a part of the organising process?
- a) Assignment of responsibilities
b) Establishing reporting relationship
c) Specialisation
d) Division of work
- viii. What make you think is the most effective leadership style that can be used during emergency situations?
- a) Democratic b) Laissez- faire c) Autocratic d) Supportive
- ix. What are the qualities that define a leader who uses laissez- faire?
- a) The group in planning and in decision making
b) Passive and puts the responsibility of decision-making to others.
c) Foster independence in team by promoting motivation and creativity.
d) Provide little autonomy and self- motivation.
- x. Which of the following statements does not highlight the relationship between planning and controlling?
- a) Planning and controlling are separable twins of management.
b) Planning without controlling is meaningless, controlling without planning is blind.
c) Planning is prescriptive, controlling is evaluative.
d) Planning and controlling are interrelated and interdependent.

Group B

Attempts any SIX question.

6×5=30

2. List the different types of organizations structure. Explain the Team structure in details.
3. What is devolution of authority? What are the advantages and disadvantages of devolution?
4. What is two factor theory of motivation. Explain in details
5. State the purpose of formal communication.
6. Explain the principles of Total Quality Management.

7. Explain the paradigm shift in structure, technology and people in the organization.
8. What is leadership? Explain the different style of leadership.

Group C*

Attempts any TWO questions.

[2x10=20]

9. State and explain the emerging challenges of management
10. Show your acquaintance with the system theory of management with its limitations.
11. Staffing function is one of the key parts of management system that ensures acquiring and maintaining a work force capable of fulfilling organizational goals. In view of this statement. Discuss the objectives and importance of staffing functions.

MODEL SET 7

Candidates are required to give their answers in their own words as far as practicable.

Attempt all the questions

[10 x1=10]

1. Circle (O) the correct answer in the following questions:

- i. The process of establishing a time sequence for the work is known as _____.
a) objective b) schedules c) procedures d) budgets
- ii. A rigid plan, no scope for discretion _____.
a) rule b) schedules c) procedures d) budgets
- iii. Specifying the manner of executing policy is known as _____.
a) objective b) schedules c) procedures d) budgets
- iv. _____ is the harmonising or synchronising of individual efforts with the purpose of achieving group goals.
a) coordination b) control c) decision making d) delegation
- v. _____ is an obligation to perform certain functions and achieve certain results.
a) responsibility b) decentralization c) centralisation d) delegation
- vi. Making assumptions for the future is called _____.
a) Making policy b) All of the options are correct
c) Setting planning premises d) Making derivative plans
- vii. The composite plan which involves setting up a long-term objective, finding a course of action, and allocation of resources is called _____.
a) Programme b) Strategy c) Procedure d) Policy
- viii. Organising doesn't include _____.
a) By whom will be done b) When will be done
c) What will be done d) How will be done
- ix. Who reports to whom is made clear by _____.
a) Organising process b) Management process
c) Planning process d) None of the above

- x. This is a joint trait of a leader which is defined as his ability to possess honesty, responsibility and maturity in the working area.
 a) Integrity b) Personality c) Intelligence d) Flexibility

Group B**Attempts any SIX question.****6×5=30**

2. What is bureaucracy Theory? Explain the contribution and limitations of Bureaucracy Theory
3. What is authority and responsibility. Give the distinctions between authority and responsibility
4. What is need hierarchy theory? Explain in details.
5. What is control? Explain the importance of control.
6. What is motivation? What are the importance of motivation?
7. Justify the importance of communication in organization.
8. What are the decision making conditions or situations?

Group C**Attempts any TWO questions.****[2×10=20]**

9. Define management. Describe the characteristics of Management
10. Matrix structure is the combination of functional and project structure. In the light of this statement. Explain the matrix structure in details. For what types of business organization need to use matrix structure for organizational structure.
11. Control is a process of monitoring activities to ensure that they are being accomplished as planned and correcting any significant system. In the light of the above statement, discuss the process and problems of control system.

MODEL SET 8

Candidates are required to give their answers in their own words as far as practicable.

Attempt all the questions**[10×1=10]**

1. **Circle (O) the correct answer in the following questions:**
 - i. Set of clear instruction in a clear and logical sequence to perform a particular task _____.
 a) rules b) programme c) policy d) methods
 - ii. Employees will be promoted on the basis of seniority is an example of _____.
 a) rules b) procedures c) policy d) methods
- ANSWER: C
- iii. Steps to be taken for selecting salespersons is an example of _____.
 a) rules b) procedures c) policy d) methods
- iv. An identified group of people contributing their efforts towards the attainment of goals is called an _____.
 a) organization b) business c) management d) department

Rules, duties and responsibilities of workers are given in writing in _____.

- | | |
|------------------------------|---------------------------|
| a) formal organization | b) informal organization. |
| c) business or organisation. | d) strategic organization |
- v. Which organisational relationship gives a greater job satisfaction and results in maximum production?
- | | |
|-----------------------------|---------------------------|
| a) formal organization | b) informal organization |
| c) business or organization | d) strategic organization |
- vi. The decision deal with novel and non-repetitive problems is _____.
- | | |
|------------------------|----------------------------|
| a) programmed decision | b) non-programmed decision |
| c) individual decision | d) non-economic decision |
- vii. The decision taken by high level of management is _____.
- | | |
|-------------------------|----------------------------|
| a) programmed decision | b) non-programmed decision |
| c) individual decision. | d) non-economic decision. |
- viii. The decision which relates to day-to-day operation of an organization is known as _____.
- | | |
|----------------------|----------------------------|
| a) major decision | b) organisational decision |
| c) personal decision | d) operative decision |
- ix. Controlling function brings the management cycle back to the
- | | |
|------------------------|-----------------------|
| a) Organising function | b) Directing function |
| c) Planning function | d) None of the above |

Group B

Attempts any SIX question.

6×5=30

2. What is administrative management Theory? Explain the contribution and limitations of Administrative management Theory.
3. What is functional structure? Explain in details:
4. Explain the concept and importance of Quality
5. What are managerial roles? Explain.
6. What are the Methods of Planning? Explain
7. Explain the concept and qualities of leadership
8. What is communication? State and explain characteristics of communication.

Group C

Attempts any TWO questions.

[2×10=20]

9. Explain the concept of manager as a 'nerve Centre'. Why this role is important to organizational performance?
10. Define motivation. State, explain and compare Maslow's need hierarchy theory of motivation and Herzberg's two factor theory.
11. Explain the concept of quality. What are the emerging issues in Quality management?